



Service

Generated on: 2026-07-01 15:24:57 GMT+0000

SAP S/4HANA | 2025 FPS01 (Feb 2026)

Public

Original content: https://help.sap.com/docs/SAP_S4HANA_ON-PREMISE/c9b5e9de6e674fb99fff88d72c352291?locale=en-US&state=PRODUCTION&version=2025.001

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Service Orders

A service order is a short-term agreement between a service provider and a service recipient, in which the service recipient orders one-off services from the service provider.

Use

You can use a service order to plan and execute services for customers. A service order contains all the information needed to plan, execute, and bill for a service such as customer details, material or service to be delivered, dates and times for service activities, and any related documents or attachments. Service orders help in the planning and scheduling of resources, monitoring service progress, and generating invoices for completed services.

Use of Items in Service Orders

A service order consists of a header and one or more items. You use items in a service order to reflect the purposes or needs of a service and to create account assignment objects so that cost and revenue from the service order can be posted to Financial Accounting. You can add items to a service order and edit them on the **Items** tab. For more information on different types of items and how to use items in service orders, see [Items in Service Orders](#).

Basic Settings

The standard transaction type of a service order is predefined in the system as SRV0. Transaction types define the characteristics and attributes of a business transaction and determine how the transaction is processed, for example, the profiles and procedures used to determine texts, business partners, statuses, and organizational data. If the standard settings do not meet your business requirements, you can configure the transaction types using the Customizing activities under **Service > Transactions > Basic Settings**. For more information, see [Business Transaction](#).

Life Cycle Statuses

A service order and each item in a service order can have the following life cycle statuses:

Life Cycle Status	Description
Open	This is the default status that all service orders or service order items have after they're created. It indicates that you're free to edit them.
In Process	You can manually set the status of a service order or a service order item to In Process to indicate that a service is taking place. For example, an executing service employee has been scheduled to visit a customer.
Released	This status indicates that you have finished planning a service and the service execution can be started. When this status is set, a service order or a service order item has been finalized and no further changes are required. It's possible to create a service confirmation against the service order that has the status Released .
Completed	This status indicates that a service has been executed. You can no longer create service confirmations based on a service order that has the status Completed .

i Note

- If a service order item is set to the status **Released** or **Completed**, it's not possible to perform the following actions:
 - Change the item category of the item.
 - Change the service contract that has been assigned to the item.

Key Features

Reference Objects

You can use reference objects in a service order at header or item level to simplify the process of entering certain data for a service order or an item. The details that has been predefined for the reference objects can be transferred from the master data to the service order or service order item. For more information, see [Reference Objects for Service Orders](#).

Credit Checks in Service Orders

You can perform credit checks in service orders. For more information, see [Credit Checks in Service Orders](#).

Multilevel Categorization

You can use the function **Multilevel Categorization** in service orders. For example, you can create a categorization for a reported defect. For more information, see [Multilevel Categorization](#).

You can also use **Auto Suggest**, which is part of the functionality of multilevel categorization, for service orders. You can create a category in a service order and assign a service order template to the category. Next time when you create a service order, the system can automatically fill the service order with the data from the service order template. For more information, see [Auto Suggest](#).

Knowledge Articles

You can obtain information from knowledge articles to solve issues related to service orders. For more information, see [Knowledge Articles](#).

Questionnaires

You can display a list of relevant questionnaires and assign questionnaires to the order depending on the transaction type of the service order.

Billing

You can release service orders for billing. After you've released the service order for billing, the system creates the required billing document requests (BDRs). For more information on billing, see [Billing](#).

You can manually schedule jobs to create billing document requests for billable service order items, billable service confirmation items, and items with ad hoc billing plan using the report **Generate BDRs for Service Orders** (CRMS4_SERVORD_BDR_GENERATE). You can find the report on the **SAP Easy Access** menu under **►Service ►Service Processes ►**.

Ad Hoc Billing for Service Orders

You can use ad hoc billing to define billing values and dates freely for fixed price service order items. For more information, see [Ad Hoc Billing for Service Orders](#).

Pricing

The system determines price information when you create or edit a service order. For more information, see [Pricing](#).

Accounting Indicator

You can use an accounting indicator in service orders with the pricing condition types DAI1 and DAI2 to influence pricing procedure and differentiate items based on discount or surcharge rates.

To set the condition types for accounting indicators, you use the activity [Set Condition Types for Pricing in Sales](#) in Customizing for [Sales and Distribution](#) under [Basic Functions](#) [Pricing](#) [Pricing Control](#) [Define Condition Types](#). To enable the transfer of the value of accounting indicators from billing to accounting, you also need to make sure that the accounting indicator is taken into account in account assignment by entering **B** in the [Rel.Acc.Assig](#) field in the [Control Data 2](#) section in the same Customizing activity [Define Condition Types](#).

Trade Compliance Check

The integration with SAP Global Trade Services (GTS) allows you to perform embargo checks, sanctioned party list screening, and legal controls for service orders and service quotations. For more information, see [Integration with SAP Global Trade Services](#).

Release of Service Orders

You can set a service order and each item in the service order to the status [Released](#) when you edit the service order. You can also use the [Release](#) button on service order header to release a service order along with all the items in the service order, provided that the service order and the items don't have any errors.

i Note

The [Release](#) button is only active when the service order is in display mode.

Rejection of Service Orders

You can reject a service order by choosing [Reject All Items](#). You can also reject single items in a service order by assigning a rejection reason to each item. You can revert the rejection when you remove the rejection reason of single items or choose [Reset Item Rejection](#) to remove the rejection for all rejected items.

You can define your own rejection reasons, group them in a rejection profile, and then assign the rejection profile to a transaction type and a sales organization for service orders. With this, you can select from the predefined reasons in a profile when you reject a service order or a service order item. You can do so with the activities in Customizing under [Service](#) [Transactions](#) [Settings for Service Transactions](#) [Settings for Rejection Reasons for Service Transactions](#).

i Note

If a service quotation has been rejected, you cannot create a follow-up service order for the service quotation. If you create a follow-up service order for a service quotation, the rejected items in the service quotation are not copied to the service order. For more information, see [Rejecting Service Quotations](#).

Business Completion of Service Orders

You can set a service order to [Business Completed](#) to indicate that no cost and revenue can be posted from the service order to Financial Accounting anymore. The [Business Completed](#) status is a prerequisite for archiving service orders.

After you have set a service order to [Business Completed](#), you can still manually set the service order to [Business Reopened](#) to revise the business completion of the service order. You might reopen the service order for business, for example, if further costs

that are related to the service order have been occurred subsequently and have to be posted to Financial Accounting and Controlling. For more information, see [Finalizing the Service Order](#).

i Note

You can only set a service order to **Business Completed** or **Business Reopened** when item-based accounting is active.

Qualification Requirements

You can define qualification requirements, such as skills, knowledge, or abilities, necessary to perform services specified in a service transaction. For more information, see [Qualifications in Service Transactions](#).

Involved Apps

[Manage Service Orders](#)

Items in Service Orders

You can learn about items in service orders in this topic.

Use

A service order consists of a header and one or more items. You use items in a service order to reflect the purposes of a service and to create account assignment objects so that cost and revenue from the service order can be posted to Financial Accounting.

The following table provides you with an overview of what types of items you can create in a service order:

Item Type	Description
Service Item/Service Product Item	You can use a service item to document customer wishes about how the service is to be performed, or to enter the planning-relevant data such as the service teams, estimated duration of performance, and dates.
External Service Item	You can use an external service item to provide details of a procured service performed by an external workforce.
Service Part Item	A spare part that is required to perform a service. It can be a spare part that is available in stock and can be picked up by a technician, or one that is not in stock and needs to be procured.
Expense Item	You can use an expense item to specify expenses that may be incurred during a service process, for example, travel costs of an executing service employee.
Sales Item	You can use a sales item to trigger the creation of a follow-up sales order.
Intercompany Service Item	You can use an intercompany service item to trigger the creation of an intercompany service order, which is a business transaction used to distribute a service between two companies. For more information, see Intercompany Service Orders .
Execution Order Item	You can add an execution order item to a service order. An execution order item is associated with a corresponding billable

Item Type	Description
	maintenance order. It is used to define the scope and terms of a service requirement as part of the Service with Advanced Execution process. For more information, see Service with Advanced Execution .
Price Item	You can use a price item in a service order to reflect a surcharge for additional service provided. For more information, see Price Items in Service Orders and Service Quotations .
Tool Item	An item within a service order that represents a tool needed during the service execution to fulfill the service and that can be reused for the next service. For more information, see Tool Items in Service Transactions .

Item Categories

When you add an item to a service order, you assign an item category to the item. An item category specifies the properties and attributes of each item. You use item categories to determine how items are processed, for example, the manner in which the items are billed.

The following table gives you an overview of which item categories you can use in a service order:

Item Type	Item Category
Service Item/Service Product Item	• SRVP: Service Item • SF0S: Fixed Price Service Item • SB0S: Service Item (Ad Hoc Billing)
External Service Item	SRVW
Service Part Item	• SRVM: Service Part Item • SF0M: Fixed Price Service Part Item • SB0M: Service Part (Ad Hoc Billing) • SPRS: Stock Transfer Service Part • SRVG: Service Part Manual Goods Issue • SRVU: Advance Shipment of Procured Service Parts • SRVV: Advance Shipment of Stock Service Parts
Expense Item	• SRVE: Expense Item • SF0E: Fixed Price Expense Item • SB0E: Expense Item (Ad Hoc Billing)
Sales Item	SRVS
Intercompany Service Item	• SVAI: Intercompany Service Item (Ad Hoc Billing)

Item Type	Item Category
	SRVI: Intercompany Service Item (Fixed Price)
Execution Order Item	MSI1
Price Item	SVPI
Tool Item	- SRVT - SB0T: Service Order Tool Item (Ad Hoc Billing) - SF0T: Fixed Price Tool Item (Order)

i Note

You can only change the item category of a service order item when the following conditions are met:

- The item has the status [Open](#) or [In Process](#).
- You have not created any follow-up documents such as billing documents for the item.

Settings

The item categories described in the overview table are predefined in the system.

If the standard settings do not meet your business requirements, you can also configure your own item categories using the Customizing activities under [Service > Transactions > Basic Settings](#).

Products and Item Category Groups

Products are used in service order items to represent which type of items you sell to customers. Item category groups are attributes in product master data that are used to control how item categories are determined in service orders. An item category group contains different item categories. You can assign an item category group to a product in the product master data. This way the item categories in the item category group can be determined for the product in a service order item.

You can use the following item category groups in products:

Item Category Group	Description
SRVP	Service Product
SRVE	Expense Item
SRVM	Service Material
NORM	Standard Item
DIEN	Execution Order Item (Service with Delivery)
LEIS	Execution Order Item (Service without Delivery)
SRVT	Service Tool

→ Remember

Make sure that the product that you enter for a service order item has the correct item category group assigned in the product master data. For example, if you want to use the product **Service Product** in an item, which represents the service item, you should assign the item category group SRVP (service product) to the product.

Key Features

You can use the following functions at service order item level:

Copy from Template

You can add predefined items to service orders by choosing **Copy from Template** when you edit items. You can create and display templates for service order items. Service order templates contain reusable sets of service-related data. They do not contain any data that describes the actual execution. When you copy the templates to service orders, only templates that have the status **Released** are displayed. For more information, see [Service Order Templates](#).

Product Proposals for Service Order Items

You can use the function Product Proposal in the items list of a service order to identify the service parts, expenses, and services relevant for a service order without entering the service part IDs. For more information, see [Product Proposals for Service Order Items](#).

Configurable Products

You can use configurable products or product variants to customize complex products with single-level variant configuration for your service processes. For more information, see [Configurable Products and Product Variants](#).

Related Information

[Service Orders](#)

Service Orders with Other Service Transactions

You can link service orders to other service transactions such as service order templates, service quotations, and service confirmations with the **Create Follow-Up** function. This allows for efficient creation of service transactions and provides a history of related transactions.

Use

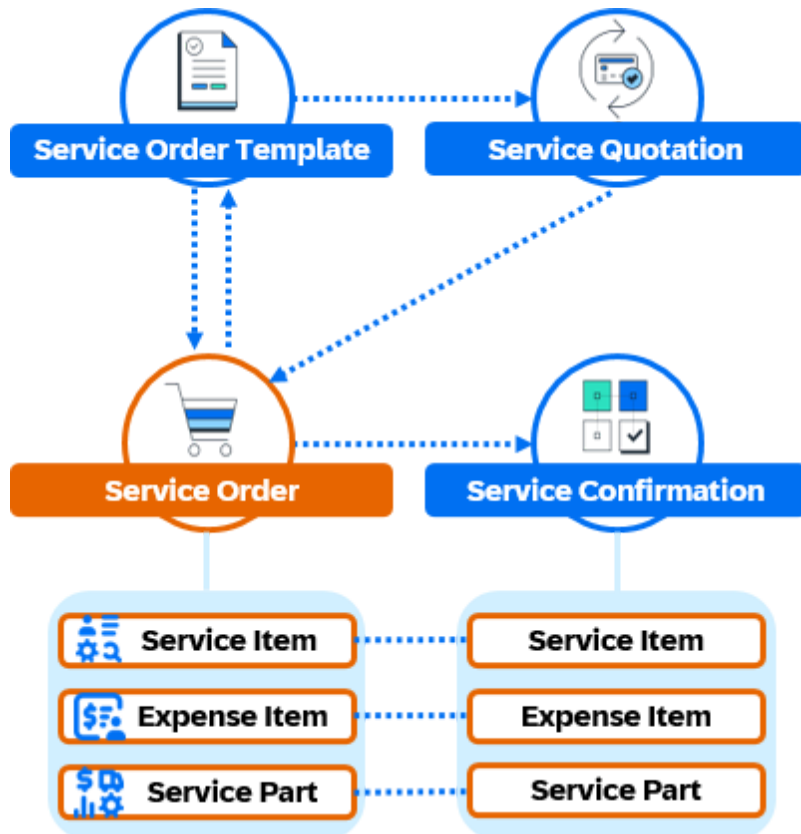
Depending on your Customizing settings, you can create a service order as the follow-up transaction of another service transaction. For example, if you create a service order as a follow-up transaction of a service order template, you can select items from the template and copy them to the service order.

You can also create other service transactions as the follow-up transactions of a service order. When you choose **Create Follow-Up** and select a follow-up transaction that you want to create for a service order, the system proposes items for selection in a popup and the item you select is automatically copied to the service order.

A follow-up transaction is a business object that refers to an existing business object. When you create a new transaction that relates to an existing transaction, you can create the new transaction as a follow-up transaction. When you do this, data from the original transaction is copied to the new transaction and the transactions are therefore linked to each other. This linking provides you with information about the history of a particular transaction. You can view this history on the **Transaction History** tab in each service transaction. This is useful for managing and tracking service-related data and documents relevant for service transactions, for example, billing documents. For more information, see [Follow-Up Transactions](#).

Example

The following diagram illustrates an example of how you can use a service order with other service transactions.



Please note that image maps are not interactive in PDF outputs.

Reference Objects for Service Orders

Use

When you create or edit a service order, you can specify a product, a functional location, or a piece of equipment as the reference object in the service order header and items.

Features

- Search for equipment and functional locations

When specifying a product, functional location or a piece of equipment as the reference object, you can search for existing products, functional locations or equipment.

Reference Object	Search Criteria
Functional location	◦ Functional location ID ◦ Functional location description ◦ Functional location category ◦ Functional location structure indicator ◦ Object type ◦ Partner function

Reference Object	Search Criteria
	- Business partner ID - Superior functional location
Equipment	- Equipment ID - Equipment description - Equipment category - Functional location structure indicator - Business partner ID - Partner function - Reference product ID - Serial number
Product	- Product ID - Product description - Product type - Product hierarchy - Configurable product - Sales organization ID - Distribution channel ID - Product type group - GTIN

i Note

Functional location, equipment, and product are mutually exclusive when you specify them as a reference object.

When you specify a functional location or equipment as the reference object and save the service order:

- The service order can be saved with errors if the equipment or functional location has been deactivated or marked for deletion.
- The service order cannot be saved if the equipment or functional location has been archived (equipment and functional locations that have already been used in service transactions cannot be archived).

• Warranty determination

If a warranty is assigned to a functional location or equipment which has been specified as the reference object in a service order, the warranty can be automatically determined in the service order.

- Main reference object

You can specify functional locations or equipment as reference objects in both the service order header and items, and these specified functional locations or equipment will be displayed in the **Reference Objects** assignment block.

In this assignment block, you can add new reference objects and specify the main reference object. In this way, you can check and view the warranty assigned to your desired product, equipment, and functional location by specifying the product, equipment, or functional location as the main reference object.

i Note

- Only the warranty assigned to the main reference object will be automatically determined in the service order header.
 - Warranty is determined on an item only when the item's service product is a part of the master warranty that is assigned to the main reference object.
 - The determined warranties cannot be changed.
- Navigate to the Fiori app to view details on equipment and functional locations

After you have defined a functional location or a piece of equipment as the reference object, the description of the functional location or equipment is automatically displayed. You can choose the description to navigate to the Fiori app to view details on the functional location or equipment.

Intercompany Service Orders

An intercompany service order is a business transaction used to distribute a service between two companies with different company codes that belong to the same organization.

Use

A company that sells a service to a customer can use an intercompany service order to distribute the service to another company. The company that receives the intercompany service order executes the service and posts the revenues from the service as costs (accounts payable) to the company that sells the service.

i Note

You can only use fixed price service items for intercompany service orders. The items that are billed based on the consumption of time and materials cannot be used for intercompany service orders.

Intercompany Item Categories

You can use the following item categories for intercompany service orders:

- [Intercompany Service Item \(Fixed Price\)](#) (SRVI)
- [Intercompany Service Item \(Ad Hoc Billing\)](#) (SVAI)
- [Intercompany Service Item Collective Accounting](#) (SRHI)

Prerequisites

To use intercompany service orders, you need to fulfill the following prerequisites:

- Activate item-based accounting in the Customizing activity [Enable Item-Based Accounting for Service](#) in Customizing for [Service](#) under [Transactions > Settings for Service Transactions > Integration](#). For more information, see [Setting Up Item-Based Accounting](#).

- Define the relevant data required for creating intercompany service orders using the Customizing activity under [Service > Transactions > Settings for Service Transactions > Intercompany Billing > Define Settings for Creation of Intercompany Service Orders](#).
- Define the business partners for sales organizations in the Customizing activity [Define Internal Customer Number By Sales Organization](#) under [Sales and Distribution > Billing > Intercompany Billing](#).
- Specify that the item category (BUS2000140) used for a fixed price service item is relevant for intercompany service orders. You can do so in the Customizing activity under [Service > Transactions > Basic Settings > Define Item Categories](#).
- Specify the receiving sales organization, which is the sales organization that receives an intercompany service order and executes a service for the sales organization that sells the service to a customer.
- Set up automatic posting to supplier account for intercompany billing with SAP Electronic Data Interchange (EDI) and set up invoice receipt via EDI. For more information on how to make these settings, see the section [Electronic Data Exchange \(EDI\) for Intercompany Service Orders](#).

Electronic Data Exchange (EDI) for Intercompany Service Orders

To send billing documents from one company code to another company code and post them to financial accounting via EDI, you need to follow the steps as follows:

1. Generate IDocs.

- If you use the output management framework, do the following: In the [Output Parameter Determination](#) app, choose **IDoc** as the channel for the output type [Billing Document](#).
- If you use the SAP S/4HANA output control (CA - GTF - OC) framework, do the following: Choose a billing type and assign the output determination procedure V40000 to the billing type. You can then create an output condition record with transaction VV31 for the output type RD04 to trigger the use of IDocs.

2. Set up automatic posting to supplier account with EDI.

To do this, perform the Customizing activity in SAP Customizing Implementation Guide under [Sales and Distribution > Billing > Intercompany Billing > Automatic Posting to Supplier Account \(SAP-EDI\)](#).

i Note

Only choose intercompany service orders as the recipients.

3. Set up invoice receipt via EDI.

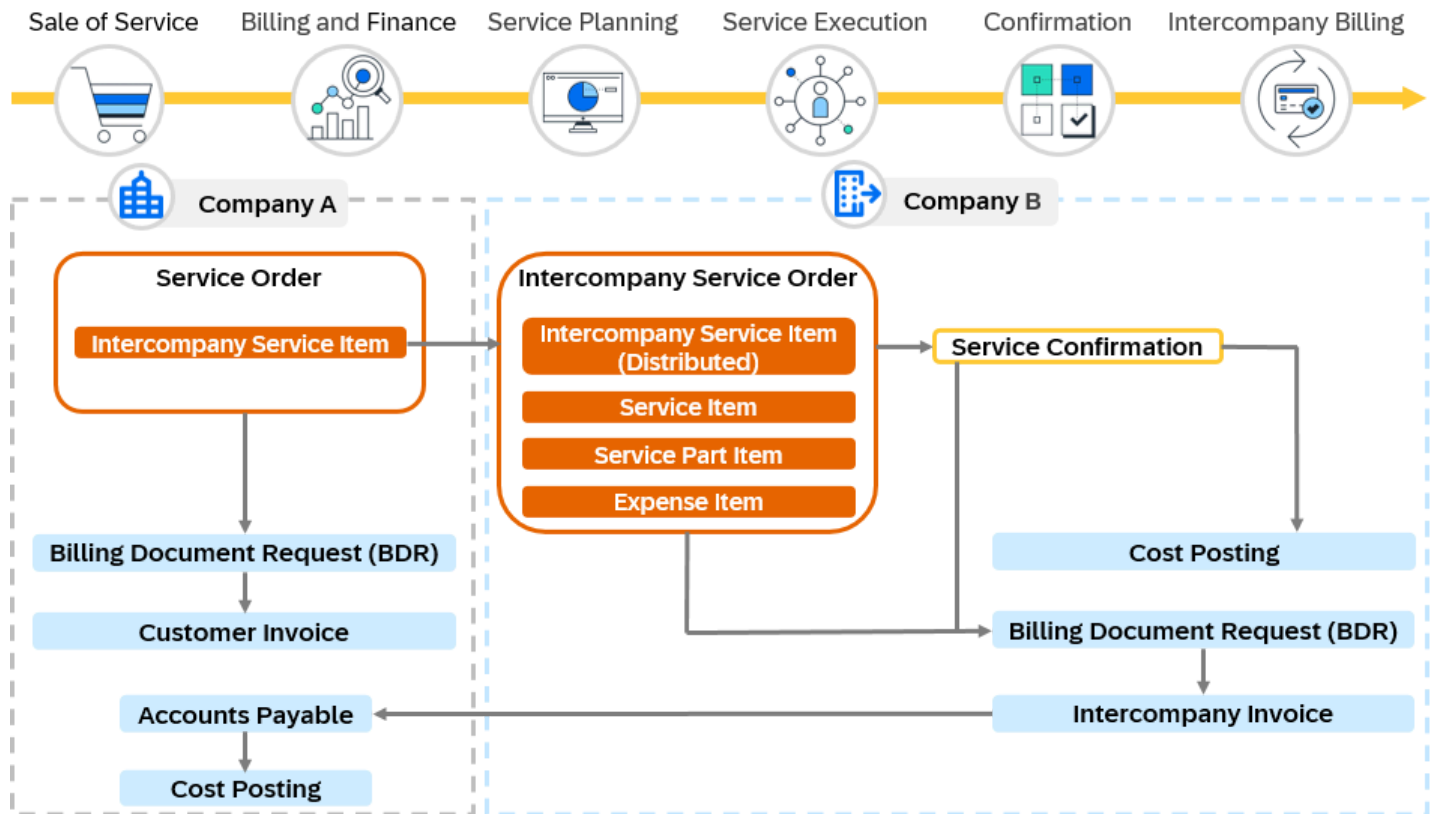
To do this, carry out the Customizing activities in the order specified under [Financial Accounting > Accounts Receivable and Accounts Payable > Business Transactions > Incoming Invoices/Credit Memos > EDI](#).

Process

The following diagram illustrates a sample process of how to use intercompany service orders.

i Note

- Company A: the company that sells and distributes a service to another company.
- Company B: the company that receives the intercompany service order and executes the service.



→ Tip

You can create more than one intercompany item in a service order. Items with the same receiving sales organizations will be created in the same intercompany service order.

Copying of Data During Intercompany Service Order Creation

In the intercompany scenarios, the transition from a service order of the company that sells and distributes the service to an intercompany service order is automated to ensure operational continuity. When a fixed price intercompany service item is released, the system creates a corresponding intercompany service order in the receiving company. During this process, all essential operational data is transferred, while sensitive or company-specific information is recalculated or redefined in the receiving organization.

The following data is **copied** from the original service order to the intercompany service order:

- Product ID and quantity including unit of measure
- Attachments (GOS) and notes
- Begin and end dates (both requested and planned)
- Reference objects (including equipment data, functional location, serial number, and the **In-House Service-Related** indicator)
- Contact person and ship-to party (including partner address data if it was changed in the transaction)
- Multi-level categorization (if profiles match)
- Configuration data for configurable service products
- Purchase order number, date, and reference field data (both for sold-to party and ship-to party)

The following data is **not copied** and is instead recalculated or newly defined in the receiving company:

- Service contract assignment

- Billing data (payment method, terms, tax type, etc.)
- Pricing data (pricing conditions are recalculated)
- Service level agreement, credit status, profit center
- Employee responsible, service employee group, executing employee
- Organizational data (redefined based on the sold-to party in the intercompany order)

Once the intercompany service order is created, key fields such as product ID and quantity become non-editable, preserving the integrity of the original request. Any changes made in the distributed item of the intercompany order do not retroactively update the source service order. The relationship between the two orders is maintained in the transaction history and process flow, providing full traceability and transparency throughout the lifecycle of the service.

This approach minimizes manual intervention and ensures that operational requirements are met during the handover of service execution.

Intercompany Reference Item Number

This is the number of the intercompany item that is used as a reference for manually entered items in an intercompany service order. You need to enter an intercompany reference item number for any items that you have entered manually in an intercompany service order.

When an intercompany service order is created, an intercompany item is added to it automatically. This intercompany item refers to the intercompany item created in the service order by the company that sells and distributes the service. When there are multiple intercompany items in the intercompany service order and you add items to the intercompany service order, you need to use **intercompany reference item number** to indicate to which intercompany item the added items belong. This ensures that the revenues from the items in an intercompany service order are posted as costs (accounts payable) to the correct account assignment objects of the intercompany items created by the company that sells and distributes the service.

i Note

If the intercompany service order contains only one intercompany item, the number of this intercompany item is entered as the intercompany reference item number automatically.

Cascade Intercompany Items

When you receive an intercompany item from a company that sells a service, you can forward this intercompany item to another company so that they can execute the service.

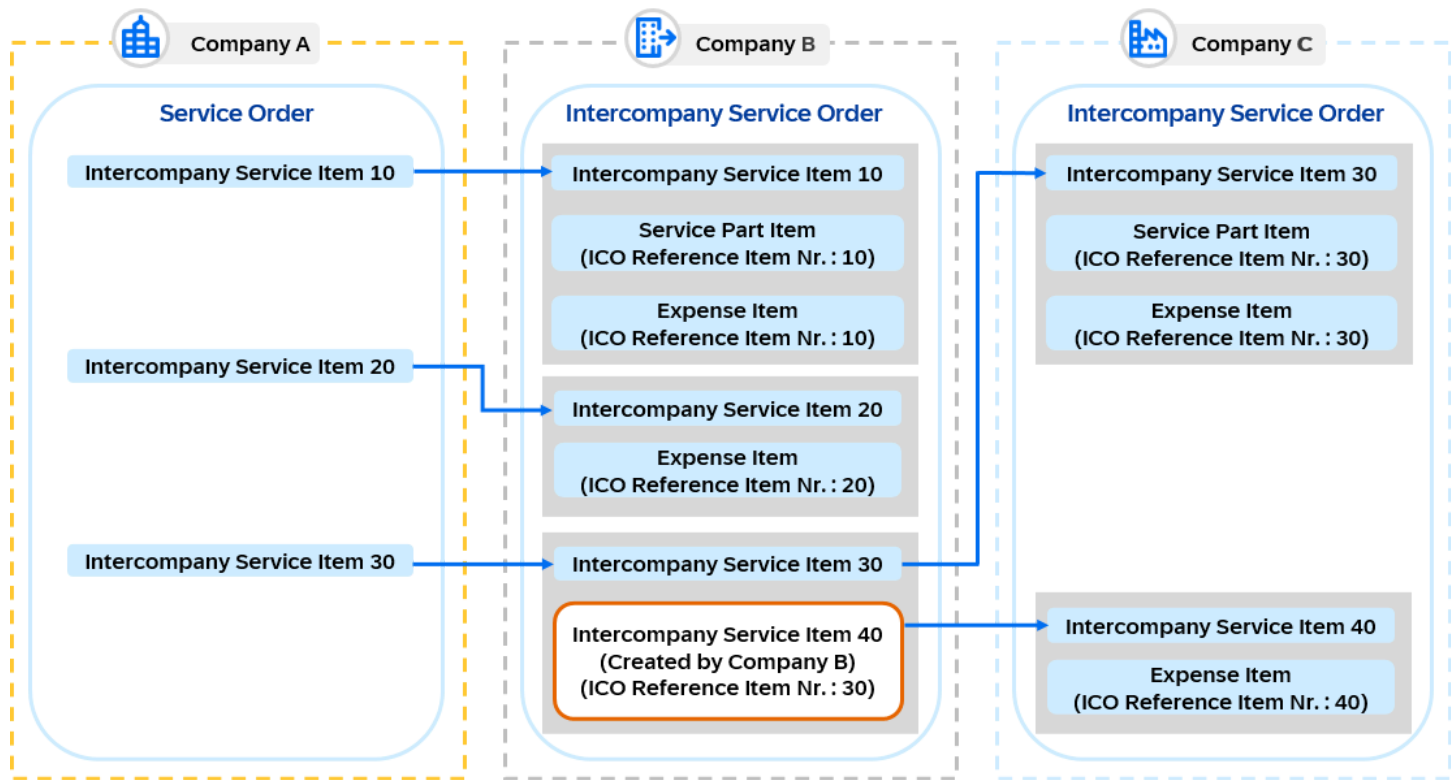
You can also add new intercompany items in an intercompany service order and distribute the new intercompany items to other companies for execution.

Example

The following diagram shows you an example of how you can cascade intercompany items and use intercompany reference item number.

i Note

- The arrows in the diagram represent the action of releasing an item.
- ICO Reference Item Nr.: Intercompany Reference Item Number



Service Hierarchy in Intercompany Service Process

You can use the concept of service hierarchy with collective accounting in intercompany service processes. The feature lets you use intercompany items as main and subitems in a service hierarchy. Costs incurred by the intercompany service order are collected under the account assignment object of the intercompany item specified as the main item of a service hierarchy. A service hierarchy can be used both in the service order of the company that sells the service and in the service order of the company that executes the service. That means that it's possible to have a service hierarchy only in the first service order, only in the second service order, or in both. Both service orders follow their respective accounting logic.

To specify an intercompany item as a main item in a service hierarchy, you use the item category **Intercompany Service Item Collective Accounting (SRHI)**.

i Note

You use SRHI in the service order of the company that sells the service. In the intercompany service order of the company that executes the service, you use **Service Item with Collective Accounting (SHSV)** and **Service Item with Collective Accounting (Fixed Price) (SHSF)**.

If you manually add items as subitems in a service hierarchy of an intercompany service order, these subitems automatically inherit the intercompany reference item number from the main item of a service hierarchy.

For more information about service hierarchy, see [Service Hierarchy with Collective Accounting](#).

Planned Costs and Revenues for Intercompany Service Order Item

Planned costs and planned revenues are now calculated for intercompany service order items in the service order. This generates more accurate financial forecasting and efficient resource allocation.

The expected cost is determined by adding together:

1. The net value of the item in the intercompany order.
2. The net value of any manually created items within the same order that reference it using the ICO reference number.

3. The net value of any sub-items within the service hierarchy of the aforementioned items in the intercompany order.

i Note

No planned costs and revenues are calculated when the item of the customer-facing service order is part of a service hierarchy (whether it is a main item or a sub-item).

Integration

Event-Based Revenue Recognition

The integration of Event-Based Revenue Recognition with intercompany service orders is supported. For more information, see [Event-Based Revenue Recognition for Service Orders](#).

Service with Advanced Execution

You can add an execution order item to an intercompany service order, release the execution order item, and trigger the creation of a maintenance order. In this scenario, the debit memos from the maintenance order are transferred from the intercompany service order to the customer-facing service order. You can view the debit memos in the transaction history of the service order and navigate to the debit memo details as well. For more information, see [Service with Advanced Execution](#).

Product Proposals for Service Order Items

You can use the function Product Proposal in the items list of a service order to identify the service parts, expenses, and services relevant for a service order without entering the service part IDs.

Use

When you choose this function, the system searches for the product or the equipment that was created as main reference object, and checks whether it has a BOM (Bill of Material) assigned or not. If an assignment exists, and if the BOM has been defined with usage **S** (Service Management), the **Product Proposal** displays the relevant service parts and the BOM items defined for the equipment. For more information, see [Maintain Bill of Material](#).

Master Data Maintenance

To use the function **Product Proposal** in a service order, you need to fulfill the following prerequisites in master data maintenance:

Create Material BOM

1. Log on to the SAP Fiori launchpad as BOM Engineer (Role: SAP_BR_BOM_ENGINEER).
2. In the app **Maintain Bill of Material**, choose **Create BOM**. Create a BOM with usage **S** and assign the relevant components (service parts and optionally service products) to it.
3. Enter BOM item category, product ID, and quantity, and then save your changes.

i Note

Use BOM item category **L** for service parts or pick-from-stock items and **N** for other items.

Create Equipment BOM

1. Log on to the SAP Fiori launchpad as Maintenance Planner (Role: SAP_BR_MAINTENANCE_PLANNER).

2. Create a technical object and choose **Equipment** as the technical object type. Assign a material ID as construction type for this equipment in the field **Structure**, or as reference product for this equipment in the field **SerData**.
3. Alternatively, you can create or change Material BOM as the construction type or reference product for this equipment in the App **Create Equipment BOM** or **Change Equipment BOM**.

Process

1. When you create a service order, enter the equipment ID or product ID that you've created as main reference object.
2. In the items list of the service order, choose **Product Proposal**. In the dialog box you can see a list where the equipment, the construction product, the reference products, and the BOM items are displayed in a hierarchy.
3. Select the products you want to use for the service order.

i Note

The quantities of the selected BOM items are copied to the service order items as well.

Execution Statuses of Service Orders

You can process service orders in systems beyond Service such as in Maintenance Management and in SAP Field Service Management. You can check the status of service execution in Service with Advanced Execution in Maintenance Management and in SAP Field Service Management.

Maintenance Management

Service is integrated with Maintenance Management in the process Service with Advanced Execution.

When you set the status of a service order to **Released**, the system creates a maintenance order as a follow-up to the execution order item. The service order is then processed in the maintenance order. The execution status shows the progress of the service order in the maintenance order.

The following execution statuses are available:

Execution Status in Service Order	What Happens in Service	System Status of the Maintenance Order
Not Available	All service order items have the status Not Available , or at least one item has the status Not Available and all other items have the status Not Relevant .	Not applicable
Execution Planning	At least one service order item has the status Execution Planning .	Created
Planning Completed	The planning process for a service has been completed.	Order Approved
In Execution	At least one service order item has the status In Execution , or at least one service order item has the status Execution Done .	Released
Execution Done	The service process has been finalized in the maintenance order.	Technically Completed

Execution Status in Service Order	What Happens in Service	System Status of the Maintenance Order
Canceled	All service order items have the execution status Not Available , or at least one item has the status Not Available , and all others have the status Not Relevant .	Do Not Execute
Not Relevant	No execution order items have been added to the service order. The service order is not processed in Maintenance Management.	Not applicable

Execution Status in Service Order Item	What Happens in Maintenance Management	System Status of the Maintenance Order
Not Available	No maintenance order has been created. The process in Maintenance Management has not yet started.	Not applicable
Execution Planning	The maintenance order has been created as a follow-up to the service order once the status of the execution order item has been set to Released in the service order.	Created
Planning Completed	The planning process for a service has been completed.	Order Approved
In Execution	In the maintenance order, the planning was completed and the service can now be performed in the maintenance order. Confirmations can also be created for the maintenance order.	Released
Execution Done	The service that has been performed in the maintenance order has been completed. The labor and material consumption have been estimated, planning has been completed, and the items have been approved in the maintenance order.	Technically Completed
Canceled	The execution order item has been rejected or the maintenance order has been manually set to the system status Do Not Execute .	Do Not Execute
Not Relevant	The execution order item has been rejected in the service order even before the maintenance order was created.	Not applicable

SAP Field Service Management

In SAP Field Service Management you execute the service in service calls.

When you set the status of the service order to **Released**, the system creates a service call and the corresponding activities. The service is then executed in the service call and in the activities. The execution status shows the progress of the service execution in the service call.

For more information, see [Integration Scenarios for Transactional Data](#).

The following execution statuses are available:

Execution Status in Service Order	What Happens in Service	Activity Status in SAP Field Service Management
Not Available	All service order items have the status Not Available , or at least one item has the status Not Available and all other items have the status Not Relevant .	Not applicable
Execution Planning	At least one service order item has the status In Planning .	Dispatching
In Execution	At least one service order item has the status In Execution , or at least one service order item has the status Execution Done .	Execution
Execution Done	The service process has been finalized in SAP Field Service Management.	Closed
Canceled	All service order items have the execution status Not Available , or at least one item has the status Not Available , and all others have the status Not Relevant .	Canceled
Not Relevant	No activity has been created. The service is not performed in SAP Field Service Management.	Not applicable

Execution Status in Service Order Item	What Happens in SAP Field Service Management	Activity Status in SAP Field Service Management
Not Available	No activity has been created. The process in SAP Field Service Management has not yet started.	Not applicable
Execution Planning	An activity has been created by the system when the status of the service order item has been set to Released . The activity is visible in the planning list or on the planning board. It is ready to be planned or is already assigned to a technician but is not yet released for the technician. The assignment of the service employee in the service order item is removed.	Dispatching
In Execution	The activity has been released to the technician and can now be viewed and completed.	Execution
Execution Done	The activity has been completed and closed by the assigned technician. All time efforts, expenses, materials, and mileage related to the activity are approved and transferred from SAP Field Service Management to Service.	Closed

Execution Status in Service Order Item	What Happens in SAP Field Service Management	Activity Status in SAP Field Service Management
Canceled	The activity has been canceled.	Canceled

i Note

Service items that have the life cycle statuses **Released** or **Completed** and do not have execution statuses are not considered for the calculation of the execution status on header level. In this case, it is considered that Service is not integrated with SAP Field Service Management.

Credit Checks in Service Orders

Perform credit checks in service orders.

Use

You can use Credit Management to minimize financial risks for your organization. To do so, you perform a credit check when you save service orders in status **Released**.

The credit check determines the creditworthiness of the payers (business partners) who are assigned to the service order items. You can check, for example, if the gross value of the items within a service order exceeds the available credit limit that has been defined in the credit segment data for the business partner (see also [Defining Credit Master Data in the Business Partner](#)).

If the payer passes the credit check, a commitment that is equal to the gross value that was checked is passed to the credit exposure in Credit Management.

If the credit check fails, the credit blocked service orders need to be approved in Credit Management. For more information, see [Perform Credit Checks in Service Orders](#).

i Note

Payers are business partners that are maintained in the partner function **Payer** under **Parties Involved**.

Prerequisites

i Note

The following settings also apply if you want to use the credit check in service quotations and to simulate the credit check for service order items and service quotation items.

Settings in Customizing

Perform the following Customizing activities in Customizing for **Financial Supply Chain Management** under **Credit Management** > **Integration with Accounts Receivable Accounting and Sales and Distribution** > **Integration mit Service** >

- **Deactivate SAP Credit Management**

i Note

If you want to use the credit check functionality, make sure that the **Deactivate Credit Check** checkbox is not selected.

- [Define Credit Control Area](#)
- [Define Credit Segment](#)
- [Assign Credit Control Area and Credit Segment](#)
- [Assign Sales Area to Credit Control Area](#)
- [Activate Credit Management in Service Transactions](#)

You can also activate the credit check for the transaction type in the [Define Transaction Types](#) Customizing activity in the [Customizing header](#) view for the business transaction category [Sales](#).

- [Activate Credit Check on Items](#)

You can also activate the credit check for the item category in the [Define Item Categories](#) Customizing activity in the [Customizing header](#) view for the business transaction category [Sales](#).

You can check whether the credit exposure categories that are provided as default values meet your business requirements. You can make your own settings in Customizing for [Financial Supply Chain Management](#) under [Credit Management](#) > [Credit Risk Monitoring](#) > [Credit Exposure Update](#) > [Define Liability Categories](#) . For more information about credit exposure categories, see [Managing Credit Exposures](#).

Settings to Perform Credit Checks

If you want to perform a credit check for a payer, make sure that the following steps are performed for the business user in Credit Management:

- Assign the Credit Management role (UKM000) to the business partner.

You have the following options:

- Assign the role to the business partner in the global credit data for the business partner.

For more information, see [Manage Credit Accounts](#).

- Perform the job [Assign Credit Management Role to Business Partners](#) in the [Schedule Credit Management Jobs](#) app.

- In the [Manage Credit Accounts](#) app, create the credit segments and define the credit limit for the business partner.

i Note

If you change the settings for the credit segments, you need to rebuild your credit management data afterwards in the [Rebuild Credit Management Data](#) app.

Note the following when you set up credit checks for business partners in Credit Management:

- If no credit segment is assigned to the sales area in a service order, no credit check is performed. The status of the service order items doesn't change.
- If the Credit Management role isn't assigned to a payer in a service order item, no credit check is performed for the service order items of this payer.

If you initially don't assign the Credit Management role to a payer but assign it after the service order items have been posted to financial accounting and invoiced, you need to use the [Rebuild Credit Management Data](#) app to update the corresponding credit exposure.

For more information, see [Rebuild Credit Management Data](#) and [Rebuild Credit Exposure](#).

Credit Check in a Distributed Landscape

You can now perform credit checks in a distributed landscape, using one system for the service transaction and a separate system for credit management. You no longer need to maintain the credit management role for the business partner in the system that processes the service transaction. A separate system can be used for all credit check-related decisions.

The BAdI method [IF_EX_UKM_R3_ACTIVATE~GET RFCDEST_FSCM](#) needs to be implemented to specify the logical destination of Credit Management.

For Financial Supply Chain Management (FSCM) APIs to perform successful credit management operations, the following steps must be completed:

- Establish connections between S/4HANA Service and Credit Management system
- Create Service definition bindings
- Define a communication user

For more information, see [Connecting Sales and Distribution in SAP S/4HANA Systems or as of SAP ERP 6.0](#).

Further Information

For more information about Credit Management, see [SAP Credit Management \(FIN-FSCM-CR\)](#).

Perform Credit Checks in Service Orders

Perform credit checks in service orders.

Use

When you save a service order with the status **Released**, the system performs a credit check to check the creditworthiness of the payers (business partners) who are assigned to the service order items.

Performing Credit Checks in Service Orders

Credit checks are automatically triggered for service order items when you change the status of the service order to **Released** and then save the service order. The system then ensures that the credit exposure that derives from the service order doesn't exceed the available credit limit for the payers.

The credit status is displayed for each item and for the service order.

→ Recommendation

For more information about the credit checks that are performed by Credit Management for service order items, see [Credit Check](#).

For more information about the credit exposures that derive from service orders, see [Managing Credit Exposures](#).

Credit Status for Service Order Items

The following statuses apply to items:

Credit Status	What It Means
Not relevant for credit check	Credit Management hasn't been enabled for service orders.
Relevant for credit check, but credit check not executed	The credit check hasn't been executed for the service order item yet.
Credit check not OK	The item has failed the credit check. The status of the item remains In Process. If the status of the item was previously set to Released, the item status remains the same. i Note This status is also set if the item has not been released in the Manage Documented Credit Decisions app in Credit Management.
Credit check OK	The item passed the credit check in one of the following ways: - You changed the status of the item to Released and saved the service order. - The service order has been rechecked in the Manage Documented Credit Decisions app in Credit Management. The status of the service order item changes to Released.
Credit released	The item has been released in the Manage Documented Credit Decisions app. The status of the service order item changes to Released.

Credit Status for Service Orders

The following statuses apply to service orders:

Credit Status	What It Means
No item is relevant for credit check	None of the items added to the service order are relevant for the credit check.
Credit check pending for all relevant items	All items that are relevant for the credit check have the credit status Relevant for credit check, but credit check not executed. The service order has the status Open or In Process.
Credit check OK for all relevant items	All items that are relevant for the credit check have the credit status Credit check OK or Credit released. The service order has the status Released.
Credit check not OK for all relevant items	All items that are relevant for the credit check have the credit status Credit check not OK. The service order has the status In Process. You can process service orders with this credit status as described in the section <i>Approving Service Orders with a Credit Block</i>.
Credit check not OK for some items	Contains items that have the credit status Credit check not OK and items that aren't credit blocked (Credit check OK and Credit

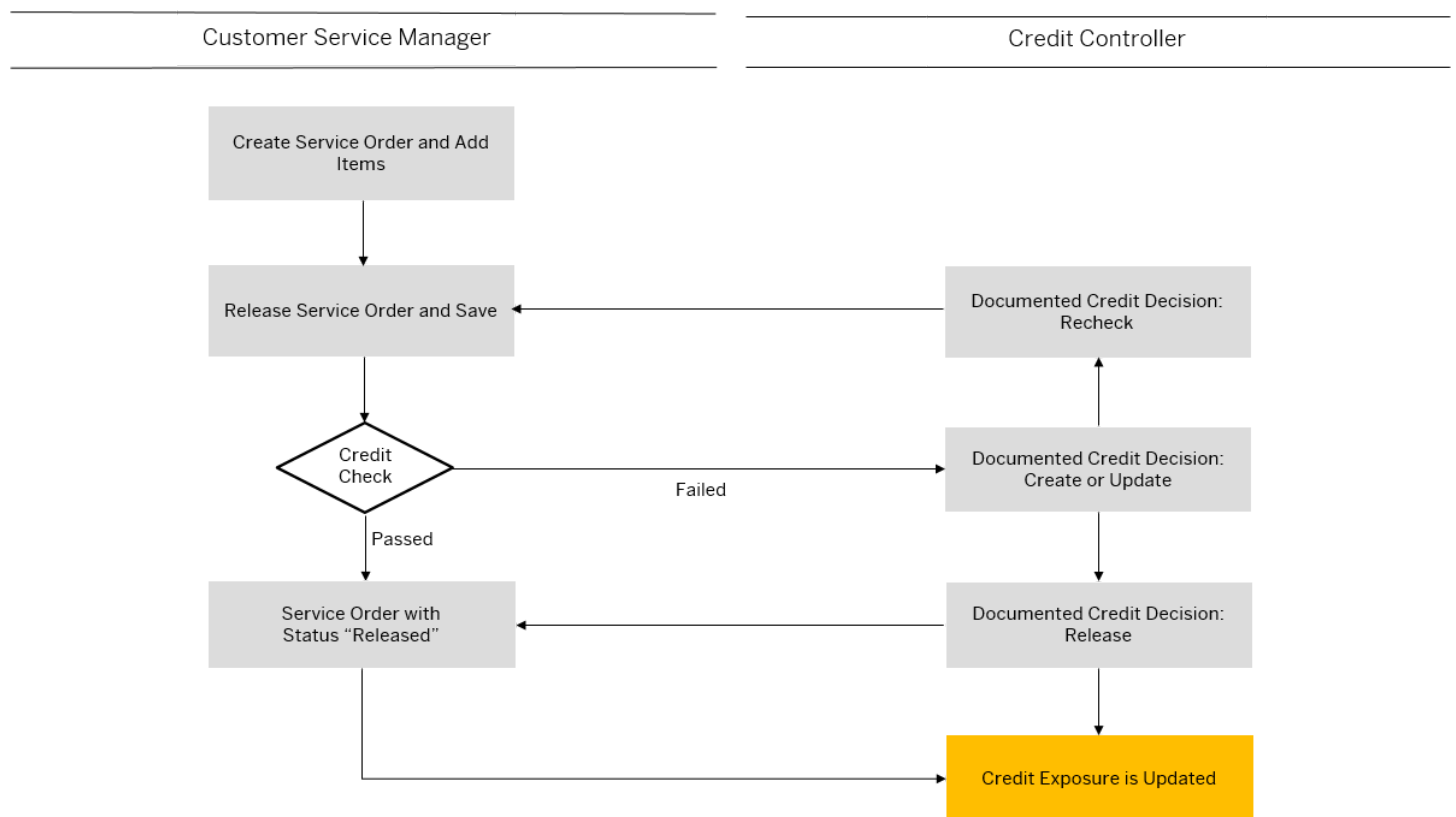
Credit Status	What It Means
	released). The service order has the status In Process. You can process service orders that are credit blocked as described in the section <i>Approving Service Orders with a Credit Block</i>.
Credit check OK for some items and pending for some items	Contains items that aren't credit blocked (Credit check OK and Credit released) and items that have the status Relevant for credit check, but credit check not executed. The service order has the status In Process.

i Note

If the service order is saved again with the status **Released**, the credit check is triggered for all items that are credit blocked from a credit check run that has previously failed. If the items pass the credit check, the credit block is removed.

Credit Check Process

The figure shows the credit check process for service orders.



Credit Checks for Sales Items

The credit check in service orders is also performed for sales items that have been added to the service order. If the sales item passes the credit check, the system creates a sales order and updates the credit exposure that is used for sales orders. If the credit check has failed for the sales item, the system does not create a sales order and you first need to approve the sales item in Credit Management (see the section [Approving Service Orders with a Credit Block](#)).

Credit Checks for Execution Order Items

Credit checks are performed for execution order items that have the billing relevance **Billing on Completion** (fixed price billing) and **Billing via Billing Plan After Release** assigned.

The system performs the credit check once the execution order item is released. If the execution order item passes the credit check, the system creates the follow-up maintenance order.

If the credit check has failed for the execution order item, the system does not create a maintenance order and you first need to approve the execution order item in Credit Management (see the section [Approving Service Orders with a Credit Block](#)).

i Note

For more information, see [Billing](#).

Simulating Credit Checks

You can use the credit check simulation to see the possible credit check result before saving the service order with the status **Released**.

The result of the simulated credit check does not change the credit status that is displayed for the service order, items and the credit exposure, or prevent further processing of the service order.

The following applies to the credit check simulation:

- The result of the simulation is displayed for the service order but not for the related items.
- The simulation allows you to check the possible credit result when the simulation is being performed. If additional changes are made before the service order is saved, the actual result of the credit check might differ from the simulation result.
- Unlike the credit check that is performed when you save the service order with the status **Released**, you can simulate the credit check for items with the status **Open** and **In Process**. If you perform the simulation for items with the status **Released**, only items with a change in the gross value are considered.
- You cannot perform the simulation for items with the status **Completed**.

Approving Service Orders with a Credit Block

If the credit check has failed for at least one service order item and the status of the service order remains **In Process**, you can approve the items in Credit Management as follows:

- Release or recheck the service order items in the [Manage Documented Credit Decisions](#) app.

The system creates cases (documented credit decisions) for service orders that contain credit blocked items. The items to be checked are grouped by the assigned payers.

- Perform the job **Service Transaction Credit Recheck** (RMS4_SRV_CREDIT_RECHECK) to repeat the credit check.

For more information, see [Check Credit for Service Transactions](#).

Once the credit recheck is passed, the status of the service order changes to **Released** and the relevant credit exposures are updated accordingly for the business partner.

Performing Credit Checks for Released Service Orders

The system automatically triggers a credit check for service orders with the status **Released** if the following applies:

- You've added service order items to the service order and then changed the status of the service order items to **Released**.

- You've changed the status of service order items with the status **Open** or **In Process** to **Released**.
- You've increased or decreased the gross value of the service order items with the status **Released**.

If the credit recheck fails for the service order items, their status remains **Released**, but their credit status is set to **Credit check not OK**.

The system doesn't perform a credit recheck for service orders with the status **Released** if the following applies:

- You've changed the status of some of the service order items with the status **Released** to **In Process**.
- You've rejected all service order items.
- You've changed the status of a service order item to **Completed**.

If the credit check fails for the service order items, you can approve the service order as described in the section [Approving Service Orders with a Credit Block](#).

Managing Credit Exposures

Use credit exposure categories to report and analyze a payer's credit situation.

Use

Credit exposure is the combined value of all items (and the corresponding credit exposure categories) that are related to a payer (business partner). You can use the [Display Credit Exposure](#) app to see an overview of the total credit exposure for a business partner according to credit segment.

The credit check only observes the total credit exposure for a business partner.

Credit Exposures for Service Orders

You can use the following credit exposure categories that are provided as default values in the [Define Liability Categories](#) Customizing activity:

Credit Exposure Category (Default)	Name of the Credit Exposure Category	Used for
100	Open Orders	Sales orders that are created as a follow-up to sales order items
110	Service Order & Service Contract	• Service order items with the status Released for which service confirmations haven't been created or • This also applies to Service with Advanced Execution • Service confirmations have been created that have a total gross value that is less than the gross value of the related service order item
200	Open Invoices	Invoices that have not yet been paid

Credit Exposure Category (Default)	Name of the Credit Exposure Category	Used for
410	Service Confirmations	Service confirmation items with the status Completed that are ready for billing
500	Billing Document Value	Invoices, credit memos, debit memos, pro forma invoices, or cancellation documents

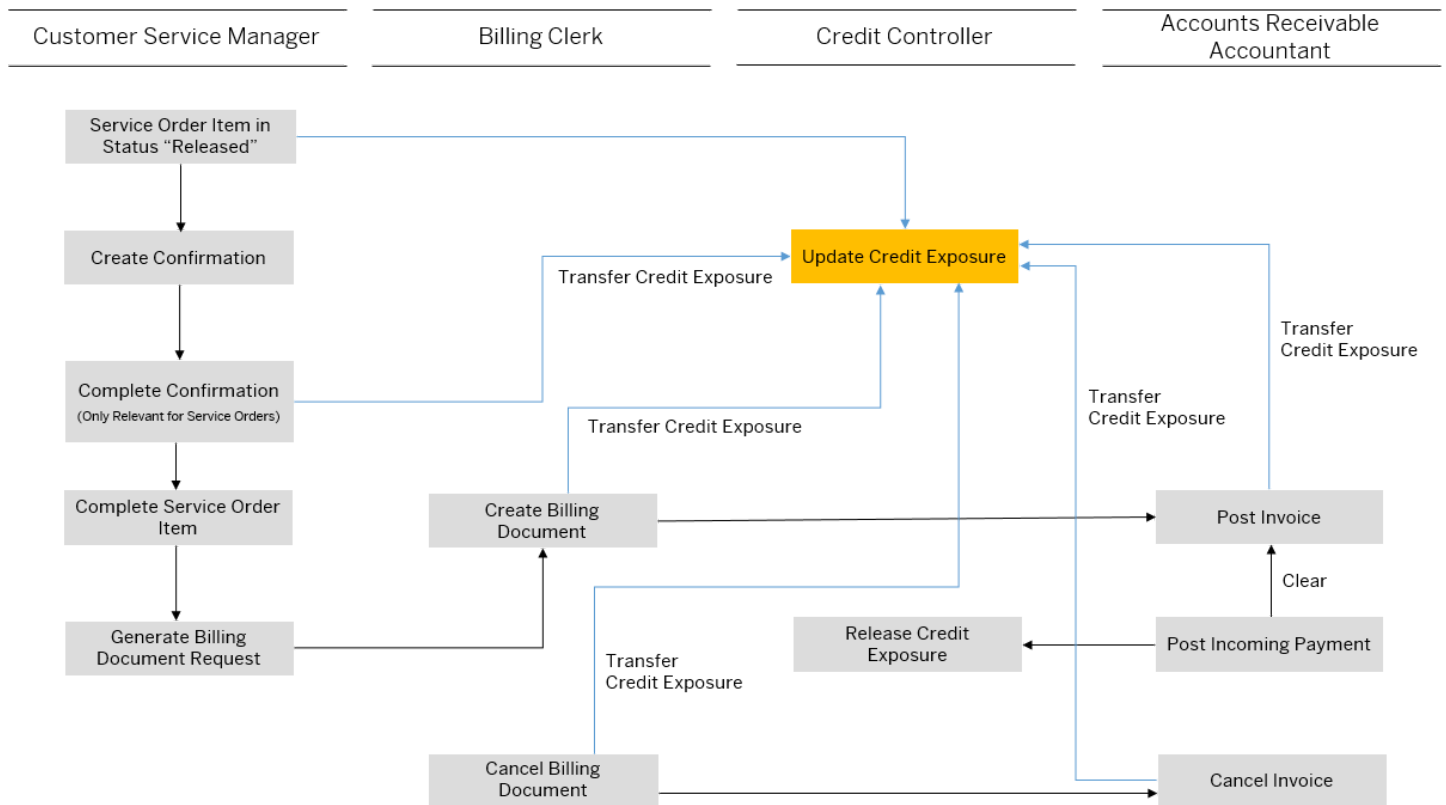
Posting Credit Exposures

If the credit check is successful, the gross value of all items is transferred to the credit exposure category **110**, which is used for service orders. If a sales order is created as a follow-up to a sales order item, the gross value is transferred to the credit exposure category **100**.

The gross value of the items is posted to the relevant credit exposure categories as follows:

Process Step	Credit Exposure Category Used (Default)
The service order items pass the credit check when being saved with the status Released .	110
The status of the service confirmation items that have been created as follow-ups for service orders changes to Completed .	Transfer from 110 or to 410
The service confirmations are cancelled.	Transfer from 410 to 110 i Note If the gross value of the service confirmation is higher than the original gross value that has been posted for the related service order item, the system uses the gross value that was originally posted for the service order item.
A billing document is created for the service confirmation items.	If the billing document is created and not posted automatically: transfer from 410 to 500 If the billing document is created and posted automatically: transfer from 410 to 200
The billing document is cancelled.	If the billing document has not been posted: transfer from 500 to 410 or 110 If the billing document has been posted: transfer from 200 to 410 or 110

Credit Exposure Processing



Managing Inconsistencies in Credit Exposure

If there are inconsistencies in the credit exposure, you can use the job [Rebuild Credit Exposure](#) in the [Rebuild Credit Management Data](#) app to rebuild the credit management data.

If Credit Management is activated in the system for the first time, but service orders exist that were created before the activation, this can lead to inconsistencies in the credit exposure: If a credit check is performed for new service orders created and no changes have been made to the existing service orders, you need to rebuild your credit management data. This ensures that the credit exposure is updated with the correct value for both existing service orders and any new service orders created.

Once you've performed the job to rebuild the credit management data, the credit exposure is cleared to zero if the following applies:

- Credit Management has not been activated (globally or for service orders).
- Credit Management has been activated, but the credit segment has not been defined for the business partner.
- Credit Management has been activated, but the Credit Management role (UKM000) has not been assigned to the relevant business partner (payer).

If credit management data is being rebuilt and a service order or a service confirmation is locked by another user, the job clears the credit exposure for that service order to zero. As soon as the service order or service confirmation is unlocked, the correct value of the credit exposure is updated.

Service Quotations

A service quotation is a legally binding offer with fixed conditions, made to a customer for the performance of services or for the delivery of service parts.

Use

Quotations provide customers with information about prices and delivery conditions before placing an order. They also enable providers to plan early for expected services orders and to get an overview of expected costs and revenues. Within the validity period of a quotation, customers have the option of agreeing to the quotation.

You can use the [Manage Service Quotations](#) app to create and process service quotations.

Depending on your Customizing settings, you can do the following:

- Create a service order from a service quotation as a follow-up transaction
- Create a service quotation as a follow-up transaction of a service order template (see [Service Order Templates](#))

Solution quotations are variants of service quotations, and enable you to create a quotation for combinations of tangible products and services. For more information, see [Solution Quotation Management](#).

i Note

The user interface for service quotations is a configuration of the component for service orders with the object type BT116_SRVQ.

Prerequisites

You can find the settings listed below in Customizing for [Service](#) under [► Transactions ► Basic Settings ►](#).

Business Transaction Customizing

The business transaction for service quotations is predefined as follows in the standard system:

- Transaction type SRVQ
- Item categories:
 - SRQP for service items
 - SRQM for service part items
 - SRQS for sales items
 - SRQE for expense items
- Fixed price item categories:
 - SFQS for fixed price service items
 - SFQM for fixed price service part items
 - SFQE for fixed price expense items
 - SBQT for tool items with ad hoc billing
 - SFQT for fixed price tool items

If the settings for these Customizing objects don't reflect your business requirements, you can copy them and modify the copies.

You also need to set up copy control to determine the follow-up transactions that are created from the service quotation. The copy control customizing is checked for every item to determine the transaction type of the follow-up transaction. If multiple transaction types are selected, the first entry applies.

Quotation Items

You define whether an item is a service quotation in the Customizing activity [Define Item Categories](#) by choosing the required item categories and the business transaction category **Sales**. In the [Customizing Item](#) view, specify in the [Relevance for Quotation](#) field whether the item is an initial quotation.

In the above Customizing activity, ensure that the value in the [Subsequ. Processing](#) field is [Quotation is copied to subsequent document](#).

Features

Service quotations contain the same type of data and have the same features as service orders (see [Service Orders](#)).

Trade Compliance Check

The integration with SAP Global Trade Services (GTS) allows you to perform embargo checks, sanctioned party list screening, and legal controls for service quotations. For more information, see [Integration with SAP Global Trade Services](#).

Specify Ad Hoc Billing in Service Quotations

In service quotations, you can assign billing plan templates to predetermine the billing conditions for individual items. This enables you, for example, to include the billing conditions defined in the billing plan template in the output document for the service quotation. When you create a follow-up service order to a service quotation to which items with billing plan templates are assigned, the system automatically generates the billing request lines in the related service order items.

For more information, such as the settings that enable billing plan templates to be assigned to service quotations, and creating billing request lines in the follow-up service orders, see [Billing Plan Templates in Service Quotations](#).

Qualification Requirements

You can define qualification requirements, such as skills, knowledge, or abilities, necessary to perform services specified in a service transaction. For more information, see [Qualifications in Service Transactions](#).

Using Service Hierarchies with Collective Accounting

i Note

You can only use service hierarchies with collective accounting if item-based accounting is active.

You can add service hierarchies with collective accounting to service quotations as main items. Sub-items that you add to a service hierarchy inherit the service contract assigned to the main item. If the service hierarchy is copied to a follow-up service order, you can use the features of the service hierarchy in the service order as described under [Service Hierarchy with Collective Accounting](#).

Using Service Quotations in the In-House Service Process

Service quotations allow you to provide a cost estimate for services that are related to the in-house service process for an in-house service object (service object). The [In-House Service Objects](#) assignment block displays the service objects and the related in-house services to which the service quotation refers. You can also assign additional service objects or remove service objects.

For more information about processing service objects in service quotations, see [In-House Service Objects in Service Quotations and Service Orders](#).

Performing and Simulating Credit Checks

In service quotations, you can perform credit checks and simulate credit checks to determine the creditworthiness of the payers (business partners) who are assigned to the related items.

The credit check is automatically performed once the service quotation is sent to the customer and the status changes to **Released**. The result of the credit check is displayed in the credit status.

i Note

The actual credit check is performed in the service order that is created once the service quotation is accepted.

If you want to know the credit check result while the service quotation is in process, you can simulate a credit check for the related items.

i Note

Unlike the credit check that is performed in service orders, the result of the checks and the simulations in service quotations doesn't change the credit exposure and doesn't block further processing of the service quotation.

i Note

Prerequisites

You can perform credit checks and credit check simulations for quotation items if the payer who is assigned to the quotation items has the credit management role assigned and credit checks have been activated for service quotations.

For more information about setting up credit checks in Customizing, see [Credit Checks in Service Orders](#).

Service Orders as Follow-Up Transactions

You can create service orders as follow-up transactions from service quotations with the following options:

- **Create Follow-Up**

When a customer accepts quotation items, you can create a follow-up service order by choosing this button and copy the accepted items into the follow-up service order. When you copy items to a service order, they're automatically marked as accepted in the quotation.

- **Accept**

You can use this button to accept a service quotation. The system then automatically creates a service order as a follow-up transaction that contains all the items from the service quotation. This function provides you with the option to accept all items in a service quotation and create a follow-up service order in one go.

i Note

- You can only choose **Accept** when the service quotation and the items in the quotation are error-free and set to the status **Released**.
- If there's more than one follow-up transaction available, you can choose the service transaction types from the popup. For example, a service order or a fixed price service order.
- After you accept the service quotation, the status of the service quotation is set to **Completed** and the status of the items in the quotation are set to **Quotation Accepted**.

- **Requote**

You can create a requote to a service quotation if you require additional service items, expense items, or service parts to perform a service, which haven't already been specified in the accepted service quotation. For more information, see [Creating Requotes](#).

Rejecting Service Quotations

You can reject service quotations or single items of service quotations. For more information, see [Rejecting Service Quotations](#).

Completion of a Quotation

Quotations are automatically set to **Completed** when:

- The validity of the quotation has expired
- All quotation items have been copied to follow-up service orders

You can also manually set the status **Completed** if the quotation is rejected by the customer.

Creating Requotes

Create requotes for service quotations that have the status **Completed** but need to be adapted subsequently.

Use

If you require additional service items, service parts, or expense items to perform a service, which haven't been specified in the accepted service quotation (status **Completed**), you can resume the quotation process by creating a requote. In the requote, you add the additional items. Having done this, you send this updated service quotation, or **requote**, to the customer.

The following terms are used when processing requotes:

- **Initial quotations** are service quotations that aren't related to a predecessor quotation or a predecessor order. When you accept an initial quotation, the system creates a service order as a follow-up.

i Note

You can't create requotes for initial quotations that have been rejected. Instead, create a new service quotation.

- **Requotes** are created as a follow-up to the initial quotation. Consequently, the service order that has been created as a follow-up to the initial quotation, is related to this requote. When you accept the requote, the system copies the items that have been added to the requote to the related service order.

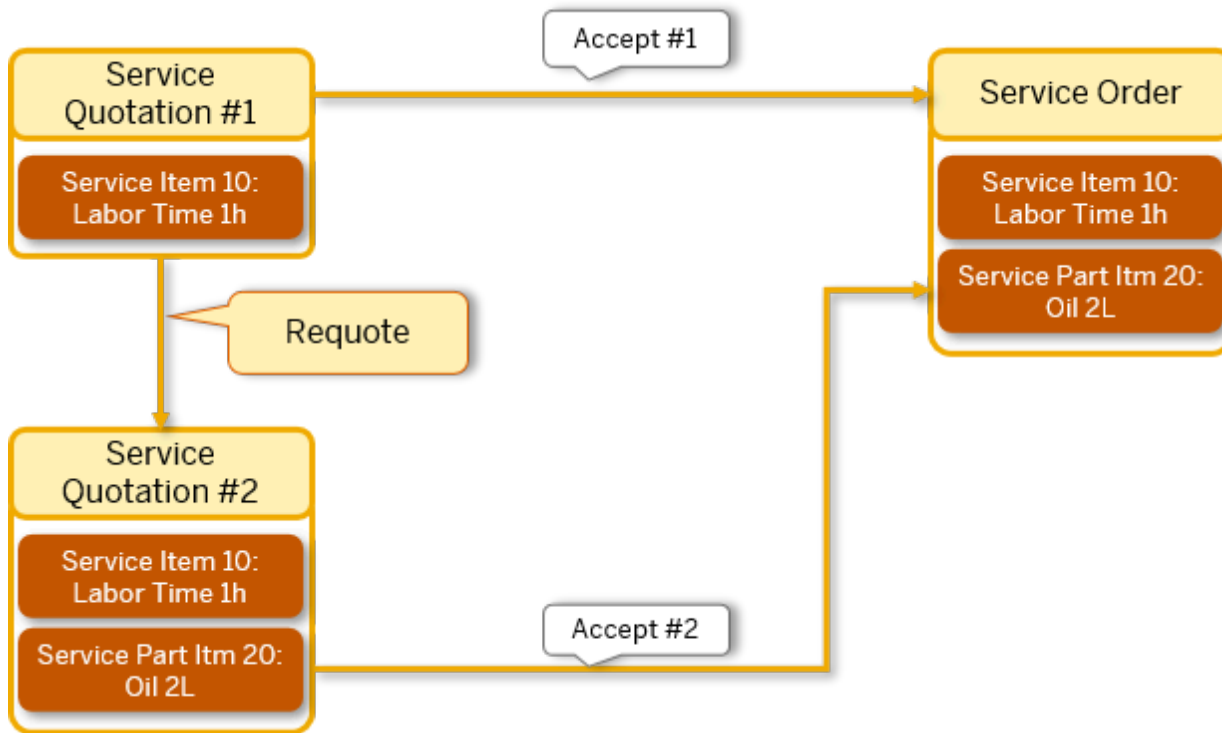
Prerequisites

You can create a requote if the following applies:

- The initial quotation has been accepted and has no predecessor service quotation or service order assigned.
- The latest quotation that can result from an initial quotation or from a requote has been accepted or rejected.

In addition, it's related to only one service order that hasn't yet been completed.

Process Overview



Example of Using Requotes

You can use requotes if you first want to plan and perform a diagnosis to figure out the basic needs of the service and then want to provide a labor and material consumption estimate to the customer:

1. Create a service quotation, that is the initial quotation, and that includes an item to perform the diagnosis.

This item can be a fixed price item.

2. Accept the service quotation.

The system creates a follow-up service order that contains the accepted item from the service quotation.

3. Plan and perform the diagnosis as defined in the items. Create service confirmations to record the actual service effort.

Do not complete the service order and do not define a service confirmation as the **Final Confirmation**.

4. Open the service quotation (initial quotation) that is displayed in the transaction history for the service order.

5. Create a requote from the initial quotation.

The requote has its own service quotation ID assigned. The accepted items of the initial quotation are copied to the requote.

The initial quotation and the related service order are displayed in the transaction history for the requote.

6. In the requote, enter the services and service parts that are required to plan and perform the service.

You can add a price item to a higher-level item that has already been completed to reflect a price surcharge for additional services to be provided.

For more information about price items, see [Price Items in Service Orders and Service Quotations](#).

i Note

Do not make changes in the service quotation items that have been copied from the predecessor quotation. Changes such as changed quantities, aren't copied to the related service order when you accept the service quotation and the status of the service quotation is set to **Completed**.

Having done this, process the service quotation as usual. When you accept the requote, the new items added are copied to the related service order. You can now plan and perform the service in the service order.

i Note

You can create additional requotes at any time until the service is completed, provided that the prerequisites for creating requotes are met.

Processing Related Service Orders

When you process the service order that is related to the initial quotation and the requote, the following applies:

- You can complete the service order if all related service quotations have been accepted or rejected, and consequently have the status **Completed**.
- If you complete the service order, you can no longer create a requote to the related service quotation . Therefore, do not define a service confirmation as the **Final Confirmation** if you're planning to create requotes.

Rejecting Service Quotations

Reject service quotations or single items of service quotations.

Use

If service quotations or single items of a service quotation have become obsolete during the quotation process, you can reject them.

You can reject service quotations and items of any status and if no follow-up service order has been created for the service quotation. If a follow-up service order already exists, you must first reject it and the existing service order items.

Rejecting Service Quotations

If you want to reject a service quotation, you have the following options:

- You can choose the button **Reject All Items**.

All items within the service quotation are rejected. The status of the service quotation and the related items changes to **Completed**.

If required, you can assign a rejection reason to the items of the service quotation.

- You can reject single items in the service quotation by assigning a rejection reason to each item to be rejected. The status of the rejected items then changes to **Completed**.

Once you've rejected all items of the service quotation, the status of the service quotation changes to **Completed** as well.

i Note

You can't create a follow-up service order for service quotations that have been rejected.

Rejecting Items in Service Quotations

If you want to reject single items in the service quotation, assign a rejection reason to the items to be rejected. The status of the rejected items then changes to **Completed**.

If you create a follow-up service order for the related service quotation, the rejected items aren't copied to the service order.

Reverting Your Decision

You can revert the rejection when you remove the rejection reason of single items or choose the button **Reset Item Rejection** to remove the rejection for all rejected items.

Once you've reverted the rejection, the status of the service quotations and/or the items changes to **In Process**.

Service Order Templates

A service order template is a service transaction used to define reusable sets of service-related data that minimize the amount of time required to create a service transaction. A service order template describes only the scope of planned services and not the actual execution.

Integration

Business Transactions

Depending on your Customizing settings, you can do the following:

- Create a service order on the basis of the service order template, as a follow-up transaction
- Create a service order template on the basis of a service order, as a follow-up transaction
- Copy items from service order templates when creating the following:
 - Service order
 - Service quotation

Prerequisites

You have configured your service order template transaction(s) in Customizing for **Service** under **Transactions** **Basic Settings**.

i Note

The user interface for service order templates is a configuration of the component for service orders with the object type BT116_SRVT.

The service order template is predefined as follows in the standard system:

- Transaction type SRVT
- Item categories:
 - SRTP for service items
 - SRTM for service part items
 - SRTS for sales items

If the settings for these Customizing objects do not reflect your business requirements, you can copy them and modify the copies.

Features

You can create hierarchies of templates by referencing a template from within another template. For more information, see [Template References](#).

Qualification Requirements

You can define qualification requirements, such as skills, knowledge, or abilities, necessary to perform services specified in a service transaction. For more information, see [Qualifications in Service Transactions](#).

Service Order Templates for Planned Recurring Service

Use

You use service order templates for planned recurring service to create service orders.

Integration

Business Transactions

Depending on your Customizing settings, you can do the following:

- Create a service order on the basis of the service order template, as a follow-up transaction
- Copy items from service order templates when creating a service order

Prerequisites

You have configured your service order template transactions in Customizing for **Service** under **► Transactions ► Basic Settings ►**.

The service order template for planned recurring service is predefined as follows in the standard system:

- Transaction type MPTL
- Item categories:
 - MPTP for service items (maintenance plan)
 - MPTM for service part items (maintenance plan)

If the settings for these Customizing objects do not reflect your business requirements, you can copy them and modify the copies.

Template References

Use

You can build hierarchies of templates by assigning templates to items in other templates. Since the templates in a hierarchy are only linked by means of reference, you only need to make changes to templates once and they are automatically available wherever referenced in the hierarchy.

❖ Example

You define a service order template for an oil change and assign it to the item “Oil Change” in the templates for a standard inspection and complete inspection. The inspection templates, therefore, implicitly contain all the template items for the oil change.

You can also reference templates in service order and service quotation items, if you have customized the item category correspondingly (see “Prerequisites” below).

Prerequisites

You have assigned the object reference profile **TEMPLATE** to those item categories from which you want to allow template references. You do this in Customizing for **Service**, by choosing ► **Transactions** ► **Basic Settings** ► **Define Item Categories** ►.

Features

If you have assigned a template to a (template) item, when you create a follow-up transaction one of the following occurs:

- The template reference is copied to the new item.
- Items from the template are automatically copied as subitems of the follow-up item.

If you want new items to be created from the referenced template, you have specified this in Customizing for **Service**, by choosing ► **Transactions** ► **Basic Settings** ► **Copying Control for Business Transactions** ► **Define Copying Control for Transaction Types** ►.

i Note

The items from all templates referenced in the hierarchy are copied as subitems on an equal level. The hierarchy of items in the template hierarchy is not reproduced.

Service Confirmations

A service confirmation is a service transaction used to report working time, expenses incurred, and service parts used while performing a service.

Use

You can use service confirmations to confirm service processes for cost postings. The actual service duration, service parts used, and expenses incurred during service execution can be used as a basis for keeping track of working time, updating stock levels for service parts, processing invoices, and capturing actual costs. For example, after a service part item in a service order has been released, a purchase requisition is triggered against this item and the relevant procurement costs are updated. You use service confirmations to confirm items from the service order and to record the actual consumption of time and materials during service execution. This way, the actual costs of a service can be posted.

Use of Items in Service Confirmations

A service confirmation consists of a header and one or more items. You use items in a service confirmation to confirm items from a service order. This in turn creates account assignment objects so that cost and revenue can be posted to Financial Accounting. You can edit items in a service confirmation on the **Items** tab. For more information on different types of items and how to use items in service confirmations, see [Items in Service Confirmations](#).

Basic Settings

The standard transaction type of a service confirmation is predefined in the system as SRVC. Transaction types define the characteristics and attributes of a business transaction and determine how the transaction is processed, for example, the profiles and procedures used to determine texts, business partners, statuses, and organizational data. If the standard settings do not meet your business requirements, you can configure the transaction types using the Customizing activities under [Service Transactions](#) [Basic Settings](#). For more information, see [Business Transaction](#).

Life Cycle Statuses

A service confirmation and a service confirmation item can have the following life cycle statuses:

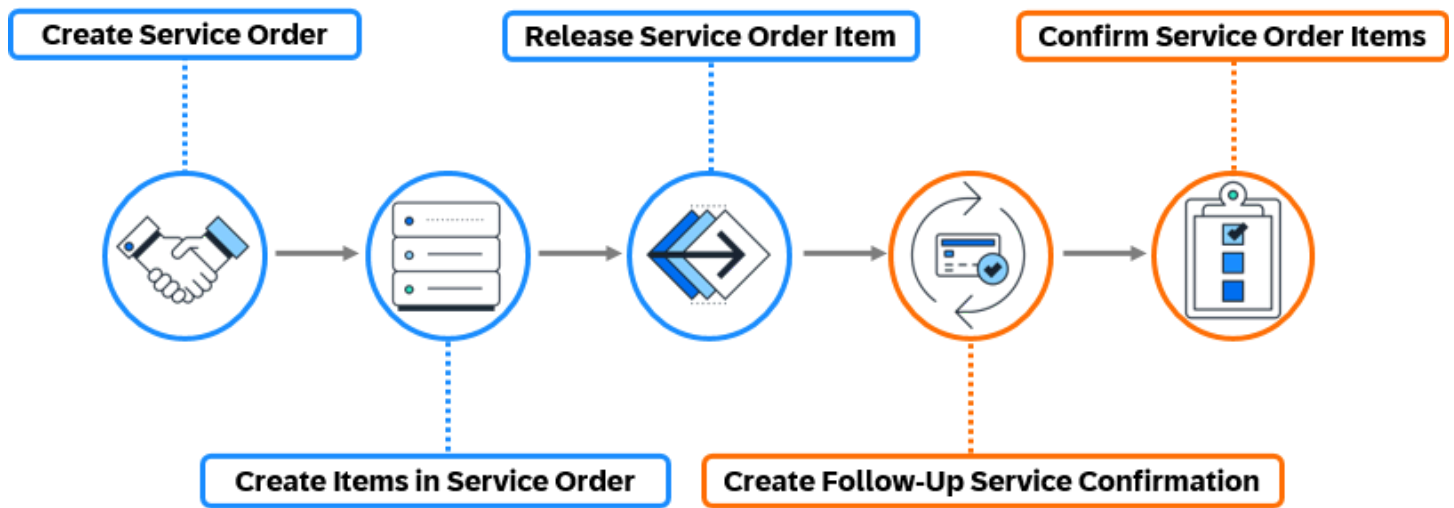
Status	Description
Open	The default status that all service confirmations have after they're created. It indicates that you can edit or cancel the confirmation.
Completed	This status indicates that no further actions are required. You can release a service confirmation for billing if the following conditions are met: - The header status of the service confirmation is Completed. - The items from the predecessor service order also have the status Completed. i Note You cannot edit a service confirmation that has this status anymore, but you can still cancel a service confirmation that has this status.

Activities

Create Service Confirmations as Follow-Up Transactions

You can create service confirmations as follow-up transactions of service orders to confirm service order items.

- Prerequisites: To create service confirmations as follow-up transactions, you must configure the respective transaction flow in Customizing. For more information, see [Follow-Up Transactions](#).
- When you create a follow-up service confirmation for a service order, the system automatically proposes the items planned in the service order for selection. The items that you select are copied to the service confirmation. The relevant data from the item, such as prices, reference objects, subjects, conditions, and working times for assignments, is also copied to the service confirmation.
- Sample process: The following diagram illustrates a sample process of creating a service confirmation from a service order.



i Note

- You can only create service confirmations for service order items that are error-free and have the status **Released**. The **Create Follow-Up** function is only active when a service order item has been set to the status **Released**.
- You need to use a separate service confirmation to confirm each service order.
- If you create a service confirmation as a follow-up, the system always suggests to take over quantity of the service order item. Any quantity that has already been confirmed will not be considered.

Create Standalone Service Confirmations

You can also create service confirmations without reference to service orders with the **Create Service Confirmation** app. For instance, you can create standalone service confirmations for unplanned services that don't relate to an existing transaction, such as additional services completed at the customer site.

i Note

You cannot create standalone service confirmations if you have activated item-based accounting.

Create Partial and Final Service Confirmations

You can create multiple service confirmations for a service order, for example, if you're required to make confirmations on a daily basis but require more than one day to complete an assignment.

- If you're working with such partial service confirmations or if multiple service confirmations are assigned to a service order, the system sets the service order to **Completed** automatically when the following conditions are met:
 - All service confirmations based on this service order have been set to **Completed**.
 - One of the completed service confirmations has been defined as the **Final Confirmation**.
 - There are no execution order items in the service order.

i Note

If the service order contains an execution order item, neither the execution order item, nor the service order, nor any other items in the service order can be set to **Completed** automatically.

- If all service confirmations have been set to **Completed** and no confirmation is defined as the **Final Confirmation**, you have to manually set the service order to **Completed**.

Cancel Service Confirmations

You can cancel a service confirmation if you have entered an incorrect service duration or a wrong service part. After you cancel a service confirmation, all the CATS postings or postings of goods issue are canceled as well. There are two ways to cancel a service confirmation:

- You can navigate to a service confirmation from the service order UI. On the service confirmation UI, choose **Cancel Confirmation** to cancel the service confirmation.
- You can also search for service confirmations. Then select the service confirmation you want to cancel and choose **Cancel Confirmation**.

i Note

If a service confirmation has the status **Completed** and has already been released for billing, you can only cancel the service confirmation after you delete or reject the billing document request (BDR) that has been created as the follow-up document of this service confirmation.

Key Features

Pricing

You can use an accounting indicator in a service confirmation item to determine different prices for products, based on absolute or percentage discounts and surcharges for services performed under warranty. You can define discounts and surcharges in pricing conditions. For more information, see [Pricing](#).

Cost Allocation

You can use an accounting indicator in a service confirmation item to determine the settlement receiver for costs. For example, if a service is covered by a warranty, you bill the quality department, but for a free-of-charge repair, you want to bill the sales department instead of the executing cost center.

Billing

You can initiate billing based on the service order items confirmed in service confirmations. For more information, see [Billing](#).

You can specify a service type and valuation type in a service product item confirmed in a service confirmation. This helps determine different rates for billing such as weekend work, overtime, travel time, senior or junior service representative.

Attachments

You can attach documents to service confirmations at header and item level. In Service, a full range of functions are available for you to manage document attachments. For more information, see [CRM Content Management \(Legacy\)](#).

Printing Service Reports

You can print reports that provide a summary of confirmed service activities using **Output** on the overview page. In Service, you can use output determination to set up the relevant output methods such as e-mail, fax, or letter. For more information, see [Output Determination](#).

Categorization

You can use categorization to define predefined content, which makes confirmation entry faster and allows automatic analysis. For more information, see [Multilevel Categorization](#).

Related Information

[Service Orders](#)

Items in Service Confirmations

You can learn about items in service confirmations in this topic.

Use

When you create a service confirmation as a follow-up transaction of a service order, you can select which item from the service order you want to confirm. Each item in a service confirmation corresponds to an item from the preceding service order. You use service confirmations to confirm items from the service order and to record the actual consumption of time and materials during service execution. This way, the actual costs of a service can be posted. For example, when you confirm a service item, time sheet entry is posted and the actual labor cost is updated.

Item Categories

The following table provides you with an overview of which item categories you can use in a service confirmation:

Item Type	Item Category
Service Item/Service Product Item	• SVCP: Service Item • SFCS: Fixed Price Service Item • SPCS: Service Item (Partial Billing) i Note The date profile SRV_RM_ITEM1 (SAP Confirmation Service Item) is assigned to the item category SVCP. This date profile contains date types such as SRVC_ACTUAL (Start of Work) and SRVC_ACT_T0 (Date Work Ends) that are used to record working time.
External Service Item	SVCW
Service Part Item	• SVCN: Service Part Item • SFCN: Fixed Price Service Part Item • SVCS: Stock Transfer Service Part • SVCG: Service Part Manual Goods Issue • SVCU: Advance Shipment of Procured Service Parts • SVCV: Advance Shipment of Stock Service Parts
Expense Item	• SVCE: Expense Item • SFCE: Fixed Price Expense Item • SPCS: Expense Item (Partial Billing)

Item Type	Item Category
Tool Item	• SVCT: Service Confirmation Tool Item • SFCT: Fixed Price Tool Item (Confirmation)

Key Features

Unplanned Items

You can add additional items that have not been created in a service order to a service confirmation using the **Unplanned Items** function. For more information, see [Unplanned Items](#).

Predecessor Items

You can add items to a service confirmation from the respective preceding service order using the **Predecessor Items** function. For example, if you have planned and created multiple items in a service order, but you have only created a service confirmation for one item and you still want to confirm other items from the service order, you can still take over other items from the service order into the same existing service confirmation. You can choose **Predecessor Items** when you edit the item list on the **Items** tab on the UI of **Service Confirmations**. When you choose **Predecessor Items**, the system automatically proposes the items that have been planned and released in the service order for selection and the item that you select is copied to the service confirmation.

Configurable Products

You can use configurable products or product variants to customize complex products with single-level variant configuration for your service processes. For more information, see [Configurable Products and Product Variants](#).

Service Rendered Date

You can use **Service Rendered Date** to record the date on which the expenses were incurred or a goods issue was posted for a stock service part. You can enter the date of the cost postings for the relevant service confirmation item types such as expense items or stock service part items. For service product items, the start of work date is used as the service rendered date. On the following UIs the service rendered date is displayed as the **Posting Date**: Material Document, Display Line Items, and Display Line Items in General Ledger.

If you enter a service rendered date, this date is used to determine how taxes are calculated for pricing-relevant items in service confirmations. When a service confirmation is released for billing, the service rendered date is used for billing document request items.

→ Remember

This date must be after the requested start date specified in the respective service order.

i Note

The service rendered date is usually a date in the past, but it can also be a date in the future. For example, as a technician, you can create a service confirmation before you execute the service. You can enter the date in the future when you'll execute the service and then record the actual time and materials consumed after you complete the service. In this case, you can ignore the warning message that asks you to check the service rendered date.

Batch Numbers for Service Parts

If you use batch-managed service parts, you can enter the batch number on the [Service Part Item Details](#) tab. Batches can only be entered at item level. Only one batch number can be entered for each item. The batch numbers entered are transferred to the goods movement posting.

Material Valuation for Service Parts

If you use materials with split valuation, you must select the valuation type on the [Service Part Item Details](#) tab in the [Material Valuation](#) field. Only one valuation type can be entered for each item. The valuation type is transferred to the goods movement posting.

i Note

The [Material Valuation](#) function is not the same as the [Valuation Type](#) that is available for the service product item and used to determine different rates for billing (such as senior or junior service representative). [Material Valuation](#) is used for costing purposes, for example, to differentiate costs of a material.

For more information about split valuation, see [Split Valuation](#).

Serial Numbers for Service Parts

You can enter the serial numbers for service parts on the [Serial Numbers](#) tab if the service parts have a serial number profile assigned with the serialization procedure MMSL in the product master data.

When you create service confirmations for service parts, you enter the serial number for each service part that is consumed. Once you set the status of the service confirmation to [Completed](#), the serial numbers are posted to the goods movement that has been created for the service part.

You control the usage of serial numbers for service parts in the [Define Serial Number Profiles](#) Customizing activity in Customizing for [Logistics - General](#):

- Assign the serialization procedure MMSL to the serial number profile. This ensures that the serial numbers are posted to the goods movement once the service confirmation is completed. For more information, see [Serial Numbers in Physical Inventory](#).
- Activate the usage of serial numbers by assigning the serial number usage 02 or 03 to the serialization procedure.

For more information about managing serial numbers, see [Serial Number Management \(LO-MD-SN\)](#).

Related Information

[Service Confirmations](#)

Unplanned Items

The [Unplanned Items](#) function enables an executing service employee to add additional items, that have not been created in a service order, to a service confirmation.

Use

Items that are created in a service order are planned items. While performing a service, an executing service employee might need to provide extra services, incur additional expenses, or use additional service parts that were not planned in the service order. In this case, you can use the [Unplanned Items](#) function to record these additional services in the service confirmation directly. When

you add an unplanned item in a service confirmation, the same unplanned item is created automatically in the preceding service order.

Unplanned items can be service items, expense items, stock service parts, or tool items.

i Note

Sales items and non-stock service parts cannot be used as unplanned items.

Unplanned items are pricing-relevant. That means that planned cost and revenue – both baseline and ongoing – can be calculated for unplanned items in a service order. This is especially useful if you use the event-based revenue recognition method **Recognize Revenue on Cost-Based Percentage of Completion (POC)**, and the unplanned items are created as subitems of the main item in a service hierarchy.

Prerequisites

To use the **Unplanned Items** function, the following prerequisites must be fulfilled:

- Activate item-based accounting in Customizing for **Service** under ► **Transactions** ► **Settings for Service Transactions** ► **Integration** ► **Enable Item-Based Accounting for Service** ►.
- Assign the item category usage UNPL (Unplanned Item) when you determine item categories in Customizing for **Service** under ► **Transactions** ► **Basic Settings** ► **Define Item Category Determination** ►.
- The service confirmation to which you want to add an unplanned item must be a follow-up transaction of a service order.
- Specify a **reference service order item** for an unplanned item in a service confirmation. The **reference service order item** must be a planned service item in the preceding service order of the service confirmation.

Reference Service Order Item

When creating an unplanned item in a service confirmation, a reference item is needed in the service order for controlling purposes. This reference item is called **reference service order item**. Only a **service product item** can be a reference service order item. When an unplanned item is created in a service confirmation and this same unplanned item is automatically created in the preceding service order, an account assignment object for the newly created unplanned service item is created.

i Note

You cannot use a price item as the **reference service order item** for an unplanned item.

Features

Service Contract Determination

You can use contract determination for unplanned items for the transaction type **Service Confirmation**. For more information on how to customize contract determination for service transactions, see [Service Contract Determination](#).

Executing Service Employee

The executing service employee cannot be determined automatically. You need to enter this manually.

Integration

The **Unplanned Items** function is integrated with SAP Field Service Management, logistics integration, and billing. However, the integration with procurement, material reservation, or credit checks is not supported.

Profit Center

The profit center of an unplanned item in the service confirmation is copied from the **reference service order item**.

Deletion of Unplanned Items

- You can delete an unplanned item in a service confirmation if the unplanned item hasn't been set to **Completed**.
- When you delete an unplanned item in a service confirmation, the corresponding unplanned item in the preceding service order is deleted automatically as well.

Apps Used in Service Order Management

You can use the apps in service order management to support your service processes.

Manage Service Orders

App ID: TBT116MSR

With this app, you can search for service orders and display the search results. You can navigate to service orders and display and edit service order details. You can also use this app to directly create service orders and process them. You can add items to service orders and edit them to detail the information needed to plan, execute, and bill for a service.

Key Features

You can use this app to do the following:

- Search for service orders using different filter criteria such as service order description, life cycle status, sold-to party, or posting date.
- Display your search results in the result list and search service orders within the result list.
- Customize the view of the result list. For example, you can personalize how many columns of data relevant for service orders are displayed in the list. You can add columns to display relevant data of service orders such as transaction type, employee responsible, or sales organization.
- Save or archive your searches.
- Navigate to service order details.
- Create service orders and add items to service orders.
- Edit all the data relevant for service order details at header level and item level.
- Trigger follow-up actions, for example, create service confirmations to confirm service execution.
- Copy service orders or reject service order items.

Supported Device Types

- Desktop

- Tablet

Situation Handling

This app uses Situation Handling to inform specific users about the following situation:

No executing service employee has been assigned to the service order item.

For more information, see [Situation Template: Missing Executing Service Employee](#).

Using automatic notifications, you can navigate to the critical service orders directly and take the appropriate action. You can also use the **My Situations** app to display your open situations in an overview table and close situation instances by setting the appropriate status manually.

For more information, see [Situation Handling](#).

Related Information

[Service Orders](#)

Create Service Order

App ID: TBT116MCR

With this app, you can create and process service orders. You can add items to service orders and edit them to detail the information needed to plan, execute, and bill for a service.

Key Features

You can use this app to do the following:

- Create service orders and add items to service orders.
- Edit all the data relevant for service order details at header level and item level.
- Display service order details and navigate to every items added to a service order and display the item details.
- Trigger follow-up actions, for example, create service confirmations to confirm service execution.
- Copy service orders or reject service order items.

i Note

You can also use the **Manage Service Orders** app to create and process service quotations. For more information, see [Manage Service Orders](#).

Supported Device Types

- Desktop
- Tablet

Situation Handling

This app uses Situation Handling to inform specific users about the following situation:

No executing service employee has been assigned to the service order item.

For more information, see [Situation Template: Missing Executing Service Employee](#).

Using automatic notifications, you can navigate to the critical service orders directly and take the appropriate action. You can also use the **My Situations** app to display your open situations in an overview table and close situation instances by setting the appropriate status manually.

For more information, see [Situation Handling](#).

Related Information

[Service Orders](#)

[Manage Service Orders](#)

Manage Service Quotations

App ID: TBT116QMSR

You can use this app to create, display, and edit service quotations.

Key Features

You can use this app to do the following:

- Search for service quotations using different filter criteria, such as the description of the service quotation, life cycle status, sold-to party, or posting date.
- Display your search results in the result list and search for service quotations within the result list.
- Customize the view of the result list. For example, you can personalize how many columns containing service quotation-relevant data are displayed in the list. You can add columns to display service quotation-relevant data, such as the transaction type, employee responsible, or sales organization.
- Save or archive your searches.
- Create service quotations and add the required items.
- Navigate to the details of the service quotation and the related items.
- Add execution order items to service quotations to trigger the planning of labor (operations) and material (components) consumption that is performed in the follow-up maintenance orders.
- Add price items to service items and execution order items as sub-items, which are used to reflect the surcharge for any additional services provided.
- Perform credit checks and simulate credit checks to determine whether the gross value of the items might exceed the available credit limit.
- Trigger follow-up actions, such as creating service orders.
- Send the service quotation to the customer.
- Modify the service quotation and send it to the customer again.

- Create a requote if additional service items or service parts are required to perform a service, which have not been specified in the accepted service quotation.
- Record whether the customer accepts or rejects the service quotation.

Supported Device Types

- Desktop
- Tablet

Create Service Quotation

App ID: TBT116QMCR

With this app, you can create and process service quotations.

Key Features

You can use this app to do the following:

- Create service quotations and add items to the service quotations.
- Edit the service quotation and the related items.
- Display the details of the service quotations and the related items.
- Add execution order items to service quotations to trigger the planning of labor (operations) and material (components) consumption that is performed in the follow-up maintenance orders.
- Add price items as sub-items to service items and execution order items that are used to reflect the surcharge for additional service provided.
- Perform credit checks and simulate credit checks to determine whether the gross value of the items might exceed the available credit limit.
- Trigger follow-up actions, for example, create service orders.
- Send the service quotation to the customer.
- Modify the service quotation and send it to the customer again.
- Create a requote if additional service items or service parts are required to perform a service and which have not been specified in the accepted service quotation.
- Record if the customer accepts or rejects the service quotation.

i Note

You can also use the [Manage Service Quotations](#) app to create and process service quotations. For more information, see [Manage Service Quotations](#).

Supported Device Types

- Desktop
- Tablet

Situation Template: Missing Executing Service Employee

Technical Name: SRVC_ORDER_MISSING_SERV_EMP

Consuming App: [Service Orders](#)

Template used for situation handling informs the service requester that no executing service employee has been assigned to the service order item.

i Note

This situation doesn't support service confirmation items.

For background information about the Situation Handling framework, see the Product Assistance for [Situation Handling](#).

Business Value

As a configuration expert for business processes (SAP business role SAP_BR_BPC_EXPERT), you can use this template for the following situation:

Customer service managers are informed automatically if the executing service employee is missing even though the requested start date is approaching for the service order item.

The manager can then instruct the Service Planner in SAP Field Service Management to assign the executing service employee, ensuring that the service is completed on time.

This standard template is useful when SAP Field Service Management integration is used, because in the non-SAP Field Service Management integration scenario, the executing service employee is assigned as a mandatory field in the service confirmation and is not updated in the service order.

Default Settings

The template comes with predefined settings. For the settings that aren't visible or self-explanatory, you can find information in the following sections. For generic information about how to configure situations based on this template, refer to the documentation of the [Manage Situation Types](#) app with which you can display and use the template.

For more information, see [Manage Situation Types](#).

Situation Trigger Object and Anchor Object

These technical settings define for which object a situation is displayed (anchor object) and the object based on which a situation is triggered (trigger object):

Trigger Type: Batch

Trigger Object: Service Order

Trigger Object View: I_SRVCORDITMMISGEXECTGEMPLSITN

Trigger Object Unique ID: ServiceDocumentItemUUID

Anchor Object: Service Order

Anchor Object View: I_SRVCORDITMEXECTGEMPLSITN

Description: The system checks for service orders and informs the manager that no executing service employee has been assigned to the service order item.

A batch job removes all situations not determined by the latest run every 24 hours, considering them resolved.

Conditions:

The condition in this standard template defines that a situation instance with the status **Open** is created and a notification is sent to the defined recipients if the system detects a service order that meets the criteria specified above.

This template provides the following filters that you can use to define your own conditions when you create situation types:

Filter	Description
Created By	The user that created the transaction.
Distribution Channel	The distribution channel defined for the service order.
Division	The division defined for the service order.
Exec. Service Emp.	The ID of the executing service employee.
Item Category	The category of the service order item or the service confirmation item for which the billing document was created.
Item Number in Doc.	The service order item number.
Sales Group	The sales group defined for the service order.
Sales Office	The sales office defined for the service order.
Sales Org. ID	The ID of the sales organization.
Sales Organization	The sales organization defined for the service order.
Service Organization	The service organization defined for the service order.
Transaction ID	The transaction ID of the service order.
Transaction Type	The transaction type of the service order. Possible values are: SV01 SV02
Responsible User ID	The ID of the service responsible employee.
Days until Requested Start Date	The difference between the requested start date and the current system date. Valid values are from 0 to 14.

For more information, see [Conditions](#).

Situation Display

When situations occur, users are informed about them by various texts, such as in-app situation messages or notifications on SAP Fiori launchpad. In this section, you can see all preconfigured texts.

For more information, see [Situation Display](#).

i Note

When you copy the template and adapt it to your needs, you can define the notification settings and change all text elements in this section. After making the changes in the original language, please remember to translate these changes into the other languages that you want to use. The template comes with translations that can be changed directly when you copy the template.

For more information, see [Translate Situation Types](#).

Recipients

In this section, you see the predefined settings for determining who is responsible for a situation instance. Based on these, you can define who receives a notification when a situation instance is triggered (if enabled) or who sees the instance in the [My Situations](#) app.

In this template, the recipient of the notification is determined using the rule **Notify responsible employee for the service order**. This means that if you have procured service parts or third-party service performers to provide a service and these part or services have not been reported as completely consumed, the employee responsible assigned to the service order is notified.

For more information, see [Recipients](#).

Situation Monitoring

Monitor Instances: Monitoring (the tracking of changes to situation instances) isn't enabled by default for situation templates. When you copy the template to create a situation type, you need to enable this option if you want to capture data for this situation type and enable the creation of [business situation events](#). You can view the data in the [Monitor Situations](#) app.

CORS

In the hub scenario, where the frontend and the backend systems are separate, the call to the reuse library from the backend to the frontend server may be blocked due to **Cross-Origin Resource Sharing (CORS)** settings. If this is the case, you need to maintain the **CORS** setting in the frontend server to allow request from the backend server in the transaction UCONCOCKPIT.

CORS should be enabled in UCONCOCKPIT transaction as per below settings:

- **Scenario:** HTTP Whitelist Scenario
 - **Context Type :** 4 - Cross-origin Resource Sharing
 - **Mode:** Active Check
1. Login to [Fiori Launchpad](#).
 2. Navigate to [Libraries](#).
 3. Open transaction UNCOCKPIT.
 4. Enable the following settings for context type Cross Origin Resource Sharing(CORS) whitelist rule:
 - HTTP Service Path: /sap/
 - Origin Host Rule: It must contain a matching of the back end server origin. (e.g. *.wdf.sap.corp)
 - Get: X
 - Head: X
 - Options: X

- Connect: X
- HTTP CORS Credentials: X
- Allowed Methods: GET,HEAD,OPTIONS,CONNECT
- Allowed Headers: X-XHR-LOGON,ACCESS-CONTROL-REQUEST-METHOD,ACCESS-CONTROL-REQUEST-HEADERS,ORIGIN,X-REQUESTED-WITH
- Exposed Headers: ACCESS-CONTROL-ALLOW-HEADERS,ACCESS-CONTROL-ALLOW-ORIGIN,ACCESS-CONTROL-ALLOW-METHODS,ACCESS-CONTROL-EXPOSE-HEADERS,X-XHR-LOGON
- Max Age: 60

i Note

Instructions may be valid for SAP or at the customer site.

For more information , please refer to [UI Integration With Situation Handling](#)

[Using the UCON Cockpit](#)

Related Information

[My Situations](#)

Service Request Management

You use Service Request Management to monitor and fulfill requests for service by your customers. Service requests can be used internally by companies where a department delivers services as well as in external customer-facing scenarios. They enable you to measure and evaluate the quality of services provided.

Service personnel or users can log service requests in the system through a service desk and then, if necessary, send them to the appropriate personnel for service delivery.

Integration

- You can use the integration with E-Mail Request Management System (ERMS). E-mails sent to the service desk through ERMS automatically generate a service request, and subsequent replies are also automatically linked to that service request. For more information, see [E-Mail Response Management System](#)
- You can connect [SAP Service Ticket Intelligence](#) to use machine learning to automatically categorize service requests.

Prerequisites

You have defined the following in Customizing for [Service](#):

- Service request-related transactions under ► [Transactions](#) ► [Basic Settings](#) ► [Define Transaction Types](#) ►. There is a predefined transaction type named [Service Request](#) (SRVR) provided in the standard system that you can use as a base.
- Settings for service requests under ► [Transactions](#) ► [Settings for Service Requests](#) ►

Features

The following features are available for Service Request Management:

i Note

These features are available for the technical object type BUS2000223 — [Customer Management Service Request](#)

- You can set up your business transactions to meet the specific requirements of Service Request Management.

❖ Example

- You can use expanded capabilities of multilevel categorization.
- You can define templates to facilitate the entry of commonly reported requests.
- You can optimize the information flow to keep your service personnel informed about the status of service requests, the customers and contacts and the reference objects.
- You can use helpful tools that enable you to monitor the service requests that are assigned to your organizations and service teams.
- You can use several functions that support you when processing service requests.

❖ Example

- Checklists for step-by-step task processing by one or more users
- Text management capabilities to provide text information for the internal and external documentation of the service request processing
- Forwarding functions that you can adapt to your requirements
- Processing log to record all processing steps
- You can monitor the processing of service requests.

❖ Example

- You can control the compliance with the agreements you have made with the partners involved in the performance of services.
- You can escalate service requests if they are not processed within a specific period of time.

Service Request

Definition

A user-related request for a defined service or for a new service.

Use

You can use this object to monitor and fulfill internal and customer service requests.

Service requests can be logged directly by users themselves or by service personnel and can, when necessary, be sent to the appropriate personnel for service delivery. They can also be monitored and analyzed to evaluate the performance of your service

Integration

Service requests can be integrated with knowledge articles. Once a solution to a problem has been found, you can create and release a knowledge article to make that solution (or workaround) available for future incidents and service requests.

More Information

[Service Request Management](#)

[Incident](#)

[Knowledge Article](#)

Incident

Definition

An event that is not part of the standard operation of a service and that interrupts or reduces the quality of that service.

The business transaction used to log such an event.

Use

You use this object to report an incident to the service desk when something in the infrastructure is not working as expected. Either the user or the service desk agent creates the incident transaction.

A service desk agent then prioritizes the incident and searches for a related knowledge article that might provide a solution or workaround. If the agent cannot find an appropriate solution or workaround, they dispatch and track the incident until a solution is found.

Within service management, incident transactions are based on the service request business object. For more information, see [Service Request](#).

Integration

Incidents can be integrated with the same objects as service requests.

More Information

[Knowledge Article](#)

Service Request Search

You can use the Enterprise Search to search for service requests using various search criteria, such as the life cycle status, sold-to party, or service team.

Use

To find a specific service request, you use the Enterprise Search in the header bar of the SAP Fiori launchpad. Before you enter any attributes in the search input field, you select the category [Service Requests](#).

Search Criteria

To search for service requests, you can use the following attributes:

- Service request ID
- Service request description
- Life cycle status
- Sold-to party description
- Employee responsible (first name and/or last name)
- Service team

You can also combine several attributes (you need to separate them by a space in the input field): For example, you can search for all service requests with the life cycle status [Open](#) that have been created by a specific employee.

Search Result

A list of service requests is displayed as a result. For the first service request, the display area includes the header and the details. To display the details of other service requests, you can expand the display area. You can also sort the search results according to your needs, export them to Excel, or share them by email.

To narrow down your search results, you can apply one or more filters, such as a life cycle status, employee responsible, or transaction type. The filters can be visualized as a list or a chart.

From the search results, you can choose [Display Service Request](#) in the display area of a service request to navigate to the selected service request. You can also use the [Search Service Requests](#) app to search for a specific service request.

Information for Key Users

Relevant Business Catalog

Your user must be assigned the business catalog `SAP_SERV_BC_EMP_ORDERS_T`.

Related Information

[Search](#)

[Searching Apps or Business Objects](#)

Item Determination

Use

The system creates an item for the service request and assigns a service product to the item based on the selected categories. The service product can be used to determine, for example, service level profiles and dates.

Prerequisites

You have made the following settings in Customizing for **Service**:

- Mapped transaction types to catalog categories in [▶▶Cross-Application Components▶▶Multilevel Categorization▶▶Assign Transaction Types to Catalog Categories▶▶](#)
- Configured item determination in [▶▶Transactions▶▶Settings for Service Requests▶▶Define Item Determination▶▶](#)

You have defined service products for categories on the UI in Service under [▶▶Service Operations▶▶Categorization Schemas▶▶](#). At the schema level, you can assign schemas to application areas. At the category level, you can assign service products to categories.

Features

Depending on the Customizing settings, the system determines the service product using multilevel categorization or by using the Business Add-In (BAI) [Product Assignment for Creation of Service Items](#). You can find this BAI in Customizing for **Service** under [▶▶Transactions▶▶Settings for Service Requests▶▶Business Add-Ins▶▶](#).

- If item determination has not been set up in Customizing and in the UI category modeler in Service, the BAI is used.
- If item determination has been set up for multilevel categorization but the service product cannot be determined based on the selected categories, the BAI is used.
- If item determination is done using multilevel categorization, the service product is assigned to the item when you save the service request.

If multiple service products are determined based on multilevel categorization, the system chooses the service product assigned to the lowest level category. If there is no service product specified for the lowest level category, the system searches for a service product at the next highest category level.

More Information

[Multilevel Categorization](#)

Business Context

Use

You can use this function to link documents, such as customer invoices, to a service request.

Integration

The business context function is integrated with the service request.

Prerequisites

You have made the settings in Customizing for **Service** under [▶▶Transactions▶▶Settings for Service Requests▶▶Define Settings for Business Context▶▶](#).

Features

You can use the business context to do the following:

- Display and maintain a list of links in service processes in Service
- Create and view links to objects in Service such as e-mail and knowledge articles
- Unlink processes and objects, depending on whether you have made the necessary settings in Customizing
- Navigate directly to linked back-end objects and processes

Objects that you want to link to the business context can be found or created in the following ways:

- Enter a known object ID to link an object in the list of **Business Context** objects.
- You can create a new object with or without dialog by selecting an object type and choosing **Create**.

Whether or not a dialog is used depends on the settings you make in Customizing or in the implementation of the object link class.

- You can search for objects in the agent inbox.
- Objects from the global data context of the IC can be automatically linked to transactions created during the current session.

If the user role does not have the object type configured for the business context, the object appears in the business context but you cannot view the object details.

Example

You are working on a service request and want to link it to a customer invoice. In the **Business Context** assignment block, you search for the invoice and confirm it. The system links the customer invoice to your service request and displays it under **Business Context**. From here, you can navigate to the customer invoice and review its details.

Setting up Service Request Transactions

Use

There are several features available that make it easier for your customers or your service teams to complete a service request.

Features

Multilevel categorization

You can use the expanded capabilities of multilevel categorization to categorize your service requests quickly and accurately and, for example, to determine service products automatically based on the selected categorization.

To use multilevel categorization, you must have made settings related to multilevel categorization under **Service > Cross-Application Components > Multilevel Categorization**

- **Define Application Areas for Categorization**
- **Assign Transaction Types to Catalog Categories**
- **Assign Item Categories to Catalog Categories**

For more information, see [Multilevel Categorization](#)

Templates

You can define templates to use for frequently reported requests.

Service Teams

You can automatically assign a service team to a service request based on a ship-to party specified in this service request. To be able to do this, you first define a service team profile using the Customizing activity [Define Service Team Profile and Assign Service Team Types](#) under [►Service ► Master Data ► Service Teams ►](#). You then assign this service team profile to the transaction type SRVR (service request) using the [Define Transaction Types](#) activity under [►Service ► Transactions ► Basic Settings ►](#).

In addition, if you use the enterprise organizational model, you have to assign partner determination procedure IT000002 to service requests using the [Define Transaction Types](#) activity. Compared to the previously used partner determination procedure IT000001, the new partner determination procedure includes the partner function [Ship-To Party](#) and excludes [Service Employee Group](#).

Processing Service Requests

Use

For the processing of service requests, several functions are available to help you to establish and optimize your internal processes and the communication with your customers.

Features

- You can customize the processing log that presents changes made to a transaction in a structured manner. For more information, see [Processing Log](#).
- You can use checklist functions, which you can use to define step-by-step processing of a task by one or more users. For more information, see [Checklist](#).

Checklist

Use

You can use checklists for step-by-step task processing by one or more users of transactions such as service requests. A checklist is a list of steps to be performed by assigned step partners. The checklist profile consists of a number of checklist-related settings, including the checklists themselves.

There are two types of checklists: simple and decision-based. Simple checklists are static and allow only a linear processing of their steps, while the flow of decision-based checklists is influenced by user input.

Integration

Checklists can be integrated with the following business transactions:

- Service request
- Service order

Prerequisites

You have completed Customizing in **Service**, under **Transactions** > **Basic Settings** > **Checklist Settings**.

Features

For processing the checklist in the appropriate assignment block, you have the following options (for both simple and decision-based checklists):

- Select the appropriate checklist from a list of all available checklists for the transaction type that you are creating or editing.
- When using simple checklists, you can insert steps manually into the selected checklist: both sequential and parallel processing of checklist steps is possible. To define a sequential processing manually, enter the previous step number.

When using decision-based checklists, adding manual steps is not possible. Also, parallel processing of checklist steps is not possible.

- Define checklist step details. For example, you can add the following information to the single checklist steps:
 - Enter work instructions and internal notes
 - Assign the parties involved
 - Schedule single steps
 - Add attachments with their properties and reference to URLs

Changes to a checklist step are logged in a change history.

- When using decision-based checklists you can trigger actions, for example, launch a new transaction, from individual checklist steps.
- Changing of the service request's attributes can result in the determination of another checklist. However, in this case, the system removes the information contained in the previous checklist from the transaction. That applies only until the checklist selection has been locked. For more information, see [Automatic Checklist Determination](#).

Processing Log

The most important actions that users perform on a service request are logged in the **Processing Log** assignment block. For example, the system logs changes to the step status or the partner. Actions that the users perform on a checklist step are logged in the change history of the checklist step.

Example

You have an employee who is moving from one office to another. To facilitate this move, several parties, for example, IT management and facilities management, must carry out specific tasks. You can create a checklist that describes each of these tasks in separate steps, and assign each step to the appropriate step partner. Those responsible can then see what tasks they need to carry out, and can indicate when each step is completed.

Customizing of Checklists

Use

In Customizing, you can define different checklists for all transactions types for which a step-by-step processing is reasonable.

Features

To configure the checklists, you have the following options:

- You can define checklist steps, simple and decision-based checklists, and checklist profiles independently. This enables you to reuse these objects for different purposes.
- You can create checklist profiles that contain a set of all checklists that you need for a special transaction type.
- You can create checklists that contain a list of steps that the responsible employee, the contact person or other predefined role shall process.
- You can create checklist steps that you can assign to different checklists. Additionally, when simple checklists are used, the steps are available in the input help of checklist steps in the checklist assignment block.

When defining checklist steps, you have the following options:

- You can define the sequence, in which the steps shall be carried out.
- You can define the objects that the users can maintain for every step. For example, you can define the following objects:

- Different text types for internal and external documentation and information.

If required, you can add fields to the checklist assignment block that contains these text types. The texts are displayed as a tooltip, so you can read the texts without opening the checklist step.

By mapping the checklist step texts with the text types of the business transaction header, you can integrate the texts in the notes assignment block. This allows you to monitor the message flow of checklist steps.

- Options to categorize the step progress, for example for reporting purposes.
- You can define rules to determine checklists automatically. For more information, see [Automatic Checklist Determination](#).
- You can define rules to determine the start and end date of the checklist steps.

For more information, see [Calculation of Checklist Step Duration](#).

- You can integrate checklists with SAP Business Workflow.

Additional Customizing of Simple Checklists

In addition to general customizing, you can use certain features and functions for simple checklists.

For editing the checklist in the appropriate assignment block, you have the following options:

- Define the sequence of the steps by changing the step number.
- Cancel steps that are not in status **Completed** and which are not set as **Mandatory** in Customizing.

When defining checklist steps, you can select whether a step should be mandatory or optional.

Additional Customizing of Decision-Based Checklists

In addition to the general customizing, you can use the following features and functions for decision-based checklists:

- Define so-called triggers and assign them to a checklist step. A trigger appears as a button on the UI and can change the status of the checklist step to **Completed**. It can also start actions, for example, sending a status report via e-mail. Actions

are defined in the Post Processing Framework.

- Add logical links to checklist steps. Logical links are either web links or links to other internal systems. You can use the Transaction Launcher to enable navigation to other systems.

Calculation of Checklist Step Duration

Use

For the steps of the checklist, you can create rules that determine the planned start and planned end dates automatically.

You can define rules to fill these dates with default values. The start date and the planned end are then determined automatically.

Features

- The system calculates the duration of the checklist steps according to the response times that you have defined in the **Response Profile** field.
- The determined dates are assigned to every step of the checklist. If required, you can overwrite these dates to schedule single steps in the time frame of the given duration. Especially for follow-on steps, manual scheduling can be helpful. The end date must not be higher than the automatically calculated value.
- The system uses the creation date of the checklist to calculate the dates.

i Note

The start date of the transaction itself does not influence the calculation of the dates in the Checklist steps.

- By editing the transaction, the system automatically recalculates the dates in the following situations:
 - When you change one of the attributes in the transaction that are used as key fields in the **Response Profile** field, for example **priority**.
 - When you change the response profile.

Automatic Checklist Determination

Use

You can use the automatic checklist determination to provide predefined checklists for transaction types such as service requests.

This feature is intended to support you during the initial phase of processing a service request.

Prerequisites

You have defined the fields for the automatic checklist determination in Customizing of **Service** under **Transactions > Basic Settings > Checklist Settings > Define Fields for Triggering Checklist Determination**.

Features

- You can define rules for the automatic checklist determination by using attributes for the transaction type, for example for the following:
 - Priority
 - Category
 - SLA response profile

Depending on the transaction details the users have entered, the system inserts the appropriate checklist or provides the users with a list from which they can select one.

- You define the rules for the checklist determination in the rule policy in the context [Checklist Determination](#) (CHECKLIST_ID).

If your rules determine more than one checklist, the checklists are listed in an input help.

- For technical reasons, the checklist selection must be locked at an early stage of transaction processing. This means that after this moment the system no longer calls the automatic checklist determination. If the rule has returned more than one checklist, the users are no longer able to change the selected checklist.

In Customizing, you can choose between the following options for determining this point in time:

- Locking the checklist after the transaction is saved for the first time.
- Locking the checklist after the first step is completed.

For more information, see Customizing activity [Transactions](#) > [Basic Settings](#) > [Checklist Settings](#) > [Define Checklist Profiles](#).

Text Management in Service Requests

Use

When processing service request transactions, you can use specific functions to add and provide text information. These functions support you in managing and monitoring the communication between the several partners involved in the process.

Reporters or specialized partners can add and change notes to describe problems or add additional information. For example, service personnel can use the notes for the following:

- Checking back on a problem
- Informing the reporters about the progress of a service request
- Providing a solution to a problem.

Prerequisites

- You have made the settings for text management in Customizing for [Service](#) under [Basic Functions](#) > [Text Management](#). In the standard system, the text object CRM_ORDERH and the text determination procedure [Service Request](#) (IT000001) are used for Service Request Management.
- The notes assignment block with ID GSTEXT is visible in the following overview pages:
 - Service Request/Incident (UI component SRQM_INCIDENT_H, View ID IncidentOV)
- If required, you have removed the other text and text log assignment blocks (UI Component ID GSTEXTEDIT).

Features

To map the particular requirements of service request management, the following functions for the **Notes** assignment block (UI Component GSTEXT) are available. For more information, see

- [Editing Notes](#)
- [Handling of Notes](#)

More Information

For more information about the standard text management, see [Text Management](#)

Editing Notes

Use

When editing notes, there are some functions available to meet the special requirements of text management in the context of service request management.

Features

Classifying texts

For editing and administration of notes, you classify texts in two ways:

- You create different text types. For example, you create text types that describe the following usages of a text:
 - Problem description
 - Solution description
 - Reply to requestor
 - Memo
 - Internal note
- You create keywords that the users can use to classify and to filter texts.

❖ Example

A user enters a deep technical analysis of a problem in a text with the type **Internal note**. Several internal notes are assigned to the problem. For this reason, the user adds the keyword **technical analysis** to that note. Now the processors of the service request are able to easily filter the list of texts to find the text that contains the technical analysis.

For technical reasons, the number of keywords that the users can assign to a note is restricted to 8. Therefore, you should consider how you want to design the usage of your text types and keywords.

❖ Example

You create different text types for the different usages and target groups of texts:

- Text types **Problem Description** and **Solution Description** to classify the communication at a special phase of the service request processing

- Text type **Internal note** for the communication between the different processors of a service request.
- Keywords **Technical Analysis** and **Test Case** to classify the content of a text type.

In Customizing, you can define the keywords for every combination of text object and text determination procedure. For more information, see Customizing for **Service** under [Basic Functions](#) > [Text Management](#) > [Define Keywords](#) .

Logging texts

For every text type, you can define in the Customizing for the text determination procedure whether it should be logged. For more information, see Customizing for **Service** under [Basic Functions](#) > [Text Management](#) > [Define Text Determination Procedure](#) . Select map **Definition of Procedure** and set the value of field **Changes (Control for Text Changes)** to P (= Log).

Text types that are not logged have the following features:

- You can only create them once for each language.
- You can change and delete them.

Text Templates

By editing a text type, you can include text templates. You can use this function to provide standardized texts for situations that occur frequently. After including a text template, you can continue editing the text and add further information.

You can create a text template by editing a text and saving it as a template. A text template requires the following attributes:

- ID and description
- Language
- Visibility level. You have to classify every text template whether it is to be used only by you or by all users (system).

Adding URLs

You can add URLs to the text body. The following URL schemas are supported and rendered as hyperlinks in the notes:

- http: and https: for adding web-sites or call of a Web application. Additional information in web-sites or by calling a Web application. The URL schemas http and https are supported for this function.
- mailto: for sending e-mails
- ftp: using the file transfer protocol

The system can automatically identify strings that you use to create a reference. To customize the URL recognition pattern, you need to develop your own class derived from class CL_CRM_TEXT_FORMAT_CONVERSION with its method SET_REGEX_PATTERNS and assign it in Customizing. For more information, see Customizing for **Service** under [Basic Functions](#) > [Text Management](#) > [Define Text Format](#) .

Handling Notes

Use

For the handling of notes the following features are available.

Features

Handling of note items

- The notes assignment contains the following basic functions:
 - Editing of texts
 - Displaying the notes history (log)

For this reason, you do not need to assign an additional text log assignment block to your service request overview pages.

- For tracking the whole notes history (log), you can sort and filter the texts that are assigned to a request by several categories such as the following:
 - Type
 - Creation Date
 - Created by
 - Keyword
 - Language

If for some notes multiple keywords are assigned, the users can enter wildcards in the custom filter to filter the notes.

- You can personalize the size of the notes assignment block quickly with the **Rows** pushbutton. This simplifies editing and displaying of long texts.
- You can simultaneously add notes even if the transaction is edited by another user (parallel text editing mode).

Hyperlinks

Even if a user is only allowed to display a note, he or she can use assigned hyperlinks to open the reference. For example, they can open a URL in the browser or in an application window.

Authorizations

With the authorization object CRM_TXT_ID, you can control which users are allowed to display or edit a text type.

Authorization object CRM_TXT_ID

Field	Description	Value
ACTVT	Activity	02Change, 03 Display
TEXTID	Text type from Customizing of Text Management	<Text type>
TEXTOBJECT	Text object from Customizing of Text Management	CRM_ORDERH or CRM_ORDERI

Visibility of Attachments

Use

You can use this function to control which user roles are allowed to display and edit documents that are attached to a business transaction.

❖ Example

For Service Request Management, you want to use this function to control the access to documents that are attached in the **Attachment** assignment block. The service requester should be able to do the following:

- Attach documents
- Open attached documents that are classified as **Information for customer**

He or she should not be able to open documents that are allowed for internal use only.

A button with a paperclip icon in the work area toolbar indicates that attachments are assigned to a service request. Choose the icon to navigate directly to the attachments assignment block.

Prerequisites

- For the authorization domain ATTACHMENT - BTX, you have defined different authorization scopes in Customizing of **Service** under ► **Transactions** ► **Basic Settings** ► **Define Authorization Scopes for attachments.** ►

In the standard, no authorization scopes are assigned to the authorization domain ATTACHMENT - BTX. If you do not define the authorizations scopes, the system does not check the authorizations. This also applies if you delete the authorization domain.

i Note

This function does not call the Business Add-In CRM_DOC_AUTHORITY that is described in Customizing of **Service** under ► **Basic Functions** ► **Content Management** ► **Business Add-Ins (BAdIs)** ► **BAdI: Authorization Check on Document Level.** ►

- You have added the authorization object CRM_AUTHSC to the required user roles (transaction PFCG).

Features

To control the access to attachments, you define in Customizing different authorization scopes for the authorization domain ATTACHMENT - BTX. By editing an attachment in the **Attachments** assignment block, the users assign the appropriate authorization scope.

❖ Example

You create the following authorizations scopes related to the different needs of the target groups:

- Information for agent
- Information for customers
- Internal message
- Confidential

In the authorization object CRM_AUTHSC, you can define for each authorization scope, which activities the users are allowed to carry out.

❖ Example

For Service Management purposes, you create the authorization scope **Information for agent**. The service requester can use attachments with this authorization scope to add additional information for the service personnel. The service requester should

be able to create, change, edit, and delete this attachment.

The following authorization levels are available with authorization object CRM_AUTHSC:

- Display: The user can open attached documents.
- Create or generate: The user can attach a document and define its authorization scope.
- Change: The user is able to change the properties of an attachment, including the authorization scope.
- Delete: The user can remove the reference to the attached document.

Authorization object CRM_AUTHSC

The following table describes the authorization object **Authorization Scope** (CRM_AUTHSC).

Field	Description
Activity	Describes, what the users are allowed to do.
Authorization domain	ATTACHMENT-BTX
Authorization Scope	Use the value help to get the authorization scopes that you have defined in Customizing.

If a user does not have the required authorization, the system works as follows:

- Display: The reference to the document is not displayed in the attachment's list.
- Create or generate, change, and delete: The authorization scope is not available in the input help of the field.

Processing Log

Use

This function provides you with an overview of all updates that have been made to a service transaction, such as changes to field values, links to other transactions, and executed actions. You can access all of this information in one area, making it easier to monitor the processing of a transaction.

Prerequisites

You have made the settings in Customizing for **Service** under **Transactions** > **Settings for Service Requests** > **Settings for Processing Log** .

Features

Processing Log

- Customizable content

You can customize the processing log to display different types of information, for example, old and new values for changed fields, who performed the update, and when the update occurred.

You can assign the required log types on the level of the transaction type and on the level of the category type.

- Default log type

You can set a default log type to be displayed for each transaction type. For example, when you open an incident, the processing log displays the change history. When you open a problem, the processing log displays executed actions.

- Navigational links

When there are changes in the form of linked transactions, attachments, or executed actions, the system provides a hyperlink to the corresponding objects.

- Assigned objects

You can log changes to the assigned service profiles or response profile.

- Checklists

You can log actions performed on a checklist, for example adding of a checklist, or changing the status of a checklist step, or automatic checklist determination.

- Authorizations

The users can display the log only for these notes or attachments that they are authorized for.

Data Determination Log

While processing a service request, you can use logs to analyze or trace the automatic determination of organizational data and partners.

- While determining organizational data, the system logs the organizational management objects by the sequence it has run through the organizational hierarchy. If you assign the user parameter CRM_ORGDETER_LOG to your user, the button **Determination Log** is available in the organizational data assignment block. By pressing the button, you can display the log.
- If changing a business transaction results in a new partner determination, the system logs the partner determination. This applies to the current session only.

If you select field **Log** in the partner determination procedure, the button **Determination Log** is available in the parties involved assignment block. In edit mode, press this button to display the log.

For more information, see Customizing of **Service** under **Basic Functions > Partner Processing > Define Partner Determination Procedure.**

Management of Objectives

Features

- Ability to classify transactions based on their urgency and impact to determine their recommended priority, which allows for accurate processing in accordance with agreed delivery times
- Escalation of service requests that sets escalation levels and dates to identify when the processing time of a service request has been exceeded. For more information, see [Escalation of Service Requests](#).
- Automatic calculation of the time spent on each stage of service request processing. For more information, see [Duration and Date Recording for Service Requests](#).
- Automatic determination of service level agreements (SLAs) to transactions based on determination procedures. For more information, see [Service Level Agreements \(SLA\)](#).

Duration and Date Recording for Service Requests

Use

This function allows you to define how to calculate how you record time spent on processing service requests, for example, an incident. You can record date information to evaluate if you are meeting the service profiles defined in the service level agreement (SLA).

Prerequisites

You have made the following settings in Customizing for **Service**:

- Defined date types and durations under ► **Basic Functions** ► **Date Management** ► **Define Date Types, Duration Types and Date Rules** ►.
- Defined status profiles under ► **Transactions** ► **Basic Settings** ► **Status Management** ► **Define Status Profile for User Status** ►.
- Assigned date types and durations to a date profile under ► **Basic Functions** ► **Date Management** ► **Define Date Profile** ►.
- Assigned status profiles and date profiles to transactions under ► **Transactions** ► **Basic Settings** ► **Define Transaction Types** ►.
- Defined date profiles for item categories under ► **Transactions** ► **Basic Settings** ► **Define Item Categories** ►.
- Assigned item categories to transaction types under ► **Transactions** ► **Basic Settings** ► **Define Item Category Determination** ►.
- Mapped the relationship between the user status, duration type, and date type under ► **Transactions** ► **Settings for Service Requests** ► **Define Settings for Durations** ►.

Features

- You can view the duration and date information for a service request in the **Service Level Agreements** and the **Dates** assignment blocks.
- You can define the durations that you want to record. There can be a one-to-one relationship between a user status and a duration or multiple user statuses can be combined together to calculate one single duration.
- The total duration is recorded. This means that if a transaction is in the same user status more than once, the duration is the sum of all these durations.
- You can set dates for date types when you change the user status of a service request. For example, the system sets the **Closing Date** of the incident when the user status is set to **Completed**.

Example

You want to record the total duration that a service transaction is in status **New**, **In Process**, and **Customer Action**. The time (duration) spent in status **New** and **In Process** represents the duration **Processing Duration**. The time (duration) spent in status **Customer Action** represents the duration **Customer Duration**. You combine the times spent in status **New**, **In Process**, and **Customer Action** to calculate the **Total Duration**.

Time Recording

Use

You can use this function to record your working time in business transactions such as service requests.

i Note

This function is not supported if you have activated item-based accounting.

Integration

The time recording function is integrated with the service request.

Prerequisites

You have assigned a service confirmation transaction type to the service request. The service confirmation transaction type will be used to automatically create a service confirmation during time recording. You can make the settings in Customizing for **Service** under **► Transactions ► Basic Settings ► Define Transaction Types ►** by doing the following:

- Select the service request transaction type and choose the business transaction category **116 - Service Process**.
- Enter the service confirmation transaction type used for time recording in the **Trans. Type Confirm.** (Transaction Type Confirmation) field.

Features

- When you enter your time spent working on a business transaction, the service product available in the transaction's item is automatically copied to the time record as read-only information. You can enter start date and time, work duration, employee responsible, as well as additional data such as service and valuation type.
- Once the first time record is saved, the system automatically creates a service confirmation as a follow-up transaction of the service request. The confirmation is created according to the copy-control settings defined in Customizing. Additional time records in the same service request will be captured as items within the existing service confirmation.
- You can enter time records at any time as long as the transaction does not have the status **Completed**.
- You can edit a time record as long as it has not been saved. After saving the time record, you must apply any necessary changes to the service confirmation to which the relevant service confirmation item is linked.

Escalation of Service Requests

Use

You can escalate service requests if they are not processed within a specific period of time. For example, incidents that are not resolved within 48 hours after the initial response can be escalated to the first escalation level.

Two different escalation levels can be used to notify the designated responsible parties that the response deadline has passed. The first escalation step can be triggered when the deadline is exceeded the first time, and the second escalation step can be triggered when the first escalation has exceeded its predefined deadline.

You use actions to control escalation and de-escalation.

Prerequisites

- You have defined an action profile in Customizing for **Service** under **Basic Functions > Actions > Actions in Transaction > Change Actions and Conditions > Define Action Profiles and Actions**.
- You have ensured that the following actions are available in each action profile:
 - IT_INCIDENT_DE_ESCALATION
 - IT_INCIDENT_ESCALATION
- You have assigned the action profile to a transaction type in Customizing for **Service** under **Basic Functions > Actions > Actions in Transaction > Assign Action Profile to the Business Transaction Type**.

Features

- Service request response times are controlled by the service level agreement (SLA) profiles assigned to the service request. For more information, see [Service Level Agreements \(SLA\)](#).
- You can schedule a batch job to escalate service requests automatically, for example, to set the escalation level and escalation dates. At the defined time, the batch job is automatically started and the action identified in the variant of RSPPFPROCESS is executed for all of the identified service requests.
- You can execute the de-escalation of a service request directly from the **Scheduled Actions** assignment block. This lowers the **Escalation Level** and removes the corresponding escalation date.

Activities

To set up the batch job to automatically escalate service requests that are exceeding their due dates, do the following:

1. Run transaction SE38.

For each action, create a variant for the program RSPPFPROCESS to identify the actions and service requests you want to include in the batch job.

- In **Application Key**, identify the range of objects that you want to include in the variant.
- Choose the relevant action profile, for example, IT_INCIDENT_MANAGMENT.
- Choose the relevant action definition, for example, IT_INCIDENT_ESCALATION
- Select the **Processing Without Dialog** checkbox.

2. For each variant (for example, escalation and de-escalation), define a batch job using transaction SM36 and specify when to run the job.

More Information

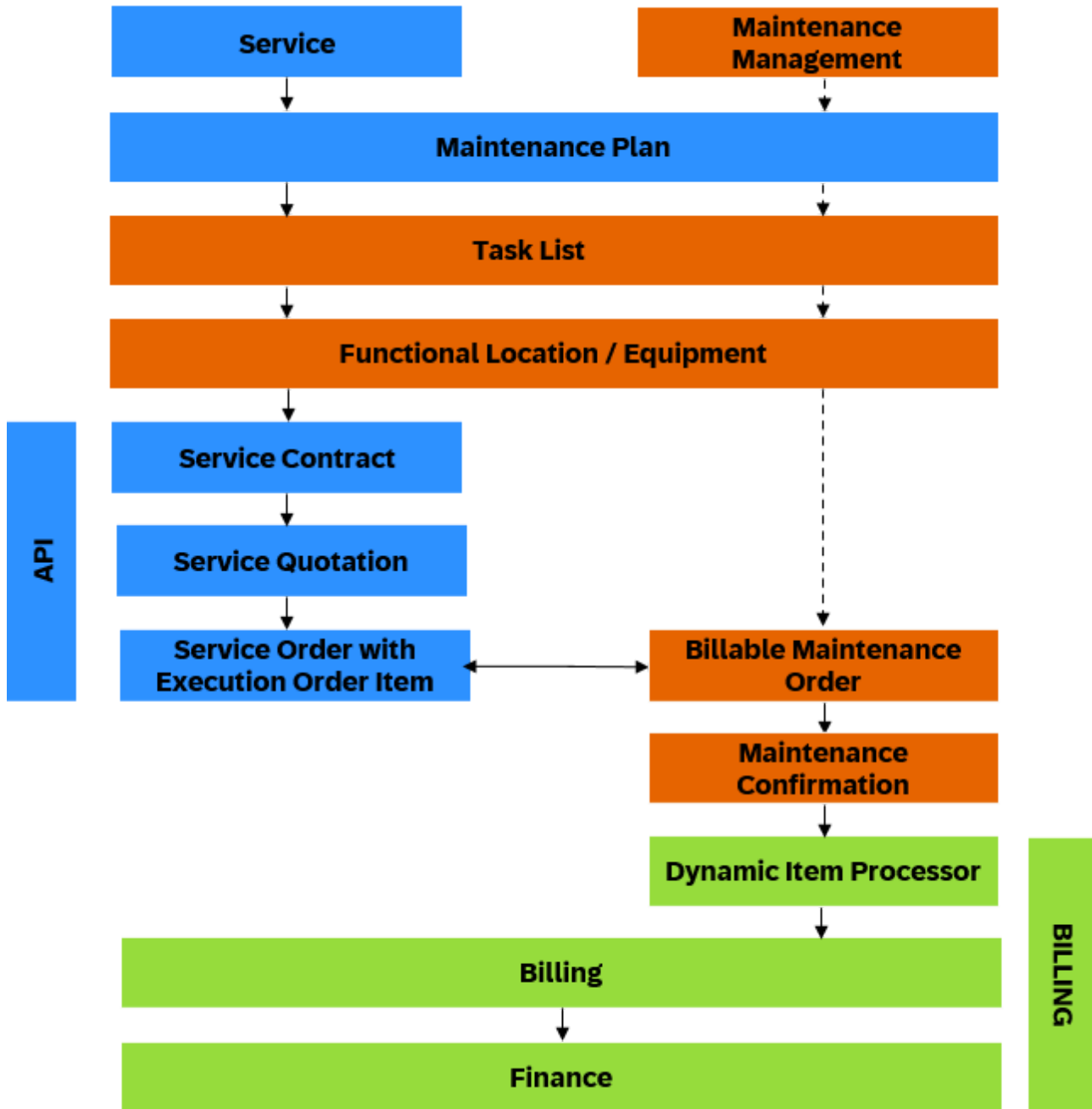
[Service Level Agreements \(SLA\)](#)

Service with Advanced Execution

Service with Advanced Execution is a process that is suitable for services which require extensive planning and execution. They generally have longer service times with a requirement to bill the time and service parts used for the service. The core characteristic of this process is the integration of the commercial aspects of Service with the planning and execution capabilities of Maintenance Management.

In the Service with Advanced Execution process both Service and Maintenance Management have specific roles to play. Service is concerned with understanding the scope of the service requirement, negotiating the price for the service, ensuring customer satisfaction, and determining profit margins. Maintenance Management is responsible for service execution planning, understanding the technicalities of the equipment to be serviced, scheduling, and executing the service as planned. This interaction between Service and Maintenance Management brings the best of both in one process.

Items created in the Service with Advanced Execution process are called **execution order items**. In corrective service, these items can be entered as part of a service quotation or a service order, with or without reference to a service contract. Execution order items can also be the result of a maintenance plan, which will facilitate preventive maintenance. Each execution order item creates a corresponding maintenance order. The maintenance order that is created enables the planning and execution of the service.



The above diagram provides an overview of the Service with Advanced Execution process. The process can be understood as interactions between elements of Service, Maintenance Management, and Billing.

Service

Service contracts are used to determine the terms of the service. A service quotation is used to provide a cost estimate for a requested service, and a service order is used to manage the service. A service quotation or service order can have multiple execution order items.

The maintenance order that is created when the execution order item is released is used to plan and execute the service. Maintenance plans in recurring service are used to schedule periodic service orders, thereby facilitating preventive maintenance.

Maintenance Management

Maintenance orders are used for the detailed planning of tasks and for monitoring their execution. The planning process involves task lists, scheduling, availability checks for material, and assignment of work centers. After the planning is completed, the actual service is performed and costs are recorded.

Billing

The billing process is carried out through the Dynamic Item Processor (DIP). A debit memo request is used to generate an invoice. The entire process works using item-based accounting in which service transaction items are recognized as account assignment objects to which costs and revenues are posted.

Use Cases

The Service with Advanced Execution process is typically suited to support the following scenarios:

Preventive Service

A service manager needs to be able to set up regular visits to the customer site based on service agreements. The service manager should be able to predefine a set of tasks that needs to be performed at different points in time to ensure the equipment is up and running.

In this way, a maintenance planner will be well informed in advance about an upcoming service so that tasks and resources can be planned accordingly.

Corrective Service

A service manager needs to be able to understand and capture customer service requirements, make a price proposal either as a cost estimate or as a final cost document. If the services requested are to be tailored to the customer's requirements, there needs to be a collaboration with the maintenance planner to come up with the cost estimate.

When the customer accepts the quotation, the service manager creates the service order, and the maintenance planner can go ahead with the planning and execution.

Billable Maintenance

Sometimes when a technician reports an issue during a regular visit or when an asset reports a malfunction, the maintenance planner is informed about the equipment not functioning correctly. The planner may not initially be aware whether the equipment is maintained by a customer or by the company. In such cases, the maintenance planner starts planning the work for maintaining the equipment. If the planner realizes that this equipment is owned by the customer, or the issue is due to customer misuse, then the planner needs to be able to bill the cost to the customer. In such a case, the maintenance planner needs to indicate that the work is chargeable so that the service manager can get involved in the process. The service manager can negotiate the price with the customer.

Prerequisites

The Service with Advanced Execution scenario can be setup and tested with the following scope items:

- 6AU (Service with Advanced Execution)

- 6F1 (Recurring Services with Advanced Execution)

Integration between Service and Maintenance Management

In the Customizing activity [Map Order Types](#) you need to maintain a mapping between transaction type and item category in Service and the maintenance order types in Maintenance Management. This mapping must be maintained to specify which order type would be used to create the maintenance order corresponding to an execution order item depending on the transaction type and item category.

► [Service](#) ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Integration](#) ► [Plant Maintenance Integration](#) ► [Map Order Types](#) ►

To map the priorities of the service order and maintenance order, you must carry out the Customizing activity [Map Priorities](#) under ► [Service](#) ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Integration](#) ► [Plant Maintenance Integration](#) ► [Map Priorities](#) ►.

To transfer information between the service order and the maintenance order, you must carry out the Customizing activity [Define Text Type](#) under ► [Service](#) ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Integration](#) ► [Plant Maintenance Integration](#) ► [Define Text Type](#) ►.

i Note

In Service with Advanced Execution, and also for Service in general, it is recommended to use service products of material type [SERV](#).

Item-Based Accounting

The Customizing activity [Enable Item-Based Accounting for Service](#) needs to be enabled to allow you to use Service with Advanced Execution. This activity can be found in the Customizing under ► [Service](#) ► [Transactions](#) ► [Settings for Service Transactions](#) ► [Integration](#) ► [Enable Item-Based Accounting for Service](#) ►.

Related Information

[Item-Based Accounting](#)

Basic Concepts

[Terminology](#)

[Execution Order Items](#)

[Billable Maintenance](#)

[Service Contracts](#)

Terminology

List of terms used in Service with Advanced Execution

- Preventive Service: Process concerned with the service of equipment on a regular basis with predefined time intervals or counter milestones. Regular service is used to keep equipment in optimum working condition and to avoid breakdowns and production downtime.
- Corrective Service: Process concerned with the service of equipment when a breakdown or failure or malfunction occurs.

- **Service Quotation:** Used to provide an estimate of the time required and the price to be paid for a particular service. The customer can accept or reject the service quotation.
- **Service Order:** Used to manage a service agreement between a service provider and a service recipient. It can be created as a standalone order, a follow-up from a service quotation, or as a call object from a maintenance plan.
- **Maintenance Order:** Used to plan and monitor execution of maintenance tasks that need to be performed.
- **Billable Maintenance Order:** Type of a maintenance order that allows billing and is used in Service with Advanced Execution.
- **Execution Order Item:** An item in a service transaction used in Service with Advanced Execution. The execution order item is generally associated with a billable maintenance order which supports in the planning and execution of the service.
- **Dynamic Item Processor (DIP):** A tool that the system uses to summarize data, such as planned costs or actual costs, into dynamic items in detailed quotation planning or resource-related billing.
- **Task list:** A sequence of individual maintenance activities that must be repeatedly performed by the service technicians.
- **Item-based accounting:** A type of controlling where service transaction items are recognized as account assignment objects to which costs and revenues are posted.
- **Distributed Execution:** Process by which the planning and execution of a service requirement is split into multiple billable maintenance orders.

Parent topic: [Basic Concepts](#)

Related Information

[Execution Order Items](#)

[Billable Maintenance](#)

[Service Contracts](#)

Execution Order Items

An execution order item is a service item that has capabilities like pricing, status management, assignment of organizational data, and reference object. It is used to define the scope and terms of a service requirement as part of the Service with Advanced Execution process.

Use

You can assign one or more execution order items to a service order template, service quotation, or service order. In general, an execution order item will be associated with a corresponding billable maintenance order. The maintenance order facilitates the planning and execution of the service.

Features

Billing Relevance

You can set the billing relevance for an execution order item depending on the type of billing scenario.

- **Not Relevant for Billing**

This billing scenario is specific for cases where the service will not be billed. For example, services under warranty or performed as goodwill.

- **Summarized Billing**

This billing scenario is specific to service with advanced execution and uses the dynamic item processor to bill services. In this scenario, a service is billed as a single main item. The billed value is determined from the aggregated cost of individual subitems together with a markup.

- **Itemized Billing**

This billing scenario is specific to service with advanced execution and uses the dynamic item processor to bill services. In this scenario, a service is billed as a list of subitems. The net price is the aggregated price of these subitems determined by pricing conditions.

- **Billing on Completion**

This billing scenario is used when the price for the service is determined from existing service contracts or pricing conditions irrespective of time and material costs.

- **Billing via Billing Plan After Release**

This billing scenario is used when billing dates are to be scheduled based on agreed milestones. This billing scenario can only be used for ad hoc billing.

Dynamic Items

Dynamic items are the data summarized according to defined criteria. Data is summarized - from defined sources, such as planned cost totals records or actual cost line items - using the dynamic item processor (DIP) and is regulated by the settings you enter in the dynamic item processor profile (DIP profile).

DIP profile is mandatory when the billing relevance is Summarized or Itemized. During billing, the DIP determines the items or lines in the debit memo request, based on the DIP profile entered in the execution order item in the service order.

When using service quotation for detailed planning and the planning is completed in the maintenance order, DIP-based subitems are entered in the service quotation under the higher-level execution order item. These subitems can be generated only on the basis of the DIP profile entered.

i Note

The DIP Profile for execution order item must have Usage **11 Quotation creation and sales pricing**. This usage also allows the creation of [Resource-Related Cost item](#).

The value of the DIP Profile can be defaulted from the predecessor document such as service contract or service quotation or from the item category customizing. It is also possible to manually enter or overwrite the default values.

Planning Plant and Service Organization

Planning plant and work center are essential in maintenance management transactions to plan and execute maintenance activities. Service organization and service team play a similar role within Service and can be treated as counterparts of planning plant and work center.

Planning plant and work center are available as additional fields in the execution order item of the service quotation and service order. These fields are mandatory in the maintenance order and are copied over from the execution order item. The planning plant and work center can be determined from the service organization unit and service employee group respectively. It is also possible to manually enter or overwrite the default values. The planning plant and work center are copied from the service quotation to the service order.

i Note

The planning plant entered in the execution order item must belong to the same company code as the sales organization of the service quotation or service order.

Maintenance Order Assignment Block

The billable maintenance order is linked with the execution order item and the maintenance order number is displayed in the Transaction History of the item.

When the billable maintenance order is created, the execution order item page displays an assignment block called **Maintenance Order**. This assignment block displays key attributes from the maintenance order as a quick reference.

Advanced Variant Configuration

You can use this function to configure execution order items in service orders. You can combine advanced variant configuration with configurable task lists to configure and manage customizable products.

To configure an execution order item, it should be marked as configurable in the material master data. You configure the product by assigning values to its characteristics. Dependencies for the characteristics, characteristic values, and the configuration profile are processed as you enter characteristic values. The characteristics can be mapped to the operations of a task list in a billable maintenance order. For more information, see [Configurable Products and Product Variants](#) and [Configurable General Maintenance Task Lists](#).

Credit Check

You can use the credit check feature for execution order items in a fixed price billing scenario. The credit check is automatically triggered when the service order status is changed to **Released** and saved. If the item passes the check, the service order has the status **Released**. If the item fails the credit check, it remains in the **In Process** status until it is released in the **Manage Documented Credit Decisions** app. For more information, see [Perform Credit Checks in Service Orders](#).

The credit check only observes the total credit exposure for a business partner. Credit exposure is the combined value of all items that are related to a business partner. For more information, see [Managing Credit Exposures](#).

For execution order items, credit check is only supported for the billing relevance **Billing on Completion**.

Intercompany Service Orders

An intercompany service order is a business transaction used to distribute a service between two companies with different company codes that belong to the same organization. Execution order items can be created as part of intercompany service orders. For more information, see [Intercompany Service Orders](#).

Integration with SAP GTS

Service with advanced execution can be integrated with SAP Global Trade Services (SAP GTS) to help you comply with global trade rules and reduce risk of non-compliance. With this integration, the creation and update of a billable maintenance order is prevented if the execution order item fails the trade compliance check. For more information, see [Integration with SAP Global Trade Services](#).

Other Aspects of Service with Advanced Execution

Service Order Templates

You can use service order templates to define reusable sets of service-related data that minimize the amount of time required to create a service transaction. A service order template describes only the scope of planned services and not the actual execution (customer and time frame). Execution order items can be added in service order templates. For more information, see [Service Order Templates](#).

Person Responsible

Person Responsible is available as an additional partner function in the execution order item of the service quotation and service order. It is an optional partner in the standard delivery. This can be determined based on the main reference object entered in the execution order item. The person responsible entered in execution order item is used to default the person responsible in the corresponding billable maintenance order. This is dependent on the Customizing in [►Service ►Basic Functions ►Partner Processing ►Map Partner Functions ►](#).

Planned Cost and Revenue

The costs for the service execution are determined from the billable maintenance order. However, the revenues of the service are determined on the service order.

The baseline planned cost is determined when the billable maintenance order is released. The baseline planned revenue in all cases, is always determined when the service order is released.

In addition, the continuous planned costs and revenues are also displayed in the fields called [Ongoing Planned Cost](#) and [Ongoing Planned Revenue](#). These fields are updated with every change in planning data of the billable maintenance order and every change of pricing data in service order.

When the billing is fixed price, the planned revenue is taken from the net value of the item. When the billing is resource-related, the planned revenue is determined from the planned cost of the billable maintenance order by using the [Get Calculated Cost](#) button.

Billing Status and Documents

Once the billing is started for the service order, then the fields [Transfer to Billing](#) and [Billing status](#) continuously reflect the latest status of the billing process. The billing documents such as Billing Document Request, Debit Memo Request and Invoice are also displayed in the [Transaction History](#) of the execution order item.

Parent topic: [Basic Concepts](#)

Related Information

[Terminology](#)

[Billable Maintenance](#)

[Service Contracts](#)

[Item-Based Accounting for Service with Advanced Execution](#)

Resource-Related Cost item

Resource-related cost items (RRCO items) help to determine the cost estimate for the service.

Use

Based on service requirements, operations and components are planned in the billable maintenance order. This planning on the billable maintenance order dynamically determines line items in the service quotation and service order as per the DIP profile on execution order item. These line items are automatically added as sub-items with item-category usage RRCO to the execution order item. Such sub-items are referred to as Resource-Related Cost (RRCO) items.

In Service Quotation

When you select [Send to Planning](#) in a service quotation for an execution order item, a billable maintenance order is created. This allows the maintenance planner to start planning for the service. When the planning is completed, the labor and material that is required for the service, is displayed in the [Items](#) assignment block of the service quotation as Resource-Related Cost (RRCO) items. The charged cost displayed in the [Price Details](#) of the RRCO item is taken from the billable maintenance order. Each RRCO

item is a subitem of the execution order item and all charged costs of statistical RRCO subitems are aggregated to the main execution order item.

The **Refresh Planning** button can be used to delete the existing RRCO items and create new ones depending on the latest changes made to the planning of the billable maintenance order and the current DIP profile maintained in execution order item.

In Service Order

When a service order is created as a follow-up from a service quotation, you have the option to copy RRCO items into the service order. However, the execution order item containing the RRCO items has to be copied as well. You also have the option to select only the execution order item and leave out the RRCO items. Furthermore, you can also select only some of the RRCO items (provided that you select the main execution order item). The unselected RRCO items will be set to the status **Completed** and will not appear in the pop-up window if the quotation is still in-process and the create follow-up function is triggered again. When you use the **Refresh Planning** button the existing RRCO items will be deleted and the updated items from the billable maintenance order will be generated.

When you add a new execution order item to a service order that has not been created as a follow-up, you can use the **Refresh Planning** button to generate the RRCO items directly in the service order.

Configuration

You need to select the item category usage **RRCO (Resource-Related Cost)** for the determination of resource-related cost items. You do this in the Customizing: **►Service ► Transactions ► Basic Settings ► Define Item Category Determination ►**

Properties

Resource-related cost items have the following properties:

- Their life cycle status mirrors that of the main execution order item and can't be changed manually.
- They can be copied from service quotation to follow-up service order.
- They will inherit the controlling object of the main execution order item.
- They can be included in pricing and planned revenue calculations.
- Their charged cost is aggregated to the main execution order item.
- They are not relevant for contract determination and can't be assigned with any service contract. They inherit the effects of the contract assignment on the main item.
- They are deleted on replanning.
- They can't be used as price items.
- They will not appear in the output control PDF file.
- They have a limited number of assignment blocks in their overview page.

Example

1. A customer has requested you to provide a quotation for the maintenance of a centrifugal pump.
2. You start by creating a service quotation with the relevant details and adding the maintenance of the pump as **Item 10** with **Execution Order Item** as the item category.
3. After setting **Item 10** to **In Process** and selecting the **Send to Planning** checkbox, you save the quotation.

4. A billable maintenance order has now been created and the maintenance planner can begin planning the service in the [Change Order \(IW32\)](#) application.
5. The maintenance planner determines that this maintenance involves the following three operations and adds it to the [Operations](#) tab in the billable maintenance order:
 - dismantling the pump – 2 hrs
 - inspecting the pump – 4 hrs
 - reassembling the pump – 2 hrs
6. Considering the age and operational life cycle of the pump, the planner determines that the bearings and the mechanical seals need to be replaced. This is added under the [Components](#) tab.
7. Once all the planning information is entered, the planner selects [Planning Completed](#). This sets the maintenance order status to [ORAI \(Order Approval in Process\)](#).
8. When you refresh the service quotation you will see that three new items have been added. [Item 20](#) represents the total time for the service, 8 hrs, [Item 30](#) represents the replacement bearings and [Item 40](#) represents the replacement seals. These three items are RRCO items of [Item 10](#).
9. You can use the resource-related cost item information together with the charged cost to provide the best price estimate for the customer.
10. When you set [Item 10](#) to [Released](#), the status of the resource-related cost items will also change to [Released](#).
11. If the customer accepts the quotation and you create the follow-up service order, you will have the option to copy the execution order item along with the resource-related cost items into the service order.

Billable Maintenance

Maintenance Management involves the inspection, maintenance, and repair of a technical object. Maintenance Management does not usually include the capability to bill a service provider. However, in certain maintenance scenarios a maintenance order needs to allow the possibility of billing. But this billing requirement comes after the maintenance order has been created. So to allow for the possibility of billing, a corresponding service order is created. Using the service order you can generate a billing document on completion of the service.

You trigger the service order creation process by selecting the [Billable](#) checkbox in the top right corner of the [Change Maintenance Order \(IW32\)](#) screen.

When the [Billable](#) checkbox is selected you can enter the sales data and service details. The fields in these sections allow you to enter details such as sold-to party, sales organization, distribution channel, service product, and quantity. Once the planning has been completed the maintenance order is released and the work can be done by the technician. After confirmation that the service is done, the order can be set to completed. The billing process can then be initiated.

i Note

The sales data and the service sections appear only when the [Billable](#) checkbox is selected. You cannot deselect the Billable checkbox once selected.

Parent topic: [Basic Concepts](#)

Related Information

[Terminology](#)

Service Contracts

Service contracts are outline agreements with business partners that define the services offered for a particular period. Service contract management has been integrated with Service with Advanced Execution to support the execution order items in service quotations and service orders.

Item-Based Accounting

Item-based accounting is supported in service contracts. Service contract items are recognized as account assignment objects to which costs and revenues can be posted. Item-based accounting must be enabled as a prerequisite for the Service with Advanced Execution process.

Contract Determination

For all execution order items, you can determine service contracts based on the product in a service contract item. The service contracts whose items contain the product that you have specified in an execution order item can be determined for your selection.

DIP Profile

In the Service with Advanced Execution process, service contracts contain an additional field of information called the dynamic item processor (DIP) profile. The dynamic item processor is a tool that the system uses to summarize data into dynamic items in resource-related billing.

How to display the DIP profile

1. Launch the WebClient User Interface and select the business role **Service Professional** (S4C_SRV_PRO).
2. Choose the **User** icon and select the **Personalize** option.
3. Choose **Personalize** Settings.
4. Choose **Enable configuration mode**.
5. Enter a service contract.
6. Choose **Billing** and select the **Show Configurable Areas** icon.
7. Choose the DIP profile from **Available Fields** and move it to the displayed area.
8. Choose **Save and Close**.

Parent topic: [Basic Concepts](#)

Related Information

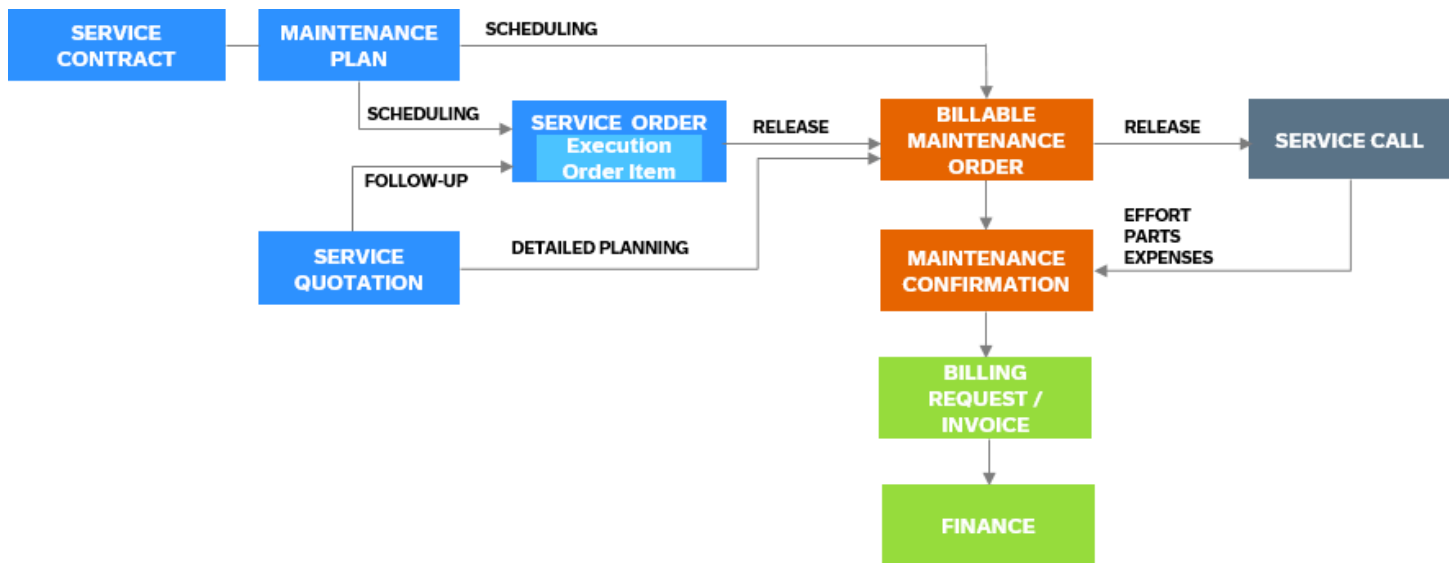
[Terminology](#)

[Execution Order Items](#)

[Billable Maintenance](#)

Processes in Service with Advanced Execution

Get an overview of the various processes in Service with Advanced Execution.



The above figure is an illustration of the Service with Advanced Execution process across Service and Maintenance Management.

The core aspect of Service with Advanced Execution is the connection between the execution order item and the maintenance order. This can be seen in the following scenarios.

- [Service with Advanced Execution in Service Quotation](#)
- [Service with Advanced Execution in Service Order](#)
- [Service with Advanced Execution in Recurring Service](#)
- [Service with Advanced Execution in Maintenance Management](#)

Service with Advanced Execution in Service Quotation

You can use Service with Advanced Execution to create a service quotation to provide customers with an estimate of the time and cost of a service.

Use

When customers want to get an estimate of the time and cost involved in the service of a technical object, you can create a service quotation to provide this information. The service quotation helps customers and service providers to plan the anticipated services at an early stage. Customers can plan their budget and their production capabilities, while service providers can plan their revenue and service resources such as labor and service parts.

You use the [Manage Service Quotations](#) app to create a quotation.

Extent of Planning

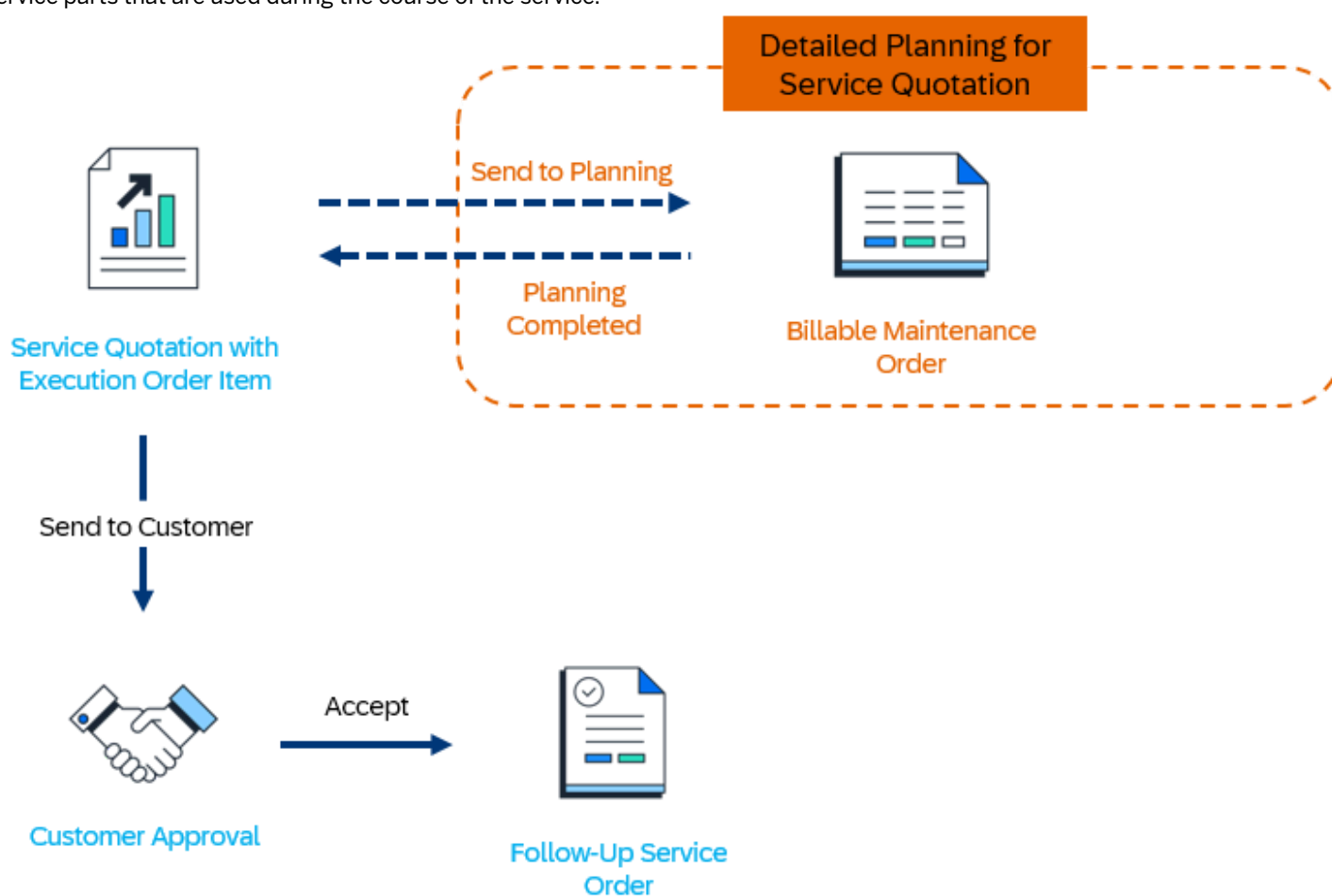
The first step to creating a service quotation is to determine the extent of planning involved in the service. For a service that requires less detailed planning, a quotation can be created and sent to the customer, who will then approve or reject the proposal. But for a service that requires extensive planning it is necessary to account for this planning in the quotation. To that effect a maintenance order is created for each item in the quotation. The maintenance order provides the information on the planning and

execution of the actual service. Once this is factored into the quotation, the quotation can be sent to the customer. For more information, see [Detailed Planning in Service Quotations](#).

When you create a quotation you can indicate the type of billing by using the **Billing Relevance** attribute. However, when creating a service quotation the billing relevance does not carry much significance, irrespective of whether detailed planning is required or not. It only offers a possibility to propose at an early stage what billing relevance can be used eventually. Depending on the billing relevance, the app that is used for billing can also be different.

Billing Relevance	Type of Billing	Billing App
Billing on Completion	Fixed price billing	Release for Billing
Summarized Billing or Itemized Billing	Resource-related billing	Create Resource-Related Billing Request: Service Order

In fixed price billing, the cost of the service is agreed upon with the help of an existing price master. Irrespective of the labor or service parts used, the cost will not change. In resource-related billing the cost of the service is determined based on the labor and service parts that are used during the course of the service.



Creating a Service Quotation

You can create a service quotation using the [Manage Service Quotations](#) app.

In the service quotation, make the following entries:

1. Enter general data such description and sold-to party.
2. Enter the sales organizational data.

i Note

This is custom documentation. For more information, please visit [SAP Help Portal](#).

Depending on customizing, automatic selection of organizational data is possible.

3. Under **Parties Involved** enter the service employee group and employee responsible.
4. Add items and select **Exec. Ordr Itm (Qtn)** (Execution Order Item (Quotation)) as the item category.
5. For each execution order item, you need to select a billing relevance. You need to enter a DIP profile if you select summarized billing or itemized billing as the billing relevance.

i Note

- You can create a maintenance order from a service quotation to facilitate the early planning of a service. For more information, see [Detailed Planning in Service Quotations](#).

Send to Customer

Once you have finalized the service quotation, you can choose **Send to Customer**. The service quotation is sent to the customer using output management. The status of the service quotation changes to **Sent to Customer**, the status of the items is set to **Released**.

Accept the Service Quotation

Once the customer agrees to the service quotation, choose **Accept** to record the customer's decision. The status of the quotation changes to **Completed**, the items accepted are set to **Completed**. A service order is automatically created with an **In Process** status for each execution order item.

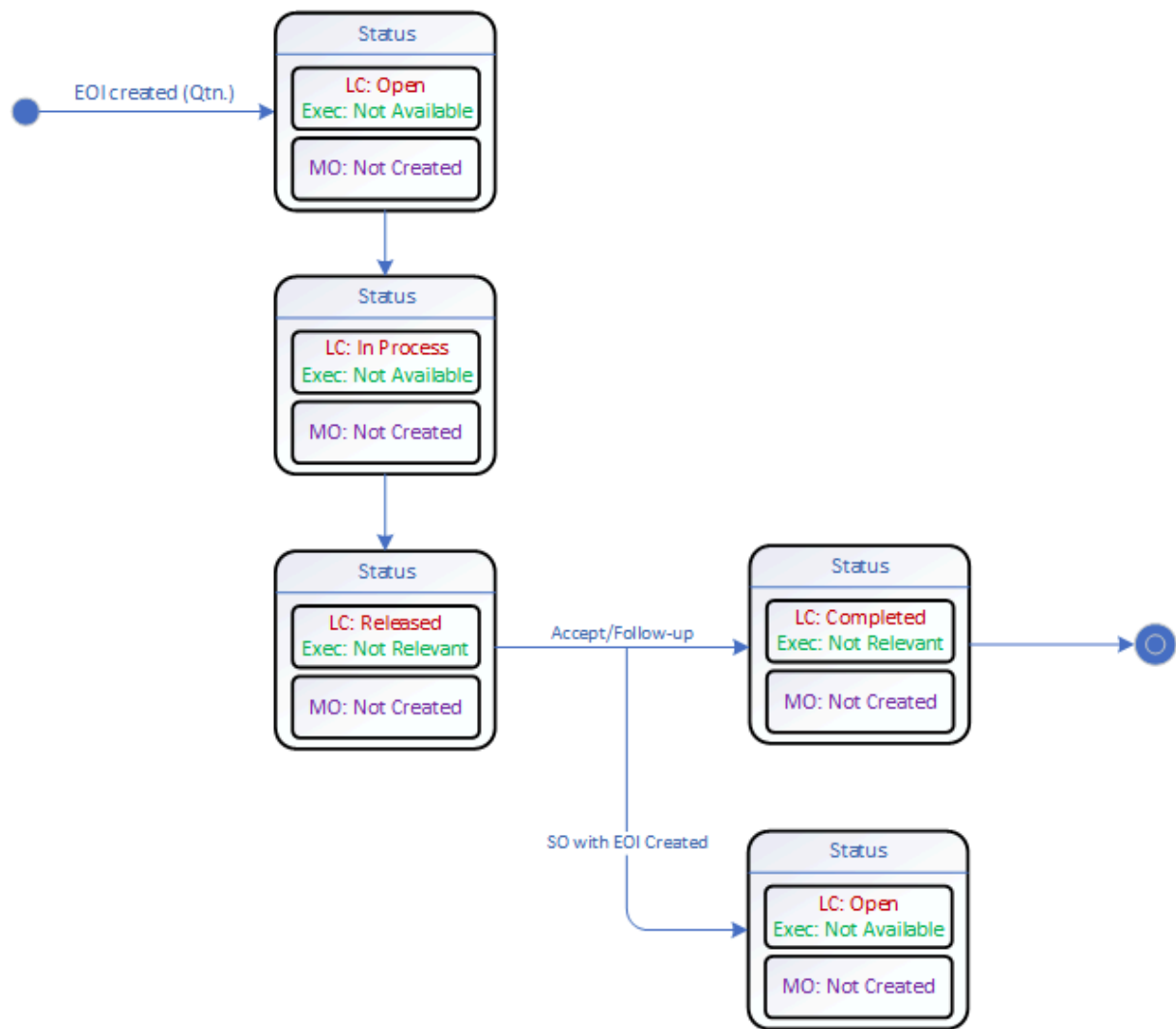
Reject the Service Quotation

If the service quotation is rejected, its status changes to **Completed** and the items are set to **Completed** and each item execution status is set to **Canceled**.

Replanning

After the service quotation is shared with the customer, there could be a need to revisit the service quotation. When such a need arises, you can change the status of the quotation from **Released** to **In Process**. This allows you to make changes without having to create a new service quotation.

Overview of Service Quotation Status



Note:
1. Accept or Create Follow-up creates SO with one EOI.

SO	Service Order
Qtn	Quotation
MO	Maintenance Order
LC	Lifecycle
Exec	Execution

Status	Description
Open	This is the default status that all service quotations get after they are created. It indicates that you are free to edit to the quotation.
In Process	In this status the quotation and its items can be edited.
Released	When the quotation items have been finalized it is set to Released .
Completed	When the service quotation has been accepted by the customer, the status is set to Completed . This status can also be manually set if the quotation is rejected by the customer.

An item execution status indicates in which stage of execution planning or execution the item actually is in. This status is important when a maintenance order has been created for an item.

Related Information

Detailed Planning in Service Quotations

Sometimes it is necessary to begin detailed planning for a service in advance. Detailed planning can provide better cost and time estimates for the actual service. The price can be set based on the cost calculations and the service quotation can be sent to the customer for approval. This is especially useful in service jobs that require a lot of time and service parts to complete. You can initiate this planning with the creation of a billable maintenance order corresponding to an execution order item.

The maintenance order enables the detailed planning for the service requirements in terms of individual tasks, operations and components and thus provides cost estimations. The process also supports an approval workflow for the planning.

Each execution order item is capable of creating its own billable maintenance order. By choosing **Plan All Items** you can create billable maintenance orders for all the items in the quotation. You can also select items for which you want to create billable maintenance orders by selecting the checkbox under **Send To Planning**.

The execution order item must be in **In Process** status to create the maintenance order. You can also trigger **Send To Planning** automatically by selecting the **Send to Planning Immediately** checkbox in the Customizing activity **Map Order Types**.

Send to Planning

The **Send to Planning** action triggers the creation of the corresponding billable maintenance order. Some data is automatically copied into the maintenance order from the execution order item in the service quotation. For example, planning plant, work center, header reference object, and object list. Partners and texts are copied based on the Customizing.

Successful creation of the billable maintenance order sets the maintenance order status to **ORAR (Order Approval Required)**. However, in case of failure to create maintenance order, errors will be raised in the service quotation and the **Send to Planning** checkbox is reset automatically. In this case, the errors can be corrected by adjusting the master data or configuration data and you can retry the Send to Planning function again.

Planning Completed

When the planning is completed, you can select the **Planning Completed** button and set the status of the maintenance order to **ORAI (Order Approval in Process)**.

With the planning completed, you can use the cost estimates to negotiate and finalize a price for the service in the service quotation. You can then either send the quotation to the customer for acceptance or could trigger an additional cycle of planning.

Re-planning

If you want to request an adjustment in the planning data, you can use the **Send to Planning** function again. The same billable maintenance order which was created before is editable again. The status of the maintenance order is now set to **ORRJ (Order Rejected)**. It is still not possible to release the maintenance order and the service quotation cannot be sent to customer since it is in planning.

The planning is adjusted based on the requirements of the service, and the **Planning Completed** button is selected again. This sets again the status of the maintenance order to **ORAI (Order Approval in Process)**.

This cycle of planning adjustments and corrections in maintenance order can be performed multiple times, until the planning data is addressing the requirements of the service and the requests of the customer.

Send to Customer

When the service quotation is sent to the customer, the execution order item status is set to **Released**. It is possible to configure the output management such that the DIP-based subitems can also be included in the service quotation document which is sent to the customer.

At this stage, the customer could request for an adjustment in the planning data or the terms of the quotation. You have the option to revoke the release of the execution order item and set the status back to **In Process**. With this, it is now possible to do one more cycle of Send to Planning, Planning Completed and Send to Customer.

If the customer rejects the service quotation, it is set to status **Canceled** and the corresponding maintenance order is set to **Do not execute**.

If the customer accepts the service quotation, it is set to status **Completed** and the maintenance order is set to status **ORAP (Order Approved)**. Simultaneously, the follow-up service order is created and the reference of the existing maintenance order is carried over into the service order execution order item so that the planning data can be retained as part of the same maintenance order.

With the status ORAP (Order Approved) on the maintenance order, the order is editable again and it is open for release. It is important to note that the Accept action on the service quotation and the subsequent setting of the ORAP status, sets the baseline planned cost.

Refresh Planning

During the planning phase of the service, the time and materials required for the service are planned in the billable maintenance order. Through the dynamic item processor (DIP) this planning information is retrieved by the service quotation as subitems of the execution order item when the maintenance order is set to **Planning Completed**. Sometimes, an error may prevent the subitems from being displayed in the service quotation or the DIP profile may need to be changed. By using refresh planning you can retrieve the planning information once more.

To use refresh planning you need to ensure that:

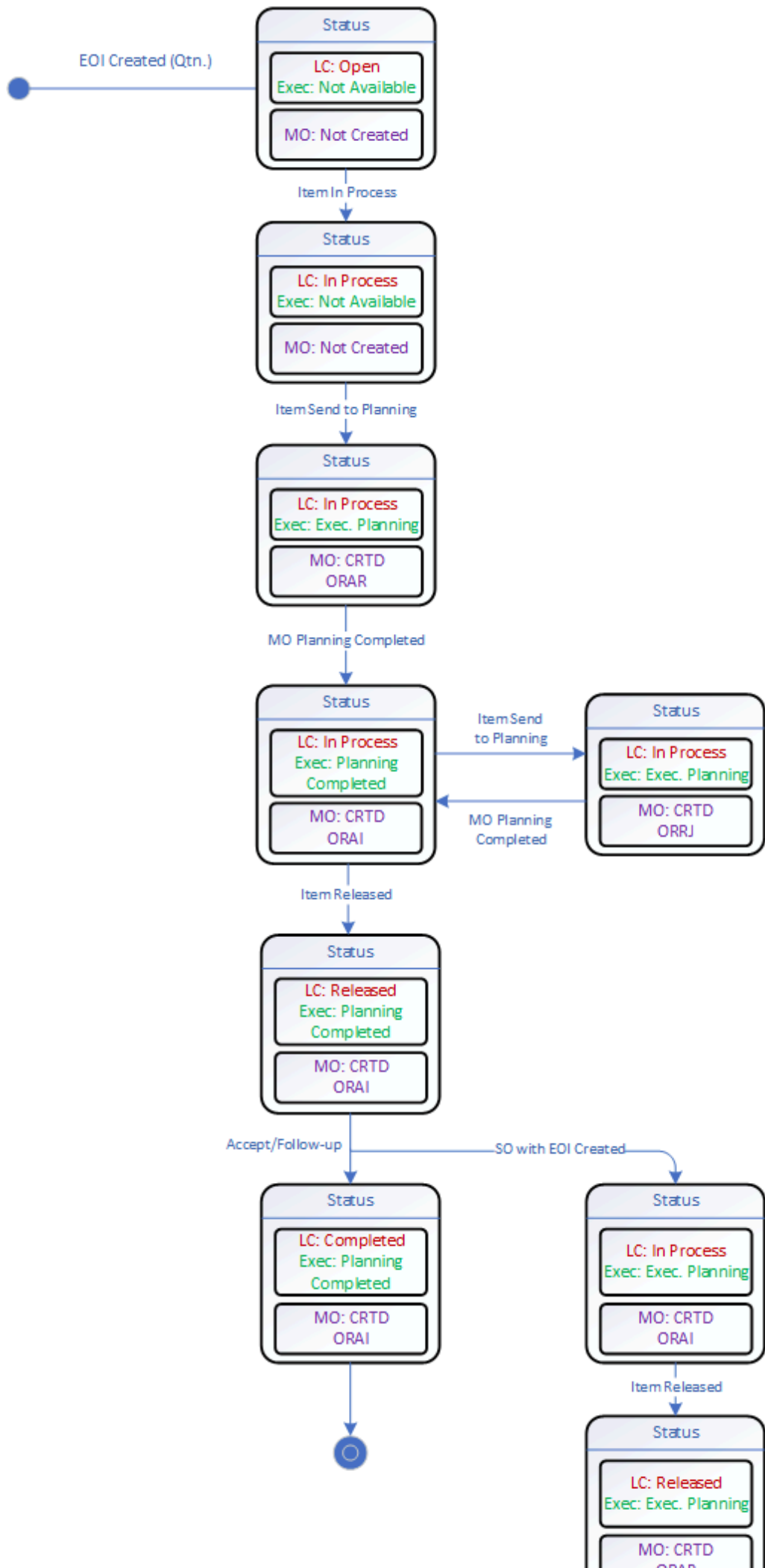
- Life cycle status of the execution order item is **In Process**
- Actual planning is completed and the execution status is **Planning Completed**

i Note

The **Refresh Planning** button is disabled when the service quotation is accepted.

Overview of Status

The following diagram and table provide an overview of the maintenance order status and the corresponding item execution status.



1. EOI cannot be released when execution status is Execution Planning.
2. Accept or Create Follow-up creates SO with one item.

SO	Service Order
Qtn	Quotation
MO	Maintenance Order
LC	Lifecycle
Exec	Execution
CRTD	Created
ORAR	Order Approval Required
ORAI	Order Approval in Process
ORAP	Order Approved

Status on Maintenance Order	Item Execution Status	What Happens in this Status
CRTD (Created)	Execution Planning	All maintenance orders are created in this default status.
ORAR (Order Approval Required)	Execution Planning	The billable maintenance order corresponding to the item has been created and the maintenance planner can start with the planning. In this status, the maintenance order can't be released, and the service quotation can't be sent to the customer.
ORAI (Order Approval in Process)	Planning Completed	The maintenance planner completes the planning and reverts to the service representative to accept or reject. In this status, the maintenance order can't be edited.
ORRJ (Order Rejected)	Execution Planning	The customer service representative rejects the planning and sends the item back for re-planning. In this status, the maintenance order can't be released, and the service quotation can't be sent to the customer.
ORAP (Order Approved)	Planning Completed	The customer accepts the service quotation. A service order is created for the item. The maintenance order can be edited or released.

Service with Advanced Execution in Service Order

You can use Service with Advanced Execution to create service orders that specify the details of a service between a service provider and recipient.

Use

You can use service orders to record information about a service that has been agreed between the service provider and service recipient. Service orders provide the necessary information regarding the service process and the cost that is incurred. Execution order items in the service order create a corresponding billable maintenance order that facilitates the planning and execution of the service.

In Service with Advanced Execution, you can create a service order:

- As a standalone service order

The service order is created in the **Open** status and can be set to **In Process**. When the execution order item is released the billable maintenance order is created.

- As a follow-up transaction, from an accepted service quotation

The service order is created in the **Open** status and can be set to **In Process**. When the execution order item is released the billable maintenance order is created.

- As a follow-up transaction, from an accepted service quotation with detailed planning

The service order is created in the **In Process** status. The billable maintenance order that was created during detailed planning of service quotation is carried over to the service order.

- As a call object in a service plan

The service order is created in the **In Process** status simultaneously with the billable maintenance order during scheduling of maintenance plan.

- As a billable maintenance order as part of billable maintenance

When a maintenance order is retrospectively marked as billable, a service order is created with an execution order item. The service order is created in the **In Process** status.

Creating a Service Order

You can create a service order using the **Manage Service Orders** app.

In the service order, make the following entries:

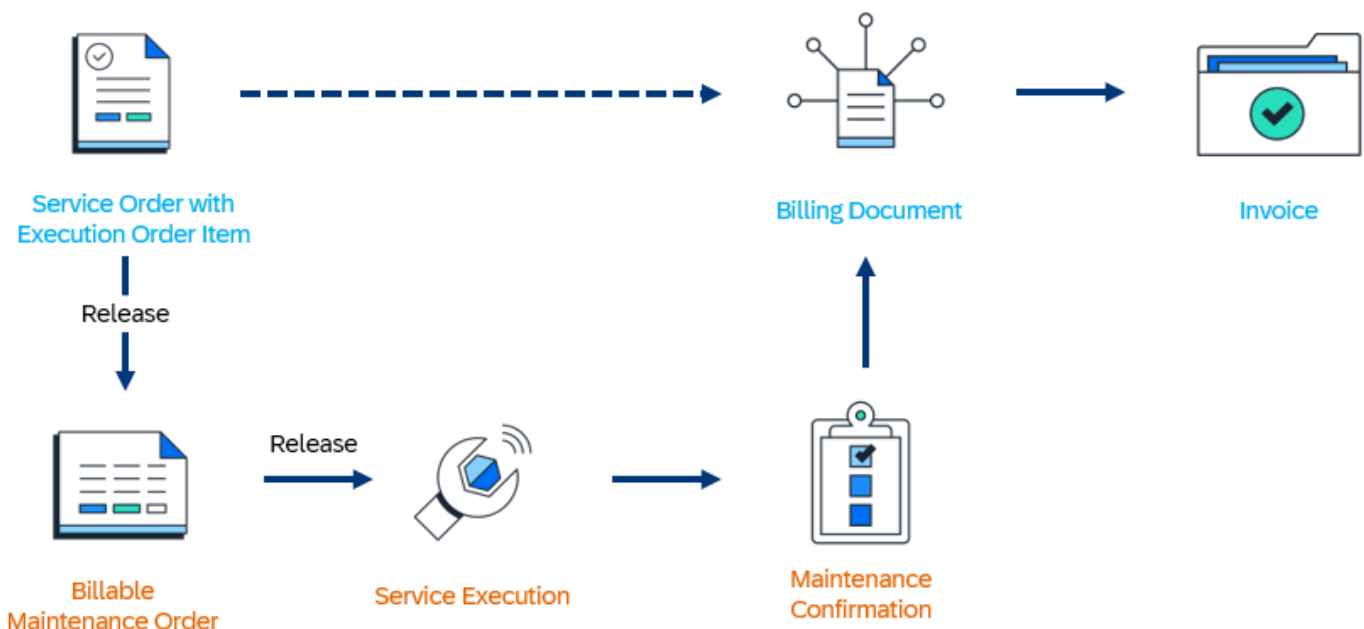
1. Enter general data such description and sold-to party.
2. Enter the sales organizational data.

i Note

Depending on customizing, automatic selection of organizational data is possible.

3. Under **Parties Involved** enter service employee group and employee responsible.
4. Add items and select **Execution Order Item** as the item category. Each execution order item creates a maintenance order. The maintenance order is used for the execution planning and execution of the service.
5. For each execution order item, you need to select a billing relevance. You need to enter a DIP profile if you select summarized billing or itemized billing as the billing relevance.

Whenever the billable maintenance order is created corresponding to an execution order item certain data is copied over from the service order to the maintenance order. This includes information such as shorts texts, planning plant, main work center, profit center, reference objects, and basic start and finish dates. Specific partners can also be entered automatically in the maintenance order based on Customizing settings.



Service Execution Planning

The actual service tasks that are to be performed usually require detailed advance planning, in order to minimize the time that the object is not productive. The more accurately you plan a task, the more efficiently it can be performed by the employees on site. This is extremely important in the case of objects that cause bottlenecks and may lead to a shutdown when they are out of order. The billable maintenance order facilitates this planning and execution.

Planning

Part of the planning involves using operations to describe the individual maintenance tasks to be performed. The operation includes the time, the work center, and other controlling information for an individual maintenance task. Another important aspect is the planning of materials. This can be a spare part or repairable spare or an activity. Materials can be reserved in a warehouse and later withdrawn by the technician or delivered to the customer. For more information, see [Planning of an Order](#).

- Scheduling

Scheduling is used to determine actual execution dates, the capacity needed to execute the order, and the date at which materials should be made available. For more information, see [Scheduling](#).

- Task Lists

Task lists describe a sequence of individual maintenance activities which must be performed repeatedly within a company. They are used as a reference and an input tool when processing orders and considerably reduce the time required during work scheduling.

The task list of the billable maintenance order is determined from configuration **OISD**. From the planning plant entered in the execution order item along with the service product, the system determines a unique task list. This task list is entered automatically in the planning data of the maintenance order. With this, the operations and components are also determined.

For more information, see [Working with Task Lists](#).

Order Release

When all the planning has been completed the billable maintenance order is released. Without releasing the order, the employees on site cannot start with the defined activities. For more information, see [Order Release](#).

Service Execution and Completion

Execution

Once the billable maintenance order is released you can move into the execution stage. Here the actual service or maintenance takes place. This stage involves the withdrawal of material, arrival of technician, actual maintenance process including replacement of parts or equipment and ends with the completion of the order. For more information, see [Execution of an Order](#).

Completion

Completion of the billable maintenance order indicates that the technician has finished the job and various confirmations have been posted. Some examples of confirmations are time completion, material completion, confirmation of measurement and counter readings, and so on.

You usually complete an order technically once the maintenance work planned in the order has been performed. The status is set to **Technically Completed**. For more information, see [Completion of an Order](#) and [Technical Completion of an Order](#).

Billing

When the technician has completed the service the billable maintenance order is set to **Technically Completed** (TECO). The technician enters the materials used and the time taken for the completion of the service. The order can then be sent for billing. Depending on the scenario the order can be billed as ad hoc, fixed price, or resource-related. For more information, see [Billing](#).

Early Creation of Billable Maintenance Order

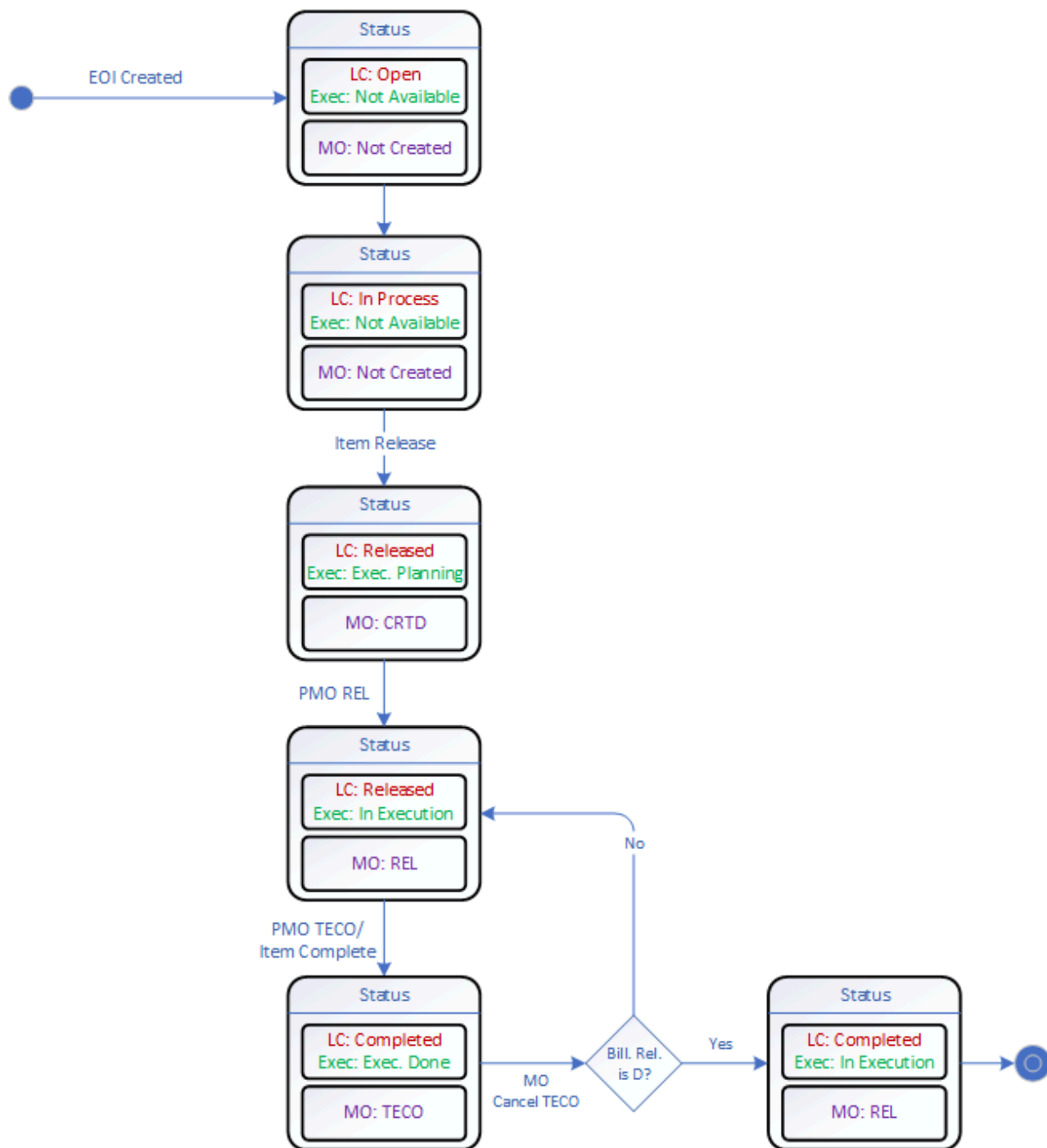
Usually, a billable maintenance order is created when the execution order item is set to **Released**. However, it is possible to create a billable maintenance order already when the execution order item is in the status **In Process**. This is particularly useful when there is no preceding document, in this case no service quotation and therefore no possibility of prior service planning. By allowing the early creation of a billable maintenance order, the maintenance planner can plan for the service execution before the actual start of service. For the early creation of a billable maintenance order the following conditions must be met:

- The execution order item must be in the status **In Process**.
- There is no corresponding billable maintenance order for the execution order item.
- There are no errors in the creation of the execution order item.

If the above conditions are met, then you can use the **Plan All Items** function or select the checkbox **Send to Planning**. You can also trigger **Send To Planning** automatically by selecting the **Send to Planning Immediately** checkbox in the Customizing activity **Map Order Types**.

Overview of Status

The following diagram and table provide an overview of the statuses that service orders can have.



Note:
1. Bill. Rel. D means Billing on Completion (Fixed Price scenario).

EOI	Execution Order Item
MO	Maintenance Order
LC	Lifecycle
Exec	Execution
DNE	Do Not Execute
CRTD	Created
REL	Released
TECO	Technically Complete
CLSD	Closed
NCMP	Not Complete
Bill. Rel.	Billing Relevance

Status	What It Means
Open	This is the default status that all service orders take after they are created. It indicates that you are free to make changes to the order.
In Process	In this status the service order and its items can be edited.
Released	When the service order has been finalized it is set to Released .
Completed	This status is automatically set when the service has been completed and the maintenance order is set to Technically Completed . The Completed status can also be set manually on the service order which will in turn set the status on the billable maintenance order to Technically Completed .

The following table provides an overview of the billable maintenance order status and the corresponding item execution status.

Status on Maintenance Order	Item Execution Status	What Happens in this Status
Created	Execution Planning	All billable maintenance orders are created in this default status.
Released	In Execution	The billable maintenance order has been released.
Technically Completed	Execution Done	This status is set when the service has been performed. The item life cycle status is Completed .
Cancel Technically Completed System status reverts back to Released	In Execution	If Technically Completed is canceled the status of the billable maintenance order reverts to Released along with the item life cycle status. The execution status reverts to In Execution from Execution Done .
Do Not Execute	Canceled	This status indicates that the service has been stopped or canceled.

If the status of a service order is changed from **Completed** to **In Process**, the maintenance order is set to **Released**.

Related Information

[Service Orders](#)

Service with Advanced Execution in Recurring Service

You can use Service with Advanced Execution to create maintenance plans for execution order items.

Use

You can use maintenance plans to generate regular services that have to occur repeatedly at certain intervals. These can be part of regular maintenance or based on usage or performance as agreed by means of a service contract. Regular maintenance helps to keep the customer's assets in optimum working conditions and minimizes the breakdown and repair costs. You can create time-based or performance-based single cycle plans, strategy plans or multiple counter maintenance plans.

Types of Maintenance Plans

- Single cycle plan

This is the simplest form of maintenance plan with or without a task list. It is defined for a one time-based or a performance-based maintenance with the same set of tasks every time. Here you specify the interval at which the maintenance plan should be executed, for example, the annual maintenance of an air conditioner.

- Strategy plan

This is used for complex maintenance cycles where the asset needs to be maintained with a different set of operations at different intervals. A maintenance strategy contains packages and general scheduling information and can be used for many maintenance plans. For example, it makes sense to use a strategy plan if different maintenance tasks for a car are due in different cycles: oil check every 1000 km, oil change every 7000 km.

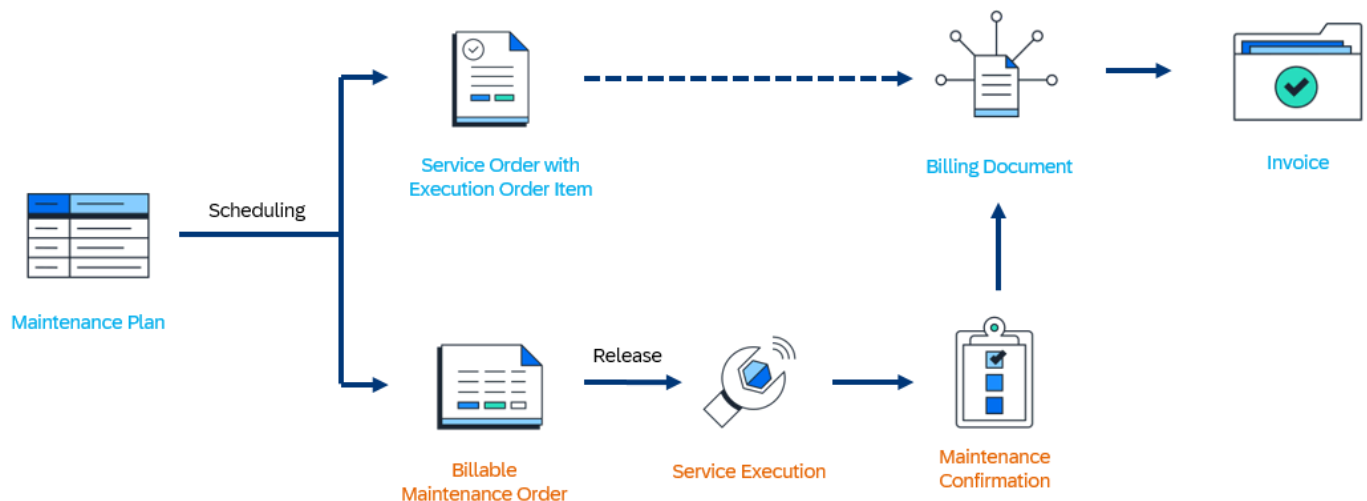
- Multiple counter plan

This is used in time-based maintenance or counter-based maintenance or a combination of both. Here you create maintenance cycles and allocate counters of equipment or functional locations with different dimensions to them. For example, maintenance of aircraft based on either flight hours, number of take-offs and landings, calendar time, or kilometers traveled.

In the Service with Advanced Execution process a service order with a billable maintenance order is created on the scheduling of a maintenance plan.

Call Object

Once you create a maintenance plan, you can schedule the plan to generate maintenance call objects when the system is due for maintenance. The call object is determined based on the maintenance plan category. In Service with Advanced Execution, service order with a corresponding billable maintenance order is used as the call object.



Creating a Maintenance Plan

You can create a maintenance plan using the **Add Single Plan**, **Add Strategy-Controlled Plan**, or **Add Multiple Counter Plan** apps depending on your requirement. Before creating a maintenance plan, ensure that you have prepared the related service contract.

In the app, make the following entries:

1. Select the maintenance plan category.
2. On the **Item** tab, enter a new item by specifying item description and technical object. In the planning data section enter service contact, service contract item, and order type.

3. Provide a name for the maintenance plan and enter the cycle. Enter a task list.

i Note

If you have created a plan without task list, you can use the [Change Maintenance Plan](#) app to assign a task list later.

4. Enter the [Scheduling Period](#) under [Maintenance plan scheduling parameters](#) to specify the number of days, months, or years over which scheduling will take place.

Scheduling a Maintenance Plan

Once you have created a maintenance plan, you need to schedule it. In the [Schedule Maintenance Plan](#) app make the following entries:

1. Enter the plan number you have created.
2. When you choose [Start](#), the scheduling list gets populated with all scheduled calls and the dates for which they have been scheduled.
3. Once you select [Save](#) the service order and its corresponding maintenance order is generated.

Mass Scheduling of Maintenance Plans

To schedule multiple maintenance plans you can use the [Mass Schedule Maintenance Plans - Service](#) app. For more information, see [Mass Schedule Maintenance Plans](#).

Call Object Generation

When the maintenance plan is scheduled, the service order and the connected billable maintenance order are generated as call objects. In the billable maintenance order you can view the task list that was entered when creating the plan. From the maintenance order you can navigate to the service order.

i Note

When the service order is created as the call object, the execution order item has the status [In Process](#).

Maintenance plans for execution order items can be created from:

- Service contract: A service contract is for a single execution order item.
- Service order template: A service order template can have more than one item. It can also be a mix of service items and execution order items.

Related Information

[Recurring Service](#)

BAdI: Reuse Service Transaction in Recurring Service with Advanced Execution

Use

This Business Add-In (BAI) is used in the Recurring Service (CRM-S4-SRV-MNP) component. You can use this BAI in Recurring Service with Advanced Execution (Maintenance plan) to reuse a service transaction. This BAI allows the addition of execution order items into an existing service transaction. You can find this BAI in Customizing under [Service > Transactions > Settings for Recurring Service > Business Add-Ins > BAI: Reuse Service Transaction in Recurring Srv with Advanced Execution >](#) .

The BAI uses the method GET_SERVICE_TRANSACTION_REUSE to determine the service transaction to which service items will be added.

Import Parameters

- IS_MPOS: Maintenance plan item details such as service contract, service transaction type, reference object and so on.
- IS_MHIS: Maintenance plan call history data such as call number, call horizon and so on.

Changing Parameters

- CV_SERVICE_TRANSACTION_GUID: Used to enter service transaction GUID based on the maintenance plan item details.
- CS_OPTIONAL_CHECK - no_service_org_check: Used as an optional check to ignore the validation of the service organization.
- CV_GEN_NEW_ORDER_IF_ERROR: If the validation for the service transaction from the BAI fails and the flag is not set, partial scheduling takes place. If the validation for the service transaction from the BAI fails and the flag is set to 'X' then the standard behavior takes over and new service order is created for the maintenance plan item.

Effect in Mass Scheduling

Mass scheduling is used to schedule many maintenance plans through background batch jobs which can be set to run in certain intervals. For each scheduling run, the system goes through maintenance plans that match the selection condition of scheduling and generates the maintenance call object for each due maintenance item. This is usually done by parallel processing where each parallel process runs multiple maintenance plans simultaneously.

During parallel processing there is a possibility of the service order getting locked. This occurs when multiple maintenance plans access the same service order. To prevent this, the BAI groups together maintenance plans that might share the same service order. Grouping is based on a combination of the sold-to party and the sales area of the service contract item that is linked to the maintenance item.

Service with Advanced Execution in Maintenance Management

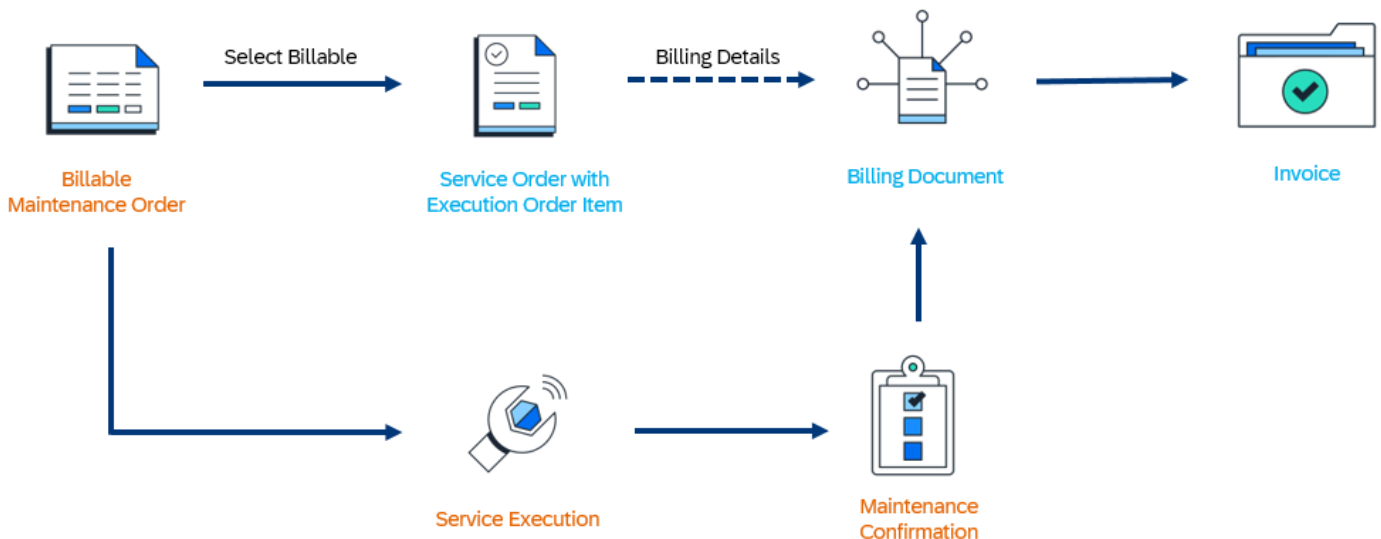
You can use Service with Advanced Execution to create billable maintenance orders to assist in the planning and execution of a service.

Use

Maintenance orders are created by maintenance planners and are used to plan and execute maintenance tasks. These tasks are generally not billed since the maintenance is performed on their own equipment. However, in some cases the customer needs to be billed. It may be due to maintenance or malfunction of the equipment attributed to a customer. At the time of maintenance order creation the planner is not aware that this maintenance needs to be billed. When the planner realizes that maintenance needs to be billed, he selects the **Billable** checkbox, which in turn triggers creation of a service order. Upon completion of the service a billing document can be created.

i Note

Enabling the **Billable** checkbox is an irreversible change.



Prerequisites

- You have made Customizing settings related to Service with Advanced Execution.
- You have maintained the required mapping between transaction type, item category, and order type in the Customizing activity **Map Order Types**.
- You have not released the maintenance order that was created.

Creating a Billable Maintenance Order

Once you have determined that the maintenance order needs to be billed, you can use the transaction **Change Maintenance Order (IW32)** to create a billable maintenance order.

1. Enter the order that needs to be changed to a billable maintenance order.
2. Select the **Billable** checkbox from the order header. The sales data and service details sections will appear when the **Billable** checkbox is selected.
3. Enter the relevant sales details:
 - Sold-to party (Mandatory)
 - Sales organization
 - Distribution channel
 - Division
 - Sales office
 - Sales group

i Note

There should be at least one sold-to party in the **Partner** tab if the sold-to party partner function is available in the partner determination procedure for the maintenance order. The sold-to party must be the same in the header and partner sections.

4. Enter the relevant service details:

- Service product (Mandatory)
- Quantity and unit (Mandatory)

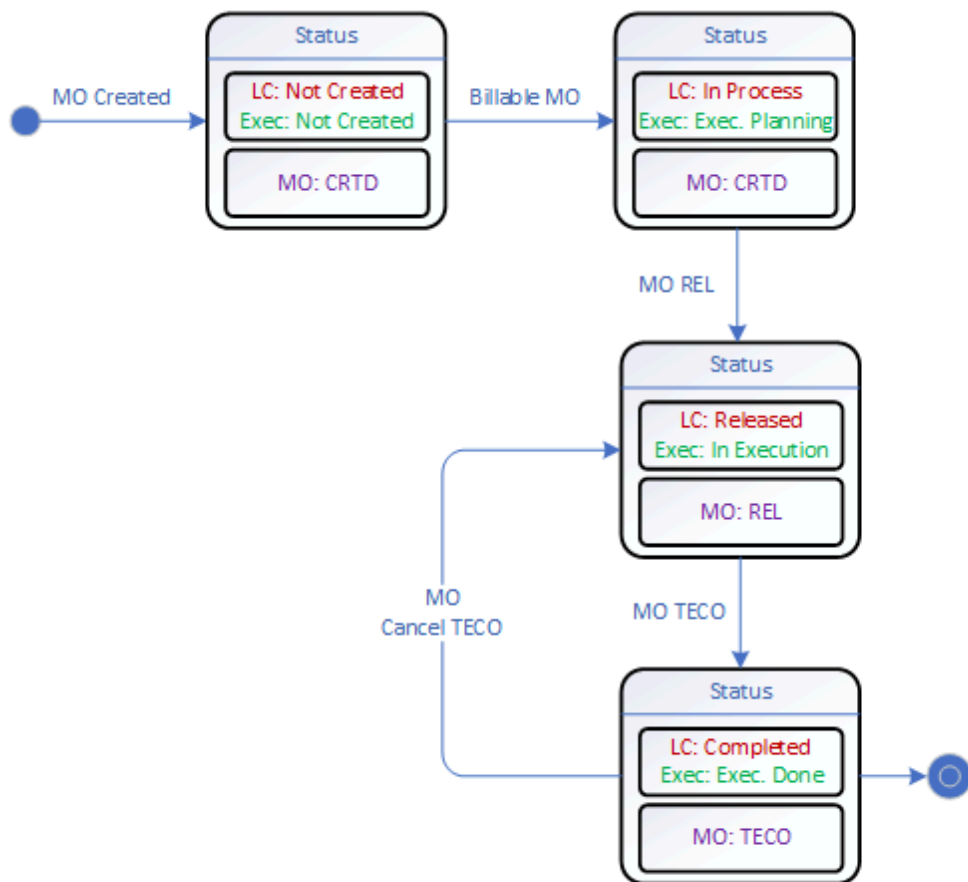
5. Click **Save** to create a billable maintenance order. The system automatically creates a service order with an execution order item

For more information, see [Specifying a Maintenance Order as a Billable Maintenance Order](#).

Synchronization with Service

After the billable maintenance order is created, a corresponding execution order item is created in a service order. There is a 1:1 relationship between maintenance orders and execution order items.

The following diagram and table provide an overview of the maintenance order status and the corresponding item execution status.



MO	Maintenance Order
LC	Lifecycle
Exec	Execution
DNE	Do Not Execute
CRTD	Created
REL	Released
TECO	Technically Complete
CLSD	Closed
NCMP	Not Complete

Status on Maintenance Order	Item Execution Status	What Happens in this Status
Created	Execution Planning	All maintenance orders are created in this default status.
Released	In Execution	The maintenance order has been released and is ready for execution.
Technically Completed	Execution Done	This status is set when the service has been performed. The item life cycle status is Completed .
Cancel Technically Completed The status reverts back to Released	In Execution	If the status Technically Completed is canceled the status of the maintenance order reverts to Released along with the item life cycle status. The execution status reverts to In Execution from Execution Done .
Do Not Execute	Canceled	This status indicates that the service has been stopped or canceled.

Related Information

[Processing a Billable Maintenance Order Created in Maintenance Management](#)

[Planning and Executing a Billable Maintenance Order](#)

Advance Shipment Using Billable Maintenance Order

You can use the billable maintenance order with this logistics process to ship service parts to a technician or customer.

Use

This logistics process enables advance shipment of service parts to the technician or customer as part of service processing.

The shipment of service parts is processed using the billable maintenance order integrated with a sales order. The billable maintenance order is also integrated with the execution order item of the service order.

The maintenance order enables a service-specific results view from a controlling perspective and improves coordination between delivery dates and deployment dates for the service technician. The sales order allows customer consignment stock to be created. As a result, parts that are actually required can be invoiced after their use has been posted to the maintenance order.

The service parts to be shipped are issued from the warehouse stock. The technician can return the service parts that were not required to the warehouse.

i Note

Billable maintenance order refers to the order that supports planning and execution of execution order item represented as a product in the service order.

Process Flow

Creating and Releasing a Billable Maintenance Order for Advance Shipment

When you create a service order for an execution order item a corresponding maintenance order is created. You can plan the service parts as components in the maintenance order to be delivered to the customer and then you can deploy a service technician.

From the maintenance order, you can create a sales order (consignment fill-up type order). You have the option of copying the service parts already planned in the maintenance order to the sales order. You can transfer service parts from the bill of material of the reference object in the maintenance order in case no components are planned in the maintenance order level.

Performing a Delivery for a Sales Order Item

After releasing the maintenance order, you create the delivery note for the sales order and post the goods issue for the delivery.

The sales order type can be configured in such a way that the delivery is created automatically for the sales order item (immediate order) when the maintenance order is saved.

Performing Completion Confirmations and Returning Service parts

After the service has been performed, the technician performs completion confirmation for the maintenance order by entering the service parts used and the working time in the maintenance order. In the completion confirmation, the technician can also change the accounting indicator.

The service technician sends back the service parts that were not used. Since the return delivery does not affect billing, it can take place before or after billing. For the return delivery of these materials, they must be included as items in the sales order.

Posting Goods Withdrawal from Consignment Stock

At the same time as completion confirmation, the goods withdrawal from consignment stock for the material that were used must be posted manually (movement type 261, special stock W). The consignment stock is then reduced according to the parts used.

Performing Resource-Related Billing

Using resource-related billing the confirmed service parts and working times are invoiced to the customer.

Creating a Sales Order for Advance Shipment

Prerequisites

When you create the sales order from the billable maintenance order, you can enter this in the user master record under the set/get parameter VORAB_AUART.

To be able to send the service parts to a service technician (instead of to a customer) when creating a consignment stock, a customer master record must be created for the service technician.

Procedure

1. In order to plan the service parts that you want to send in advance to the customer, choose **Components**.
2. The component overview is displayed.
3. Enter the service parts, the quantities, and the item category (for example, stock material). Material master records with sales data must be created and have prices maintained for the service parts.
4. If you have entered a reference object in the order header, you can select the service parts to be sent by clicking on the icon **Structure list** and copying them into the component overview.

5. Select the service parts that you want to ship in advance and click on the icon [Advance shipment](#).
6. The dialog box for entering the advance shipment parameters is displayed.
7. You can overwrite the sales order type. Enter a shipping condition and an accounting indicator where necessary and choose [Continue](#).
8. In the next dialog box that is displayed, you can:
 - Perform an availability check
 - Perform a completeness check
 - Make changes to the items proposed from the billable maintenance order (these changes are not copied to the maintenance order)
9. Choose [Back](#) to return to the billable maintenance order and save it.

Result

The system saves the sales order together with the billable maintenance order. You can use the document flow function to see the assignment of maintenance order to the sales order.

Both the maintenance order and the individual service parts are assigned the system status [Advance shipment occurred](#). This means that you can use this status to search for billable maintenance orders and components.

The system also sets automatically the indicator "Non-MRP-relevant" for the service parts. This is because the materials are copied into the sales order and are planned there.

Advance Shipment Status

You can use the [Advance shipment status](#) function in the component overview or detail screen in the billable maintenance order to obtain the current status overview of the sales order. You can see all system and user statuses of the sales order at a glance. This enables you to display the required status information from the billable maintenance order without having to call up the sales order or the delivery note. Furthermore, the current delivery status of the individual components is also recorded here. This means you can use this function to track the delivery dates of the service parts with the deployment date of the service technician.

Distributed Execution

You can use Distributed Execution to create multiple billable maintenance orders for an execution order item.

Use

Distributed execution allows you to break down the execution of a service requirement into multiple smaller parts. By creating multiple billable maintenance orders for the same execution order item, you can plan and execute each part of the service separately.

Consider a piece of equipment that requires a service due to maintenance or breakdown. If the equipment can be broken down into separate components that can be serviced independently by different teams and at different times, you can create a billable maintenance order for each component. You can also create multiple billable maintenance orders for a scenario where you want to separate each aspect of the service using a separate order. For example, a maintenance order for inspection, another one for service execution, a separate order for clean-up and testing and so on.

The breakdown of the planning and execution into multiple maintenance orders are, in general, carried out by the maintenance planner.

Prerequisite

You have selected the checkbox **Allow Multiple Maintenance Orders** for item **BUS2000116** (execution order item). You do this in Customizing under ► **Service** ► **Transactions** ► **Basic Settings** ► **Define Item Categories** ►.

i Note

Distributed execution is only possible with a service order. Other Service transactions are currently not supported.

Process Flow

When you have enabled the creation of multiple maintenance orders, the next step is to create a service order with a corresponding billable maintenance order. Alternatively, you can also start by creating a maintenance order and selecting the **Billable** checkbox to create a corresponding execution order item in a service order.

Once you have created the initial link between Service and Maintenance Management, you can use distributed execution in one of the following three ways:

1. Copy

You can use this feature in the **Create Maintenance Order (IW31)** app to create a billable maintenance order by using an existing billable maintenance order with similar data as reference. You can choose the data you wish to copy over by selecting the appropriate checkbox. This feature is useful for maintenance tasks that are repetitive.

2. Create follow-on

You can use this feature in the **Create Maintenance Order (IW31)** app to create a billable maintenance order as a follow-on of an existing billable maintenance order. You need to select the checkbox **Create Follow-On Order** and enter the reference order number. This can be used in scenarios where the defects reappear or unexpected issues develop after execution of the initial maintenance order.

3. Create suborder

You can use this feature in the **Create PM Sub-Order (IW36)** app when a particular maintenance requirement needs to be broken down into smaller parts. A hierarchical relationship exists between the initial billable maintenance order (superior order) and the suborders. A Superior maintenance order is mandatory to create suborders and the superior order can be set to technically completed only after all its suborders are set to technically completed. Also when resource-related billing process is started for the superior order, the system automatically aggregates costs from its suborders.

You need to select the **Service** checkbox in the **Create Maintenance Order (IW31)** app or the **Create PM Sub-Order (IW36)** app to create multiple billable maintenance orders. This will ensure that the new maintenance order is a billable maintenance order. Additionally, the service order and execution order item reference are carried into the new billable maintenance order.

The new billable maintenance orders and suborders can be viewed in the **Maintenance Orders** assignment block in the execution order item overview page of the service order.

Restricting Synchronization

If the checkbox **Allow Multiple Maintenance Orders** is not selected, only one billable maintenance order can be created corresponding to the execution order item. In addition, changes made in the service order are copied over to the billable maintenance order. Similarly, changes made in the billable maintenance order are copied over to the service order. This is applicable for fields like planning plant, work center, reference objects, and so on.

However, if the checkbox has been selected to enable multiple maintenance orders, then the synchronization of fields as described above does not take place. The only exception to this is the accounting indicator and profit center fields.

Selecting the checkbox allows more flexibility during the planning and execution of the service. Sometimes with multiple billable maintenance orders the planner will want to change attributes and differentiate the maintenance orders as required. For example, if there are three maintenance orders, and these need to be executed at three different work centers, in this case the maintenance planner can change the work center in the various maintenance orders.

Related Information

[Distributed Execution](#)

Billing

Get an overview of the billing process in Service with Advanced Execution.

Billing represents the final processing stage for a service transaction in Service with Advanced Execution.

Types of Billing

- Not Relevant for Billing

This type of billing is used when the customer is not being billed for the service. Goodwill, under warranty, free-of-charge service may be some of the business decisions for the use of this billing type.

- Ad Hoc Billing

This type of billing is used to define billing values and dates freely for fixed price execution order items. Billing document requests (BDRs) are generated for fixed price items with long duration that need to be billed on specific dates and not just when the execution order item is completed.

- Fixed Price Billing

In fixed price billing, the service is billed at a predetermined price irrespective of the actual cost incurred during the service. It is based on the price master data of the service product. The price master data (along with discounts) is maintained in Sales.

- Resource-Related Billing

Here customers are billed for the resources used (for example, labor, material, travel) to carry out services. Resource-related billing can be either itemized billing or summarized billing.

Related Information

[Billing](#)

Ad Hoc Billing

You can use ad hoc billing for execution order items in Service with Advanced Execution.

Use

Ad hoc billing is a structured method of billing that doesn't conform to periodic or milestone billing plan rules, but instead enables you to define billing values and dates freely.

Ad hoc Billing Plan

You can use an ad hoc billing plan to define the dates on which billing is to occur and the value that is to be billed. For each billing date, the system generates one billing request line. In addition, you can optionally specify the settlement period that an individual billing request line covers.

Billing of Remaining Net Value

This feature enables the system to automatically generate a billing request line with all the remaining net value of the execution order item. The billing date of this billing request line is set to the current date by default.

Prerequisites

1. You have made the necessary settings for ad hoc billing plans in Customizing under **Service > Transactions > Basic Settings > Billing Plan > Define Billing Plan Types**. In the **BilPlanUse** field, choose **Ad hoc Billing**.
2. The billing plan **Ad hoc Billing** is assigned to the relevant item category. You do this in Customizing under **Service > Transactions > Basic Settings > Billing Plan > Assign Billing Plan Type to Item Category**. The predefined Customizing settings include the transaction type SRVO and the item category MS11.
3. You have maintained the billing relevance for the execution order item as **Billing via Billing Plan After Release**.

Process Flow

Service Order

1. Using the **Create Service Order** app create an execution order item with the billing relevance **Billing via Billing Plan After Release**.
2. On setting the above billing relevance, the **Billing Plan** assignment block automatically appears.
3. Within the **Billing Plan** assignment block you can choose the billing date and the billing value. You can then view the remaining value to be billed.
4. Save the service order.

Related Information

[Ad Hoc Billing for Service Orders](#)

Fixed Price Billing

You can use fixed price billing for execution order items in Service with Advanced Execution.

Use

In fixed price billing, the completed service is billed based on the price master data of the service product. The price master data (along with discounts) is maintained in Sales. This price is proposed for the service order.

The billing relevance used for fixed price billing is always **Billing on Completion**. When the execution order item status is **Completed**, a billing document request (BDR) can be created through the **Release for Billing** app.

With the **Release for Billing** app, you can release maintenance service order items for billing. After you have released the items, the you can create the billing document requests (BDR).

i Note

If an item belongs to a service bundle, you must release all items of the bundle simultaneously.

Process Flow

1. Open the [Release for Billing](#) app.
2. Enter the service order ID and search.
3. From the results, select the service order ID and then choose **Release for Billing**.

i Note

Once you choose [Release for Billing](#), the service order disappears from the search results.

4. Open the [Create Billing Documents](#) app and search using the service order ID.
5. Select the order from the search result and choose **Create Billing Documents**.
6. The invoice is generated.

Related Information

[Billing of Service Transactions](#)

Resource-Related Billing

You can use resource-related billing for execution order items in Service with Advanced Execution.

Use

The price for customer-specific services is not always defined in a service contract as a fixed price nor can it always be determined using standard pricing.

Typical examples of this are:

- External plant maintenance
- Specific services, such as consulting

In these cases customers are billed for the resources used (for example, personnel, material, travel) to carry out services. In resource-related billing, the system creates dynamic items from the expenditure information on the maintenance order.

Resource-related billing can be either [itemized billing](#) or [summarized billing](#). They are identified with the billing relevance field attribute at the item level.

- Itemized billing

A service is billed as a list of subitems. The net price is the aggregated price of these subitems determined by either pricing conditions or by their costs. Itemized billing can have multiple debit memo requests depending on number of visits, change in tasks, requirement of additional parts etc. Itemized billing also supports event-based revenue recognition.

- Summarized billing

A service is billed as a single main item. The net price is determined from existing pricing conditions of the main item or as the aggregated cost of individual subitems. There is only one debit memo request created at the end of the service. Summarized billing does not currently support event-based revenue recognition.

Resource-related billing is carried out in the [Create Resource-Related Billing Request - Service Order](#) app.

i Note

You can use the transaction code **DP92** to launch the app from SAP GUI.

DIP Profile

The dynamic item processor (DIP) profile is mandatory in the case of resource-related billing. During billing, the dynamic item processor determines the items or lines in the debit memo request, based on the DIP Profile entered in the execution order item in the service order.

The value of DIP Profile can be defaulted from the predecessor document like service contract or service quotation or from the item category customizing. It is also possible to manually enter or overwrite the default values. For more information, see [Dynamic Items](#).

i Note

The DIP Profile entered in the service order must have Usage **1 Billing**.

Prerequisites

In Customizing

You have maintained a dynamic item processor (DIP) profile.

In the Service Order

- You have created one or more execution order items in the service order that are supported by a corresponding maintenance order.
- The billing relevance is selected either as Itemized billing or Summarized billing.
- You have maintained the following data in the service order
 - Customer
 - Sales organization
 - Distribution channel
 - Division
- You have entered a DIP profile for the execution order item.
- You have released the maintenance order and booked actual costs.

Process Flow

1. Open the [Create Resource-Related Billing Request - Service Order](#) app.
2. Enter the service order ID and the execution order item. Alternatively, you can use the help to search for a service order or item. Then select **Create**.

3. The system determines the items in the expenditure view using the DIP profile and actual cost from the maintenance order.
The prices are determined using Sales and Distribution Pricing. The required pricing procedure is determined using the document category specified in the DIP profile.
4. You can set the quantity or the amount to be billed. You can also reject or postpone by quantity or amount.
5. Select the **Billing Request** button to generate a debit memo request (DMR).

Related Information

[Event-Based Revenue Recognition for Service with Advanced Execution](#)
[Resource-Related Billing](#)

Apps Used in Service with Advanced Execution

You can use the following apps in Service with Advanced Execution.

Apps Used in Service Quotations and Service Orders

App Name
Manage Service Quotations
Manage Service Orders

Apps Used in Recurring Service

App Name
Create Maintenance Plan (IP01)
Change Maintenance Plan (IP02)
Display Maintenance Plan (IP03)
Create Maintenance Item (IP04)
Change Maintenance Item (IP05)
Display Maintenance Item (IP06)
Schedule Maintenance Plan (IP10)
Add Single Plan (IP41)
Add Strategy-Controlled Plan (IP42)
Add Multiple Counter Plan (IP43)
Mass Schedule Maintenance Plans - Service

Apps Used in Maintenance Management

App Name
Change Maintenance Order (IW32)

App Name
Display Maintenance Order (IW33)
Change PM Orders (IW38)
Enter PM Order Confirmation (IW41)

Apps Used for Billing

App Name
Release For Billing
Create Resource-Related Billing Request: Service Order (DP92)
Mass Creation of Resource-Related Billing Requests: Service Order (DP92A)

Create Resource-Related Billing Request

App ID: DP92

With this app, you can create debit memo requests for the time and service parts spent on the service.

Key Features

This app provides the following key features:

- Generate a debit memo request for an execution order item. Other service order items are not currently supported. Can also generate credit memo request.
- Items are determined using DIP profile and actual costs are determined from billable maintenance order. Prices are determined using Sales and Distribution Pricing.
- Supports only billing relevance type summarized billing and itemized billing.
- Supports the billing of multiple billable maintenance orders corresponding to a single execution order item.

Supported Device Types

- Desktop
- Tablet

Mass Creation of Resource-Related Billing Requests

App ID: DP92A

With this app, you can process multiple service orders and execution order items to create one or more billing requests.

Key Features

You can use this app to do the following:

- Create credit memo or debit memo requests in bulk that are due over a period for a set of service orders using a combination of sales organization, distribution channel, division or sold-to party.
- Run individual billing request lines of a service order and decide whether to bill now, bill later, or reject the billing request line.
- Save billing lines in an extract that are periodically processed to create debit memo or credit memo requests.
- Process billing requests in the background while working on other tasks in the foreground.
- Process multiple billing requests at the same time to reduce the overall processing time.

Supported Device Types

- Desktop
- Tablet
- Smartphone

Case Management

Case Management enables you to consolidate, manage, and process information about a complex problem or issue in a central collection point, the case. Within a case, you can group diverse information, such as business partners, transactions, products, and documents. This information can reside in different physical systems.

Use

You can also use Case Management to process problems and issues that involve multiple processing steps or multiple processors. Case Management therefore supports the processing and communication flow between organizational units and helps you to increase processing efficiency.

Case Management contains various [functions and features](#) that are applicable to different scenarios.

SAP provides the framework for processing issues in different industries and scenarios for Case Management:

- Incidents relating to complex or expensive products for which you provide technical services
- Issues in the utilities industry, such as power outages
- Insurance claims cases
- Services in government agencies

SAP provides predefined case types to support the following scenarios:

- General incident tracking
- Disputes regarding customer invoices and payments in the telecommunications industry

Prerequisites

You have made all the necessary settings in Customizing under  **Service**  **Case Management** .

SAP provides standard Customizing settings that enable you to run Case Management with few or no modifications. Depending on your business processes and system landscape, however, you may need to make further settings.

The section **Enhanced Customizing** contains more specific settings if you have special requirements. For example, define a case closing profile, or set up case archiving.

You only need to maintain the **Special Settings** section under **Enhanced Customizing** in exceptional cases. To do this, you require assistance from an experienced consultant.

If you want to use reference categories, you must have already set up [Multilevel Categorization](#). You do this in the Customizing for Service under **►Cross-Application Components ► Multilevel Categorization ►** and you need to perform additional tasks in the WebClient UI for the IC Manager role under **Process Modeling**.

For details on how to set up Case Management, see the documentation of the individual Customizing activities under **►Service ► Case Management ►**.

Features

- Linking existing business objects to a case: You can link business object types to a case, for example:
 - Business partner
 - Products, objects, installed base components
 - Sales and service business transactions
 - Problems and solutions from the knowledge base
- Assigning and displaying the involved parties: When you assign or link involved parties with a case, you can assign partner functions and specify the main business partner. In Customizing you can define partner determination procedures using a Business Add-In (BAI) for each case type and assign partner functions only using the existing BAI. Furthermore, you can define a minimum and maximum number of partners that must be entered for the various partner functions for each case type.
- Linking electronic documents, such as forms and policies, to a case: Content Management functionality enables you to structure documents, and link archived documents. For more information, see [Content Management](#) and [Document Management](#).
- Linking activities to a case: From Case Management you can directly access activity management. You can create multiple activities to plan and track the required processing steps for a case. You can assign an activity to a person responsible who is not necessarily the case processor. Processors can generate a list of activities for which they are assigned as person responsible. You can also link existing activities to a case. For more information, see [Activity Management](#).
- Creating and linking actions: You can create and link actions from within a case. For example, you can define actions for sending e-mail messages, creating and linking service orders. Actions allow you to process cases more efficiently.
- Entering case notes: You can enter notes if you want to record case-specific information. These notes are always assigned directly to the case and facilitate communication between agents. Case notes are stamped with the name of the user who created the note and the date and time of creation. You can categorize notes by assigning them a note type. You can also make a note of the reason for linking objects to a case next to the objects themselves.
- Change history: Changes to a case are automatically recorded in a log, with a user, date and time stamp. This change history helps you to track changes to case attributes, linking of business objects to a case, and unlinking of business objects in a case.
- Status management: You can assign customer-defined statuses to cases. You define the system status in Customizing where you specify which status changes are permitted.

- Using case categories: You can categorize cases and analyze the data. You define categorization in using [multilevel categorization](#).
- Using reference categories: You can assign a reference category for multilevel categorization at case header level, which allows you to control case processing. You can use reference categories to include various business objects and functions in your case processing. You can use activity templates that are linked to categories to create additional activities and to link them to the case. You can use document templates that are assigned to categories to create documents within a case. By assigning extended attributes, you can define a time version of the reference category.
- Case hierarchies: You can manage associated cases in a hierarchy. To do this, you need to define subordinate cases.
- Creating new cases: You can create a new case from an existing one and use it as a template, or you can choose to create a new case type from a list of case types. When you copy a case, a case of the same case type is created.

Case

Definition

A structured collection of information related to an issue, incident, or problem that requires processing. The processing flow of a case is defined by activities and processors that are assigned to the case.

Use

Cases allow you to group related objects in a central collection point that serves as a single point of access for processing an issue. You can create a case when you establish that an issue extends beyond a single business partner or transaction, or involves many processing steps.

Structure

A case can contain the following information:

- Case attributes

These include case type, case category, priority, status, processor, employee responsible, cause, escalation reason, creation date, and change date.
- Case notes
- Parties involved
- Case category
- Business objects:
 - Transactions
 - Products, objects, installed base components
 - Sales and Service business transactions
 - Attachments
 - **SAP Contract Accounting** documents

You can enable users to link additional business objects by extending the record model for the case. Record models are part of **SAP Records Management**, on which generic SAP Case Management and Case Management in Service are based.

- Electronic documents
- Activities
- Log of changes made to the case
- You can show the relationships between cases by defining a hierarchy of cases. In addition to the existing hierarchical case structures (subcases, higher-level cases), you can [link cases](#) which are related, but do not form a hierarchy.

Example

You receive recurring complaints about a product defect and decide to create a case to monitor these. You then link the affected case business partners and product involved to the case, and also the complaints documents already created to record complaints about the product. You also create an initial activity **Perform technical analysis**, write a note describing the reason for opening a case, and assign the case to a processor.

Case Authorizations

You can use various authorizations in Case Management to control user access to case information. You set up authorizations by defining values for authorization objects in authorization profiles and then assigning them to users or user groups via roles.

You maintain authorization objects for user or roles in the SAP menu ([Architecture and Technology](#) > [System Administration](#) > [User Maintenance](#) >).

For more information about authorizations, see [SAP Authorization Concept](#).

The following authorization objects are specific to Case Management. In addition to these, you also require the SAP Records Management authorization objects. These are also included in the predefined roles. For a description of the SAP Records Management authorization objects, see [The Authorization Concept for Working with Records Management](#) (and subsequent sections) and [The Authorization Concept for Customizing](#).

S_SCMG_CAS (Case Management: Case)

This object defines authorizations for entire cases. It contains the following authorization fields for which you can define values to control authorization:

- CASETYPE: Case type
- SCMG_ACT: Case processing activity, for example, create, change, or delete
- SCMG_KEY: Key for authorization check

This field is a user exit that is relevant for authorization checks using the Business Add-In (BAI) SCMG_AUTHORITY_S (method DETERMINDE_KEY). In this field, you can enter a current value for a field of an application. The BAI generates a new key each time the authorization check is run.

This field is only checked if this BAI is implemented.

- SCMG_LVL: Authorization level

This field controls user access and processing activities dependent on the authorization level assigned to a case.

The values that you can enter for this field are specified in the Case Management Customizing activity [Basic Settings](#) > [Determine Permitted Values for Case Attributes](#) > [Create Values for Authorization Levels](#) > . At runtime, the system checks whether the user has authorization for the authorization level currently entered in the case. If the user does not have the correct authorization level, they cannot execute the activity.

- SCMG_ROLE: Role of the user in relation to the case (see above)

This field controls user access and processing activities dependent on the user's current role as defined in the case header data.

You can enter the value P (Processor) or R (Responsible) for this field. Note that the value C is only relevant for [SAP Dispute Management \(FIN-FSCM-DM\)](#).

Case processors have the role [Processor](#) or [Responsible](#) if they are currently named in the case header as in the processor or person responsible respectively. At runtime, the system checks which, if either, of these roles the case processor currently has. If the user does not have authorization for a role or is neither a processor or person responsible, they cannot execute the activity.

- SPS_ID: Element type within a case

S_SCMG_FLN (Case Management: Field-Level Authorization)

This object defines authorizations for the attribute fields in a case header. It contains the following authorization fields for which you can define values to control authorization:

- ACTVT: Activity ([Change](#), [Display](#))
- CASETYPE: Case type
- SCMG_FIELD: Attribute field

This field controls whether a user can display or change a case attribute.

- SCMG_ROLE: Role of the user in relation to the case (see above)
- SPS_ID: Element type within a case

S_SCMG_STA (Case Management: Status)

This object defines authorizations to change the case status. It contains the following authorization fields for which you can define values to control authorization:

- ACTVT: Activity ([Change](#))
- CASETYPE: Case type
- ORDER_NO: Status

The status that you enter here is the user-defined status, as defined in the Case Management Customizing activity [Extended Customizing](#) [Create Status Profile](#).  If you select values for this field using F4 Help, you can also see the details of the relevant status profile.

- SPS_ID: Element type within a case

S_SCMG_TXT (Case Management: Notes)

This object defines authorizations for case notes. It contains the following authorization fields for which you can define values to control authorization:

- ACTVT: Activity ([Add or Create](#), [Change](#), [Display](#))
- CASETYPE: Case type
- SPS_ID: Element type for case notes

- TEXTID: Text ID

The values that you can enter for this field are specified in the Case Management Customizing activity [Extended Customizing](#) [Create Text Profile](#) [Create Text IDs](#). At runtime, the system checks whether the user has authorization for the relevant type of text to determine whether the user can perform the desired activity for the text.

Document Management

Use

Case Management supports a variety of functions to make documents in electronic formats available in cases. When you create a case, you can upload attachments from your hard drive or you can link to documents on your hard drive. All documents that are linked to cases are physically saved in Content Management. You can always access and manage the documents by using Content Management or SAP Product Lifecycle Management.

Features

Records

You can use the case record structure to link record documents to the model nodes provided in the record. The assigned record model predefines the respective object (record documents or other business objects such as business partners) for each model node. You can edit the list of records linked to the case and the type of record.

Each record document occupies its own instance node in the record hierarchy. Changes to a document are saved as a new version of the document in Content Management.

Documents

In Case Management, you can process case documents using an unstructured collection of case documents, or you can also create folders or structures of folders to flexibly structure existing documents. You can even work with document templates from Content Management. You can upload these from the Content Management system or set a link to them.

Activities

You can assign activities with case notes to a case. A notes document is attached the case but does not appear in the record structure. It is a separate element.

Attachments

In Case Management, you can upload documents from SAP PLM, or from the Content Management system. If you want, you can attach the document as a link or append the actual document to the case. If you upload a document from Content Management and you edit the document, it is saved as a new version. You can also upload a document from a local hard drive and attach it to the case. If there is an existing attachment linked to the case with the same name, you can overwrite it with the most current one from your hard drive. However, in this case no version management occurs.

You can upload a URL as an attachment. It is then a link to any valid external or internal Web site.

Extensibility for Case Management

There are two options for extensibility in Case Management:

- Field extensibility

- Table extensibility

Features

Field Extensibility

Key users use the [Custom Fields and Logic](#) app to extend fields.

Business Context

There are two business contexts for fields:

- Case attribute extension
- Case note attribute extension

All of the standard data types available in the [Custom Fields and Logic](#) app can be used for field extension in Case Management.

Table Extensibility

You can use the Application Extensibility Tool (AET) to extend tables. For more information on the AET, see [Application Enhancement Tool](#).

Related Information

[Key User Extensibility](#)

[Custom Fields App and Custom Logic App](#)

Enhanced Case Management in Service - Features

There is an expanded set of functions available in Case Management. The functions cover:

- [Automatically created case notes](#). A case note can now be created automatically to inform a case processor about events that occurred and affect the case. You can define additional attributes for your case notes to store additional information.
- [Reassignment of business partners in linked objects](#). If you transfer an ongoing case to another case processor or department, all outstanding case business partners and their associated activities are automatically transferred with the case. You can use the delivered Business Add-In (BAI) to influence this behavior.
- [Configuring recurring tasks](#). Enhanced Case Management provides functions that enable you automate recurring tasks and activities that are part of a case. You can also define the parameters required to repeat these tasks and activities. For example, there may be a report that must be run weekly over the life of an open case to monitor its progress.
- [Sending e-mail directly from a case](#). You can now send an e-mail directly from within a case that includes one or multiple attachments.
- [Linking related cases](#). Cases are normally structured in a hierarchy of higher-level cases and sub-cases. You can now link related cases that are not part of the same hierarchy to obtain a different overview.
- [Defining case closing profiles](#). You can create profiles that define how cases with open dependent transactions are to be closed. These profiles allow you to handle different transaction types and to define system behavior dependent on the transaction type.
- [Case archiving](#). You can define checks before archiving a case, archive (write) cases to the database, and delete cases from the database once they have been archived. You can also call up archived cases if you want to review an archived cases for

Configuring Recurring Tasks

Use

You can create recurring tasks in Activity Management. Recurring tasks are actions that a user executes in a defined period of time on a regular basis. You can use a recurring task, for example, if a user needs to make a weekly report for an activity that is meant to run over a long period of time.

A default recurring task is part of the standard delivery, however, you can create your own recurring tasks. To see how the default is configured and use it as a reference for your own configuration, see the explanation of the process in Customizing under ► [Service](#) ► [Transactions](#) ► [Settings for Activities](#) ► [Configure Recurring Tasks](#) ►.

Prerequisites

A recurring task must have the following date fields:

- Start Task Date
- Recurrence, determining the frequency at which a task is to be created
- Recurrence Rule, calculating when the next task is to be created
- Next Recurrence
- Recurrence End
- Completion Date

Activities

There are various activities you need to perform to create a repetitive task:

1. Create a date profile.
2. Create an action profile to determine the action the system performs when the task is executed.
3. Define the conditions under which the action is executed.
4. Set a callback using the function module CRM_CMG_SET_COMPLETION_DATE.
5. Define a transaction type.

Displaying and Editing Time-Dependent Data

Use

You can display and edit time-dependent information related to the primary person of a case without having to navigate away from the main information to the business partner.

Time-dependent information refers to any historic case information that gives an overview of changes over time. The primary person is the person who is directly related to this case, such as the claim applicant.

This information is important to determine eligibility and the amount of a certain benefit. By showing all case-related information, which was valid at a point in the past, the case processor can understand what decisions were made based on different assumptions.

Features

A case processor can display and edit the nationality, income, name, and date of birth of the primary person.

A case processor can also identify if:

- overlapping records exist, for example, whether a specific point in time two or more records for the same data set existed
- gaps have occurred, for example, if data changes to the applicant's status have occurred that require a recalculation of benefits

Changes to data in the past can be restricted as such changes could invoke complex re-calculation of benefits.

Example

In some instances, information changes regarding the primary person associated with the case. For example, a person receives a state benefit, but there has been a change to the marital status, income earned, place of residence, or any other factor that affects the eligibility of a claimant. If changes occur that must be documented, you can make them directly in the application.

Sending E-Mail from a Case

Use

You can send e-mail when working in a case and attach relevant documents from the case. When the e-mail is sent, an automatic link to an activity can be set for the case. For example, if an e-mail is sent to a customer stating that action will be taken regarding a problem in order to correct it, a case processor could assign an activity such as a service ticket or a follow-up call.

This function appears in the [Activities](#) assignment block and the [Record](#) assignment block.

Features

In the [Case Details](#), when you click [Create E-Mail](#), a new page opens allowing you to send an e-mail. You can:

- Create and send a new e-mail automatically
- Specify references regarding the e-mail such as the account, contact, employee responsible for handling this e-mail and an additional reference, for example, if it is a lead, an activity or an opportunity
- Specify the priority (high, low, or normal) and the sensitivity (normal or private)
- Upload attachments related to the case from your PC or network
- Use plain text or HTML formats and format the appearance of the e-mail (HTML only)
- Create and use an e-mail signature for plain text or HTML formats

Reassigning Business Partners in Linked Objects

When a business partner at the case level is changed, business partners in linked objects of a case can be automatically reassigned. For example, if a new case processor or a different department takes over the responsibility for an existing case, the former case processor or department are also replaced by the new processor in all dependent, open activities.

i Note

You must use this Business Add-In (BAdI) implementation to be able to reassign business partners.

This behavior is applicable to business partners for all partner functions and all linked objects, which have their own business partners.

In Customizing, you can use a BAdI to influence how the business partners are reassigned. Choose **►Service ►Case Management ►Process Partners ►BAdI: Reassign Business Partners in Linked Objects ►**.

The BAdI contains the following methods:

- **GET_PARTNER_FCT_TO_CHANGE:** Receives the list of partner functions that can exist in the case; includes only the partner functions with the categories Sold-to, Person Responsible, and Organization Responsible
- **GET_OBJECT_TYPE_TO_CHANGE:** Returns the list of object types to process for changes to business partners; includes only the object types Activity Task, Business Activity, Service Contract, and Service Transaction
- **PROCESS_BPCHG_IN_CASE:** You can use this method to change the processor attribute of a case when the person responsible in the partner function is changed.
- **CHECK_BEFORE_BPCHG_FOR_TX:** You can use this method to decide whether or not specific, open, dependent objects of the case are updated.

i Note

The default implementation contains coding for only the first two methods.

Case Notes

Use

You can store information in the form of case notes within case processing. You can manually create case notes. The system also supports event-controlled, automatic generation of predefined case notes.

A case note consists of a short text and a long text. If you don't create a long text, the system uses the short text as long text information when you save a note. You can define text IDs in Customizing to categorize cases note according to their content.

i Note

Users who create or change case notes aren't recorded with a time stamp. Case notes are saved language-independently. The language specification for case notes is purely informative.

Prerequisites

- You've made the Customizing settings for case notes under **►Service ►Case Management ►Basic Settings ►Create Text Profile ►**.
- You've defined the relevant text profile for each case type in Customizing for Service under **►Case Management ►Basic Settings ►Define Case Types ►**.

Features

Automatic Creation of Case Notes

The RFC-enabled function module CRM_CMG_ADDNOTES supports the automatic creation of case notes. These automatic case notes are either triggered in relation to events within case processing or by applications that are connected to Case Management. For more information, see [Configuring Automatically Created Case Notes](#).

The module supports the following possible sources for automatic case notes:

- Case notes from T100 messages

In this connection, you can make full use of the features of messages, for example, the use of up to four message variables or language-dependency (see Maintaining Messages).

- Case notes from applications or systems

The import parameters IV_TDTITLE and IV_LONGTEXT enable you to transfer short and long texts from other Service applications. You can also use Remote Function Calls to import short and long texts from other systems.

The Business Add-In (BAI) CRM_CMG_NOTES_ADD supplements the function module. It enables you to control the automatic creation of case notes in relation to the case type.

Configuring Automatically Created Case Notes

Use

You can customize your system so that case notes are automatically created. You can either use T100 messages, or you can integrate a text editor if you wish to support manual input. You are responsible for implementing an appropriate text editor for this purpose.

T100 messages are already fully integrated in SAP systems and support functions such as multilingual capability and using variables.

Process

The following is the sequence of the activities you need to perform or consider when configuring automatically created case notes using T100 messages.

1. Create T100 messages

Create the required case notes as T100 messages in the customer name space.

i Note

To support multilingual case notes, create the additional language versions for each message (see [transaction SE63](#)).

2. Create text IDs

In Customizing for Service, choose  **Case Management**  **Extended Customizing**  **Create Text Profile**  **Create Text IDs**.

Create the text IDs you require for the text object SCMG_CASE. The text IDs are displayed as text types on the user interface for case notes and classify a case note according to its content.

Example

The standard system contains the predefined profile DEMO SAP with the text IDs [Description](#), [Internal Note](#), [Concluding Remark](#), and [Reply](#).

i Note

To support multiple languages, create the required language versions of the text IDs using the SAP translation tool (see [transaction SE63](#)).

3. Create text profiles and assign the text IDs they contain

In Customizing for Service, choose [Case Management](#) > [Basic Settings](#) > [Create Text Profile](#) > [Create Text Profile](#). >

4. Assign text profile to case type

In Customizing for Service, choose [Case Management](#) > [Basic Settings](#) > [Define Case Types](#), and determine the relevant text profile for each case type. You can use the same text profile for each case type (n:1).

5. Call the function module CRM_CMG_ADDNOTES

Implement the call of the function module CRM_CMG_ADDNOTES at the desired point in time of an application. You can do this using a corresponding BAdI implementation or BRF event. To find out which exit you can use for this purpose, refer to the documentation for the respective application.

More Information

[Case Notes](#)

Customizing Case Note Attributes

Use

You can customize case note attributes to modify the standard structure and fields, or you can add two customer-defined fields. This is done using the standard structure delivered with Case Management CRMT_CMG_NOTES_GEN.

i Note

You can currently only add two fields to the user interface. If you create more than two attributes, the additional ones will not be displayed.

Activities

1. Run transaction SE11.
2. Select the [Data type](#) radio button, enter **CRMT_CMG_NOTES_GEN** and click [Change](#).
3. Under the [Components](#) tab create the attribute extensions and save your work.

More Information

[Case Notes](#)

[Configuration of Automatically Created Case Notes](#)

Defining a Case Closing Profile

The case closing profile lets you define how open, dependent transactions are to be handled that are in a case that is to be closed. In addition, you can define case closing profiles for specific business transactions that may differ from standard business transactions. After a case is closed, you can extract the case from the database and [archive the information](#).

The case closing profile contains a default reaction for all open transactions and reactions for open transactions of a specific transaction type. The default case closing profile is S_CLACT (**Do not close case if open transactions exist**). The case closing profile influences processing before the save when a case is closed (performing checks), and after the save when a case is closed (automatic closing). Depending on the profile and the case type, different system actions occur when the case is closed.

The action CRM_CMG_CASE_CLOSE is included as an example in the demo case action profile CRM_CMG_DEMO.

In Customizing choose ► [Service](#) ► [Case Management](#) ► [Basic Settings](#) ► [Define Case Closing Profile](#) ►.

After you have created the profile, you can assign it to a case type under ► [Service](#) ► [Case Management](#) ► [Basic Settings](#) ► [Define Case Types](#) ►.

Additional Activity

There is an Business Add-In (BAI) available that you can use to define additional tasks for the case closing profile. Tasks can be performed on the case or its dependent actions either at the check level or when processing the automatic close function. In Customizing choose ► [Service](#) ► [Case Management](#) ► [Basic Settings](#) ► [BAI: Define Additional Tasks for Case Closing Profile](#) ►.

Archiving Cases

Use

After cases have been closed or completed, you can archive them. There are several Business Add-Ins (BAIs) you can use to set up case archiving.

In Customizing choose ► [Service](#) ► [Case Management](#) ► [Extended Customizing](#) ► [Case Archiving](#) ►.

Using the available activities, you can:

- Define custom checks that run before a case is archived so that only cases meeting specific criteria are archived
- Archive (write) cases to the database directly
- Delete cases from the database once they've been archived

Central to the data archiving process is the archiving object. In Case Management, this object is called **SCMG**. For more information about the archiving process, see ► [Service](#) ► [Case Management](#) ► [Extended Customizing](#) ► [Case Archiving](#) ► [Case Data Archiving Process](#) ►.

Prerequisites

An SCMG object must also fulfill the following conditions to be archivable:

- The business transactions related to the case are in status **Closed** or **Completed**
- All the lower-level objects of the case are already archived

Linking Related Cases

You can link related cases that don't form a hierarchy. Normally in Case Management, cases are structured in a hierarchy where they're independent from each other. However, cases can contain similar data or features or have a common element. As a result you can link these cases to gain a better overview of related cases.

You can display all related cases and from the overview page, you can edit the list to add, modify, or delete a related case, as well as give a reason for linking it.

You can't set a link to cases that have been archived. However, you can edit nonarchived, related cases. To edit information, click the hyperlink associated to the case, which brings you to the active case information where you can edit it.

Multilevel Categorization in Case Management in Service

Use

Case Management in Service supports multilevel categorization, allowing you to assign cases to a category of a multilevel categorization schema. Such a category is called a "reference category".

For general information about multilevel categorization, see [Multilevel Categorization](#).

Integration

Multilevel categorization in Case Management in Service uses the Multilevel Categorization application. If you want to use document templates assigned to a reference category, you must first define document template profiles in Content Management. You can assign a profile to one or more categories in Service on the WebClient UI. Select the role **Service Professional** then goto ► [Home](#) ► [Service Operations](#) ► [Categorization Schemas](#) ► under [Search](#) or [Create](#).

If you want to use activity templates assigned to a reference category, you must define activity templates in Activity Management under [Sales](#). You can then assign the templates to one or more categories.

Activities

Setting up multilevel categorization for Case Management in Service consists of the following activities:

- [Creating Multilevel Categorization Schemas for Case Management](#)

The information provided here is specific to creating multilevel categorization schemas for case management.

- [Deploying Categorization Schemas for Case Management in Service](#)
- [Assigning Linked Objects](#)

More Information

[Using Multilevel Categorization Schemas in Case Management in Service](#)

Creating Multilevel Categorization Schemas for Case Management

Use

In Case Management in [Service](#), a multilevel categorization schema lets you do the following:

- Create a case using a case type that is assigned to a deployed categorization schema
- Assign a reference category to a case. Reference dates present different views on your categorization schema, depending on the validity periods of the single categories.
- Use activity templates assigned to this category to create new activities within the case
- Use the document templates of the document template profile assigned to this category to create new documents within the case

You create multilevel categorization schemas on the WebClient UI in [Service](#). Select the role [Service Professional](#) then goto ► [Home](#) ► [Service Operations](#) ► [Create](#) ► [Categorization Schemas](#) ►.

i Note

A categorization schema can be assigned to several case types, but a specific case type can be assigned to only one categorization schema.

You can perform your Customizing for [Service](#) by choosing ► [Cross-Application Components](#) ► [Multilevel Categorization](#) ►.

Deploying Categorization Schemas for Case Management in Service

Use

Only categorization schemas with status [Released](#) , [Deployed](#) or [Active](#) can be used in Case Management in [Service](#).

Integration

Multilevel categorization allows for changes to deployed categorization schemas to be reflected in different schema versions. Case Management in [Service](#) requires a new schema version whenever an existing categorization schema version undergoes changes to any of the following:

- Hierarchy structure, for example, adding new categories
- Validity intervals of single categories, that is, deactivation of categories by setting a new [valid to](#) date for a single category
- Linked objects of a category, depending on the setup of the authorization profile for authorization object **CRM_CATEGO**

New categorization schema versions cannot be released with a past [Valid From](#) date/time. When a new categorization schema version is released, the [Valid To](#) date/time of the currently valid schema version is adjusted to the [Valid From](#) date/time of the new schema version.

When a case is first assigned to a reference category, it is bound to the currently valid categorization schema version. For more information see the section [Setting a Schema Reference](#) in [Categorizing Cases in Case Management](#).

Defining Extended Attributes

Use

When you create categorization schemas in the WebClient UI in [Service](#) you can define extended attributes, such as validity periods, for categories.

Activities

- Defining validity periods

You can define validity periods on a single category level, making it possible to specify validity intervals that are independent of the technical validity period of the categorization schema. This functionality allows flexible activation of categories independent of the release time of the schema.

- Inheritance concept of validity periods

The validity period of a category is passed on to its subcategories. Additionally, it is possible to locally define category values. Therefore, the effective validity period of a single category is always a combination of its inherited values and the locally defined values. In the case of conflicting values, the more restrictive value is the valid one.

- Deactivation of validity periods

A category can be removed from a current schema version in the following two ways:

- Create a new schema version and delete the category from the new schema version.
- Create a new schema version and deactivate the category by changing its **valid to** date.

A deactivated category can no longer be used in newly created cases that have a reference date higher than the deactivation date. For more information, see the section [Maintaining a Reference Date](#) in [Categorizing Cases in Case Management in Service](#).

A deactivated category remains available for searching.

- Defining additional extended attributes

You can define category attributes for customer-specific requests.

Assigning Linked Objects

Case Management currently supports assigning activity template linked objects to categories. You can assign one or more activity templates (business activities or tasks) to a category to enable creating new activities.

Subcategories inherit the activity templates of their parent categories. You can also assign additional activity templates on a subcategory level.

For more information on Customizing settings for Multilevel Categorization in [Service](#), choose [Cross-Application Components](#) [Multilevel Categorization](#).

Using Multilevel Categorization Schemas in Case Management in Service

Use

You can use multilevel categorization to set up reference categories for use in Case Management.

The main business reason for assigning a reference category to a case is process control.

Prerequisites

- You have set up a categorization schema in the WebClient UI in [Service](#).
- You have assigned the categorization schema to a corresponding case type.
- You have released and deployed the categorization schema.

Activities

The following are either integral or possible activities associated with case processing when using multilevel categorization schemas:

- Categorizing the case to reference a specific schema version. For more information see [Categorizing Cases in Case Management in Service](#).
- [Searching for Cases by Reference Category](#).

Example

- A case worker can use the activity templates that are assigned to a reference category when creating new activities within the case.
- Assigning reference categories to cases enables companies to produce hierarchical reports based on the multilevel reference categories. The reports provide an accurate picture of how many cases have been created in the different business areas, as represented by the reference categories.

Categorizing Cases in Case Management in Service

Use

The first time a case is categorized, it receives a fixed date/time reference, which cannot be changed, to a specific schema version. This prevents inconsistent case data resulting from schema changes, such as new schema versions.

Before assigning a reference category to a case, you must maintain a reference date.

Both the [General Data](#) page block and the result lists contain a [Reference Category](#) field for assigning a reference category to a case. The field always displays the category label.

The fields that are displayed in the [General Data](#) page block are the result of Customizing. The result lists that are produced by search operations let you do blueprint personalization to define which available fields to display. Through blueprint personalization, you can also display the [Ref. Category ID](#) field.

Activities

- Setting a schema reference

The following table illustrates how the schema reference mechanism allows you to categorize cases that were created before a multilevel categorization schema was set up for Case Management in [Service](#).

Categorizing cases using the schema reference function

Prior to MLC schema	After MLC schema
1. You create a new case. The reference date/time of the schema is (internally) set to the point in time when the case is created. 2. You save the case without assigning a reference category to it. No schema reference date/time is set for this case.	1. You reactivate an existing case. The reference date/time of the schema is (internally) set to the point in time when the case is reopened. 2. You assign a reference category to the case. 3. You save the case. The schema reference date/time is persisted in the database; a case worker cannot change it in the future. Dependent linked objects can now be used during case processing.

- Maintaining a reference date

You can use the [Reference Date](#) field to predate a case to the past.

❖ Example

Public Service Scenario

A citizen calls the local municipality office to apply for sickness benefits. The reference date of this case is not the date of the call but the date when the citizen first became ill, two weeks earlier. The case worker at the municipality office can create a case whose reference date is in the past.

When assigning a reference category to this case, the case worker can only assign a category that has an effective validity period containing the reference date. These categories might be different from the categories that are valid when the citizen calls the municipality office.

The [Reference Date](#) field is initially set to the date when a case is created; however, the reference date can be changed during case processing.

- Assigning a reference category

You assign a reference category to a case in one of the following two ways:

- Manually entering either a category ID or a category label in the [Reference Category](#) field (combined input field). A wildcard input is currently not supported.

The category ID is checked first because it is always unique in a hierarchically structured categorization schema.

- Using F4-help (only available for the [Reference Category](#) field)

When you call the F4-Help for the [Reference Category](#) field, the case type and the reference date are transferred to the F4-Help to define the limits of the search.

The F4-help provides two different kinds of search:

- Structure Search

Clicking [Go](#) without entering any further search criteria for the [Ref. Category ID](#) and [Reference Category](#) fields, displays a hierarchical tree that reflects the valid schema version for the current case. You can expand the different hierarchy levels and search for the category.

Selecting the corresponding category line in the hierarchy transfers the category information back to the Case Management application.

■ Category Search

If you already know the label or parts of the ID of the category you want to assign to your case, you can use either the [Reference Category](#) or [Ref. Category ID](#) field to do a wildcard search on the categories contained in the schema version that is valid for the current case.

Clicking [Go](#) displays a flat list of the categories that fit your search criteria.

Selecting the category that should be assigned to the case transfers the category information back to the Case Management application.

Changing the reference date within the F4-help produces a result list that contains only categories **that** are valid for that date. After you choose a category, the reference date is also transferred back to the Case Management application.

Whenever a reference category-relevant field—the reference category itself or the reference date—is changed, a categorization check is executed to see whether or not the entered/deleted values invalidate the case. For example:

- A category ID is entered manually but the category is not valid for the entered reference date
- The reference date is changed but the assigned reference category is no longer valid for this reference date
- A category value has been entered without first specifying a reference date

Searching for Cases by Reference Category

Use

You can search for cases by specifying the [Ref. Category ID](#) field as the selection criteria within the [Advanced Search](#) tool of Case Management in [Service](#).

In addition, selecting the checkbox [Incl. Subcategories](#) lets you search for cases assigned to any sub-categories of the category defined in the [Ref. Category ID](#) field.

Caution

If the number of subcategories is significant, the search may not be executed within a reasonable time frame.

The [Ref. Category ID](#) field supports an F4-help, which searches for the reference category that should be used for the case search. This F4-help is similar to the F4-help described in the section [Assigning a Reference Category](#) in [Categorizing Cases in Case Management in Service](#).

Only currently valid schema versions are used to search for the category; old schema versions are not displayed here.

Using Additional Object Types with Case Management in Service

Use

The case record model specifies which object types can be linked to a case and how these objects are structured. You must assign case record models to a case type. In Customizing, choose [Service](#) > [Case Management](#) > [Extended Customizing](#) > [Special Settings](#) > [Define Processes](#) > [Define Process-Dependent Parameters](#).

You can specify per case type which object types are used and how these object types are structured. You use case records to define the usage and structure of object types. Before you can use a case record, however, you need a **case record model**.

Depending on the business process and the scenarios, the case record model has a different structure and different object types.

The service provider and the element type are central to SAP Records Management and record models in Service.

- **Service provider:** For each business object type that you want to integrate into this framework, you must define and implement a service provider. If a service provider already exists for an object/element – for instance, there's a service provider for business object types – you can reuse the service provider. You therefore don't have to implement a separate service provider for each business object type.
- **Element type:** The element type is a form of service provider and is the starting point for users to integrate elements/objects into the case record. A service provider can only be used when at least one element type for this service provider exists. Each element type belongs to only one service provider. A service provider can have any number of element types. You create and edit element types in Customizing.

Example

In standard Customizing, the service provider SCMG_SP_BOR is delivered. It supports the integration of business object types into the framework of SAP Records Management. For this service provider, the element type CRM_BUSINESS_PARTNER is shipped with the standard delivery. When you use this element type, it specifies that only business partners can be integrated into the framework. You can use an element type to specify that objects from external systems can be integrated.

The elements of a case record model can be hierarchical, with the following elements available:

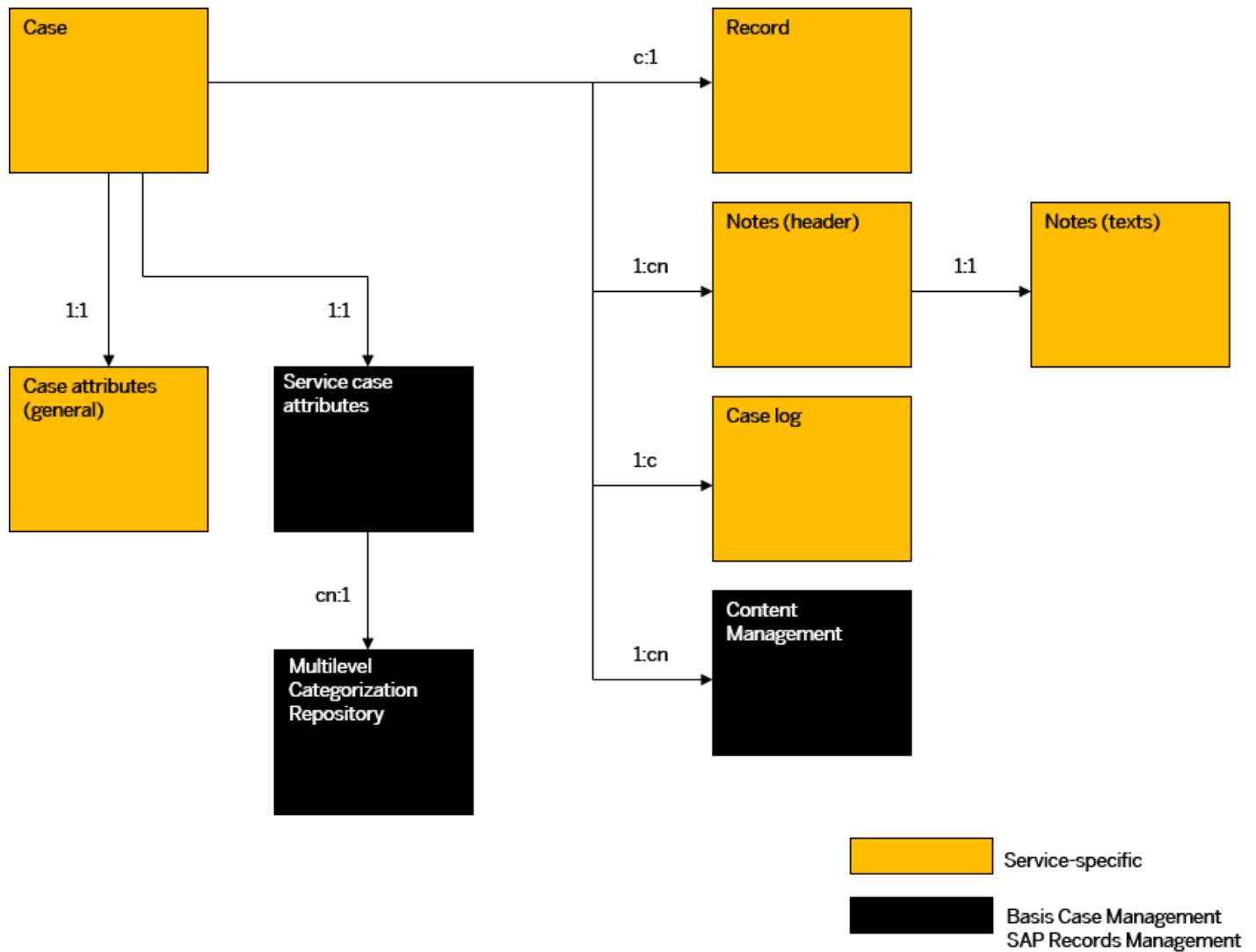
- **Structure node:** Structure nodes act as headers for other nodes and enable you to structure case records. You can't assign elements to structure nodes.
- **Model node:** You can assign elements in case records to model nodes. Model nodes act as placeholders for the objects that can be assigned later. For each model node, you specify the possible object types (for example, business partner, product) and how many of these objects can be assigned.
- **Instance node:** An instance node already has an element/object assigned to it in the record model. There can only be one instance node in a case record. Using instance nodes, you can define information that must be provided in every case record.

More Information

[Creating a Custom Case Record Model](#)

Architecture of Case Management in Service

Case Management in Service is an extension of Basis Case Management, which in turn uses the SAP Basis Records Management. The following diagram illustrates the specific components of a case in Service.



Case Management in Service – Architecture

The [Case](#) in Service consists of attributes such as ID, short text, status, priority, and other attributes, as well as of several subcomponents. These subcomponents are the case record, case notes, case log, and the integration into the document management system.

Case Attributes

A case consists of attributes from the Basis case and can be enhanced with Service attributes. Basis Case Management allows for other applications, for example, Case Management in Service, to add application-specific attributes. To be able to use this feature, you must set an additional table in the attribute profile in Case Management Customizing. In Service, the table [CRMD_CMG_CASE_AT](#) is used for this task. As an additional attribute in Service, only the attribute [Short text in upper case](#) is used. This additional attribute causes upper case short texts to be stored twice – once in a Basis case table and once in the Service table. This feature enables non-case-sensitive searches.

Creating a Custom Case Record Model

Context

The purpose of this procedure is to create a custom case record model on the basis of existing business object types. You create a custom case record model by copying the default element types. Element types are defined in the registry of SAP Records Management.

Procedure

1. Open the registry using transaction **SRMREGEDITC** and choose **S_AREA_CMG**.
2. Create a new element type for the case record itself:
 - a. Expand the application area **S_AREA_CMG**.
 - b. Expand **SCMG_SP_CASE_RECORD** and right-click on the element type **CRM_SPS_CASE_RECORD** and choose **Copy** from context menu.
 - c. In the **Copy To** field, enter an element type ID and a short description. Save your data.

i Note

Using this procedure, you can copy other element types. You can find the element types for the business object types under the service provider **SCMG_SP_BOR**. For each business object type that you want to use, you have to create a separate element type – preferably through copying. After you perform these steps, you can create the case record model using the Records Modeler.

3. Run transaction **SCASE** and choose the Records Management System (RMS) ID. For Case Management in Service, use **CRM_CMG_DATA**.
4. Open the folder **Record and Case Record Models** and right-click the entry **Modeler for CRM Case Records**, and then choose **Create**. A case record model with the default entry **Worklist** as the structure node and **Worklist Element** as the model node now appears in the work area.
5. Change these default entries and create new nodes to set up the model for your own requirements by choosing the **Worklist Element**. The element details are displayed on the right of the screen. For structure nodes, you only define the description. For a model node, you must maintain the following properties:
 - **Description**, for example, **Service Order**
 - **Element Type**: By setting the element type, you determine which object types you can insert into the case record at this point, and can then also be controlled. After you save, the element types created under Step 2 can be used.
 - Number of objects that can be integrated: **At Least** and **Maximum** fields. If an undetermined number of objects can be integrated into a case, enter an asterisk (*) in the field **Maximum**.
 - **Relationship**: This value specifies the relationship between the element/object and the case record. To select the relation, use the F4 help.
 - Attribute **ANCHOR**: This attribute is used to insert objects/elements into a case by using the Basis Case Management API. Since Case Management in Service uses this API, you must maintain this attribute. Use the same value as the one for the corresponding relation. For example, the ID of the relation service order is **CS**, so the corresponding attribute **ANCHOR** must have the value **CS** as well.
6. After the case record model has been created, save your work, and then activate the hierarchy nodes.
7. Release the case record model by clicking the **Model** button, then from the list choose **Change Status** and select **Status Released**. Enter a short text for the case record model as well as a unique ID. You can generate a unique name by clicking the icon at the end of the field.
8. Now, you insert the model within the element type that you created. You do this in the Records Management registry. Run transaction **SRMREGEDITC** and choose **S_AREA_CMG**.
9. Right-click on the element type that you created previously, and in the context menu choose **Change**. Under **Connection Parameter Values**, choose the parameter **MODEL_ID** and click **Change**. Enter **CRM_CMG_DATA** and then press **Enter**. Search for the case record model you created using the F4 help, choose it from the list and confirm the entry. Save the settings.
10. To be able to use the case record model, you must assign both the record model and the element type you created to the case type that you want to use. You do this in Customizing under **Service > Case Management > Basic Settings > Define Case Types**.
11. After creating a new element type, you assign it to the process **C_01**. You do this in Customizing under **Service > Case Management > Extended Customizing > Special Settings > Define Processes > Define Process-Dependent Parameters > Assign Element Types and Business Object Types to a Process**.

Results

You've created a custom case record model and element type that you can use in Case Management.

Integrating Business Object Types With the Case Record Model

Use

You can integrate additional business object types with Case Management in Service. To do this, you must create a new element type.

Prerequisites

You can create element types in Records Management. To create a new element type, you must perform the steps relating to element types described in [Creating a Custom Case Record Model](#).

i Note

Objects and/or elements are integrated with a case using relations or anchors. For each object type, a separate relation exists, for example, the relation and anchors C1 for the business object type **Business Partner**. The same also applies to new object types. As creating new relations is a modification, we recommend that you use the relations X1, X2, X3, X4, X5, X6, X7, X8, and X9.

Process

1. Run transaction **SRMREGEDIT**.
2. Choose **S_AREA_CMG**.
3. Right-click on the service provider **SCMG_SP_BOR** and choose **Create Element Type**.
4. Enter an element type ID and a short text and then press .
5. Select whether the object is local or transportable and save your work. If it's transportable, enter a transport package.
6. Right-click on the element type you created and choose **Change**.
7. Under the **Connection Parameter Values** tab, set the values for **BOR_OBJECT_TYPE** to the desired business object, and leave the setting for **LOGICAL SYSTEM** as **NONE**.
8. Go to the **Classification** tab and maintain the parameter IDs using the values for **TYPE** and **RMS_ID**.
9. Save your changes.

Result

You've created and integrated the element type. To use the new business object type in cases, you must create a new model node for the element type in the case record model you're using.

Additional Customizing Configuration for Case Management in Service

The following is a list of Customizing configuration activities using BAdIs that you can use to enhance Case Management in Service.

- Integrating customer-specific validations using BAdIs SCMG_VALIDATE_C and SCMG_VLDT_BFR_STR_C
- Specifying default field values when creating a case. There are two scenarios: Creating a case from a case management application using BAdI SCMG_INITIALIZE_C, or creating a case from a business transaction using BAdI CRM_CMG_ON_CREATION
- Storing additional link attributes using the enhancement spot /CRMS/CMG_REC_ATTR

For more information in Customizing, choose ► [Service](#) ► [Case Management](#) ► [Extended Customizing](#) ► [Special Settings](#) ► [Define Processes](#) ► [Business Add-Ins \(BAdIs\)](#) ►.

Integrating Multilevel Categorization

Use

Multilevel categorization enables you to individually design the categorization and integration of specific functions in your applications. Case Management in Service supports the multilevel categorization function, making it possible to assign a case to a category of a multilevel categorization schema.

More Information

[Multilevel Categorization in Case Management in Service](#)

[Multilevel Categorization](#)

Additional BAdIs for Case Management in Service

Use

There are various additional enhancements and Business Add-Ins (BAdI) that you can use for Case Management in Service.

Features

- Inserting new object types in a case record
- Initializing case attributes
- Changing a case before the check
- Additional checks to validate cases and case attributes
- Triggering an event after saving a case

In the Customizing choose ► [Service](#) ► [Case Management](#) ► [Extended Customizing](#) ► [Special Settings](#) ► [Define Processes](#) ► [Business Add-Ins \(BAdIs\)](#) ►.

Postprocessing Actions for Cases

Using actions (in the Post Processing Framework) you can schedule and start any postprocessing steps depending on the case type, as well as the various scheduling and start conditions. Examples of postprocessing steps are sending e-mail or starting workflows. You can also stipulate that a specific ABAP coding is executed.

Actions

The actions you can perform are:

- Creating an action profile with action definitions; in the action definitions, you define the action characteristics, such as the processing time or whether the action is modifiable in the document (case)
- Defining conditions for the action definitions (scheduling and start conditions)
- Assigning the action profile to the desired case type
- Assigning the partner determination for actions using a Business Add-In (BAI)

For more information, in Customizing choose [Service](#) > [Case Management](#) > [Extended Customizing](#) > [Process Actions](#) >

Interaction Center

The interaction center provides multichannel collaboration and communication tools to enable efficient and consistent customer service.

Interaction center (IC) agents can handle inbound and outbound service transactions using the phone, e-mail, fax, chat, and letter communication channels. They can process business transactions, such as service orders and enhance their productivity by using alerts and a knowledge search. All relevant account information is available to them in the IC, such as account data, service order status, and product-related information.

The IC provides agents with the tools they need for the processes that comprise their daily work. Generally, the IC supports the following processes:

- Communication process

This business process involves the communication between an IC agent and a business partner (customer, employee, and so on) in real time, for example using the telephony channel. Since IC agents can also work asynchronously, processing e-mails, service requests, or workflow items from the agent inbox, the communication process is not necessarily performed at all times.

- Interaction process

The interaction process is the central IC process, which often encompasses the communication process and the business transaction process, and is performed whenever an agent uses the IC to communicate or interact – either directly with a business partner or asynchronously by working on a business transaction belonging to a business partner.

The interaction process starts in the interaction mode with either the identification and confirmation of the business partner or the selection of a business transaction, e-mail, workflow, or other item belonging to a business partner. The interaction process includes all steps that the agent takes to work on a customer request. The interaction process can optionally result in the creation of one or more business transactions.

- Business transaction process

Some simple customer or employee requests can certainly be resolved by providing a short answer or other information, not making use of a business transaction. However, frequently an interaction will result in one or more follow-on business transactions, such as a service request. In such cases, each business transaction is linked to the interaction record, providing a connection between the business process and the interaction process.

To start the interaction center, you use the WUI transaction and select a corresponding business role, such as [Service IC Agent](#).

Related Information

[Interaction Center WebClient](#)

[E-Mail Response Management System](#)

[Integrated Communication Interface](#)

Interaction Center WebClient

Use

The interaction center (IC) WebClient is a thin-client, highly optimized desktop application for interaction center agents. It provides a comprehensive set of interaction center (IC) functions and supports various processes required to answer customer queries and work in corresponding business transactions.

Features

- [Account Identification](#)

A customer search function that agents use to identify customers in the system. You can set up this function to identify customers even before agents accept contacts, based on automatic number identification (ANI) or the address of an incoming e-mail.

- [Account Overview](#)

An overview that agents use to get complete information for an entire account, including individual account details, notes, roles, planned activities, and business partner relationships.

- [Interaction Record](#)

A customer-focused overview of the current interaction and customer history.

- Interaction History

A comprehensive interaction history that provides one view of a customer. Search and overview of business transactions for a customer. Agents can use various search criteria to find the business transactions they are looking for. Agents can record data both during and immediately after an interaction.

- Knowledge Article Search

Interaction center agents can search for knowledge articles, which can include documents, links to Web sites, and screenshots, to answer customer queries.

- [Agent Inbox](#)

A central worklist that allows agents to process incoming and outgoing mails, planned activities, and other work items.

- [E-Mail](#)

An interface that agents use for processing incoming and outgoing e-mails, as well as completing unfinished e-mails.

- [Chat](#)

An interface that agents use for processing incoming chat sessions.

- [Scratch Pad](#)

An electronic notepad that agents use to record miscellaneous information during an interaction. Notes from the scratch pad can be imported into fields in the IC WebClient.

- [Alerts](#)

Functions that combine system resources intelligently to guide agents during customer interactions.

- Queue Status

A feature that agents use to quickly display the channels and queues they are assigned to, based upon configuration in the communication management software.

Specifics for Business Transactions in the Interaction Center

Use

This topic summarizes the specifics that apply to the processing of business transactions in the interaction center (IC).

Features

- The information from a confirmed account is automatically copied to any newly created business transaction.
- The business transactions that you use in the IC have a tiled layout. Alternatively, you can configure an overview layout for these business transactions in Customizing for the UI framework.

Settings

- Business transaction profile in IC

You can make the following settings in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Business Transaction](#) > [Define Business Transaction Profiles](#) :

- Define which transaction types are available to the agent to create business transactions, and which transaction type is to be the default
- Define whether dialog boxes are allowed and displayed automatically (automatic display is only supported by the interaction record)
- Define the transaction type for transaction data and the additional business transactions separately

- Automatic routing

Set up automatic routing. For information about the necessary settings in the rule modeler, see [Interaction Center WebClient](#) > [Business Transaction](#) > [Define Automatic Routing](#) .

Sales Transactions

Get an overview on how to create and change sales orders and sales quotations, display item proposals, as well as display and print sales transactions.

To use the sales transactions, you use the WUI transaction and select a corresponding role, such as [Sales Professional](#).

[Entering Sales Documents on the WebClient UI](#)

Entering Sales Documents on the WebClient UI

Use

You use this function to create and change sales orders and sales quotations.

You carry out your presales activities such as Opportunities and Activities on the WebClient UI, and then, for example, process a rush order without having to exit.

This function is of particular use to sales employees who work mainly with the WebClient UI and therefore do not require the full range of functions offered by Sales Quotation and Order Management. For example, sales assistants can use this function to create simple sales orders.

Prerequisites

You can directly create and change a sales document on the WebClient UI once you have carried out the following tasks:

- You have assigned the relevant sales documents to a profile in Customizing under ► [Service](#) ► [Transactions](#) ► [Settings for Sales Transactions on WebClient UI](#) ► [Define Profiles for Sales Transactions](#) ►.
- You have defined copy control of transaction types in Customizing for [Service](#) under ► [Transactions](#) ► [Settings for Sales Transactions on WebClient UI](#) ► [Copy Control for Opportunities and Sales Transactions](#) ►.
- If you want to use payment cards in sales orders, you have made settings in Customizing under ► [Sales and Distribution](#) ► [Billing](#) ► [Payment Cards](#) ►.

Features

When you maintain a sales document using the WebClient UI, the system enters data directly in SAP. This means that you can decide yourself which UI you use to process the sales order.

Creating Sales Quotations and Sales Orders as Follow-Up Transactions from an Opportunity

The sales representative or sales assistant has determined a potential sales opportunity, and has therefore created an opportunity. If the customer then requests a quotation, the sales employee can immediately create a sales quotation. To do so, the sales employee opens the opportunity that he or she created earlier, and creates the sales quotation as a follow-up transaction from the opportunity.

When you do so, the following data is copied to the sales quotation:

- Business partners
- Sales organization
- Distribution channel
- Division
- Sales office
- Sales group

- Product configuration
- Items
- Product
- Quantity
- Notes
- Unit

You can only display the following fields:

- Product
- Product ID
- Transfer
- Item number

You can navigate to the sales document using the transaction history in the follow-up activity.

Offering Alternative Items in Sales Quotations

In addition to the item requested by the customer, the sales employee can also offer an alternative item in the sales quotation. In order to offer the alternative item, when the sales employee copies the standard item and the alternative item from the opportunity to the sales quotation, the sales employee must make sure that the alternative item comes directly after (by item number) the standard item in the list of items.

Creating a Follow-Up Activity from a Sales Document

You can create a follow-up activity from a sales document. You can then navigate to the sales document using the transaction history of the follow-up activity. The system shows the follow-up activity in the following:

- Transaction history of sales orders and sales quotations
- Document flow shown in SAP GUI (transactions VA02, VA22, and VA42)

Correcting Errors

The system handles errors according to [Sales and Distribution](#) rules. This means that if errors occur, you cannot save the sales order. You can only continue processing the sales order after the errors have been corrected.

i Note

The assignment of digital payment cards is currently not supported on the WebClient UI.

Activities

Certain restrictions apply to this function. Therefore, it should only be used by sales employees who work with sales documents occasionally.

Example

You create a sales quotation as a follow-up transaction for an opportunity. The system copies data directly from the opportunity to the sales quotation.

Item Proposals in Sales Transactions

Use

- Item proposals are used as input help in sales orders and quotations.
- Item proposals are available on the WebClient UI for Service, including the Interaction Center (IC).

Prerequisites

You have set up the item proposals in sales transactions in Customizing by choosing **► Service ► Transactions ► Settings for Sales Transactions on WebClient UI ► Product Proposals ►**. Under the node **Past Orders**, you make the settings for item proposals from previous orders.

Features

Item proposals are displayed directly in the **Items** assignment block.

You can automatically display item proposals, or manually trigger the display. To manually trigger the display of proposals, set the indicator **Past Orders on Request** in the Customizing of product proposals, in the activity **Assign Method Schema to Transaction Type**.

Item proposals appear at the end of the item list as suggested items, that is without item numbers. You can transfer the suggested items as final order items, by entering an order amount. Only then can the system carry out pricing and an availability check for the items.

- **Item Proposals from Past Orders**

The system proposes items from the previous sales transactions for this customer, that is products and order quantities.

- You can define the transaction types and the number of transactions in a certain time frame that the system draws from to determine proposed items, for example the last 10 sales orders from the past three months.
- You can define how the system calculates the proposed quantities: from the last order, or as an average quantity from the previously defined number of orders.
- You can select the required transactions using a popup, or the system can immediately (after you have entered the customer data) display all products from the past transactions.

Searching for, Displaying and Printing Sales Transactions on the WebClient UI

Searching for and Displaying Sales Transactions on the WebClient UI

If you want to provide your customer with information on the current status of a sales order that has already been created, you can use the **Search** function to locate transactions quickly, and forward the necessary information to the customer immediately. In addition, this function allows you to enter changes requested by the customer quickly and easily in the sales transaction.

Displaying sales transactions related to the relevant account

You can display sales transactions, that is, sales quotations and sales orders related to the relevant account on the [Account](#) overview page, under the corresponding assignment blocks. This allows you to gain an overview of all existing sales transactions for a specific customer, for example, in preparation for a customer visit.

Sales representatives can also create new sales transactions, edit or delete existing sales transactions from these assignment blocks. When you create a new sales transaction from here, the [Sold-to Party](#) field is automatically filled with the account from the [Account](#) overview page. Once you save and return to the [Account](#) overview page, the newly created sales transaction appears in the relevant assignment block.

Data retrieval parameters are delivered for each of these assignment blocks, which enable you to filter which data you want to display on the corresponding assignment blocks. For example, you may only want to display sales orders with posting dates for last month.

The following data retrieval parameters are available for the corresponding assignment blocks:

- Quotations: [Posting Date](#), [Transaction Type](#), [Valid From/To](#)
- Sales Orders: [Posting Date](#), [Transaction Type](#)

For more information on data retrieval parameters, see [Configurable Data Retrieval](#).

Printing Sales Transactions

You can print sales transactions, that is, quotations and sales orders from the WebClient UI. This allows sales representatives to preview, save or print sales transactions in order to check them, share them with their team, or file them to their documents. It is not intended for sales representatives to send transactions directly to their customers; this will continue to be done from the standard [Sales and Distribution](#) UI.

Once you have saved the sales transaction on the WebClient UI, you can use the [Preview Output](#) pushbutton for a print preview with a PDF viewer. You can also save locally or print out the preview using a PDF viewer. To print sales transactions, you can choose the output type set up in SAP [Sales and Distribution](#), including PDF-based print forms, SAPscript and Smart Forms.

For more information, see PDF-Based Print Forms

Specifics for Interaction Center: Communication Management Software

Use

Communication management software (CMS) refers to software products that manage communication channels and handle individual contacts with customers through different channels, such as phone, e-mail, and chat. The Integrated Communication Interface (ICI) allows the CMS to communicate with the Interaction Center WebClient.

For more information about ICI, see [Integrated Communication Interface](#).

For information about the features that are available in the telephony channel, see [Communication Management Software](#).

For information about the features that are available in the chat and e-mail channels, see [Chat](#) and [E-Mail](#) in the Interaction Center WebClient documentation.

Prerequisites

- You have made the settings in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Define Communication Management Software Profiles** ►.
- Optionally, you have made the settings for checking the continued communication between the communication session, the CMS, and the Web browser in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Define Heartbeat Profile** ►.

In this Customizing activity, you can make settings for the following checking methods:

- Heartbeat without watchdog
- Heartbeat and watchdog
- CMS session availability

Business Process Push

Use

You can use this function to push business processes to external routers or communication management software (CMS) for distribution to interaction center (IC) agents based on their real-time availability. For example, the following business processes can be pushed:

- Business transactions, such as service orders
- Faxes and letters received through SAPconnect or the ArchiveLink scenario

The ID of the business process and a parameter that specifies the type of business process are transferred to the external router or CMS.

In addition, for each business transaction, you can define a set of additional parameters to represent, for example, status and priority. In order for the business process to be pushed and prioritized correctly in the CMS, you define rule policies to set the values for the parameters before these values are transferred to the CMS.

Prerequisites

- You have made the settings for business process push in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Business Process Push** ► **Define Settings for Business Process Push** ►.
- You have assigned parameters to those agent inbox item types that you want to be pushed to external routers or CMS in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Business Process Push** ► **Assign Parameters for Business Process Push** ►.
- You have defined rule policies to set the values for the parameters. For more information, see [Rule Policies](#).
- You have made the settings for pushing faxes in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Settings for Asynchronous Inbound Processing** ► **Define Receiving E-Mail/Fax Settings** ►.
- You have made the settings for pushing letters in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Business Process Push** ► **Define Letter Settings** ►.

Features

Business transactions, such as service orders, are automatically transferred to an external router or CMS if the following conditions are met:

- The business transaction is saved and has ended, or is saved and escalated.
- The values of the parameters that have been defined in Customizing for the business transaction correspond to the values of the parameters that are set in the rule policies.

For example, the business transaction must have status **In Process** and priority **High** to be pushed to an external router or CMS.

The business transaction is locked and the current employee responsible for the business transaction is removed. No changes to the business transaction and its status are possible until the transaction has been pushed to an IC agent.

When an IC agent is alerted that an inbound business transaction is routed, the business transaction ID and type are shown on the screen. The IC agent can either accept or reject the business transaction. The IC agent who accepts it is assigned as an employee responsible and can process that business transaction and change its status.

- Rule policies

You define rule policies to set the values for the additional parameters for business transactions and cases before these values are transferred to the CMS. You define rule policies in the **Order Routing** and the **Intent Driven Interaction (IC WebClient)** contexts using attributes, conditions, and actions.

Example

You can create the following rule policy in the **Order Routing** context:

Conditions:

If Priority Is Not Equal To Very High

Actions:

- Set business process push parameter (Item Type = SALES ORDER; Parameter Name = STA
- Send business process push parameter

You can create the following rule policy in the **Intent Driven Interaction (IC WebClient)** context:

Conditions:

If Current Event Equals Order Was Saved

Actions:

- Set business process push parameter (Item Type = SALES ORDER; Parameter Name = PRI
- Set business process push parameter (Item Type = SALES ORDER; Parameter Name = STA

Conditions:

If Current Event Equals Interaction Ended

Actions:

Send business process push parameter

- Pushing of faxes and letters received through SAPconnect or the ArchiveLink scenario

If the Customizing settings for pushing faxes and letters have been made, faxes and letters received through SAPconnect or the ArchiveLink scenario are pushed to IC agents based on their real-time availability. The number at which the fax arrives is the indicator for pushing the fax.

An IC agent is alerted that an inbound fax or letter is pushed. The IC agent can then either accept or reject it. Once the IC agent accepts the fax or letter, the [Identify Account](#) page and the [Fax Preview](#) or [Letter Preview](#) is displayed.

ERMS E-Mail Push

Use

You can use this function to push e-mails to the communication management software (CMS) for distribution to an interaction center (IC) agent.

When e-mail is routed to an agent, the [Accept](#) and [Reject](#) context area buttons allow the agent to decide whether or not to accept the inbound message. If the agent accepts the e-mail, the system searches for the business partner based on the e-mail address. The agent can then use standard e-mail functions in the e-mail editor view.

Prerequisites

You have specified [Inbox](#) as the e-mail provider for the e-mail profile in Customizing for [Service](#) under [IC WebClient](#) [Basic Functions](#) [Communication Channels](#) [Define E-Mail Profiles](#).

Activities

- Create rules in the [ERMS](#) context to push e-mail to the CMS for distribution

❖ Example

You can create the following rule policy in the [ERMS](#) context:

Condition:

If E-Mail Sender Equals jdoe@customer.com

Action:

Push E-Mail to IC WebClient Agent

- Configure a CMS system ID that points to a CMS system that supports action routing

Create an entry for the CMS system ID ERMS_ACTION. You do this in the [SAP Easy Access](#) menu under [Service](#) [Interaction Center](#) [Interaction Center WebClient](#) [Administration](#) [Communication Management Software](#) [Interface Settings](#) [Maintain Communication Management Software Connections](#).

More Information

[E-Mail Response Management System](#)

Queue Status

Use

The queue status, located in the bottom right hand corner of the interaction center, allows you to quickly display the channels and queues you are assigned to.

Features

The queue status:

- Displays the current date and time (default display) or queue.

If queue information is available, then an arrow is shown after the date. You can click on the arrow to show which queue the current incoming contact was sent to.

Clicking on the arrow again returns the current date and time.

- Displays a tool tip with a quick view of multichannel information.

The queue status tooltip displays the channels you are working in, for example, e-mail, chat, and telephony, specific phone numbers and e-mail addresses from which you can receive inquiries, and all queues to which you are currently assigned.

The queue status tool tip is only available if the connection to the relevant system is currently available.

- Displays the [Agent Dashboard](#).

If you click on the queue status, the agent dashboard is displayed.

Agent Dashboard

Use

You can use the agent dashboard to view information relevant to your current interaction center (IC) session. It is useful when you need more information about what is displayed in the [queue status](#). You can access the agent dashboard by clicking the queue status at the bottom right corner of the browser window.

Features

The agent dashboard displays the following information:

- Your user ID and business role
- Channels and queues

You see all channels you are working in, for example, e-mail, chat, and telephony, specific phone numbers and e-mail addresses from which you can receive inquiries, and the queues to which you are assigned.

This information is only available if:

- There is a connection to a communication management software system, and
- The communication management software passes queue and channel information to the interaction center.
- System information

System information is used primarily by system administrators for troubleshooting. It includes:

- Information on the domain in which the servers are operating (for example, wdf.sap.corp)
- Name and description of the communication management software being used

Specifics for Interaction Center: Toolbar

Use

The toolbar provides an interface between your communication management software (CMS) and the Interaction Center (IC) WebClient. The toolbar contains pushbuttons and icons for communication functions that apply to the various communication channels.

You can modify the content of the toolbar to better meet your business requirements. For example, you can specify that different sets of pushbuttons are available in the toolbar depending on the connection of an agent to the CMS and the status of the agent's interaction with a contact.

Integration

The following processes are examples of how you can use the pushbuttons in the Interaction Center toolbar:

- Communication process

For a phone call, you start the process by choosing the pushbutton **Dial** or **Accept**, and you end the process by choosing the pushbutton **Hang Up**.

- Interaction process

Includes account maintenance and the interaction record. Additionally, business processes such as sales and service can also be handled in the interaction center. You start the process by confirming the account and you end the process by choosing the pushbutton **End**.

- Communication process and interaction process

The two processes can take place simultaneously and do not necessarily start and end in the same order, for example in the following cases:

- Multiple phone calls

You can dial and end multiple phone calls during the same interaction. This occurs, for example, when a phone line is disconnected during a customer interaction and needs to be dialed again.

- Multiple interactions

Multiple interactions can happen during a single phone call. If the agent receives and accepts the next call whilst wrapping up the previous interaction, they can end the first interaction by choosing the **End** pushbutton despite already being in the next active phone call.

Prerequisites

- Your system administrator has set up the CMS and ensured that the user names there match those in the IC. For more information, see [Specifics for Interaction Center: Communication Management Software](#) and the documentation for the CMS you are using.
- You have made the settings for the toolbar in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Toolbar Profiles**.

Features

You use the toolbar to work in different channels. The system determines the pushbuttons that appear in the toolbar, depending on the following:

- The communication channel in which you are currently working

- Customizing settings

Example

Pushbuttons in the IC toolbar

Channel	Name	Definition
Universal (available for all channels)	Accept	Accepts the contact. In telephony, Accept is similar to picking up the receiver.
	Reject	Sends the contact back to its original queue.
	Transfer or Blind Transfer (in the telephony toolbar)	In telephony, puts caller on hold while you transfer the call to another agent's extension without first consulting with that agent.
	End	Ends the interaction (indicating that the agent has finished wrap up activities), saves all objects created or changed during the interaction, and clears customer information from the screen and from the business data context.
	Hang Up	Similar to hanging up the receiver. Ends a phone call that was started with Dial or Accept .
	Clear Interaction	Discards the interaction record and resets the account identification. i Note Discarding other transactions such as sales or service transactions is not possible in this way. Depending on the process status, this may be done by choosing Cancel if it is part of the transaction toolbar. If you want to clear the entire context of the session including the business data context, you have to choose the Clear Interaction pushbutton followed by the End pushbutton.
	Reset CTI	Resets the context area of the browser session to synchronize it with the communication session state. The different states can for instance be caused by network issues.
Telephony	Dial Pad	Displays a dial pad that looks like the numbers on a telephone, which you can use to make a call.
	Hold	Puts the active call on hold.

Channel	Name	Definition
	Toggle	Switches between a call on hold and an active call, automatically placing the active call on hold and vice versa.
	Retrieve	Takes a call off hold and makes it active once again.
	Consult	Puts customer call on hold while you call another agent. After you are finished consulting with the other agent, the agent call is dropped and the customer call is active once again.
	Warm Transfer	Puts caller on hold while you consult with another agent before transferring the call to the agent's extension.
	Conference	Puts customer on hold while you call another agent. Then you activate both calls at the same time so that all three parties can participate in a discussion.
	Wrap Up	Indicates to the system that you need time to wrap up the call. Wrap up time ends when you choose the End pushbutton.
E-Mail	Pushbuttons specific to the e-mail function, such as Reply , New , or Forward , appear on the e-mail screen dynamically, as and when they are required.	Represent standard e-mail functions.
Chat	Leave	Finishes the chat.

Work Modes

Name	Use
Ready	When you choose Ready , the CMS system sends incoming contacts to you.
Not Ready	When you choose Not Ready , the CMS system does not send incoming contacts to you.

Scratch Pad

A temporary workspace in the interaction center that you use as an electronic notepad to store miscellaneous information during an interaction.

Customers usually provide a lot of information during the first few minutes of conversation, before you have a chance to access the relevant transaction. Most agents have separate notepads to take notes and then enter the data into the system later.

The scratch pad replaces the need to write manual notes. It allows you to electronically store information so it can be copied into specific functions that require this data.

The scratch pad allows you to focus on recording and listening to what the customer is relaying. Customers do not have to hold their conversation until you follow through the sequence of the interaction, gather the preliminary information, such as contact information, and then navigate to the appropriate area in the IC WebClient to record the information. This also saves customers from having to provide information more than once.

The scratch pad is located in the top left corner of the IC, under the title bar. It is available throughout the interaction, and can be copied into different IC functions, such as the interaction record or knowledge search by choosing [Import Scratch Pad](#). All of the data on the scratch pad is imported, but you can still edit, add, or delete it, if necessary.

Data remains on the scratch pad for the entire interaction until you either select [End](#) from the toolbar, or select [Clear](#). Closing the scratch pad does not delete the contents.

Alerts and Messages

Use

These functions display texts on your page to alert you about items or situations that require your immediate attention. There are three types:

- Alerts

Alerts are predefined by system administrators or Interaction Center (IC) managers to be triggered by certain events in the system.

- Messages

There are three types of messages: informational, warning, and error.

Features

- Placement on page

Alerts appear in the alerts area and system messages appear in the system messages area.

- Alerts

- If more than two alerts appear, an arrow appears, which allows you to page through alerts in pairs.
- All alerts disappear when you choose [End](#).

Example

Alerts

Marcus is a customer service representative at an interaction center. He accepts a call from Ms. Anna Glenn and confirms her identity in the system. The following alert appears:

Service order 00225

Marcus asks Ms. Glenn if she is calling regarding this order. She says that she would like to know the status of the service order. Marcus simply chooses the link to the service order and can inform Ms. Glenn of its current status.

Creating Alerts

Use

You can create alerts in that use system information to provide agents with information they may need during customer interactions.

i Note

No preconfigured alerts are delivered. Interaction center managers define alerts in the [Alert Editor](#) and assign them through rule policies. Alerts are created independent of any rule policy context.

Alerts can only be triggered in the [Intent-Driven Interaction](#) context. This context allows you to automate specific interaction center functions, for example, triggering and terminating alerts, via business rules. Interaction center alerts are not relevant for other rule policy contexts.

Prerequisites

- You have completed the Customizing for [Service](#) under [Interaction Center WebClient](#) > [Additional Functions](#) > [Intent-Driven Interactions](#) > [Service Manager](#) > [Configure Alerts](#).
- You have completed the Customizing for [Service](#) under [Interaction Center WebClient](#) > [Additional Functions](#) > [Intent-Driven Interaction](#) > [Define Intent-Driven Interaction Profiles](#).
- If you want to use user-triggered alerts, you have selected this alert type in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Basic Functions](#) > [Communication Channels](#) > [Define Communication Processing Profiles](#).

Features

The following types of alerts are available in the IC:

- Real-time alerts

With these alerts, the system continually polls the server to check for alerts.

- User-triggered alerts

With these alerts, the system waits until the user triggers an event to perform a server round-trip to check for alerts.

This type of alert allows you to show alerts without using server polling and can be used with asynchronous processing.

When creating alerts, you can select a theme that affects the appearance of alerts by changing, for example, the font style, font color, and the alert icon. You can also preview the theme. Once you select a theme, an example text is displayed with the theme applied.

Themes must be defined in Customizing before they are available for selection when creating an alert. You can define themes in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Additional Functions](#) > [Define Themes for Alerts](#).

Broadcast Messaging

Use

Interaction Center (IC) supervisors can send messages to IC agents in real time using broadcast messaging. They can also create and maintain distribution lists for broadcast messaging and display sent messages. IC agents can receive messages, but cannot send messages.

Prerequisites

You have made the settings for broadcast messaging in Customizing for [Service](#) under ► [Interaction Center WebClient](#) ► [Additional Functions](#) ► [Define Broadcast Messaging Profiles](#) ►.

Features

- Distribution lists

To create or maintain distribution lists, click [Maintain Distribution Lists](#). Broadcast messaging profiles contain an organizational ID. When you create distribution lists, the names that appear in the input help for [Add User](#) come from the organizational unit entered in the selected broadcast messaging profile. You can also copy an existing distribution list, delete a user from a distribution list, or delete a distribution list. The default distribution list is the one that appears first alphabetically.

To share distribution lists, the owner of the list can flag other users within the list as administrators. As an administrator, you can add and remove users from the list, use the list to send broadcast messages, and you will be able to view any broadcast messages that were sent to this list in the [Sent Log](#).

- Themes

You can define themes for broadcast messaging in Customizing for [Service](#) under ► [Interaction Center WebClient](#) ► [Additional Functions](#) ► [Define Themes for Broadcast Messages](#) ►.

Once a theme has been defined in Customizing, it is available for selection when creating a broadcast message. The theme affects the appearance of broadcast messages by changing, for example, the font style and color. You can also preview the theme. Once you select a theme, an example text is displayed with the theme applied.

Interaction Record

Use

The interaction record is a screen in the interaction center (IC) that contains data for the current interaction with an identified business partner. IC agents can log current interactions in this screen, and postprocess them if necessary.

Interaction with a customer always includes an interaction record, which can then be linked to another business transaction, such as a service process. The interaction record can, however, also be saved without creating another business transaction.

i Note

From a technical point of view, the interaction record is created as a business transaction.

Once an agent has identified an account, the system navigates to the interaction record. In the interaction record, the agent logs the interaction, records central data from the call, and postprocesses the interaction when the call is finished.

To increase system performance at high volume call centers, you can use the lean interaction record. The lean interaction record is a modified version of the full interaction record.

Prerequisites

- You have defined a date profile and assigned this to your transaction type to use interaction time recording in Customizing for **Service** under ►► **Basic Functions** ► **Date Management** ► and ►► **Transactions** ► **Basic Settings** ► **Define Transaction Types** ►.
- You have activated the rescheduling view and optionally enabled automatic text entry for call rescheduling in Customizing for **Service** under ►► **Interaction Center WebClient** ► **Business Transaction** ► **Define Business Transaction Profiles** ►.
- **Using Activity Reasons:**
 - You have made the settings for activity reasons in Customizing for **Service** under ►► **Transactions** ► **Settings for Activities** ► **Define Activity Reasons** ►.

i Note

The interaction reason corresponds to the activity reason.

- You have assigned the **Reason for Activity** subject profile in your transaction type to the **Business activity** transaction type in Customizing at the header level.

You can make settings for transaction types in Customizing for **Service** under ►► **Transactions** ► **Basic Settings** ► **Define Transaction Types** ►.

- **Using Multilevel Categorization:**

You have made the following settings in Customizing for **Service**:

- Activate categorization for the interaction record under ►► **Interaction Center WebClient** ► **Business Transaction** ► **Define Business Transaction Profiles** ►.
- Make the necessary settings under ►► **Transactions** ► **Basic Settings** ► **Define Transaction Types** ►.
- Define the subject profile for the category modeler under ►► **Basic Functions** ► **Catalogs, Codes and Profiles** ►.
- Make the necessary categorization settings for the interaction record under ►► **Interaction Center WebClient** ► **Business Transaction** ► **Define Categorization Profiles** ►.

See [Multi-Level Categorization](#) for more information.

- If you want to use the lean interaction record instead of the full interaction record, you have assigned a business transaction profile that contains transaction type 0100 to the business role. You do this in Customizing for **Service** under ►► **UI Framework** ► **Business Roles** ► **Define Business Role** ►.

Activity Clipboard

Use

Interaction Center (IC) agents can use this function to view a list of the business objects involved in a current or past interaction. It is located in the [Interaction Record](#) or the navigation area.

i Note

The activity clipboard in the interaction record visualizes past and current interactions.

The activity clipboard in the navigation area visualizes current interactions.

The objects that were created or changed during those interactions are listed including navigation links.

Integration

This function is integrated with:

- Interaction Record
- Interaction History
- Business data context

In the Interaction Center (IC), the activity clipboard is one view of the business data context.

Prerequisites

Depending on where you want the activity clipboard to appear, you have to make different settings in Customizing:

Activity Clipboard in the Interaction Record

- You have made the necessary settings in Customizing to define the display of the activity clipboard so that the business partner appears in the activity clipboard list. Your system administrator can either use the default activity clipboard profile, or define which objects are relevant for linking and should appear in the activity clipboard. You do this in Customizing for [Service](#), by choosing ► [Interaction Center WebClient](#) ► [Basic Functions](#) ► [Define Activity Clipboard Profiles](#) ►.
- Both the programming of the business object itself and your Customizing settings must be in place for a business object to appear in the activity clipboard.

Activity Clipboard in the Navigation Area

You have made additional settings in Customizing for [Service](#):

- Define a direct link group of the type EE under ► [UI Framework](#) ► [Technical Role Definition](#) ► [Define Navigation Bar Profile](#) ► [Define Direct Link Groups](#) ►.
- Assign the direct link group to a navigation bar profile under ► [UI Framework](#) ► [Technical Role Definition](#) ► [Define Navigation Bar Profile](#) ► [Define Navigation Bar Profiles](#) ► [Assign Direct Link Groups To Nav. Bar Profile](#) ►.
- Assign the navigation bar profile to your business role profile under ► [UI Framework](#) ► [Business Roles](#) ► [Define Business Role](#) ►.

Features

If a business object is accessed repeatedly, it only appears once on the activity clipboard, in the order in which it was first accessed.

- By default the data is displayed in three columns, however, in Customizing you can choose the number of columns to be displayed with the maximum being five columns. The first column contains the name of the object or activity. Your system administrator can define what information appears in the other columns in Customizing for the activity mentioned above. By default, the [Object ID](#) and the [Process Type](#) are displayed.
- You can use the links to navigate to objects that appear in the activity clipboard, such as leads or sales tickets.

Account Identification/Account Fact Sheet

Use

IC agents can use this part of the interaction center (IC) to identify accounts in different ways, such as by name, company, or product, and to obtain an overview of an account and contact. If an account does not yet exist in the system, or if it must be updated, IC agents can also create and change accounts from this screen. In addition, they can quick create an account with just the basic data.

Prerequisites

- You have set up one of the following account identification profiles:
 - The standard account identification profile in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define Account Identification Profiles**.
 - The profile that enables confirmation of multiple business partners at once in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define Account Identification Profiles for Multiple Business Partners**.
- In your account identification profile, you have made settings for the following:
 - Contact attached data (if you want to use this feature)
 - Configurable context area (if you want to use this feature)
 - Embedded object components
 - Search scenario
 - Search results filtered for account roles and contact person roles
 - Default contact type for contact searches using search fields for contact communication details (**Telephone**, **E-Mail**, **Fax**)
 - International address versions (if you want to use this feature)
 - Employee search (if you want to use this feature)

- If you are using contact attached data (CAD), you have mapped the contact attached data from the communication management software to account identification using an XML file.

You have created a function profile in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Profiles for Contact Attached Data**.

Additionally, if you want to use CAD attributes when creating alerts or rules, you have enabled the relevant CAD attributes for fact gathering. For more information, see Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Enable CAD Attributes for Fact Gathering**.

- If you are using international address versions, you must have activated them. You do this in Customizing for **Service** under **SAP NetWeaver** > **Application Server** > **Basis Services** > **Address Management** > **International Settings** > **Activate International Address Versions**.

Mapping from logon language to address version is available in the default implementation of Business Add-In ADDR_LANGU_TO_VERSION, or an implementation exists that maps the logon language to that version.

- If you are using employee search, you have defined employees as business partners.
- If you are using the standard account identification profile, you have defined any object components that you want to be displayed in the upper right area of the screen in Customizing for **Service** under **Interaction Center WebClient** > **Customer-Specific System Modifications** > **Define Object Components**.

If you are using the account identification profile for multiple business partners, you can define object components within the search tabs you define in Customizing for **Service** under **Interaction Center WebClient** > **Master Data** > **Define**

Account Identification Profiles for Multiple Business Partners

- You have defined an interaction history profile in Customizing for **Service** under ► **Interaction Center WebClient** ► **Master Data** ► **Define Interaction History Profiles**.
- If you are using the briefcase, you have made the necessary setting in Customizing for **Service** under ► **Interaction Center WebClient** ► **Master Data** ► **Define Briefcase**.
- If you are using the enhanced search, you have made the necessary setting in Customizing for **Service** under ► **Master Data** ► **Business Partner** ► **Search** ► **Activate Wildcard Search for Accounts and Contacts**.
- If you want to ensure that all rules and events that are triggered when the **Business Partner Confirmed** (BPConfirmed) event is raised are also triggered when a call is transferred, you have added the **Business Partner Confirmed for Call Transfer** (BPConfirmed_Trans) event to the relevant rule policy in the rule modeler. Once you have done so, an account that is confirmed by an agent is confirmed automatically if the call is transferred to a second agent, and the system does not create a new interaction record when the call is transferred. For more information about creating rule policies, see [Rule Modeler](#).

Features

- Automatic number identification (ANI)

If your telephony switch system has ANI, the system can identify accounts even before IC agents accept contacts.

- Business context data (BCD)

When IC agents confirm an account or product, the system displays all relevant information for the account or product in the activity clipboard. This information is also recorded in the BDC so that it can be reused by other functions.

For more information, see [Activity Clipboard](#).

- Contact Attached Data

When IC agents confirm an account, the system can incorporate contact attached data from the communication management software (CMS). This information is displayed on the contact attached data screen.

You can also enable contact attached data attributes for fact gathering so that these attributes can be used when creating alerts or rules. Using these attributes, you can, for example, retrieve a service request number from the CAD attributes of an incoming call and set up a rule that displays an alert providing this information to the IC agent once the account has been confirmed. The IC agent can then click the service request number in the alert to navigate directly to the service order. When using CAD attributes for this purpose, you must ensure that the data that is passed has the same CAD group name and CAD attribute name that have been defined in Customizing.

- Configurable Context Area

The context area provides background information for the current interaction. If you would like to display different account information in the context area, ask your system administrator to configure the context area accordingly.

- Embedded Object Components

If you are using the standard account identification profile, you can replace the upper right area of the **Account Identification** screen with your own views that enable you to search for business partner-related business objects. You can define object components so that the agent can search for:

- Objects based on the confirmed business partner
- Business partners based on the confirmed objects

If you are using the account identification profile for multiple business partners, you can define object components within the search tabs you have defined.

- International Address Versions

If international address versions are activated in your system, you can use the default character set or the international character set to search for business partners (provided a character set is defined for your logon language).

The system automatically searches in both the default character set and the international character set. For example, in Japan, agents can search for the caller's name using the furigana version of the name. As a result, all names with the same pronunciation are displayed, irrespective of the character set.

International address versions are available for all countries for which you have maintained an international address version.

- Search Scenario

You can customize search fields and search views that appear as tabs on the **Account Identification** screen. For each search, you can define the title of the tab and the type of search (for example, business partner or object component).

You can also define the sequence of relationship types that appear in the search criteria.

- Filter search results for account roles and contact person roles

You can set the system to filter search results for account searches by account roles and by contact person roles.

- Define default contact type for searches using the contact communication data search criteria

For contact searches using the fields for contact communication details (**Telephone, E-Mail, Fax**), you can set a default contact type in Customizing that determines whether the search criteria are used to search for a contact of the type account, contact person, or both. IC agents can change this setting in the WebClient UI.

- Use contact name in account search

When searching for accounts in the IC, entering data in both the contact name fields and the account fields will generate a combined search result and provide more precise results.

i Note

You cannot search for accounts using the account search fields in combination with the search fields for contact communication details (**Telephone, E-Mail, Fax**). The contact communication details take priority over search criteria entered in the other search fields. If one of the search fields for contact communication details is filled, the system performs a contact search based on the communication details only.

- Employee Search

You can configure your interaction center to service internal employees. IC agents can use the employee search to look up employee details such as cost center, personnel area, employee group, and employee subgroup. They can also look up telephone numbers and e-mail addresses, or, when they receive a telephone call or e-mail, the system can identify the employee by ANI or from the incoming e-mail, and display the employee's details.

- Multiple Business Partners

You can configure your interaction center so that IC agents can confirm more than one business partner at once. You can also define a list of partner functions that can be used to confirm business partners and determine the sequence in which partner functions are displayed. You can then map the partner functions of confirmed business partner to the existing partner functions in business transactions and the interaction record.

- Account Fact Sheet

The account fact sheet gives an overview of the confirmed account. You can personalize views of the fact sheet. For more information, see [Fact Sheet](#).

- Briefcase

You can set up a briefcase that contains a collection of briefing cards that give you an overview of relevant information, such as addresses and contacts. Each briefing card can be assigned to more than one briefcase.

- Account Quick Create

You can quick create an account from every business object such as an opportunity or an activity. When you save the new account, the data you have entered in the fields for the business object is automatically filled in for the new account.

- Enhanced Search

You can search for an account and contact with the following search criteria:

- Region by Country

The search criterion refines your search for accounts by filtering the correct region of the respective country.

- E-Mail, Telephone, and Website

You can enter an asterisk (*) at the beginning or the end of a search string to perform a wildcard search. For e-mail, you can also put the asterisk within a search string.

Chat

Use

Interaction center (IC) agents can use the chat function to process incoming chat messages. The chat function in the IC includes standard chat functions, such as joining chat sessions, sending messages, and transferring chat sessions. Agents can transfer chat sessions to other queues or agents. Agents can also see additional details such as chat subject, location where the customer initiated the chat, and shopping cart items. Further features are described below.

Integration

Chat is integrated with the following IC functions:

- Toolbars

Agents use the channel-specific toolbar that is available for telephony, e-mail, and chat. The toolbar options change dynamically depending upon what channel is selected.

- Interactive scripting

Agents can use scripts to guide them through the chat session. The script is available directly next to the chat text area, so agents can easily and quickly access the information without navigating to a different screen.

- Knowledge search

Agents can use the knowledge search to find solutions to problems or other issues reported by customers.

- Telephony

Agents can remain in the chat channel and make a short phone call by using **Dial** in the toolbar.

- Interaction record

Depending on Customizing settings, the chat transcript can be saved and later accessed in the activity clipboard of the interaction record.

Prerequisites

- The Integrated Communication Interface (ICI) is configured (see [Integrated Communication Interface](#)).
- You have defined the chat profiles in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Define Chat Profiles** ►.
- You have assigned chat profiles to agents through business roles.
- If you want to use standard responses, you have defined the standard response group in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Define Standard Response Groups** ►.
- If you want to add additional knowledge bases for use with chat, you have made the settings for knowledge base implementation in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Define Knowledge Base Implementation** ►.
- If you want a list of available queues when transferring chat sessions, [Communication Management Software](#) settings are defined.

Features

- Account identification

The customer search is automatically started when an incoming chat session is routed to an available agent. The customer's e-mail address or other business partner details are used to determine if customer data exists. If the account is confirmed, then chat transcripts can be saved to the confirmed business partner's interaction record. For more information, see [Account Identification/Account Fact Sheet](#).

- Standard responses

IC agents can insert standard responses into the chat text. They can choose from a predefined list of commonly used responses, or can use the standard response screen to search and preview other responses.

- Knowledge base integration

You can define which knowledge bases IC agents can access while using the chat function. You can, for example, integrate knowledge bases such as interactive scripts, knowledge articles, mail forms (standard responses), solution database (problems and solutions), or chat transcripts.

- Multiple chat sessions

IC agents can process multiple chat requests simultaneously to maximize efficiency.

i Note

For more information about settings for multiple chat, contact your communication management software (CMS) vendor.

When you use multiple chat, several parallel HTTP sessions are created on the server in Service. This affects the sizing on the server side.

If you are using multiple chat, the client switch scenario is not supported.

- Web shop integration

IC also supports chat in Live Web Collaboration in Web Channel. You can integrate IC with your Web shop and with Internet Customer Self-Service (ICSS) so that customer requests for assistance via chat are forwarded directly to the interaction center.

E-Mail

Use

You can use this function to process incoming and outgoing e-mails. This function is integrated with the following functions in the interaction center (IC):

- **Toolbar**

IC agents use the channel-specific toolbar, which is available for telephony, e-mail, and chat. The toolbar options change dynamically depending upon what channel is selected.

- **Account identification**

Upon receipt of an e-mail, the customer search is automatically initiated based upon the **From** e-mail address. For more information, see [Account Identification/Account Fact Sheet](#).

- **Information search**

Results from the knowledge search can be added to e-mails as attachments or as text in the e-mail message. For more information, see [Knowledge Articles](#).

- **Agent inbox**

IC agents can access e-mail through their inbox and process them. For more information, see [Agent Inbox](#).

- **Interaction record**

IC agents have access to previously sent or saved e-mails in the confirmed business partner's interaction record. For more information, see [Interaction Record](#).

- **Telephony**

IC agents can make a phone call by using the **Dial** function in the application toolbar while remaining in the e-mail channel.

Prerequisites

- You have made settings to display fields (for example, date and from fields) and editor buttons in Customizing for **Service** under **UI Framework > Technical Role Definition > Define Parameters**. If you make fields visible, IC agents can personalize by choosing to show or hide those fields. If you set no parameters, the system shows all fields.
- You have made settings for e-mail in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels**. For example, you have made the following settings:
 - You have defined toolbar properties for e-mail in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Toolbar Profiles**.
 - You have defined standard response groups in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Standard Response Groups**.
 - You have defined outgoing e-mail address groups in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Outgoing E-Mail Address Groups**.
 - You have defined e-mail profiles in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles**.

Features

- **Standard responses**

IC agents can reuse use predefined text. For more information, see [Standard Responses](#).

- Multiple incoming e-mail addresses

Interaction centers can use more than one incoming e-mail address. Each address is assigned to a group of IC agents. IC agents can be assigned to more than one address.

- Multiple outgoing e-mail addresses

Interaction centers can also use more than one outgoing e-mail address. IC agents are assigned to one default outgoing e-mail address based upon their e-mail profile as defined in Customizing.

- Unfinished e-mails

IC agents can process unfinished e-mails if they need to deal with another transaction before completing an e-mail response.

Smart Responses

Use

Smart responses are [standard responses](#) with the ability to include additional dynamic information not limited to simple business partner attributes or system information.

Text is populated with business values and inserted into the e-mail content at runtime. This reduces the amount of time agents have to spend searching for and compiling information in a response. Agents are unaware if the response they choose is a standard response or a smart response.

Example

A customer sends an e-mail to the interaction center asking what his account balance is. After the agent confirms the business partner and initiates a reply to the e-mail, the agent selects a standard response. Variables such as account balance and account number are populated with the business partner's values in the response.

i Note

Interaction center agents cannot insert smart responses manually into an e-mail using the [Standard Response](#) dropdown.

Prerequisites

- You have defined mail forms in the WebClient user interface. You can access this function in the role of an interaction center manager under [Knowledge Management](#) > [Mail Forms](#).
- You have defined [Multilevel Categorization](#). (Not relevant for Auto Acknowledgements)
- You have defined the [rule policy](#) with the [Send Auto Acknowledgement](#), [Auto Respond](#), and/or [Auto Prepare](#) actions.
- You have assigned mail forms to actions in the WebClient UI or to categories through multilevel categorization.
- You have defined the data source classes in Customizing for [Service](#) by choosing [Interaction Center WebClient](#) > [Customer-Specific System Modifications](#) > [Smart Responses](#) > [Define Data Source Classes](#).

Features

Like standard responses, you can insert placeholders in the mail form that can be filled in by data related to the e-mail or to the business partner of the e-mail. You can define how the values for business and transactional data are retrieved and placed in mail forms. You can use data from either the fact base attributes or from the business data context (BDC).

Smart responses can be sent automatically to customers with or without agent intervention based upon multilevel categorization. They can be inserted into the e-mail text or attached to the e-mail. For more information, see the actions and parameters table in [ERMS Rule Policies](#) for the **Auto Respond** and **Auto Prepare** actions. They also can be used in auto acknowledgements and response proposals.

Standard Responses

Use

IC agents can use standard responses to help automate e-mail processing. Standard responses use predefined text configured from mail forms that are available to IC agents in the e-mail editor. Standard responses provide consistent handling and messaging to customers regardless of which IC agent processes the e-mail.

On the e-mail editor screen, two assignment blocks offer standard responses.

1. The **Standard Responses** assignment block provides a list of standard responses from four different sources:
 - Standard response groups as defined in the **Define Standard Response Groups** Customizing
 - Standard responses are available in the e-mail editor when you select a relevant category
 - Standard responses from a manual search done by the IC agent
 - Standard responses that the IC agent has saved in their favorites.
2. The **Standard Response Tree** assignment block displays standard responses based on the assigned multilevel categorization schema. Standard responses are also available in chat text. For more information about e-mail and chat, see [E-Mail](#) and [Chat](#). Multilevel categorization and Customizing settings help determine which standard responses are available.

Prerequisites

- You have defined standard response groups in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Standard Response Groups**.
- You have assigned the standard response groups to e-mail profiles in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define E-Mail Profiles**.
- If you want to use standard responses for chat, you must assign the standard response groups to chat profiles in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Chat Profiles**.
- If you want to determine the default language for your standard response texts, you must make settings in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **BAdIs for Communication Channels** > **BAdI: Language Determination for Standard Response Texts**.
- If you want to filter the standard responses to limit their use across channels, you must make settings in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **BAdIs for Communication Channels** > **BAdI: Filter for Standard Response Texts**.
- If you want to use the **Standard Response Tree** assignment block, you must have your e-mail editor profile to the EMAIL_PROFILE parameter for the e-mail application area. You can choose any categorization schema without the restriction of using an email profile when you use the CRM_IC_STD_RESP_SCHEMA business add-in (BAdI) definition, within enhancement spot CRM_IC_STD_RESPONSE. These standard responses will be in addition to those defined in the **Define Standard Response Group** Customizing.

Features

IC agents can do the following:

- Perform a search and preview standard response texts before choosing a standard response
- Choose the language of standard responses after searches
- Sort and filter standard responses
- Manage favorite standard responses
- View the source of standard response proposals (for example, from favorites or searches)

You can use variables in the mail form to insert placeholders for business partners. When IC agents choose the standard response or insert the standard response into an e-mail, the system fills these placeholders with values. For example, you can populate the customer's name in the salutation.

More Information

- [Auto Suggest](#)
- [Smart Responses](#)
- [Actions and Parameters](#)
- [ERMS Rule Policies](#)

Fax and Letter

Use

Interaction center (IC) agents can create, edit, and receive fax messages and letters in the interaction center. IC agents can also use an internal HTML editor to process faxes and letters.

Prerequisites

- You have made settings to display fields (for example, subject, date, and from fields) and editor buttons in Customizing for **Service** under **UI Framework** > **Technical Role Definition** > **Define Parameters**. If you make fields visible, IC agents can personalize by choosing to show or hide those fields. If you set no parameters, the system shows all fields.
- If you would like the **To** field of fax messages to be filled automatically, you have maintained the business partner (BP) data with a fax number that includes the country code.
- If you want to use business process push (BPP) with fax and letter, you have to specify the communication management software (CMS) connection by executing the program on the **SAP Easy Access** screen, under **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Communication Management Software** > **Interface Settings** > **Maintain Communication Management Software Connections**. For more information about BPP, see [Business Process Push](#).

Caution

You must enter the following values:

- Under **Comm. Mgmt Software System ID** enter **FAX_LETTER**.
- Under **Connection** enter the RFC connection to your CMS system.

Features

Agents can search for incoming fax messages and letters in the agent inbox and display them. They can also view old fax messages or letters for a confirmed account in the interaction record.

i Note

Once a fax has been sent or a letter has been finished, it can no longer be edited.

Using the HTML editor, IC agents can:

- Create, send, and print fax messages and letters
- Display information for the current interaction, such as:
 - Reason
 - Priority
 - Status
 - Name
- Add and display attachments
- Change the default header data for fax messages and letters

Activities

Creating Letters

Before IC agents can process letters in the interaction center, you must upload them to the interaction center by doing the following:

1. Scan the letters.
2. Save them to your hard drive.
3. Go to transaction oawd.
4. Expand the node and double-click the highlighted line that appears.
5. Select **Drag & Drop**.
6. In the dialog box that appears, drag and drop the relevant files from your local drive.

Agent Inbox

Use

You can use the agent inbox (inbox) to call up e-mails, business transactions, and various other business objects for display or processing. The inbox is a central worklist that the entire team can use to work on incoming objects.

Prerequisites

- You have made the basic settings required to use the inbox. For more information, see [Basic Settings for the Agent Inbox](#).

- If you want to display an alert showing the number of incomplete e-mails linked to a particular account, you have created a rule policy and a rule to set up the alert. The rule policy must include the **Business Partner Confirmed** (BPConfirmed) event and the rule must include the **Show Number of Incomplete E-Mails** (AH_NO_CUST_MAIL) action.

For more information about creating rule policies, see [Rule Modeler](#) and [ERMS Rule Policies](#).

Features

Business Roles

You can use the inbox in business roles with the following profile types:

- **IC WebClient Business Role**
- **CRM WebClient Business Role**

i Note

The following inbox functions are not available for business roles with the profile type **CRM WebClient Business Role**:

- Interact
- Link
- Next item
- Displaying e-mails, fax messages, and letters

Inbox Items

You can retrieve, access, and process the following items in the inbox:

- Business transactions in Service

You can call up business transactions, such as service orders.

- Workflow work items

You can process SAP workflow work items, including those that you have customized for use in the inbox. These work items can originate from the current system (SAP S/4HANA or SAP S/4HANA Cloud Private Edition) or from another SAP system.

- E-mails, fax messages, and letters

- Outbound correspondence

The inbox supports outbound e-mails, fax messages, and letters created in the interaction center (IC) and linked to an interaction record. The inbox also supports outbound e-mails created by the E-Mail Response Management System (ERMS).

- Inbound e-mails, fax messages, and letters

E-mails and fax messages that were received via SAPconnect appear as workflow work items. Inbound letters are also available in the inbox as work items.

A workflow forwards inbound e-mails to the appropriate group of agents.

- Multiple incoming and outgoing e-mail addresses

Agents can be assigned to various incoming addresses. All items sent to the agent's various addresses appear in the same inbox. Agents can also be assigned to more than one outgoing e-mail address. A default address is proposed, or the agent can select a different address.

- ERMS routing

Depending on the rule policies you define, you can use a general task for an ERMS decision to allow e-mail routing to all agents or you can specify individual agents or agent groups to allow e-mail routing only to those agents.

- Number of incomplete e-mails linked to confirmed account

You can set up the inbox so that once an account is confirmed, an alert is displayed that shows the number of incomplete e-mails linked to the confirmed account. IC agents can click the alert to navigate directly to the inbox, where only the incomplete e-mails from the confirmed contact person of the account are displayed. If no contact person has been confirmed, only e-mails from the e-mail address linked with the account are listed in the inbox.

Inbox Search

For information about the inbox search features, see [Inbox Search](#).

Inbox Result List

For information about the inbox result list features, see [Inbox Result List](#).

Basic Settings for the Agent Inbox

Use

System administrators use this process to set up the agent inbox in the Interaction Center (IC) WebClient.

Prerequisites

- The SAPconnect interface is configured.

For more information about SAPconnect, search for "SAPconnect (BC-SRV-COM)" on the SAP Help Portal at <http://help.sap.com>.

- E-mail addresses are created.
- If you want to edit scanned letters in your agent inbox, you have defined an ArchiveLink scenario.

For more information, see the following Customizing activities under **SAP NetWeaver > Application Server > Basis Services > ArchiveLink**:

- **Basic Customizing > Edit Document Classes** and **Edit Document Types**
- **Customizing Incoming Documents > Workflow Scenarios > Assign Document Type to a Workflow**

Process

Step	Description
Create business partner master data for each employee, with an employee role.	Maintain all inbound e-mail addresses that the agent should be responsible for. Assign a user ID.

Step	Description
Maintain settings for recipient distribution	In the SAP Easy Access menu, choose ► Service ► Interaction Center ► Interaction Center WebClient ► Administration ► Agent Inbox ► Maintain Recipient Distribution ►.
Maintain automatic workflow Customizing	In Customizing for Service , choose ► Basic Functions ► SAP Business Workflow ► Maintain Standard Settings for SAP Business Workflow ►. For more information, see SAP Business Workflow in the Interaction Center.
Activate event linkage for workflow template WS 14000164 and assign agents for e-mail handling	In the SAP Easy Access menu, choose ► Service ► Interaction Center ► Interaction Center WebClient ► Administration ► Agent Inbox ► Assign Agents for Fax and Letter Processing ►. For more information, see Assigning Agents for E-Mail Handling .
Forward to employees/groups	In the SAP Easy Access menu, choose ► Service ► Interaction Center ► Interaction Center WebClient ► Administration ► Agent Inbox ► Forwarding ► Define Recipient Profile for Forwarding ►. Use this transaction to define the content of the dropdown box Forward To in the result list of the agent inbox.
Define quick searches	In the SAP Easy Access menu, choose ► Service ► Interaction Center ► Interaction Center WebClient ► Administration ► Agent Inbox ► Define Quick Searches ►. Use this transaction to define the content of the dropdown box Quick Search in the agent inbox search.
Maintain Customizing activities	In Customizing for Service , choose ► Interaction Center WebClient ► Agent Inbox ►. For more information, review the Implementation Guide (IMG) documentation for each activity.

More Information

[Agent Inbox in the Interaction Center WebClient](#)

Assigning Agents for Fax and Letter Processing

Use

System administrators use this procedure to make the following settings for the agent inbox (inbox) in the Interaction Center (IC) WebClient:

- Activate event linking.

Workflow can define how incoming fax messages and letters are routed and processed. Activating event linking activates the workflow you defined and allows it to be used for the agent inbox. You only have to activate event linking once.

- Assign agents for fax and letter processing.

i Note

The incoming e-mails are routed using E-Mail Response Management System (ERMS). For more information, see [ERMS Rule Policies](#).

Procedure

Activating Event Linking

1. In the **SAP Easy Access** menu, choose **Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Assign Agents for Fax and Letter Processing**.
2. In the **Event Linkage** column, choose **Activate event linking**.
3. Under **IC WebClient: Inbound Process AUI** expand the workflow template WS 14000164 (**IC: Mail Inbox Handling**).
4. Select the icon in the **Activate/deactivate** column.

Once selected, the icon should display as **Activated**.

Result

Workflow is activated for the agent inbox.

Define Rules for Fax and Letter Processing

Use

System administrators follow this procedure to define routing rules for incoming fax messages and letters in the agent inbox (inbox) in the Interaction Center (IC) WebClient.

The IC WebClient inbox uses workflow to route incoming fax messages and letters to agents.

Additional routing rules use business partner attributes to route e-mails to agents who specialize in particular areas (see [Additional Routing Rules for Fax and Letter Processing](#)).

i Note

The incoming e-mails are routed using E-Mail Response Management System (ERMS). For more information, see [ERMS Rule Policies](#).

Process

As a system administrator, you define routing rules in the following steps:

1. Create rules.
2. Assign rules to agents or grouping of agents.

Creating Rules

1. In the **SAP Easy Access** menu, choose **Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Rules for Fax and Letter Processing**.

2. Choose **Create responsibility**.

3. Enter the following:

- Object abbreviation
- Description
- Start date
- End date

4. Choose **Continue**.

5. Enter a priority.

6. Enter values for each business partner field that you want to use to further define routing.

To enter more than one range for the same field, select the field and choose **New row**.

Enter * (asterisk) for the fields that you do not want to define.

7. Save your data.

Assigning Rules to Agents

1. On the **Responsibilities for Rule** screen, select the rule that you created.

2. Choose **Insert agent assignment**.

3. Select an object type.

4. Search for and select one or more objects of the type that you specified in the previous step.

5. Choose **Create**.

If you want to delete an assignment, select a rule on the **Responsibilities for Rule** screen and choose **Delete agent assignment**.

Example

You want to use the business partner language Spanish as an additional routing field. You make the following settings:

Name	From	To
Address	*	<no entry>
Country	*	<no entry>
Language	ES	<no entry>
Name	*	<no entry>
Region	*	<no entry>
Postal code	*	<no entry>

Additional E-Mail Routing Rules

Use

The Interaction Center (IC) WebClient agent inbox uses workflow to route incoming mails to agents. In the standard workflow, after a customer sends an incoming mail to the interaction center, the system retrieves the address to which the customer sent the mail. It determines which agents are assigned for processing incoming mails for that address. This determination is based on the e-mail addresses entered in the employee's business partner master data.

In addition, you can use certain business partner data to define which agents or groups of agents should process the mails (for example, to route e-mails to agents who specialize in particular areas). The routing rules can be assigned to object types such as users, positions, and organizational units.

Prerequisites

Routing rules for e-mails have been defined and assigned to agents or groups or agents (see [Defining Routing Rules for E-Mails](#)).

Features

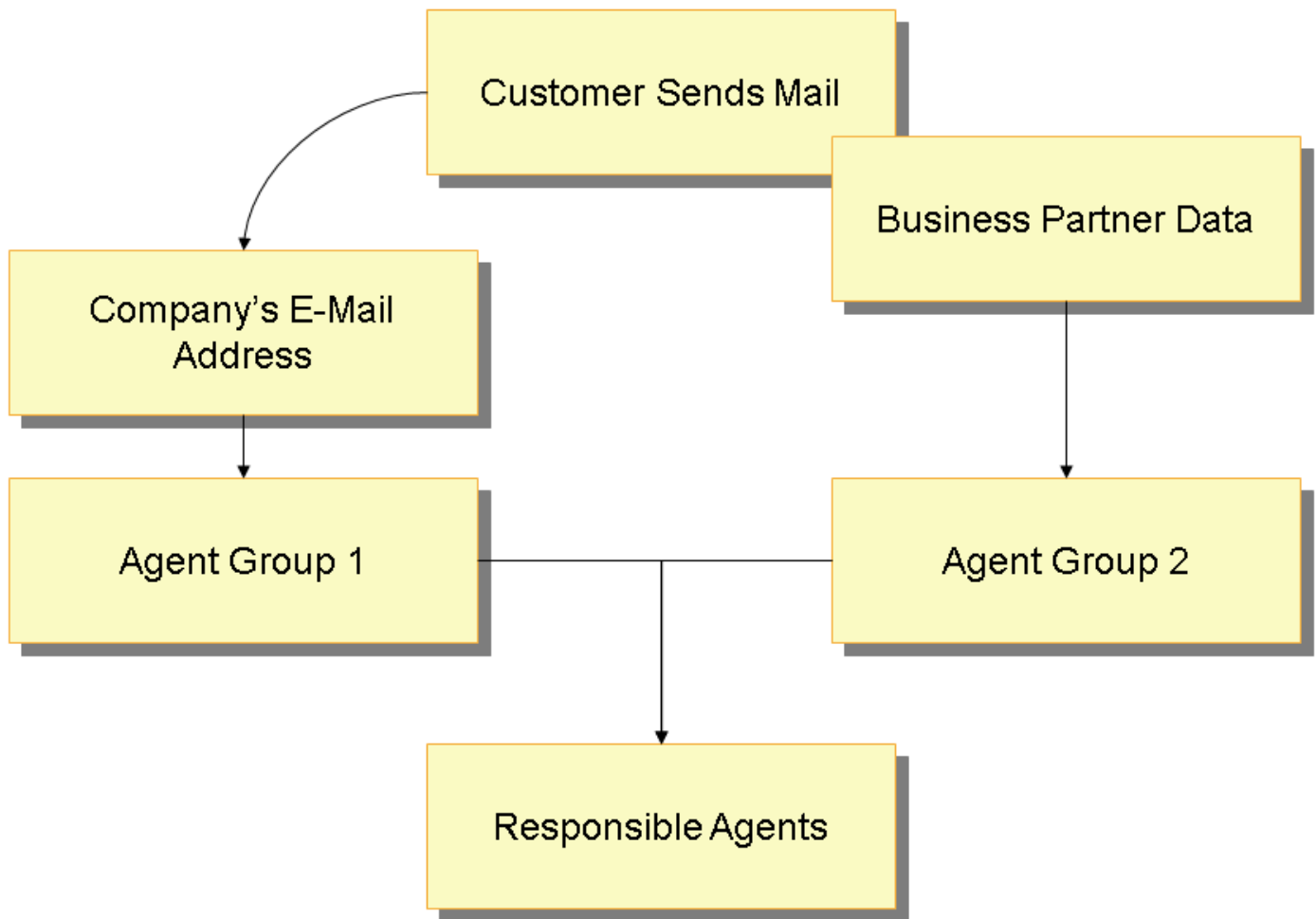
The following business partner data can be used to define how mails are routed:

- Address
- Country
- Language
- Name
- Region
- Postal code

To use other fields, you need to create a new rule. From the [SAP Easy Access](#) menu, choose ► [Architecture and Technology](#) ► [ABAP Workbench](#) ► [Development](#) ► [SAP Business Workflow](#) ► [Definition Tools](#) ► [Rules for Agent Assignment](#). ► [Create](#) ►. The [Responsibilities](#) tab allows the creation of rule definitions and assignments to agents.

Activities

The figure below shows how the system determines which agents process incoming mails and is followed by an explanation:



1. With additional routing rules, the system tries to determine the business partner by the customer's e-mail address. If the business partner is determined, attributes from the business partner master data can be used to further define which agents process the mail.

i Note

If the business partner cannot be determined, the additional routing rules are not used. In this case, agents who are assigned to process mails for that incoming address or number receive the mail (see [Defining Routing Rules for E-Mails](#)).

2. The system checks which agents are assigned to the incoming mail address (**Agent Group 1**), as specified in the inbound distribution.
3. The system checks which agents are assigned to process the specific business partner attributes (**Agent Group 2**) if the business partner address or number is found. This depends upon the defined rules and which agents are assigned to them.
4. The routing depends on how many agent groups exist:

- o Two agents groups

The mail is routed only to agents who are in both of the groups (**Agent Group 1** and **Agent Group 2**). If no agents are found in both groups (agents in **Agent Group 1** and **Agent Group 2**, but none belonging to both groups) the mail is routed to all agents in each group.

- o One agent group

If no agents are in **Agent Group 1**, the mails are routed to agents in **Agent Group 2**. Alternatively, if there are no agents in **Agent Group 2**, the mails are routed to agents in **Agent Group 1**.

Example

Routing works as follows when two agent groups exist:

1. Mr. Smith sends an e-mail to service@company.com.

Julie and David (**Agent Group 1**) are assigned to process mails sent to this e-mail address.

2. The system identifies Mr. Smith's business partner data based upon his outgoing e-mail address.

3. The system determines that an additional rule can be used based upon Mr. Smith's country (**US**).

Michael and Julie (**Agent Group 2**) are assigned to an e-mail rule for US customers.

4. The system determines that only Julie should receive the incoming mail because she is the only agent in both of the agent groups.

If Julie was not in both of the groups, then all of the agents, David, Michael, and Julie, would receive the incoming mail.

Define Recipient Profile for Forwarding

Use

Use this procedure to define to which person(s) you may forward items for further processing from the agent inbox in the Interaction Center (IC) WebClient. Inbox items may be forwarded to an individual employee or a group.

Prerequisites

Business partners specified as recipients must be defined.

Procedure

In the **SAP Easy Access** menu, choose **Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Forwarding > Define Recipient Profile for Forwarding**.

Quick Searches

Use

You can define searches that can be used as options for finding business objects in the agent inbox (inbox).

Prerequisites

Features

You can define quick searches in the **SAP Easy Access** menu, under **Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches**.

SAP delivers one Quick Search (0001) that is defined to select **My Open Items** sorted first by priority, then by main category. You can define additional quick searches based on your own selection criteria.

You can use an asterisk (*) as a wildcard to specify the title of a business object that has been derived from the business transaction.

You can also define a default quick search for each business role in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Define Inbox Profiles** ►.

Note that the inbox supports the following time periods for quick searches: creation period, change period, and due period.

Example

You can define quick searches to retrieve item groups such as:

- All overdue items
- All items for which you are responsible
- All high-priority items

Inbox Search

Use

The agent inbox (inbox) search provides a broad range of search attributes and search features that you can use to tailor the inbox search according to your business requirements.

Prerequisites

You have set up and configured the inbox. For more information, see [Agent Inbox in the Interaction Center WebClient](#).

Features

Search Attributes

For information about the available search attributes, see [Search Attributes](#).

Search View: Standard and Advanced

The standard search provides a fixed arrangement of search fields that the user cannot change.

The advanced search provides selectable search attributes, the multiple use of search attributes, and the use of saved searches.

You can specify whether you want to use the standard search or the advanced search. You do this in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Define Inbox Profiles** ►.

Lean Search

During a lean search, the system determines only the required data and does not store any data. It only determines the object attributes that are relevant for the search result list. This enables a faster search and faster processing times when working with the result list. The remaining attributes are determined when the object is used, for example, when the object is opened in the display mode or in the edit mode.

Predefined Searches

Quick Searches

You can create predefined search queries that appear as selection criteria in the **Quick Search** field. For example, you can define a search that agents can use to retrieve all high-priority items. For more information, see [Quick Searches](#).

Quick searches are assigned to business roles through the inbox profile.

Saved Searches

The saved searches function is only available in the advanced search. Saved searches are user-defined.

Search Attribute Validity Check

The following attribute validity check prevents unclear search results:

If a main category cannot be combined with a selected search attribute, the main category is excluded from the search. For example, a search involving the main category **E-Mail** in combination with the **Checklist ID** search attribute is not valid, and the search is not performed.

This system behavior also applies if only one main category has been selected or if no main category has been selected. In the latter situation, the validity check is performed for all main categories that are defined for the corresponding inbox profile in Customizing, and the search is performed for all main categories that have passed the validity check.

Search Traceability

A search processing log consolidates various detailed search messages, such as which search attribute is not supported by which main category, in a structured way. The processing log can be accessed using a corresponding button in the result toolbar.

In addition to the entries in the processing log, the following general messages are displayed on the UI, as appropriate:

- Search has been only partially performed
- Search could not be performed

This enables the user to understand the search behavior and to tailor future searches accordingly, or to contact the system administrator, as required.

Predefined Inbox Search Tables

You can use predefined inbox search tables that contain the inbox search attributes and inbox result attributes that are relevant to specific inbox item types. Predefined inbox search tables are available for the following inbox item types:

- **Inbox Item: OneOrder**
- **Inbox Item: Workitem - Inbound Email, Fax and Letter**
- **Inbox Item: Outbound Correspondence**

For information about the prerequisites and features, see [Predefined Inbox Search Tables](#).

Search Attributes

Use

You can use various search attributes and search attribute combinations to locate the required agent inbox (inbox) items.

Prerequisites

For information about the necessary settings for the inbox search, see [Inbox Search](#).

Features

Available Search Attributes

Area	Search Attribute	Function
Predefined searches	Quick Search	Predefined search criteria (for example, My Open Items) You can configure these searches on the SAP Easy Access screen under ► Interaction Center ► Interaction Center WebClient ► Administration ► Agent Inbox ► Define Quick Searches ►. You must assign the relevant quick searches to the corresponding inbox profile, and you can define a default quick search. You do this in Customizing for Service under ► Interaction Center WebClient ► Agent Inbox ► Define Inbox Profiles ►.
	Saved Searches	The saved searches function is available in the advanced inbox search.
Inbox item	Main Category	Inbox item type as defined in the following Customizing activities in Customizing for Service under ► Interaction Center WebClient ► Agent Inbox ►: ► Inbox Search Definitions ► Define Item Types for Searches ► • Define Inbox Profiles
	Object ID	ID of the inbox item
	Description	Description of the inbox item
Partner information	Created By	User who created the inbox item
	Account	Account for which the inbox item has been created
	Assigned To	Assigned agent or agent group To use the Assigned To search attribute to search for business transactions and cases, the relevant user or group must be maintained as a business partner in the system. The business transaction and case search is based on the business partner in combination with the inbox partner mapping (inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). The search for work items (inbound e-mails, fax messages, and letters), outbound correspondence (outbound e-mails, fax messages, and letters), social media posts, and workflow work items is based on the assigned users. Assigned To: Me The search returns the inbox items that are assigned to the current user. The search returns the following results: • Business transactions and cases to which the current user is assigned as the employee responsible • Work items for which the current user is the employee responsible (assigned agent) • Work items that are available for processing by the user and have not yet been picked by another user

Area	Search Attribute	Function
		- Outbound correspondence that was last changed by the user - Social media posts for which the current user is the employee responsible (assigned agent) Assigned To: My Groups The search returns the inbox items that are assigned to the selected groups. The search returns the following results: - Business transactions and cases to which the selected groups are assigned as the groups responsible - Work items to which the users of the selected groups are assigned as the processor - Work items that are available for processing by the users of the selected groups - Outbound correspondence that was last changed by users of the selected groups - Social media posts to which the users of the selected groups are assigned as the processor Assigned To: <no entry> For work items, the Assigned To: My Groups search is performed. Note that the search for workflow work items always returns the workflow work items assigned to the current user. Therefore, the Assigned To search attribute is not relevant for workflow work items.
	Employee Responsible	Assigned employee responsible The specifics for the Assigned To: Me search attribute also apply to the Employee Responsible search attribute. The Employee Responsible search attribute cannot be combined with the search attribute Assigned To.
Time	Creation Period Change Period Due Period	Predefined periods of time for searching inbox items (for example, Last 3 days) In the standard search, you can use the respective From/To fields to enter a user-defined date or date range. You can define the time periods in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Time Periods for Searches. You can define which time periods are available for the respective search attributes in the inbox search. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.
	Created On Changed On Due On	Date or date range These search attributes are only available in the advanced search.
Importance	Status	Inbox status, such as Open, In Process, Completed You can define the inbox statuses in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Inbox Status

Area	Search Attribute	Function
		 You must map the system statuses and user statuses of the inbox item types to the inbox statuses. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Item Status to Inbox Status .
	Priority	Inbox priority, such as High, Medium, Low You must map the inbox item type priorities to the inbox priorities. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Map Item Attributes to Inbox Attributes > Map Item Priorities to Inbox Priorities .
Multilevel categorization	Category ID	ID of category used in multilevel categorization For more information, see Multilevel Categorization.
Checklist	Checklist ID	Not applicable
E-mail addresses	Sender Address Recipient Address	E-mail addresses that can be used to search for the following inbox items: - Work items (inbound e-mails, fax messages, and letters) If you are not using the predefined inbox search table for work items, entries in the Sender Address field are restricted to 20 characters (use an asterisk in case of longer e-mail addresses). - Outbound correspondence (Outbound e-mails, fax messages, and letters)
Sorting	Sort By	Sort attributes by which the search result is sorted A maximum of two sort attributes can be used for each search.
Generic functions	Clear	Clears the search attribute values
	Reset	Sets the inbox search criteria and the inbox result list back to the initial state: - The Quick Search field is cleared. - The search criteria is set back to the predefined combination of search fields (relevant to the advanced search only), and the field values are cleared. - The Maximum Number of Results field is set back to the default. - The result list is cleared, and the sorting and the filtering are set back to the initial state. Note that personalization settings for the result list and the selected display type (table or tree) are not set back.
	Maximum Number of Results	Maximum number of search results You can define the maximum number of search results to improve the search performance, for example. You can define the default values and range of user entries for the Maximum Number of Results field. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles .

Search Attributes not Enabled for Multiple Use:

- Category ID**

- **Assigned To: Me** and **Assigned To: My Groups** cannot be combined with other **Assigned To** search criteria. You cannot for example perform a search using **Assigned To: Me** in combination with **Assigned To: My Groups**.
- **Sort By**

Predefined Inbox Search Tables

Use

The standard system contains predefined inbox search tables, which contain the inbox search attributes and inbox result attributes relevant for a specific inbox item type. You can improve the search performance by using predefined inbox search tables. For each of the following inbox item types, a separate table is available:

- **Inbox Item: Workitem - Inbound Email, Fax and Letter**

Inbox search table for work items (CRMD_AUI_WI_IDX)

- **Inbox Item: Outbound Correspondence**

Inbox search table for outbound correspondence (CRMD_AUI_OUTCORR)

i Note

For outbound correspondence, the predefined inbox search table is mandatory for both of the abovementioned inbox item types.

- **Inbox Item: Inbox One Order item**

For this inbox items type a predefined search table is not required because the search is optimized already. However, in case of Customizing changes, this inbox item type must be updated. For more information, see the *Features* section.

Prerequisites

General

- You have defined the business transactions and work items that you want to use in the inbox in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Inbox Search Definitions** ► **Define Item Types for Searches** ►.
- You have assigned partner functions for use in the inbox in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Map Item Attributes to Inbox Attributes** ► **Map Business Transactions to Responsible Employees and Groups** ►.
- You have defined the inbox status mapping in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Map Item Attributes to Inbox Attributes** ► **Map Item Status to Inbox Status** ►.

Inbox Search Table for Work Items

- You have made the settings in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Settings for Asynchronous Inbound Processing** ► **Assign Standard Tasks to Communication Methods** ►.
- You have made the settings in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Define Inbox Profiles** ► **Work Item Table** ►.

Inbox Search Table for Outbound Correspondence

- If you want to adjust the data record of an outbound correspondence item before it is stored in the inbox search table for outbound correspondence, you make the settings in Customizing for [Service](#) under [Interaction Center WebClient](#) [Agent Inbox](#) [Business Add-Ins](#) [BAdI: Outbound Correspondence Database Update](#).

Features

Data Load

To use the predefined inbox search tables, you first need to upload the data for the relevant main categories. You do this in the SAP GUI, on the [SAP Easy Access](#) screen under [Interaction Center](#) [Interaction Center WebClient](#) [Administration](#) [Agent Inbox](#) [Process Data for Inbox-Specific Searches](#) (transaction CRM_IC_INDEX_LOAD).

Only the main categories that are mapped to the corresponding inbox item types in Customizing activity [Define Item Types for Searches](#) are available in the program [Process Data for Inbox-Specific Searches](#).

i Note

If you change any of the Customizing settings detailed in the [Prerequisites](#) section, you must update the corresponding inbox search table accordingly.

In this case, you must also update the entries for the inbox item type [Inbox One Order Item](#) to fill various header fields in the corresponding business transaction header table, for example CRMS4D_SERV_H for service orders.

Process Data for Inbox-Specific Searches: "Inbox Item Type"

- Choose the inbox item type ONEORDER [Inbox One Order Item](#) to update the following header fields in the corresponding business transaction header table, for example CRMS4D_SERV_H (only in the case of Customizing changes):
 - INBOX_STATUS
 - INBOX_ASSIGNED_TO
 - INBOX_GROUP_ASSIGNED
 - INBOX_ACCOUNT
 - INBOX_CONTACT_PERSON
- Choose the inbox item type WORKITEM [Inbox Workitem](#) to process the data for the inbox search table for work items. The upload is required initially and if the Customizing settings under *Prerequisites* are changed.
- Choose the inbox item type OUTCORRESP [Outbound Correspondence Item](#) to process the data for the inbox search table for outbound correspondence. The upload is required initially and if the Customizing settings under *Prerequisites* are changed.

Process Data for Inbox-Specific Searches: Displayed Figures

- [Source Entries](#) (Not available for outbound correspondence)

The number of entries in the tables, which the system uses to update the corresponding predefined inbox search table

- [Target Entries](#)

The number of entries in the predefined inbox search table that is being updated

Process Data for Inbox-Specific Searches: Processing Options

- [Update](#)

Loads the data for the selected main categories from the source tables to the corresponding inbox search table

Note that for the selected main categories, all previous entries in the corresponding inbox search table are removed and new entries are provided.

- **Remove**

Removes the selected main categories from the corresponding inbox search table

- **Calculate**

Updates the displayed figures

- **Reset Status**

Resets the status to **Not Started**

You can use the **Reset Status** function if you have canceled a job and the status of the corresponding main category is still **In Process**.

The functions above are also available if multiple main categories are selected, unless one of the main categories cannot be processed due to the status.

Process Data for Inbox-Specific Searches: Availability of the “Update” and “Remove” Functions

The following table provides an overview of the availability of the **Update** and **Remove** functions and the corresponding status:

Context	Status of Corresponding Main Category	Update	Remove
One of the selected main categories is no longer supported according to the settings in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Item Types for Searches	Not Supported	Not enabled	Enabled
One of the selected main categories is already included in another data load.	In Process	Not enabled	Not enabled
You have canceled a job and reset the status to Not Started using the Reset Status function.	Not Started	Enabled	Enabled
The initial data load for one of the selected main categories has not yet been performed.	Not Started	Enabled	Enabled
One of the selected main categories has already been removed from the inbox search table.	Not Started	Enabled	Enabled
The data for one of the selected main categories has	Completed	Enabled	Enabled

Context	Status of Corresponding Main Category	Update	Remove
been uploaded to the inbox search table.			

Delta Update

The automatic delta update is limited to the main categories that you have uploaded, or are in the process of uploading, to the relevant search tables:

- Inbox search table for work items
 - When a work item is saved
- Outbound correspondence
 - Outbound e-mails
 - When an e-mail is saved, sent, or deleted
 - Outbound faxes and outbound letters
 - When a fax or letter is saved or deleted and the session is ended
- For outbound correspondence, you are not required to upload the data for the relevant main categories to enable the delta update.
- The system status is used to calculate whether a business transaction is overdue.

Specifics for the Inbox Search Table for Outbound Correspondence

The system fills the following fields from the interaction record using your settings in Customizing for **Service** under **Interaction Center WebClient** > **Business Transaction** > **Assign Partner Functions to Business Transactions** :

- ACCOUNT
- CONTACT

If more than one interaction record is assigned to an outbound correspondence item, the interaction record that the fields are supplied from (during data load) is undefined.

During data load, the system loads the outbound correspondence items linked to an interaction record during an interaction center session or during ERMS processing. For more information, see [Interaction Record](#).

Caution

If you delete outbound correspondence items from the inbox search table, you can use the **Process Data for Inbox-Specific Searches** transaction to restore items that are linked to an interaction record. However, you cannot use this transaction to restore items that are not linked to an interaction record. For example, you cannot restore outbound e-mails sent by ERMS when they were not created with an interaction record.

Inbox Result List

Use

The agent inbox (inbox) result list displays the inbox search results in table or tree format and provides the functions required to process the inbox items.

Prerequisites

- You have set up and configured the inbox. For more information, see [Agent Inbox in the Interaction Center WebClient](#).
- If applicable, you have configured the result toolbar in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Define Inbox Toolbar Profiles** ►.

Features

Result List in Tree Format or Table Format

You can display the result list in tree format or table format:

- The tree format provides context information, such as related business transactions.
- The table format provides better performance, sorting and filtering functions, and the use of table charts.

You can define whether the result list is displayed in tree format or in table format as a default. You do this in Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Define Inbox Profiles** ►.

The user can switch between the tree format and table format in the inbox, as required.

Maximum Number of Hits

You can define the maximum number of search results.

For more information, see Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Define Inbox Profiles** ►.

Advanced Warning for Due Date

You can define an advanced warning for each inbox item type, with reference to the due date.

For more information, see Customizing for **Service** under ► **Interaction Center WebClient** ► **Agent Inbox** ► **Inbox Search Definitions** ► **Define Item Types for Searches** ►.

Configurable Result Toolbar

You can configure the result toolbar according to your business needs for interaction center (IC) business roles (business role profile type **IC WebClient Business Role**) and for other core business roles in Service:

- Result toolbar buttons

You can define the following for each button:

- Description
- Label
- Tooltip
- Icon
- Availability of button for use in IC business roles, in other business roles for core or both (providing the button function is available in the corresponding business role)

You can also add user-defined buttons.

- Result toolbar profiles

You can define toolbar profiles for IC business roles and for other core business roles. You can define the following for each profile:

- Which buttons are available
- Button order
- Button grouping

You can group buttons using separators.

- Whether the label, icon, or both are displayed for the button

If you do not define toolbar profiles for your business roles, the system uses default profiles.

Available Toolbar Buttons and Corresponding Functions:

Button Description	Button Function
Complete	The statuses of the selected inbox items are set to Completed. The Complete button is only supported for the following inbox items: • Work items (inbound e-mails, fax messages, and letters) • Workflow work items Whether a workflow work item can be completed depends on the corresponding workflow task. • Outbound text messages
Delete	The selected inbox items are deleted. The Delete button is only supported for the following inbox items: • Work items (inbound e-mails, fax messages, and letters)
Display	The system navigates to the details page for the inbox item in the display mode: • No interaction record is created. This also applies if the user switches from the display to the edit mode from the object. (For workflow work items and work items (e-mails, fax messages, letters), switching from the display mode to the edit mode is not supported.) • The account is not confirmed and not published in the context area. When switching to the edit mode from the object, the system defines the user who is currently logged on as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups) and confirms the business partner. The Display button is not supported for work items (inbound e-mails, fax messages, and letters) for business roles with the profile type WebClient Business Role.
Edit	The following steps are performed: • The system navigates to the details page for the selected inbox item in the edit mode. • The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible

Button Description	Button Function
	Employees and Groups). - No interaction record is created. - The business partner is confirmed and published in the context area. If other business transactions are opened during the editing process, always the business partner of the business transaction that was most recently opened is confirmed and published in the context area. - The business transaction is linked to the activity clipboard. - All subsequently opened business transactions are also linked to the activity clipboard, unless the user chooses the End button. Choosing the End button clears the activity clipboard. - If the user chooses Interact, all business transactions that were linked to the activity clipboard during the editing process are linked to the interaction record. - Editing during an interaction If the user edits other business transactions during an interaction, the business partners for the other business transactions are not confirmed or published in the context area. The Edit button is only supported for business transactions.
Forward	Defines the forwarding target (employee or group) as the employee responsible or group responsible, respectively When the user chooses the Forward button, a dropdown menu is displayed that contains the recipients that you have defined in the recipient profile for forwarding that is assigned to the inbox profile. If the recipient profile contains more than 10 entries, the 10th item of the button dropdown menu is a More entry that launches an input help. The input help contains all remaining recipients that you have defined in the recipient profile for forwarding. The forwarding function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Forward (Extended)	Defines the forwarding target (employee or group) as the employee responsible or group responsible, respectively When the user chooses the Forward button, a dropdown menu is displayed that contains the following entries (maximum of 10 entries): - The recipients that you have defined in the recipient profile for forwarding that is assigned to the inbox profile - The More entry that launches an input help The input help contains the remaining recipients that you have defined in the recipient profile for forwarding. - The Other Employees entry that launches the employee search - The Other Service Teams entry that launches the business partner search The extended forwarding function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Interact	The following steps are performed by the system: - The system navigates to the details page of the inbox item. - The account and contact are confirmed and published in the context area.

Button Description	Button Function
	• An interaction record is created. • The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). • In the case of work items (e-mails, fax messages, letters) and workflow work items, the system navigates to the account identification where a preview of the work item or workflow work item is displayed. The Interact button is not supported for the following inbox items: • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages) • Workflow work items The Interact button is not supported for business roles with the profile type WebClient Business Role.
Link	If the user selects a second inbox item during an interaction and chooses the Link button, the second inbox item is linked to the current interaction record. The Link button is not supported for the following inbox items: • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages) • Workflow work items The Link button is not supported for business roles with the profile type WebClient Business Role.
Next Item	The system navigates to the next inbox item that is available for processing and locks the item for other users. Once the user has chosen the Next Item button, the button is deactivated until the user chooses the End button. If the Next Item button is used for outbound correspondence (outbound e-mails, fax messages, and letters), the system navigates to the next outbound correspondence item that is in process and to which the current user is assigned as the employee responsible. In this case, the outbound correspondence item is not locked for other users. The Next Item button is not supported for business roles with the profile type WebClient Business Role.
Preview (Toggle)	Displays a preview of the currently selected e-mail, fax message, and letter The preview is displayed below the table chart area. The preview does not provide follow-up buttons, such as Reply and Forward.
Reference	Sets a reference to the currently selected inbox item If the user chooses the Reference button, a dropdown menu is displayed that contains the relevant entries from the activity clipboard. If the user chooses an object from the dropdown menu, the currently selected inbox item is referenced from the object (the inbox item is added to the object's business context). Note that the Reference button only works if the BUSINESS_CONTEXT function profile is assigned to the business role. In Customizing for the business context, the following is defined: • The inbox item types to which a reference can be added Only these inbox item types are available in the dropdown menu of the Reference button. • Which inbox item types can be added as a reference

Button Description	Button Function
	For more information, see Customizing for Service under Transactions > Settings for Service Requests > Define Settings for Business Context. The Reference function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages) and for business roles with the profile type WebClient Business Role.
Reference (Extended)	Unlike the dropdown menu for the Reference function, the dropdown menu for the Reference (Extended) function contains entries from the Recent Items. The last entry in the dropdown menu is the Other Transactions entry that launches a transaction search. Apart from this, the Reference and Reference (Extended) functions are identical. The Reference (Extended) function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages) and for business roles with the profile type IC WebClient Business Role.
Refresh	The system performs the most recent search again, based on the selected search criteria or based on a quick search. The Refresh button enables the user to conduct the most recent search again without expanding the Search Criteria view and choosing the Search button, or selecting the quick search again.
Reserve	The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). The Reserve button is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Reset Reservation	Resets the reservation of the inbox item and removes the employee responsible Only the employee responsible assigned to the inbox item can reset the reservation of the inbox item. The Reset Reservation function is not supported for the following inbox items: • SAP ERP business transactions • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Reset Reservation (Extended)	Resets the reservation of inbox items on behalf of other users The Reset Reservation (Extended) function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages).
Separator	The separator is used to group toolbar buttons in the result toolbar profile.
Search Processing Log	Provides access to the search processing log. For more information, see Inbox Search (Search Traceability section).
Show Table View (Toggle)	Displays the result list in table format
Show Tree View (Toggle)	Displays the result list in tree format

Multiple Selection of Inbox Items

The following functions are available when multiple inbox items are selected (unless the function is generally not available for one of the selected inbox items):

- [Delete](#)

- [Reserve](#)
- [Reset Reservation](#)
- [Complete](#)
- [Link](#)
- [Forward](#)
- [Reference](#)

The following functions are not available when multiple inbox items are selected:

- [Edit](#)
- [Display](#)
- [Interact](#)
- [Preview On](#)

Business Transaction Routing

Use

You can use this function to forward business transactions and cases in the interaction center (IC) manually or automatically.

Prerequisites

You have made the necessary settings for automatic routing in the BSP application rule editor.

For more information, see Customizing for [Service](#) under [Interaction Center WebClient](#) > [Business Transaction](#) > [Define Automatic Routing](#).

Features

You can forward business transactions and cases to agent groups as well as to individual agents. Both manual and automatic routing are available:

- Manual routing

You can manually enter a responsible employee or a responsible group as the business partner.

The transaction or case can be selected in the agent inbox of the assigned responsible employee as an open item for processing.

- Assigning a responsible employee

You can either enter an employee directly or using a search.

If you start a search and a responsible group is assigned, the system searches for agents in the responsible group. If there is no responsible group assigned, the system takes all agents into account during the search.

- Assignment of a responsible group

You can enter the responsible group directly or using a search. If you start a search and a responsible employee is assigned, the system searches for all organizational units to which this employee is assigned in the organizational model. If no responsible employee is assigned, the system searches for all organizational units that are assigned to the business role. The organizational units are displayed in the search results according to the structural authorization of the agent.

- Automatic routing

You trigger automatic routing by choosing [Escalate](#).

You can include up to five of the following attributes in the routing:

- Priority
- Status
- Confirmed product
- Confirmed customer
- Categorization
- Transaction type
- Business role

You can also push business transactions and cases to external routers or communication management software (CMS) for distribution to IC agents based on their real-time availability. For more information, see [Business Process Push](#).

E-Mail Response Management System

Use

E-Mail Response Management System (ERMS) is a tool for managing large amounts of incoming e-mail. Instead of routing all incoming e-mails into one queue, ERMS provides services for automatically processing and organizing incoming e-mail. There are several automated activities that reduce the need for manual intervention by interaction center agents, thus substantially increasing efficiency and processing accuracy, by helping agents to process e-mail in less time and allowing them to provide the same quality response regardless of agent skill level. ERMS provides tools for agents to efficiently and consistently respond to messages, and for managers to administer, monitor, and report on the whole e-mail process.

SAP delivers preconfigured ERMS settings and functions that you can extend using custom coding.

Integration

- Organizational Management

Defines how e-mails are routed and to what groups or organizations and how they are going to handle and process e-mails. You can route e-mails to an organizational unit or responsible group, a position, or even an individual user. The e-mails are routed to the agents' inboxes where they are processed.

- SAP Business Workflow

Enables automated processing and routing of e-mails. For more information, see [Workflow for ERMS](#).

- SAPconnect

Receives inbound and sends outbound e-mails. Once e-mails are received, SAPconnect creates SAP office documents, which are then processed by SAP Business Workflow.

- Multilevel Categorization
- Machine Learning

You can connect ERMS with [SAP Service Ticket Intelligence](#) to automatically categorize incoming e-mails.

- E-Mail Response Management System Analytics

Allows you to generate reports and analyze your ERMS processing on historical data. You can analyze historical ERMS e-mail volume, handling time, response time and service level.

Features

Manager Features and Tools for Productivity

The WebClient UI is available for modeling rules and categories, and for monitoring and viewing analytics.

- [Rule Policies](#)
- Category Modeler
- [ERMS Simulator](#)
- [E-Mail Workbench](#)

Agent Features and Tools for Productivity

- [Agent Inbox in the Interaction Center WebClient](#)

Central location in which agents receive and respond to e-mails. E-mails are integrated with other communication channels.

- [E-Mail Threading](#)
- Automatic Business Object Creation

Interaction records, along with appropriate interaction reasons, and service tickets can be created and associated to incoming e-mails automatically. For more information, see the [Create Interaction Record](#) and [Create Service Ticket](#) actions detailed in the actions and parameters table in [ERMS Rule Policies](#).

Automated Features

- [ERMS Responses](#)
- E-Mail Filtering

Certain incoming e-mails that are not relevant for agent processing, such as out-of-office replies, spam, or bounced e-mails, can be filtered or deleted based upon the e-mail subject or content.

- [Rules](#)

These identify and determine the automated processing of e-mails.

- E-Mail Routing

Incoming e-mails can be routed to particular queues or agent groups that are most capable of responding to the issue in the e-mail. For example, you can route e-mails that contain foreign languages to queues in which agents are fluent in that language.

- Content Analysis and Fact Gathering

The system analyzes attributes and content regarding each incoming e-mail and stores it in a fact base, enabling the use of the data for making decisions in respect to categorizing, routing, and automatic responses.

- [E-Mail Escalation](#)

E-mails can be escalated if they are not processed within a specific period of time. For more information, see [Defining E-Mail Escalation](#).

- Web Forms

In addition to accepting incoming e-mails, it is also possible to accept information created from a web form on a website.

- Support of Multiple Incoming and Outgoing E-Mail Addresses

You can define multiple incoming e-mail addresses. You can also define multiple outgoing addresses that agents can change when they are responding to e-mails.

More Information

[Setting Up ERMS](#)

ERMS Rule Policies

Use

You can create rule policies in the e-mail response management system (ERMS) context to handle and evaluate incoming e-mails.

Rules are evaluated every time an incoming e-mail is received if the policy in which they belong is invoked during rule processing.

To create ERMS rules and policies, you use objects configured in the ERMS repository.

Prerequisites

- You have defined attributes, actions, and action parameters in Customizing for **Service** under ► **E-Mail Response Management System** ► **Define Repository** ►.
- You have created authorization groups in Customizing for **Service** under ► **E-Mail Response Management System** ► **Define Repository** ►.
- You have assigned the authorizations groups to authorization object CRM_ERMS_P in **Role Maintenance** (transaction PFCG).
- You have assigned the authorization groups to policies in the WebClient UI in Service. You can see (and in some cases select) authorization groups in the WebClient UI under **Rule Policy Details**, after you select a policy (the top node of the policy structure). The possible values are the entries you maintained in the Customizing activity above.

Activities

- Select attributes

Most attributes are self explanatory. Only those needing further explanations are listed below.

Attribute	Use
ERMS: Content Analysis Accuracy	A value between 0 and 1 describing the accuracy of the content analysis result. Accuracy = 1 indicates the result is highly focused. All matching categories form a single path in the category hierarchy. Accuracy = 0 indicates the result is contradictory. At least two matching categories point to different paths in the category hierarchy.
ERMS: Best Minus Next Best Score	Distance in score between the category with highest score and the category with second highest score. Each matching category has a score which is a value between 0 and 1 describing how good the content-queries of the category match the given textual content.
ERMS: Number of Top Categories	Number of matching categories that all have the highest score.
ERMS: Best Category Score	Highest score among all matching categories.
ERMS: Top Scoring Category	Matching category with the highest score.
Escalation Date	Date when the escalation of an e-mail occurs. The date is calculated based upon either rule evaluation, Customizing, or service level agreements (SLA) at the time the e-mail arrives.
Escalation Time	Time when the escalation of an e-mail occurs. The time is calculated based upon either rule evaluation, Customizing, or SLA at the time the e-mail arrives.
Escalation Type	Depends on the property METHOD of the service UT_ESCALDETERM. If the METHOD is set to 1, the Escalation Type is SLA. If the METHOD is set to 2, the Escalation Type is determined as the shortest time out of one of the following: rule evaluation, Customizing, or SLA.
Response Time Duration (Hours)	Used for escalations. Represents the initial response time in hours.

- Set actions and parameters

This covers the standard delivered actions and their corresponding parameters.

Action	Use	Parameters
Add Attribute Value	Adds a value to a multi-value attribute	Attribute The attribute that a value is added to. Value The value that is added.

Action	Use	Parameters
Auto Prepare	Same as Auto Respond, except the response is not sent automatically. The agent can review the content first, make any necessary changes, and then send the response. This action is triggered when an agent selects Reply to an incoming e-mail.	Mail Form Name of the mail form that determines the content of the outgoing e-mail. Outgoing E-Mail Address Address used for the outgoing e-mail's sender address. If you do not use this parameter, the incoming e-mail address that the e-mail was originally sent to, will be used. Category Schema Schema to determine which Top Scoring Category is used. Inline Determines whether standard responses and solutions tied to the category are inserted directly into the e-mail content instead of as an attachment. If both are inserted into the e-mail content, then the standard responses are shown first in the order they are attached to the category. The solutions are then added, also in the order they are attached to the category.
Auto Respond	Sends an automated response with proposed answers and/or solutions, to an incoming e-mail, without agent intervention. The solutions are inserted into the e-mail as attachments and are determined from the Top Scoring Category of the incoming e-mail. Not to be confused with the Send Auto Acknowledgement action.	Category Schema Mail Form Outgoing E-Mail Address See descriptions above from Auto Prepare. Create Interaction Record Creates an interaction record, after the response is sent, which includes both the incoming and the outgoing e-mails. Route To (On Exception) Organizational unit that the incoming e-mail is sent to if no solutions are attached to the Top Scoring Category. Inline See description above from Auto Prepare.
Create Interaction Record	Creates an interaction record in the background, only if there is a business partner associated to the incoming e-mail address. The incoming e-mail is linked to interaction record, so the original e-mail can be viewed at a later time, if necessary.	Description Description saved in the interaction record.

Action	Use	Parameters
Create Service Ticket	Creates a service ticket in the background, only if there is a business partner associated to the incoming e-mail address. The incoming e-mail is linked to service ticket, so the original e-mail can be viewed at a later time, if necessary.	Description Description saved in the service ticket. You can configure the transaction type (for example, TSRV or TSVO) for this service ticket. You do this in Indirectly Called Services and Properties in Customizing for Service by choosing E-Mail Response Management System > Service Manager > Define Service Manager Profiles.
Delete E-Mail	Deletes an incoming e-mail. Can be combined with the Create Interaction Record action, so the interaction record shows an e-mail was received and was also deleted.	(No parameters)
Forward E-Mail		Forward To Forward From
Invoke Policy	Temporarily suspends the processing of the current policy and continues with the processing of the specified policy. After the specified policy is finished processing, the action stops processing the original policy, unless the processing was stopped previously by the Stop Further Rule Processing action. This action can be nested, that is a policy that is invoked may invoke other policies. If a policy has multiple variants, then the active variant is invoked.	Policy Name of the rule policy that is invoked
Push E-Mail	Allows e-mail routing through the communication management software instead of directly going to an inbox. E-mails that are pushed to agents using this action are also routed to the Agent Inbox by the ERMS default routing, unless another routing destination is specifically maintained. Therefore, we recommend directly specifying a Route E-Mail action after the Push E-Mail action, allowing all pushed e-mails to be monitored by a supervisor or special organizational unit, and avoiding incorrect routing of push e-mails to the default routing that are already completed.	Organizational Object
Route E-Mail	Directs the incoming e-mail to an agent or group of agents. The e-mail is then accessible from the agent inbox.	Organizational Object You can enter an agent's system user name or an organizational unit. If you enter an organizational

Action	Use	Parameters
		unit, then the e-mail can be processed by any agent who belongs to that organizational unit.
Route to Case Processor	Directs the incoming e-mail to the agent who is assigned as the case Processor in the Case Overview , only if the incoming e-mail is related to an existing case and the case number is still in the e-mail text.	Route To (On Exception) Organizational unit that the e-mail is routed to if there is no case associated to the incoming e-mail or if there is no processor assigned to the case.
Route to Service Ticket Responsible	Directs the incoming e-mail to the agent who is assigned as the Employee Responsible in the Service Header , only if the incoming e-mail is related to an service ticket and the service ticket number is still in the e-mail text.	Route To (On Exception) Organizational unit that the e-mail is routed to if there is no service ticket number associated to the incoming e-mail or if there is no employee responsible assigned to the service ticket.
Send Auto Acknowledgement	Sends an automatic notification that the e-mail was received.	Mail Form Outgoing E-Mail Address See descriptions from Auto Prepare . Create Interaction Record Creates an interaction record to record that an acknowledgement was sent after receiving the e-mail. (The outgoing e-mail acknowledgement text is not tied to the interaction record.)
Set Attribute Value	Allows you to set an attribute value in the fact base if certain conditions are met. This attribute can then be used in rule evaluation or passed on to actions.	Attribute Value
Show Number of Incomplete E-Mails	Searches for incomplete e-mails linked to an identified account.	(No parameters)
Stop Further Rule Processing		(No parameters)

- Set rule policy authorizations

You can define your own groupings for authorization, for example, by department or by organization.

Authorization Field	Description	Values
ACTVT	Activity	The following values affect authorizations for policies: 01: Create or Generate 02: Change 03: Display 06: Delete 43: Release (For releasing draft rules)

Authorization Field	Description	Values
ERMS_AUGR	Authorization Group	The values you defined for the authorization groups in Customizing activity Define Repository .
ERMS_CTXT	Context	The context in the WebClient UI in Service as defined in Customizing activity Define Repository .

More Information

[Rule Modeler](#)

[Rule Policies](#)

[Rules](#)

[Conditions](#)

[Actions and Parameters](#)

[Conflict Resolution](#)

[Rule Policy Authorizations](#)

ERMS Simulator

Use

The **ERMS Simulator** allows an interaction center (IC) manager to simulate the results of new or changed rules contained in rule policies that are used for e-mail handling. IC managers also see a detailed view of the resulting actions and categorizations for those e-mails. An IC manager can run the simulation in the background and have the result sent to an e-mail address.

The functions in the **ERMS Simulator** are available in the WebClient UI.

Prerequisites

- You have created [rule policies](#).
- You have set up ERMS and already processed incoming e-mails, using ERMS.

More Information

[Setting Up ERMS](#)

E-Mail Workbench

Use

The **E-Mail Workbench** is an overview of e-mail processing in an interaction center, which you can also use as a means to mass process e-mails.

You can access the **E-Mail Workbench** in the WebClient UI by choosing **IC Manager > E-Mail Workbench**.

You can perform manual operations for e-mails, such as searching, routing, forwarding, re-categorizing, and deleting e-mails. You can also view the specific rules that were invoked for a particular e-mail and process [escalated e-mails](#).

You can view and in some cases edit the following elements of an e-mail:

- E-mail text
- Attachments
- [Rules](#) for processing the e-mail
- Associated categories, that you can add or remove. Automated processing can assign categories to e-mails as determined from Multilevel Categorization. You can change these categories if the assignment is not accurate or complete.
- The Fact Base
- Related e-mails

Prerequisites

- You have [set up](#) and actively used ERMS to process incoming e-mails.
- You have set **UT_ERMS_REPLICAT** as your ERMS service manager profile for **Directly Called Services**. You do this in Customizing for **Service** by choosing **E-Mail Response Management System > Service Manager > Define Service Manager Profiles**.

ERMS Responses

Use

There are several types of E-Mail Response Management System (ERMS) responses you can use to automate e-mail processing. This automated processing saves agents time in compiling and searching for information and ensures consistent handling and messaging to customers. You can use mail forms to configure the e-mail content for the responses.

Prerequisites

- You have defined mail forms in the WebClient user interface. You can access this function under **Knowledge Management > Mail Forms**.
- You have defined multilevel categorization. (Not relevant for Auto Acknowledgements)
- You have defined a rule policy with the **Send Auto Acknowledgement**, **Auto Respond**, and/or **Auto Prepare** actions.
- You have assigned mail forms to actions in the WebClient UI or to categories through multilevel categorization.

Features

The following is a list of responses you can determine for ERMS:

- Auto Acknowledgement

When an e-mail is received, the system can send an automatic acknowledgment informing the sender that the e-mail was received. If the business partner is identified, you can use business partner attributes, such as first and last name, to personalize the response.

- Auto Response

The system can send a solution or a response automatically, without intervention from agents, based upon the e-mail content.

- Auto Prepare

Response proposals are similar to auto responses. However, instead of automatically sending the response, the response is reviewed by agents for accuracy. Agents can send, modify, or delete the proposal.

More Information

[ERMS Rule Policies](#)

E-Mail Threading

Use

You can use e-mail threading for service requests and incidents to link e-mails to the associated business transaction, keeping all communication and related e-mails together for easy access to the information, regardless who handles the business transaction. E-mail threading also enables the e-mails to be routed to the person responsible for that business transaction if the customer sends the e-mail back to the interaction center.

Mail forms are used to create a tracking text.

Prerequisites

- You have created tracking text in WebClient UI under **Knowledge Management > Mail Forms**. You can add the **Email Response Management** context attribute to the **Text Element**.

The **Service Request Tracking Text** mail form field is available as a variable for service requests and incident numbers.

If the variable is not available in the mail form, you have to manually add the CRMS_ERMS_SCENARIO_FIELDS structure to the **ERMS** context in Customizing. You do this in Customizing for **Service** by choosing **Basic Functions > Communication Channels > Maintain Attribute Contexts for Mail Forms**.

- You have enabled tracking text insertion by maintaining the **Tracking Text Form** field and selecting the **Track Text on Reply** indicator in Customizing for **Service**. You do this by choosing **Interaction Center WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles**.
- You have defined rules using the **Route to Service Request Responsible** action. For more information, see the actions and parameters table in [ERMS Rule Policies](#).

Features

- E-Mail Tracking Thread

The threading is enabled by the use of tracking text you define. You use a variable that is filled at runtime with the service request number to insert in the tracking text. This variable allows e-mails to be linked to business transactions.

- Tracking Text Insertion

The system inserts the tracking text in the e-mail content when an e-mail is created, either manually or automatically through automated processing, or when the e-mail is replied to. If a service request or incident is automatically created for an incoming e-mail from the auto acknowledgement action, you can link both the incoming e-mail and the outgoing reply to the service request or incident.

- **Thread Detection and Routing**

When customers reply to e-mails, the system determines whether tracking text is inserted. The incoming e-mail can be associated automatically to the related business transaction. Attributes from these transactions, such as the service request number, can be used during rule evaluation. When you create rule policies, you can create rules that route the e-mail to the agent who is currently assigned to the business transaction.

Example

❖ Example

You create a mail form with the following as the tracking text.

```
***** DO NOT DELETE *****
```

```
{SrvReqNo:[8000006814]}
```

```
***** DO NOT DELETE *****
```

You define a rule, "If Service Request Responsible Is Not Equal To "", Then Route to Service Request Responsible to forward the e-mail".

More Information

[E-Mail Threading Process](#)

E-Mail Threading Process

Use

This process describes e-mail threading using service tickets as the relevant business object.

Prerequisites

You have defined [e-mail threading](#).

Process

1. The customer sends an e-mail.
2. Either an agent sends a manual response or the system sends an automatic response.

Manual Response

- a. The agent processes the e-mail and creates a service ticket.
- b. The agent replies to the e-mail.

- c. The system uses the mail form you entered in the agent's e-mail profile. It inserts the mail form text and tracking text into the outgoing e-mail. The system populates the variable with the actual service ticket number that was just created.

Automatic Response

- a. The e-mail response management system processes the e-mail automatically and creates a service ticket, as defined through an [action](#).
 - b. The system sends a reply to the customer as defined in an action, such as **Send Auto Acknowledgement**.
 - c. The system uses the mail form that is defined as a parameter in the **Send Auto Acknowledgement** action. The mail form includes the service ticket tracking variable. When the system sends the reply, the variable is populated with the service ticket number that was just created.
3. The customer replies to the e-mail and sends it back to the interaction center.
 4. The fact gathering services scan the e-mail and pick up the tracking text and store this in the fact base during automated processing.
 5. The e-mail is associated to the service ticket.
 6. The e-mail is routed to the agent who is processing the service ticket through a [rule](#) that uses the **Route to Service Ticket Responsible** action.

E-Mail Escalation

Use

E-mails can be escalated if they are not processed within a specific period of time, for example, e-mails that are not processed within 24 hours after the initial receipt are escalated. You can define the time periods that determine the escalation and you can be notified and view escalated e-mails as soon as the escalation occurs. From reviewing escalated e-mails, you can reroute the e-mail or decide on another action to speed up processing.

Integration

The SAP Business Workflow is used for the escalation process. In [Workflow for ERMS](#), the escalation times you define are used as the **Latest End** date and time.

Prerequisites

You have [defined e-mail escalations](#).

Features

Escalation Definition

Escalated e-mails are determined when defined initial response times are not met. The initial response time is the time between an e-mail is received in the system until the time that an agent processes the e-mail.

Initial response time is calculated by one of the following:

- [Service Level Agreements](#) (SLA)

You can set a deadline for escalation by using a service contract. The system uses the service contract that is assigned to the business partner and reads the duration time from the response profile associated with the service contract.

- [Rules](#) and [actions](#)

In the WebClient user interface, you create a rule using the **Set Attribute Value** action with the value **Response Time Duration** to determine at what point an e-mail is considered escalated. You can also use some fact base attributes as additional criteria, for example, if a customer is a high priority and valued customer with a specific system status, and sends an e-mail with a request, but the e-mail is not processed within a specific number of hours as determined in the rule, the e-mail is escalated.

- E-Mail Addresses

You can determine an escalation time based upon the receiving e-mail address.

Escalation Determination

You can choose which method is used to determine the initial response time. You can either use the:

- Service Level Agreement
- Combination of SLA, rules and actions, and e-mail addresses

The system uses the definition with the shortest initial response time out of all three methods.

Escalation Process

See [E-Mail Escalation Process](#).

Escalation Notification

You are notified of a new escalated e-mail by receiving an e-mail. Mail forms are used to create the notification text.

Escalation Monitoring

After receiving the escalation notification, you use the overview in the [E-Mail Workbench](#) to monitor the number of escalated e-mails per organizational unit and navigate directly to the individual e-mail for further action.

E-Mail Escalation Process

Use

This process describes the e-mail escalation process.

Prerequisites

You have [defined e-mail escalations](#).

Process

1. An e-mail is received in the interaction center (IC).
2. Automated processing begins.
3. The initial response time (IRT) is determined by the escalation definition.

4. If the IRT limit is reached, the e-mail becomes escalated and the workflow escalation process starts.
5. The notification e-mail is sent to the IC manager.
6. The manager accesses the [E-Mail Workbench](#), views the situation and takes further actions with the escalated e-mail.

More Information

[E-Mail Escalation](#)

Defining E-Mail Escalation

Prerequisites

You have authorization to read mail forms.

Context

You use this procedure to define and setup [E-Mail Escalation](#).

Procedure

1. Define the method used for determining initial response times.
 - o Service Level Agreements
 - a. Run transaction CRMD_SERV_SLA.
 - b. Maintain a response profile and assign a response time. For **Name of Duration**, select **Duration Until First Reaction**. The response profile is associated to a service contract. For more information, see [Service Level Agreements \(SLA\)](#).
 - o Rules and Actions

Define a rule in the WebClient user interface and use the **Set Attribute Value** action with the **Response Time Duration** attribute.
 - o E-Mail Addresses

Define the escalation time period for each receiving e-mail address in Customizing for **Service** by choosing **Interaction Center WebClient** > **Agent Inbox** > **Settings for Asynchronous Inbound Processing** > **Define Receiving E-Mail/Fax Settings**.
2. Assign the method used for determining initial response times and assign the fact gathering service.
 - a. In Customizing for **Service** choose **E-Mail Response Management System** > **Service Manager** > **Define Service Manager Profiles**.
 - b. Select your service manager profile.
 - c. Under **Directly Called Services**, find the Service ID RE_RULE_EXEC and add the following services underneath to ensure that they are executed in the correct order with the attribute **Invocation Order**. If you do not use Rules and Actions to determine your initial response times, the order of the services could be invoked before RE_RULE_EXEC:
 - i. Under **Directly Called Services**, add the Service ID FG_ESCALTIME if it is missing. You can maintain the property values using the F1 field help.

- ii. Under **Directly Called Services**, add or select Service ID UT_ESCALDETERM and choose **Properties**.
- iii. In the **Property ID** field add or select **METHOD**, and in the **Property Value** field enter the value. Refer to the F1 field help for e-mail escalation values.

i Note

Do not include the service UT_SEND_ESCL as part of your **DEFAULT** service manager profile.

3. Customizing of the notification mail.

- a. In the WebClient UI, define the mail form used for notifying managers of the e-mail escalation. This is done in the interaction center (IC) manager role under **Knowledge Management > Mail Forms**. Insert escalation specific placeholders by clicking **Attributes** in the mail form and adding those specific escalation fields (for example, escalation date, time and type, sender's name and e-mail address and the e-mail subject).

If these escalation objects are not available, ensure that object CRMS_ERMS_ESC_FIELDS has been added and configured to the correct context in the mail form. In Customizing for **Service** choose **Basic Functions > Communication Channels > Maintain Attribute Contexts for Mail Forms**.

- b. Assign the mail form and the receiver's and sender's e-mail information to the service manager profile.

- i. In Customizing choose **Service > E-Mail Response Management System > Service Manager > Define Service Manager Profiles**.
 - ii. Select service manager profile ERMS_ESCALMSG and maintain the property values of the service UT_SEND_ESCL under **Directly Called Services**. Refer to the F1 field help for e-mail escalation values.
- c. In Customizing choose **Service > E-Mail Response Management System > Service Manager > Assign Service Manager Profiles**. Assign the service manager profile ERMS_ESCALMSG to the object **ObjectID** 15107965 without entering the **Address/Number**.
- d. In transaction SWU3 under **Maintain Runtime Environment**, maintain the **Schedule Background Job for Workflow Deadline Monitoring** and specify the interval until next deadline check.

Automated Processes and Architecture

Rule Processing Services

Use

Rule processing, or rule invocation, services is one of the four types of services found in the E-Mail Response Management (ERMS) service manager. There are standard delivered rule processing services available. For more information, see Customizing documentation for **Service >> E-Mail Response Management System > Service Manager**.

Features

Rule processing services evaluate a set of predefined rules and determine a set of actions that need to be executed during rule processing.

Rules consist of conditions and actions that are triggered if the conditions evaluate to true. The rule processing services support rules and nested rules. Nested rules are only processed if the condition in the higher-level rule is true.

Rule processing starts with the first rule defined within a given policy and continues sequentially with all subsequent rules in exactly the order defined in the rule modeler. All actions that are triggered during rule evaluation are collected and passed back to the service manager. For each of these actions, the service manager determines the appropriate action handler and calls these handlers accordingly.

Special rule processing

The **Stop Further Rule Processing** action can immediately interrupt rule processing. If this action is triggered, only that actions that have executed to that point are passed to the service manager.

The **Invoke Policy** action is executed by the service manager after the current policy is completely processed. It is not executed immediately.

Content Analysis

Use

This function identifies the categories that best match specific content and makes automatic category suggestions.

Content analysis is used in the E-Mail Response Management System (ERMS) to analyze the content of inbound e-mails. It uses predefined search queries that are assigned to categorization schemas, to determine suitable categories.

Integration

Function	Description
Category Modeler	You define content queries for content analysis in the category modeler. If necessary, you can reset the content analysis in the page area Application Areas for each application area. You do this by choosing Reset Content Analysis.
E-Mail Response Management System (ERMS) - automated processing	ERMS manages a high volume of inbound e-mails. This system partially automates routing and responding to e-mails. The automated processing of e-mails in the ERMS uses content analysis to evaluate conditions assigned to categories. The following ERMS components relate to content analysis: • Fact gathering • Rule operators Matches Category and Rule Attributes
Interaction Center	Content analysis provides suggested categories based on the content of e-mails received in the interaction center and selected for processing in the agent inbox. These categories are then displayed in the relevant fields of the following IC-specific business transactions: • Service order • Service request • Interaction record

Prerequisites

You have made content analysis settings in Customizing for [Service](#) by choosing [Cross-Application Components](#) [Multilevel Categorization](#) [Define Application Areas for Categorization](#).

Features

- Automated categorization

Category suggestions are made automatically using content queries.

- Language detection

Content analysis also supports automatic language detection for textual content.

More Information

[E-Mail Response Management System](#)

Fact Base

Use

The fact base is an E-Mail Response Management System (ERMS) component that serves as a data container for any information that is related to an inbound e-mail and can be relevant during rule evaluation.

Integration

Fact gathering services gather data relevant to the inbound e-mail and add the data to the fact base.

The rule engine runtime retrieves data from the fact base and uses it for rule evaluation and processing.

Features

The fact base stores information that is relevant for exactly one instance of an inbound e-mail. Such information may include e-mail content, business partner information, and the incoming e-mail address to which the e-mail was originally sent.

The fact base is organized as an XML document. Its structure is flexible to include any tree structure and content, depending upon your business requirements.

Updates and read access to the fact base occurs through XSL transformations (XSLT) and XPath queries. These standardized methods are platform- and programming-language independent. This provides maximum flexibility if you need to extend the fact base, for example, for data residing in legacy systems.

Overview Report

This report provides a real-time snapshot of the activity and current status of the [E-Mail Response Management System \(ERMS\)](#). It gives you an overview of the volume of incoming e-mails, which you can display by organizational unit, e-mail address, or Web form ID.

To use this monitoring function, you must be actively using ERMS. To use this function, you must be running Service, and you must activate the necessary BSP service.

Performance Report

You can use this report to monitor the overall effectiveness of the [E-Mail Response Management System](#) (ERMS) in your interaction center. This report provides a real time overview of key performance indicators such as volume, average handling time, average response time, and service level. You can sort this information by organizational unit or by e-mail address.

To use this monitoring function, you must be actively using ERMS.

Workflow for ERMS

Use

This workflow is used for the E-Mail Response Management System (ERMS) in the Interaction Center.

ERMS allows automated processing of inbound e-mails received via SAPconnect. E-mails can be automatically responded to or routed to an agent group for manual processing. An inbound communication triggers the workflow template **WS00200001**, which controls the work item processing and fulfills several tasks.

The following system tasks are the most important:

- Determining the responsible agent group and routing the work item to that group
- Deleting and stopping further manual processing

For a complete list of tasks, see the workflow template **WS00200001** in the workflow definition tools.

Prerequisites

ERMS configuration activities are completed.

Process

1. The incoming e-mail is received.
2. Task TS00207915 **ERMS Rule Execution** invokes the service manager.

The service manager provides capabilities (necessary to process e-mails) such as fact gathering, content analysis, and rule processing.

3. Depending on the outcome of the service manager invocation, the system continues processing the work item by one of the following steps:

- Delete

You can determine when the system deletes an item by using a rule, for example, in cases of junk e-mail (spam). Task TS66107918 **ERMS Delete E-Mail** removes the office document. The workflow is ended.

- Do not process further

You define when processing is stopped in either a rule or in one of the services in the service manager.

- Route

Task TS00207914 **ERMS Decision** forwards the work item to an agent or agent group as selected by the processor determination.

The processor determination decides which agents are responsible for a work item. Task TS00207914 determines which agents or agent groups are eligible for processing e-mails. The actual determination of which agent or agent group receives the work item is defined in rules.

Technical Execution

The following technical information is useful if you wish to customize your own workflow:

- Workflow template

The actual procedure is implemented in the workflow template WS00200001 (**ERMS1**).

- Triggering events for the workflow template

The event **MailReceived** of Business Object Repository (BOR) object ERMSSUPRT2 is maintained as a triggering event in the template WS00200001.

- Processing methods

The following table explains how methods of BOR object ERMSSUPRT2 are used in the corresponding tasks of the workflow template WS00200001:

Method	Use
RECEIVE	Receives the mail from SAPconnect
RULEEXECUTION	Raises the event MailReceived and triggers workflow

Setting Up ERMS

Use

This procedure describes the basic settings needed for [E-Mail Response Management System](#).

Prerequisites

Your IT administrator has created e-mail addresses for receiving incoming e-mails.

Procedure

Mandatory and Recommended Steps

Step	Description
Create an organizational model.	In Customizing for Service under Master Data > Organizational Management > Organizational Model > Create Organizational Model .
Maintain workflow standard settings.	In Customizing for Service under Basic Functions > SAP Business Workflow > Maintain Standard Settings for SAP Business Workflow .

Step	Description
Define receiving e-mail addresses.	In the SAP Menu, choose Interaction Center E-Mail Response Management System Settings Define Receiving E-Mail Addresses/Fax Numbers .
Assign receiving e-mail addresses to ERMS workflow.	In the SAP Menu, choose Interaction Center E-Mail Response Management System Settings Maintain Recipient Distribution .
Assign agents for e-mail handling.	In the SAP Menu, choose Interaction Center E-Mail Response Management System Settings Assign Agent for E-Mail Handling . Choose Assign Agents . Under workflow ERMS 1 either: - Use a general task for ERMS decision to allow e-mail routing to all agents, depending on the rules, or - Specify individual agents or agent groups to allow e-mail routing only to those agents, depending on the rules.
Activate event linking. Incoming e-mails can use workflow to define how mails are routed and how they are handled. Activating event linking activates the workflow you defined and allows it to be used for ERMS.	In the SAP Menu, choose Interaction Center E-Mail Response Management System Settings Assign Agent for E-Mail Handling . Choose Activate Event Linking . For New Recipient , choose possible entries and select SAP Object Instance . Use business object repository (BOR) Object Type ERMSSUPRT2 (ERMS Support 2) . When this BOR object type is invoked, it triggers workflow WS00200001 (ERMS1). For more information, see Workflow for ERMS .
Define outgoing e-mail addresses.	In the SAP Menu, choose Interaction Center E-Mail Response Management System Settings Maintain Sender E-Mail Addresses .
Define service manager profiles.	In Customizing for Service under E-Mail Response Management System Service Manager Define Service Manager Profiles . Choose Default and double click Directly Called Services to set the DEF_ROUTING for Invocation Order 50 to an organizational unit.
Assign service manager profiles.	In Customizing for Service under E-Mail Response Management System Service Manager Assign Service Manager Profiles .
Create ERMS policies and rules	In the Interaction Center, choose Process Modeling Rule Policies . Choose E-Mail Response Management System as the Context .
Assign standard tasks to communication methods	In Customizing for Service under Interaction Center WebClient Agent Inbox Settings for Asynchronous Inbound Processing Assign Standard Tasks to Communication Methods . Choose INT as the communication method for Object ID 207914 .
Define e-mail templates and e-mail forms.	In the Interaction Center, choose Knowledge Management Mail Forms .

Optional Steps/Extensions of Default Customizing

Step	Description
Define additional services. You can define services that are available in the service manager. This activity determines how all necessary services for evaluating policies are invoked. You should only	In Customizing for Service under E-Mail Response Management System Service Manager Define Services .

Step	Description
execute this activity if you want to change the default settings or create new services.	
Extend the ERMS repository.	In Customizing for Service under E-Mail Response Management System Define Repository .
Define additional e-mail profiles.	In Customizing for Service under Interaction Center WebClient Basic Functions Communication Channels Define E-Mail Profiles .
Define catalogs, codes, and subject profiles. Catalogs are resources which help ensure the uniform use of terms	In Customizing for Service under Basic Functions Catalogs, Codes and Profiles .
Define additional categorization profiles. This influences which categorization schema(s) are displayed in the corresponding business transactions, and the behavior of the auto suggest function.	In Customizing for Service under Interaction Center WebClient Business Transaction Define Categorization Profiles .
Define reporting for e-mail statuses and events.	In Customizing for Service under E-Mail Response Management System Reporting .
Delete reporting data.	In the SAP Menu, choose Interaction Center E-Mail Response Management System Utilities Delete Reporting Data .

Integrated Communication Interface

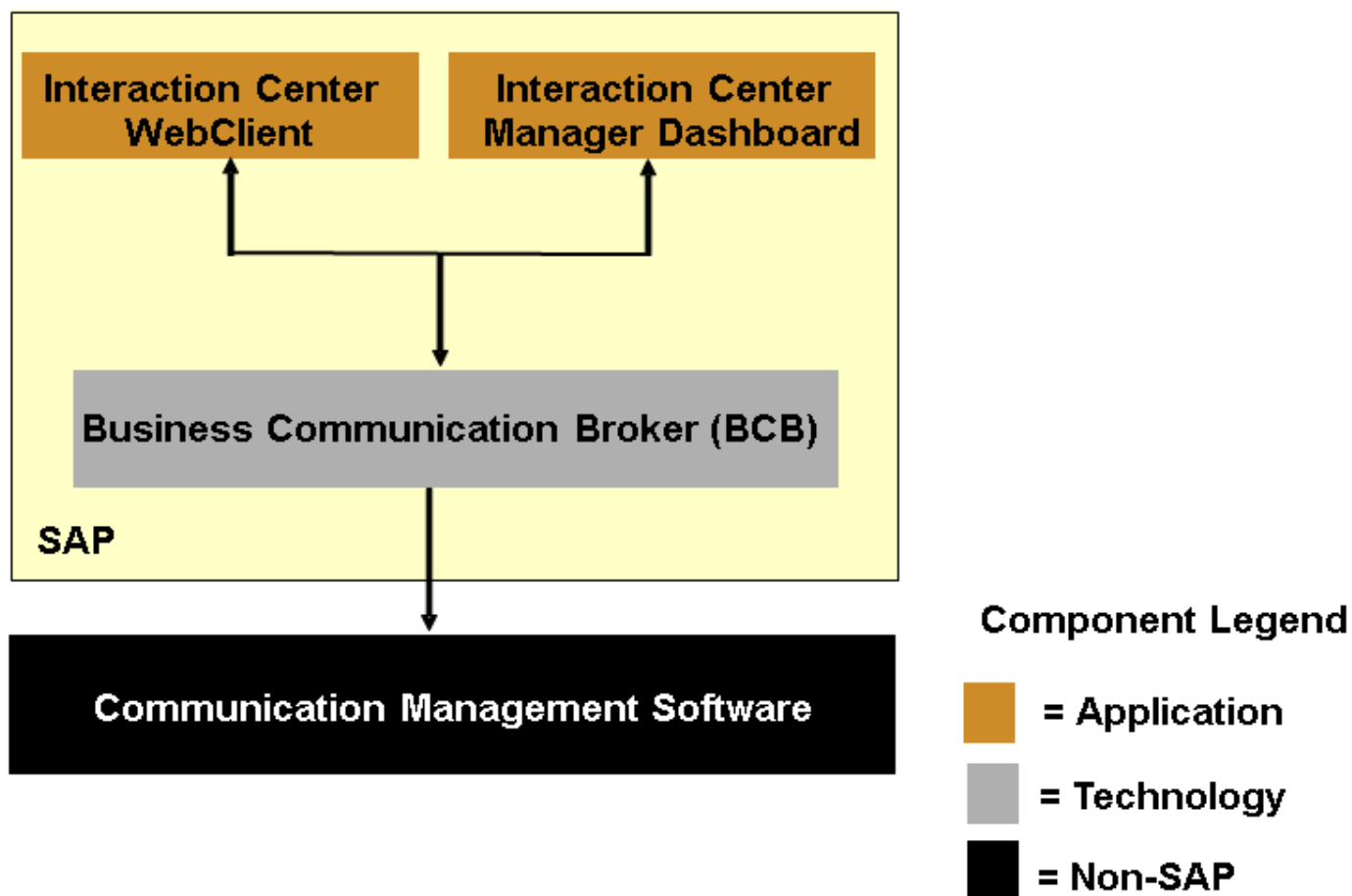
Use

The Integrated Communication Interface (ICI) supports the integration of non-SAP communication products with SAP components that use communication services. The ICI is intended in particular for scenarios involving multiple communication channels such as telephony, e-mail, and chat, but it can also be used in single-channel scenarios. It enables activities such as an agent accepting an inbound phone call or deflecting an inbound chat request, or a supervisor monitoring a queue. The ICI is only one of several components required to implement communication scenarios of this kind.

i Note

An example scenario is the integration of communication management software systems with the interaction center (IC), specifically the IC WebClient and IC manager dashboard. For illustration purposes, this example is used throughout this documentation.

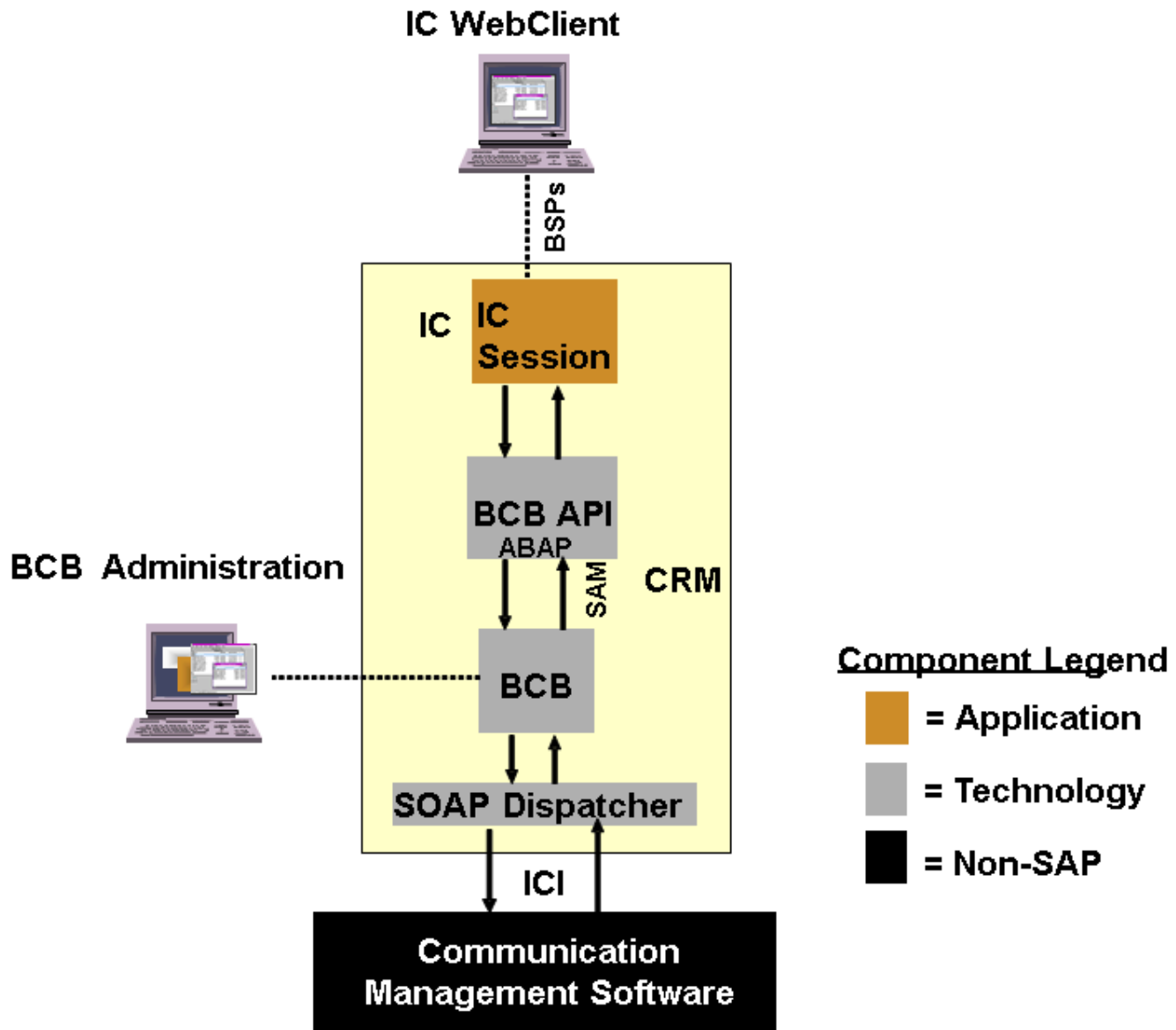
The graphic below represents a simplified view an example scenario without detailing the interfaces. The scenario is made up of functionality from three distinct sources – an application component, a technology component, and an external (non-SAP) component.



Integration

The ICI is based on Simple Object Access Protocol (SOAP) and Extensible Markup Language (XML); they are non-proprietary, Web-oriented, open interface technologies. HTTP is used at the transport level.

The system architecture required to connect the interaction center with external communication management software using the ICI is made up of several components. The system architecture of the interaction center example scenario in a live environment is shown below:



Live System Architecture

BCB Components

The BCB, BCB API, and SOAP dispatcher components enable the CMS. The messages passed between the application component and the external component are passed via the ICI.

i Note

CMS Customizing for IC WebClient is done using transaction CRMM_BCB_ADM.

Application-Specific Components

The application-specific components are linked to the ICI-specific components via the Business Communication Broker Application Programming Interface (BCB API). This is the API for application customizing.

In the interaction center example scenario, the application consists of the IC WebClient, which provides an ABAP-based user interface for interaction center agents.

The IC server sends actions to the communication management software via the ICI-specific components and receives events back from the communication management software also via the ICI-specific components. The actions might include agent logon,

initiate outbound phone call, or remove chat line from chat session. The events might include new inbound phone call, or new posting in chat session.

External Components

These are non-SAP communication systems. In this scenario, the external product is the communication management software, which is a multichannel communication product. This is responsible for merging requests from different media, routing requests to specific agents or agent groups, queuing requests, signaling inbound requests to the IC, and so on. The external products communicate with SAP via the ICI using SOAP and XML.

Certification

A certification process is implemented to verify the connectivity of external communication products with SAP using the ICI. An external product does not have to support the ICI in full. Partial certification is possible to enable the connection of single-channel communication products, such as telephony products.

Features

Many application components use external communication services. Integrating an external communication product with an application component using the ICI can provide:

- Integration of multiple communication channels such as phone, e-mail, and chat
- Active reporting of new, changed, or completed phone calls, e-mails, or chat sessions (known generically as items)
- Monitoring and statistics

i Note

Multichannel management (that is, the synchronization and serialization of incoming communications) is not part of this SAP functionality. The ICI is a server-to-server interface. It does not support direct communication between the SAP client and the external communication product.

Situation Templates in Service

Get an overview of situation templates available in Service, together with the apps that display the situations. You can filter them according to various criteria.

i Note

For more information about the Situation Handling framework in SAP S/4HANA, see the Product Assistance for [Situation Handling](#).

Situation Templates in Service

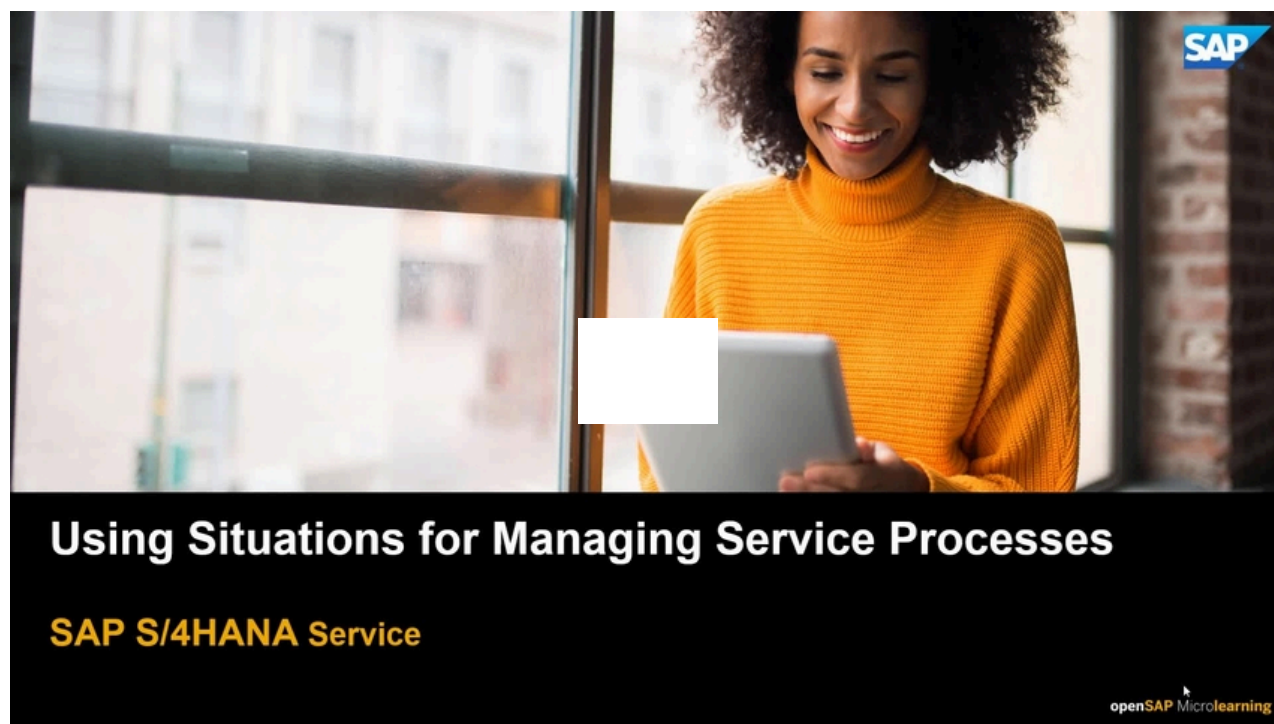
Area or Topic	Solution Capability	Situation Template	Situation Template ID	Situation Template Type	Data Context	App Name	SAP F
Service	In-House Service	Situation Template: In-House Service Not Confirmed	SRVC_DELAY_IN_IHR_CONFIRMATION	Standard object-based	No	Manage In-House Services	F4025

Area or Topic	Solution Capability	Situation Template	Situation Template ID	Situation Template Type	Data Context	App Name	SAP F
Service	In-House Service	Situation Template: Processing of In-House Service Overdue	SRVC_DELAY_IN_IHR_PROCESSING	Standard object-based	No	Manage In-House Services	F4025
Service	In-House Service	Situation Template: Follow-Up Step of Precheck Pending	SRVC_IHRITM_PRECHECK_DELAYED	Standard object-based	No	Manage In-House Service Objects	F7695
Service	Service and Maintenance Plan	Service Maint. Plan Mass Schedule Issue	SRVC_MPLAN_MASS_SCHEDULING_ISSUE	Standard object-based	No	Mass Schedule Maintenance Plans - Service	F4925
Service	Service Contract Management	Service Contract Due to Expire Soon	SRVC_EXPIRING_SERVICE_CONTRACT	Standard object-based	No	Service Contracts	TBT11 (WebC UI ID)
Service	Service Contract Management	Service Contract Item Due to Expire Soon	SRVC_EXPIRING_SRVC_CONTR_ITEM	Standard object-based	No	Service Contracts	TBT11 (WebC UI ID)
Service	Service Contract Management	Service Contract Item Not Released	SRVC_CONTRACT_ITEM_RELEASE_DUE	Standard object-based	Yes	Service Contracts	F3763
Service	Service Contract Management	Service Contract with Zero Billing Value	SRVC_CONTR_WITH_EXP_BILL_VALUE	Standard object-based	No	Service Contracts	TBT11 (WebC UI ID)
Service	Service Contract Management	SrvcContr Item Blocked for Billing	SRVC_CONTRACT_ITEM_BILLG_BLOCKED	Standard object-based	Yes	Service Contracts	F3763
Service	Service Contract Management	Price Check for Automatic Renewal	SRVC_CONTR_AUTO_RENEW_PRICE	Standard object-based	No	Service Contracts	F3763
Service	Service Contract Management	Service Contract GTS Blocked/Pending	SRVC_CTR_GTS_BLOCKED_OR_PENDING	Standard object-based	No	Service Contracts	F3763
Service	Subscription Order Management	Subscription Contract Item Due to Expire	SOM_EXPIRING_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F6193
Service	Subscription Order Management	Subscription Contract Item Grace Period	SOM_GRACE_PERD_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F6193

Area or Topic	Solution Capability	Situation Template	Situation Template ID	Situation Template Type	Data Context	App Name	SAP F
Service	Subscription Order Management	Subscription Contract Item Distribution	SOM_DISTR_ISSUE_SUB_CONTR_ITEM	Standard object-based	No	Manage Subscription Contract Lifecycle	F6193
Service	Service Order Management	Trade Compliance Check Blocked/Pending	SRVC_GTS_BLOCKED_OR_PENDING	Standard object-based	No	Manage Service Orders	TBT11 TBT11
Service	In-House Service	Situation Template: Stock Inconsistency for Service Object	SRVC_IHRITM_PROD_LOGS_ERROR	Standard object-based	No	Manage In-House Service Objects	F7695
Service	In-House Service	Situation Template: Inconsistency in Logistics Information for Service Object	SRVC_IHRITEM_EQUIP_LOGS_ERROR	Standard object-based	No	Manage In-House Service Objects	F7695

Using Situations for Managing Service Processes (English Only)

Watch this video at learning.sap.com.



Related Information

[Use Cases for Situation Handling](#)

Data Management in Service

Data management includes data archiving, blocking, unblocking and deleting personal data, as well as data destruction.

[Data Archiving in Service](#)

[Blocking, Unblocking, and Deletion of Personal Data in Service](#)

Related Information

[Data Management in SAP S/4HANA](#)

Data Archiving in Service

Data archiving is used to remove mass data that is no longer required but must be kept in a format that can be analyzed from the database. This section includes archiving objects for Service.

For more information about the archiving objects that are used for leads, opportunities, and activities, see [Data Archiving in Sales](#).

Prerequisites for Archiving Service Transactions

Depending on the Finance integration scenario, you should consider the following when archiving service orders and service contracts:

- If you use item-based accounting, the service orders and service contracts to be archived must be set to **Business Completed**.
- If you use the integration with account assignment manager, the service orders and service contracts to be archived must be set to **Completed**.

Regardless of the Finance integration scenario used, all other transactions, such as service requests, service quotations, and service confirmations, must be set to **Completed** to be archived.

Related Information

[Data Archiving](#)

[Data Management in Global Trade Management](#)

[Item-Based Accounting](#)

[Finalizing the Service Order](#)

[Finalizing the Service Contract](#)

Archiving Service Contracts Using CRM_SRCONT

You can use archiving object CRM_SRCONT to archive service contracts.

Tables

CRM_SRCONT archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_SRCONT can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_SRCONT is delivered with the following programs:

Program	Name
Write	CRM_ARC_SRCONT_SAVE
Delete	CRM_ARC_SRCONT_DELETE_2
Preprocessing	CRM_ARC_SRCONT_CHECK_2

When archiving service contracts, you must use these programs or custom variants. For information on creating variants, see [Variants for CRM_SRCONT](#)

Prerequisites

The following prerequisites must be fulfilled before you can archive a service contract:

- The service contract has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up service orders and service confirmations are archivable.
- If item-based accounting has been enabled: The service contract has been set to **Business Completed**.

For more information about item-based accounting, see [Item-Based Accounting](#).

The following checks have been carried out:

- You've executed an archivability check for service contracts by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service contracts that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to process smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_SRCONT ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_SRCONT, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance, SC for service contract.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization the rule applies to.
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service contract was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object CRM_SRCNT assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers an implementation of the following BADIs for CRM_SRCNT:

- [BAdI: Tables Used by an ILM Object](#) (BADI_DTINF_ILM_OBJ_TABLES)
- [BAdI: Archiving Information of Tables](#) (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note [2939146](#) .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity [Maintain Purposes](#) (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity [Personalize Data Collection via Profiles](#) (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_SRCNT, there is at least one active archive information structure that contains the following selectable field:
 - [Transaction ID](#) (object_id)

This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SC that is provided in the standard system.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRM_SRCONT archiving object, you must first check the settings in Customizing for [Service](#) under [Transactions](#) > [Data Archiving](#).

In the activity [Configure Archiving Preselection](#), you define parameters for the check and preprocessing programs. In the activity [Define Residence Times](#), you define the residence time of service contracts.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Contracts Archived With CRM_SRCONT

You have the following options for displaying archived service contracts:

- From within [Archive Administration](#) (transaction SARA), you can search for archived service contracts using the read program CRM_ARC_SRCONT_READ
- You can also use the [Archive Information System](#) (transaction SARI) to display archived service contracts by entering the archiving object CRM_SRCONT and the archive infostructure SAP_CRMS4_SC

In both cases, the system provides a range of criteria that enable you to select the service contracts that you want to display.

Variants for CRM_SRCONT

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRM_SRCONT.

Defining Write Variants

The write variant contains the parameters for the service contracts that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service contracts that you want to archive.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

Here you can enter a short text that helps you identify and better find the archiving session in [Archive Management](#).

Defining Preprocessing Variants

The preprocessing program CRM_ARC_SRCNT_CHECK_2 determines archivable service contracts.

You can define variants for this program using the following parameters:

- Transaction Type

You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Requests Using CRM_INCDNT

You can use archiving object CRM_INCDNT to archive service requests.

Tables

CRM_INCDNT archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_INCDNT can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_INCDNT is delivered with the following programs:

Program	Name
Write	CRM_ARC_SRQM_INCIDENT_SAVE
Delete	CRM_ARC_SRQM_INCIDENT_DELETE_2
Preprocessing	CRM_ARC_SRQM_INCIDENT_CHECK_2

When archiving service requests, you must use these programs or custom variants. For information on creating variants, see [Variants for CRM_INCDNT](#).

Prerequisites

The following prerequisites must be fulfilled before you can archive a service request:

- The service request has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up service orders and service confirmations are archivable.

The following checks have been carried out:

- You've executed an archivability check for service requests by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service requests that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to process smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_INCDNT ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_INCDNT, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance, SRVR for a service request.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization the rule applies to.
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service request was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object CRM_INCDNT assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BADIs for CRM_INCDNT:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note [2939146](#) .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_INCDNT, there is at least one active archive information structure that contains the following selectable field:
 - **Transaction ID** (object_id)

This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRM_INCDNT that is provided in the standard system.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRM_INCDNT archiving object, you must first check the settings in Customizing for **Service** under **Transactions** > **Data Archiving**.

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of service requests.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Requests Archived With CRM_INCDNT

- From within Archive Administration (transaction SARA), you can search for archived service requests using the read program CRM_ARC_SRQM_INCIDENT_READ
- You can also use the Archive Information System (transaction SARI) to display archived service requests by entering the archiving object CRM_INCDNT and the archive infostructure SAP_CRMS4_SRVR

In both cases, the system provides a range of criteria that enable you to select the service requests that you want to display.

Variants for CRM_INCDNT

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction **SARA**, you can create variants of the write and preprocessing programs available for **CRM_INCDNT**.

Defining Write Variants

The write variant contains the parameters for the service requests that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program **CRM_ARC_SRQM_INCIDENT_CHECK_2** determines archivable service requests.

You can define variants for this program using the following parameters:

- Transaction Type

You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Orders Using CRM_SERORD

You can use archiving object CRM_SERORD to archive service orders.

Tables

CRM_SERORD archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_SERORD can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_SERORD is delivered with the following programs:

Program	Name
Write	CRM_ARC_SERORD_SAVE

Program	Name
Delete	CRM_ARC_SERORD_DELETE_2
Preprocessing	CRM_ARC_SERORD_CHECK_2

When archiving service orders, you must use these programs or custom variants. For information on creating variants, see [Variants for CRM_SERORD](#).

Prerequisites

The following prerequisites must be fulfilled before you can archive a service order:

- The service order has the status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up service confirmations are archivable.
- If item-based accounting has been enabled: The service order has been set to **Business Completed**.

For more information about item-based accounting, see [Item-Based Accounting](#).

The following checks have been carried out:

- You've executed an archivability check for service orders by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service orders that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_SERORD ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_SERORD, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance, SRVO for a service order.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization the rule applies to.
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.

Condition	Description
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service order was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object CRM_SERORD assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BADIs for CRM_SERORD:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note [2939146](#) .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_SERORD, there is at least one active archive information structure that contains the following selectable field:
 - **Transaction ID** (object_id)

These fields are used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SRV0 that is provided in the standard system.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRM_SERORD archiving object, you must first check the settings in Customizing in **Service** under **Transactions**  **Data Archiving** .

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of service orders.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Orders Archived With CRM_SERORD

You have the following options for displaying archived service orders:

- From within **Archive Administration** (transaction SARA), you can search for archived service orders using the read program CRM_ARC_SERORD_READ.
- You can also use the **Archive Information System** (transaction SARI) to display archived service orders by entering the archiving object CRM_SERORD and the archive infostructure .

In both cases, the system provides a range of criteria that enable you to select the service orders that you want to display.

Archiving Service Orders in Hierarchical Settlements

It is possible to archive service orders even when one or more items are part of a financial hierarchy with a service contract and when hierarchical settlement is used. It is, however, required that all service contracts involved are already set to **Business Completed**.

If you revoke a completed service contract, you can also revoke any completed follow-up service orders associated with it, provided they are within the same financial hierarchy. The following applies when doing so:

- Lower levels in the hierarchy cannot have a status that precedes the status of the higher levels.
- If the service contract is archived, the associated service orders and PM orders must also be archived.
- If the service contract is **Business Completed**, the associated service orders and PM orders must be at least **Business Completed** as well. They cannot be reopened for business.

Variants for CRM_SERORD

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRM_SERORD.

Defining Write Variants

The write variant contains the parameters for the service orders that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

Here you can enter a short text that helps you identify and better find the archiving session in [Archive Management](#).

Defining Preprocessing Variants

The preprocessing program CRM_ARC_SERORD_CHECK_2 determines archivable service orders.

You can define variants for this program using the following parameters:

- Transaction Type

You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving In-House Services Using CRMS4_REPA

You can use archiving object CRMS4_REPA to archive in-house services.

Tables

CRMS4_REPA archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRMS4_REPA can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRMS4_REPA is delivered with the following programs:

Program	Name
Write	CRMS4_ARC_REPA_SAVE
Delete	CRMS4_ARC_REPA_DELETE
Preprocessing	CRMS4_ARC_REPA_CHECK

When archiving in-house services, you must use these programs or custom variants. For information about creating variants, see [Variants for CRMS4_REPA](#).

Prerequisites

The following prerequisites must be fulfilled before you can archive an in-house service:

- The in-house service has the status **Complete**.

- The residence time is fulfilled.

The following checks have been carried out:

- You've executed an archivability check for in-house services by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). You can use the program to run several check jobs in parallel, which is especially useful when archiving mass data. During the archivability check, in-house services that meet the archivability criteria, are assigned the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case, parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the ILM object CRMS4_REPA as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

You can select several of the following condition fields and put them into a specific order for the ILM object CRMS4_REPA. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type to which the rule applies, for instance, REPA for in-house service.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization to which the rule applies.
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, to which the rule applies.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you've selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the in-house service was last changed. The available time offsets are the end of the month, quarter, or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object CRMS4_REPA assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data for a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BADIs for CRMS4_REPA:

- [BAdI: Tables Used by an ILM Object](#) (BADI_DTINF_ILM_OBJ_TABLES)
- [BAdI: Archiving Information of Tables](#) (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note [2939146](#) .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity [Maintain Purposes](#) (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in the Customizing activity [Personalize Data Collection via Profiles](#) (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRMS4_REPA, there's at least one active archive information structure that contains the following selectable fields:
 - [Contact Person](#) (CONTACT_PERSON)
 - [Employee Responsible](#) (PERSON_RESP)
 - [Ship-To Party](#) (SHIP_TO_PARTY)
 - [Sold-To Party](#) (SOLD_TO_PARTY)

These fields are used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_REP1 that is provided in the standard system.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRMS4_REPA archiving object, you must first check the settings in Customizing for [Service](#) under [Transactions](#)  [Data Archiving](#) .

In the activity [Configure Archiving Preselection](#), you define parameters for the check and preprocessing programs. In the activity [Define Residence Times](#), you define the residence time for in-house services.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying In-House Services Archived With CRMS4_REPA

You have the following options for displaying archived in-house services:

- From within **Archive Administration** (transaction SARA), you can search for archived in-house services using the read program CRM_ARC_REPA_READ.
- You can also use the **Archive Information System** (transaction SARI) to display archived in-house services by entering the archiving object CRMS4_REPA and the archive infostructure SAP_CRMS4_REPA.

In both cases, the system provides a range of criteria that enable you to select the in-house-services that you want to display.

Archiving of Attachments

For more information about archiving attachments using the Document Management Service (DMS), see [Archiving Document Info Records Using CV_DVS](#).

Variants for CRMS4_REPA

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRMS4_REPA.

Defining Write Variants

The write variant contains the parameters for the in-house services that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type
You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the in-house services that you want to archive.
- Transaction ID
You can define ranges or individual IDs that you want to include or exclude in archiving runs.
- Changed on
You can define a date range. All archivable transactions that are edited during the time range you define, are included.
- Test mode
In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).
- Production mode
In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.
- Detail log
In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

Here you can enter a short text that helps you identify and better find the archiving session in [Archive Management](#).

Defining Preprocessing Variants

The preprocessing program CRMS4_ARC_REPA_CHECK determines archivable in-house services.

You can define variants for this program using the following parameters:

- Transaction Type

You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose [No success messages](#), no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Service Confirmations Using CRM_SRVCON

You can use archiving object CRM_SRVCON to archive service confirmations.

Tables

CRM_SRVCN archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose [Database Tables](#). You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_SRVCN can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_SRVCN is delivered with the following programs:

Program	Name
Write	CRM_ARC_SRVCN_SAVE
Delete	CRM_ARC_SRVCN_DELETE_2
Preprocessing	CRM_ARC_SRVCN_CHECK_2

When archiving service confirmations, you must use these programs or custom variants. For information on creating variants, see [Variants for CRM_SRVCN](#).

Prerequisites

The following prerequisites must be fulfilled before you can archive a service confirmation:

- The service confirmation has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- Any follow-up transactions are archivable.

The following checks have been carried out:

- You've executed an archivability check for service confirmations by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, service confirmations that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_SRVCN ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_SRVCN, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (OBJECT_TYPE)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance, SVC1 for a service confirmation.
Sales Organization (SALES_ORG_SD)	Use this field to define the sales organization the rule applies to.
Country/Region Key (BS_COUNTRY_OF_VKORG)	Use this field to define a country/region.
Organizational Management Object (BS_CRM_ORGUNIT)	Use this field to define the sales organization created in Service, the rule applies to.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the service confirmation was last changed. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object CRM_SRVCN assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BADIs for CRM_SRVCN:

- [BAdI: Tables Used by an ILM Object](#) (BADI_DTINF_ILM_OBJ_TABLES)
- [BAdI: Archiving Information of Tables](#) (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note [2939146](#) .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity [Maintain Purposes](#) (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity [Personalize Data Collection via Profiles](#) (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_SRVCN, there is at least one active archive information structure that contains the following selectable field:
 - [Transaction ID](#) (object_id)

This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_SRVC that is provided in the standard system.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRM_SRVCN archiving object, you must first check the settings in Customizing for [Service](#) under [Transactions](#) > [Data Archiving](#).

In the activity [Configure Archiving Preselection](#), you define parameters for the check and preprocessing programs. In the activity [Define Residence Times](#), you define the residence time of service confirmations.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Service Confirmations Archived With CRM_SRVCN

You have the following options for displaying archived service confirmations:

- From within [Archive Administration](#) (transaction SARA), you can search for archived service confirmations using the read program CRM_ARC_SERORD_READ
- You can also use the [Archive Information System](#) (transaction SARI) to display archived service confirmations by entering the archiving object CRM_SRVCN and the archive infostructure SAP_CRMS4_SRVC

In both cases, the system provides a range of criteria that enable you to select the service confirmations that you want to display.

Variants for CRM_SRVCN

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRM_SRVCN.

Defining Write Variants

The write variant contains the parameters for the service confirmations that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service confirmations that you want to archive.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

Here you can enter a short text that helps you identify and better find the archiving session in [Archive Management](#).

Defining Preprocessing Variants

The preprocessing program CRM_ARC_SRVCON_CHECK_2 determines archivable service confirmations.

You can define variants for this program using the following parameters:

- Transaction Type

You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Knowledge Articles Using CRM_KA

You can use archiving object CRM_KA to archive knowledge articles.

Tables

CRM_KA archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_KA can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

CRM_KA is delivered with the following programs:

Program	Name
Write	CRM_ARC_KA_SAVE
Delete	CRM_ARC_KA_DELETE_2
Preprocessing	CRM_ARC_KA_CHECK_2

When archiving knowledge articles, you must use these programs or custom variants. For information on creating variants, see [Variants for CRM_KA](#).

Prerequisites

The following prerequisites must be fulfilled before you can archive a knowledge article:

- The knowledge article has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.

The following checks have been carried out:

- You've executed an archivability check for knowledge articles by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, knowledge articles that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_KA ILM object as part of SAP Information Lifecycle Management. In transaction **IRMPOL**, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object CRM_KA, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Business Transaction Category (BUKRS)	Use this field to define the leading business transaction category.
Transaction Type (PROCESS_TYPE)	Use this field to define the transaction type the rule applies to, for instance KNAR for a knowledge article.
Country/Region Key (BS_CRM_COUNTRY_OF_ORGUNIT)	Use this field to define a country/region.

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The time reference is the date when the knowledge article was last changed. The available time offsets are the end of the month, quarter, or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object CRM_KA assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers implementations of the following BAdIs for CRM_KA:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note [2939146](#) .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity [Personalize Data Collection via Profiles](#) (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_KA, there is at least one active archive information structure that contains the following selectable field:
 - [Employee Responsible](#) (PERSON_RESP)

This field is used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_CRMS4_KNAR that is provided in the standard system.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRM_KA archiving object, you must first check the settings in Customizing for [Service](#) under [Transactions](#) ► [Data Archiving](#) ►.

In the activity [Configure Archiving Preselection](#), you define parameters for the check and preprocessing programs. In the activity [Define Residence Times](#), you define the residence time of knowledge articles.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Knowledge Articles Archived With CRM_KA

You have the following options for displaying archived knowledge articles:

- From within [Archive Administration](#) (transaction SARA), you can search for archived knowledge articles using the read program CRM_ARC_KA_READ
- You can also use the [Archive Information System](#) (transaction SARI) to display archived knowledge articles by entering the archiving object CRM_KA. You must create an archive infostructure for this purpose. For more information, see [Providing Archive Infostructures for Data Access](#)

Variants for CRM_KA

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for CRM_KA.

Defining Write Variants

The write variant contains the parameters for the activities that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

Here you can enter a short text that helps you identify and better find the archiving session in **Archive Management**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_KA_CHECK_2 determines archivable knowledge articles.

You can define variants for this program using the following parameters:

- Transaction Type

You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

Archiving Cases Using SCMG

You can use archiving object SCMG to archive cases.

Tables

SCMG archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose **Database Tables**. You can display the relevant tables in the lower part of the screen.

Archiving Classes

SCMG can trigger further archiving classes to write additional data, for example change documents, into the archive.

Programs

SCMG is delivered with the following programs:

Program	Name
Write	CASE_WRITE_PROGRAM
Delete	CASE_DELETE_PROGRAM
Preprocessing	CASE_PRE_PROGRAM

When archiving cases, you must use these programs or custom variants. For information on creating variants, see [Variants for SCMG](#).

Prerequisites

This is custom documentation. For more information, please visit [SAP Help Portal](#).

The following prerequisites must be fulfilled before you can archive a case:

- The case has status **Complete** or **Not Relevant**.
- The residence time is fulfilled.
- All follow-on transactions related to the case must be archivable.

The following checks have been carried out:

- You've executed an archivability check for cases by scheduling the cross-archiving-object check program using **Check/Delete** in **Archive Administration** (transaction SARA). Using the program you can run several check jobs in parallel, which is especially useful for archiving mass data. During the archivability check, cases that meet the archivability criteria, receive the status **Archivable**.
- Alternatively, you've scheduled an archivability check using the archiving-object-specific check program. You schedule this check in **Archive Administration** using the **Preprocessing** function. In this case parallel processing isn't possible, which makes this option less suitable for the processing of mass data. Only use it to archive smaller volumes of data or to test the archivability check.

ILM-Related Information for the Archiving Object

You can use this archiving object with the SCMG ILM object as part of SAP Information Lifecycle Management. In transaction **IRMPOL**, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

For ILM object SCMG, you can select several of the following condition fields and put them into a specific order. These fields define your rule structure:

Condition	Description
Case Type (CASE_TYPE)	Use this field to define the case type.
Area ID (AREA_ID)	Use this field to define the area the rule applies to, for instance case management.
RMS ID (RMS_ID)	Use this field to define the records management system ID. The RMS ID is a logical grouping of all objects that have been created for a specific business unit.
System Status (STATUS)	Use this field to define the case status the rule applies to, for instance, Completed .

Once you have selected the appropriate condition fields, you can create rules on the basis of these fields. You can define residence and retention periods for a specific number of days, months, or years. These periods have a time reference and a time offset. The available time references are the end date of the case as well as the month, quarter or year of the end date. The available time offsets are the end of the month, quarter or year. You can also choose not to define a time offset.

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object SCMG assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To

help you get started with the data modeling process, SAP delivers implementations of the following BADIs for SCMG:

- **BAdI: Tables Used by an ILM Object** (BADI_DTINF_ILM_OBJ_TABLES)
- **BAdI: Archiving Information of Tables** (BADI_DTINF_ARCHIVE_DETAILS)

For more information, see SAP Note [2939146](#) .

Before you can start a search for personal data related to this ILM object, the following prerequisite must be fulfilled:

- In the Customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP), assign the ILM object to a purpose for which the data was collected and processed.

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object SCMG, there is at least one active archive information structure that contains the following selectable fields:
 - **Case Key** (CASE_GUID)
 - **Customer** (FIN_KUNNR)

These fields are used by the IRF to determine which data should be fetched from the archive.

You can use the archive information structure SAP_FSCM that is provided in the standard system.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the SCMG archiving object, you must first check the settings in Customizing for **Service** under **Transactions**  **Data Archiving** .

In the activity **Configure Archiving Preselection**, you define parameters for the check and preprocessing programs. In the activity **Define Residence Times**, you define the residence time of cases.

Authorizations Objects

You need the following authorization objects:

Activity	Required Authorization Object
Write	S_ARCHIVE
Delete	S_ARCHIVE
Read	S_ARCHIVE
Preprocess	S_ARCHIVE

Displaying Cases Archived With SCMG

You have the following options for displaying archived cases:

- From within [Archive Administration](#) (transaction SARA), you can search for archived cases using the read program CASE_READ_PROGRAM
- You can also use the [Archive Information System](#) (transaction SARI) to display archived cases by entering the archiving object SCMG and archive infostructure SAP_CRM_CMG.

In both cases, the system provides a range of criteria that enable you to select the cases that you want to display.

Related Information

[BAdis for SCMG](#)

Variants for SCMG

When you schedule an archiving run, you must either use an existing variant or create a new one. In transaction SARA, you can create variants of the write and preprocessing programs available for SCMG.

Defining Write Variants

The write variant contains the parameters for the cases that you want to archive as well as settings for the archiving run.

SAP delivers the following parameters:

- Transaction type

You can define which types of transaction you want to archive. It's possible to select multiple values. A write variant contains the parameters for the service orders that you want to archive.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Changed on

You can define a date range. All archivable transactions that are edited during the time range you define, are included.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

In this field, you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

- Archiving Session Note

Here you can enter a short text that helps you identify and better find the archiving session in [Archive Management](#).

Defining Preprocessing Variants

The preprocessing program CASE_PRE_PROGRAM determines archivable cases.

You can define variants for this program using the following parameters:

- Transaction Type

You can define which types of transaction you want to archive. It's possible to select multiple values.

- Transaction ID

You can define ranges or individual IDs that you want to include or exclude in archiving runs.

- Test mode

In the test mode, the program is only simulated, meaning that no actual changes take place on the database (for example, no data is written or deleted from the database).

- Production mode

In this mode, the data is physically processed, according to your selection criteria and if applicable, additional archiving-object-specific criteria. For example, during the delete phase data is removed from the database by the delete program.

- Detail log

Using this field you can decide whether a detail log is to be generated in addition to the summarized log during the execution of the program. The summarized log contains each message only once. For each message, the number of affected objects is listed as well as an example of an object. The detail log contains all processed objects including the corresponding messages. If you choose **No success messages**, no success messages are written to the detail log.

i Note

Only generate detail logs if you're processing a small number of objects, or, for example, in the test mode. Otherwise, the program can terminate due to a memory overflow.

- Log Output

Here you can determine, whether the log is output into the list (spool for background programs), to the application log, or to both.

BAdis for SCMG

You can enhance the archiving functions provided for cases using Business Add-Ins (BAdi).

Business Add-In	Use
ARC_SCMG_CHECK	You can use this BAdi to define custom checks that must be met before a case can be archived.

Business Add-In	Use
ARC_SCMG_WRITE	You can use this BAdi to write additional data to the archive that isn't directly associated with the case.
ARC_SCMG_DELETE	You can use this BAdi to delete additional data that isn't directly associated with the case.

For more information on the BAdis, see the system help in Customizing under [Service](#) > [Case Management](#) > [Extended Customizing](#) > [Case Archiving](#) > [BAdis for Case Archiving](#).

You can access the BAdIs using transaction SE18.

Archiving of Subscription Product-Specific Data Using SOM_PROD

You can use archiving object SOM_PROD to archive subscription product-specific data.

Tables

SOM_PROD archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose [Database Tables](#). You can display the relevant tables in the lower part of the screen.

Archiving Classes

SOM_PROD may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose [Archiving Classes Used](#).

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before subscription product-specific data can be archived:

- The related product master is marked for deletion.
- No drafts exist for the subscription-product specific data.
- The product is not used in any subscription order/contract.

ILM-Related Information for the Archiving Object

You can use this archiving object with the SOM_PROD ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- PRODUCT: Product number

The following time references are available:

- CREATION DATE: Date when the product was created
- LAST_CHANGE_DATE: Date when the product was last changed

For more information, search for SAP Information Lifecycle Management in the relevant version of [SAP S/4HANA](#) on the SAP Help Portal.

Performing Application-Specific Configuration

There is no specific Customizing for the SOM_PROD archiving object.

Defining Preprocessing Variants

There is no preprocessing program for SOM_PROD.

You can define variants for this program using the following parameters:

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

This program analyzes the subscription product-specific data for archiving and then writes the data to an archive file. The program selects all subscription product-specific data that fits in with the selected products. You can execute the report in either test or production mode. Data, however, is written to the archive file in production mode only.

In the [Comment for Archive Run](#) field, you can enter your own comment for the archive run.

You can archive subscription product-specific data only if the following prerequisites are met:

- The related product master is marked for deletion.
- No draft exists for the subscription product-specific data.
- The product is not used in any subscription order or subscription contract.

A write variant contains the parameters for the subscription product-specific data that you want to archive.

SAP delivers the following parameters:

- Product

This is the product number for the related product.

The write program is CRMS4_SOM_PROD_ARCHIVE_WRITE.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The delete program CRMS4_SOM_PROD_ARCHIVE_DELETE for SOM_PROD deletes the subscription product-specific data based on an existing archive. The program first reads the subscription product-specific data belonging to an archiving run from the archive and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program CRMS4_SOM_PROD_ARCHIVE_READ displays the archived subscription product-specific data. This data must be archived and deleted. The selected product IDs are displayed using ALV. You can navigate to the details of each product by double clicking on the relevant record.

You can select the archived data by using the following selection parameters:

- Subscription product-specific data UUID
- Product ID
- Key of archive file

Defining Postprocessing Variants

There is no postprocessing program for SOM_PROD.

Defining Reloading Variants

There is no reloading program from SOM_PROD.

Dependencies of SOM_PROD

Subscription orders and subscription contracts must be archived before you can archive subscription-product specific data.

The related material master (archiving object MM_MATNR) can only be archived after the subscription-product specific data is archived.

Authorizations for SOM_PROD

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	SOM_PROD
ACTVT: 01 - Add or create ARCH_OBJ: SOM_PROD APPLIC: IS	S_ARCHIVE

Enhancements for SOM_PROD

The SOM_PROD archiving object offers the following Business Add-Ins (BADIs):

BAdI / User Exit	Description	Used to
ARC_SOM_PROD_CHECK	Archiving subscription product-specific data: Check archivability	Check if an object can be archived

BAdI / User Exit	Description	Used to
ARC_SOM_PROD_WRITE	Archiving subscription product-specific data: Write additional table	Write additional tables for archiving

Displaying Subscription Product-Specific Data Archived with SOM_PROD

You can use program CRMS4_SOM_PROD_ARCHIVE_READ to display archived and deleted subscription product-specific data.

Archiving of Allowance Definition Groups Using SOM_ADG

You can use archiving object SOM_ADG to archive allowance definition groups.

Tables

SOM_ADG archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose [Database Tables](#). You can display the relevant tables in the lower part of the screen.

Archiving Classes

SOM_ADG may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose [Archiving Classes Used](#).

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before an allowance definition group can be archived:

- Status of the allowance definition group must NOT be **In Preparation**.
As long as the allowance definition group has this status, it can be deleted directly.
- No draft exists for the allowance definition group.
- The allowance definition group, to be deleted, is not used in any subscription product-specific data.

ILM-Related Information for the Archiving Object

You can use this archiving object with the SOM_ADG ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- GROUP_ID: Allowance Definition Group ID

The following time references are available:

- CREATION_DATE: Date when the allowance group was created.

- **LAST_CHANGE_DATE:** Date when the allowance group was last changed.

For more information, see [SAP Information Lifecycle Management](#).

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see [Setting Legal Holds](#).

Performing Application-Specific Configuration

There is no specific Customizing for the SOM_ADG object.

Defining Preprocessing Variants

There is no preprocessing program for SOM_ADG.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

A write variant contains the parameters for the allowance definition group that you want to archive. This program covers the following features:

- The analysis of allowance definition groups with respect to archivability.
- Writing allowance definition groups to an archive file based on the analysis in the previous step.

The following database tables are included:

- CRMS4_SOM_ADG
- CRMS4_SOM_ADGT
- CRMS4_SOM_AD

Additionally, change documents related to the archived allowance definition group are also archived.

SAP delivers the following parameters:

- Allowance Definition Group ID

The write program is CRMS4_SOM_ADG_ARCHIVE_WRITE.

Defining Delete Variants

The program CRMS4_SOM_ADG_ARCHIVE_DELETE is used to delete the allowance definition groups based on an existing archive.

The program first reads the allowance definition groups belonging to an archiving run from the archive, and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program CRMS4_SOM_ADG_ARCHIVE_READ displays the selected allowance definition groups using ALV. You can navigate to the details of each allowance definition group by double clicking on the relevant record.

You can select the archived data by using the following selection parameters:

- Allowance Definition Group UUID
- Allowance Definition Group ID
- Key for Archive File

Defining Postprocessing Variants

There is no postprocessing program for SOM_ADG.

Defining Reloading Variants

There is no reloading program for SOM_ADG.

Dependencies of SOM_ADG

Authorizations for SOM_ADG

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	CRM_ISX_AD
ACTVT: 01 ARCH_OBJ: SOM_ADG APPLIC: IS	S_ARCHIVE

Enhancements for SOM_ADG

The SOM_ADG archiving object offers the following Business Add-Ins (BAdIs):

BAdI / User Exit	Description	Used to
ARC_SOM_ADG_CHECK	Archiving Allowance Definition Group: Check Archivability	Check if an object can be archived.
ARC_SOM_ADG_WRITE	Archiving Allowance Definition Group: Write Additional Table	Write additional tables for archiving

Displaying Allowance Definition Groups Archived with SOM_ADG

You can use the program CRMS4_SOM_ADG_ARCHIVE_READ to display archived and deleted allowance definition groups.

Archiving Subscription Orders Using CRM_PRV0

You can use archiving object CRM_PRV0 to archive subscription orders (BUS2000265) in transaction SARA.

Tables

CRM_PRV0 archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose [Database Tables](#). You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_PRV0 may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose [Archiving Classes Used](#).

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a subscription order can be archived:

- The subscription order status must be set to Completed at the header level.
- The residence time (number of days after the last change to the business transaction) must have elapsed.
- Subsequent documents (such as tasks and complaints) must already be archived and deleted from the system.
- If you have set up ILM integration for the CRM_PRV0 ILM object, your ILM rules must also allow archiving.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_PRV0 ILM object as part of SAP Information Lifecycle Management. In the transaction IRMPOL, you can create policies for residence time and retention rules, depending on the available policy category. Here, you can also see the available time reference and the existing condition fields. You can decide which of these will be used to define your rule structure. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- OBJECT_TYPE - Business Transaction Category
- PROCESS_TYPE - Business Transaction Type
- BS_CRM_COUNTRY_OF_ORGUNIT - Country of Org Unit

The following time references are available:

- LAST_CHANGE_DATE - Last Changed On

For more information, see [SAP Information Lifecycle Management](#).

Enablement for Information Retrieval

The ILM object CRM_PRV0 assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for CRM_PRV0:

- BAdI Implementation BADI_CRM_SUBSCR_ILM_OBJ_TABLES of BAdI: Tables Used by an ILM Object (BADI_DTINF_ILM_OBJ_TABLES)
- BAdI Implementation SOM_CRM_PRVO_ARCH_DETAILS of BAdI: Archiving Information of Tables (BADI_DTINF_ARCHIVE_DETAILS)

This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note [2939146](#) (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_PRVO, there is at least one active archive information structure that contains the following selectable fields:
 - Field name (SOLD_TO_PARTY)

These fields are used by the IRF to determine which data should be fetched from the archive.

- Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection.
- You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRM_PRVO archiving object, you must first check the settings in Customizing under **SAP Customizing Implementation Guide > Service > Transactions > Data Archiving**.

Defining Preprocessing Variants

The preprocessing program CRM_ARC_PRVO_CHECK_2 for CRM_PRVO checks the prerequisites discussed above so that subscription orders can be archived.

You can define variants for this program using the following parameters:

- Transaction Type

Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction **SARA**.

The write program **CRM_ARC_PRVO_SAVE** for **CRM_PRVO** writes the data to the archive file. The program contains the parameters for the subscription orders that you want to archive.

SAP delivers the following parameters:

- Transaction Type

Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

- Changed On

Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction **SARA**.

The delete program **CRM_ARC_PRVO_DELETE** for **CRM_PRVO** deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

SAP delivers the following parameters:

- Transaction Type

Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Changed On

Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Read Program Variants

The read program **CRM_ARC_PRVO_READ** for **CRM_PRVO** reads data from the archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction **SARI**, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

i Note

The infostructure must be activated before the data is archived.

While SAP delivers the infostructure **SAP_CRMS4_PRVO** and field catalog **SAP_CRMS4_PRVO**, you can maintain your own infostructure and field catalogs.

Archiving Subscription Contracts Using CRM_PRVC

You can use archiving object CRM_PRVC to archive subscription contracts (BUS2000266) in transaction SARA.

Tables

CRM_PRVC archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose [Database Tables](#). You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_PRVC may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose [Archiving Classes Used](#).

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a subscription contract can be archived:

- The subscription contract statuses must be set to Completed and Finished at the header level
- The residence time (number of days after the last change to the business transaction) must have elapsed.
- All linked subscription orders must be archived and deleted from the system.
- The FI-CA provider contracts that correspond to the subscription contracts must be archived and deleted from the system. For more information about archiving a FI-CA provider contract, see Archiving of Provider Contracts (FI_MKKCAVT) on the [SAP Help Portal](#).
- Subsequent documents (such as tasks and complaints) must already be archived and deleted from the system.
- Sharing Contracts can only be archived when no more linked Shared Contracts exist, i. e. Shared Contracts have to be archived and deleted before the Sharing Contracts they refer to.
- If you have set up ILM integration for the CRM_PRVC ILM object, your ILM rules must also allow archiving.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM-PRVC ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- OBJECT_TYPE - Business Transaction Category
- PROCESS_TYPE - Business Transaction Type
- BS_CRM_COUNTRY_OF_ORGUNIT - Country of Org Unit

The following time references are available:

- LAST_CHANGE_DATE - Last Changed On

For more information, see [SAP Information Lifecycle Management](#).

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see [Setting Legal Holds](#).

Enablement for Information Retrieval

The ILM object CRM_PRVC assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for CRM_PRVC:

- BAdI Implementation BADI_CRM_SUBSCR_ILM_OBJ_TABLES of BAdI: Tables Used by an ILM Object (BADI_DTINF_ILM_OBJ_TABLES)
- BAdI Implementation SOM_CRM_PRVC_ARCH_DETAILS of BAdI: Archiving Information of Tables (BADI_DTINF_ARCHIVE_DETAILS)

This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note [2939146](#)  (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_PRVC, there is at least one active archive information structure that contains the following selectable fields:
 - Field name (SOLD_TO_PARTY)

These fields are used by the IRF to determine which data should be fetched from the archive.

- Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection.
- You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRM-PRVC archiving object, you must first check the settings in Customizing under **SAP Customizing Implementation Guide > Service > Transactions > Data Archiving** .

Defining Preprocessing Variants

The preprocessing program CRM_ARC_PRVC_CHECK_2 for CRM_PRVC checks the prerequisites discussed above so that subscription contracts can be archived.

You can define variants for this program using the following parameters:

- Transaction Type

Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The write program CRM_ARC_PRVC_SAVE FOR CRM_PRVC writes the data to the archive file. The program contains the parameters for the subscription contracts that you want to archive.

SAP delivers the following parameters:

- Transaction Type

Here, you can restrict the transaction types of the business transactions to be analyzed for archivability.

- Transaction ID

Here, you can restrict the transaction IDs of the business transactions to be analyzed for archivability.

Changed On

Here, you can restrict the 'changed on' data of the business transactions to be archived.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction SARA.

The delete program CRM_ARC_PVC_DELETE for CRM_PRVC deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

Defining Read Program Variants

The read program CRM_ARC_PRVC_READ for CRM_PRVC reads data from archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction SARI, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

i Note

The infostructure must be activated before the data is archived.

While SAP delivers the infostructure SAP_CRMS4_PRVC and field catalog SAP_CRMS4_PRVC, you can maintain your own infostructure and field catalogs.

Archiving Master Agreements Using CRM_PRVMA

You can use archiving object CRM_PRVMA to archive Master Agreements (BUS2000267) transaction SARA.

Tables

CRM_PRVMA archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose [Database Tables](#). You can display the relevant tables in the lower part of the screen.

Archiving Classes

CRM_PRVMA may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose [Archiving Classes Used](#).

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a master agreement can be archived:

- The header status must be set to Completed.
- The residence time (number of days after the last change to the business transaction) must have elapsed.
- All subscription orders, subscription contracts, and solution quotations that refer to the master agreement must be archived.

ILM-Related Information for the Archiving Object

You can use this archiving object with the CRM_PRVMA ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- PROCESS_TYPE - Business Transaction Type

The following time references are available:

- LAST_CHANGE_DATE - Last Changed On
- CONTRACT_END_DATE - Contract End Date

For more information, see [SAP Information Lifecycle Management](#).

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see [Setting Legal Holds](#).

Enablement for Information Retrieval

The ILM object CRM_PRVMA assigned to this archiving object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers the following implementations for CRM_PRVC:

- BAdI Implementation BADI_CRM_MA_ILM_OBJ_TABLES of BAdI: Tables Used by an ILM Object (BADI_DTINF_ILM_OBJ_TABLES)
- BAdI Implementation SOM_CRM_PRMA_ARCH_DETAILS of BAdI: Archiving Information of Tables (BADI_DTINF_ARCHIVE_DETAILS)

This BAdI is used to retrieve archived information for the IRF.

For more information, see SAP Note [2939146](#)  (S/4HANA).

Before you can start a search for personal data related to this ILM object, the following prerequisites must be fulfilled:

- Assign the ILM object to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

If you also want to enable the retrieval of personal data stored in archive files, perform the following additional steps:

- Create a profile for the data collection in Customizing activity **Personalize Data Collection via Profiles** (transaction DTINF_MAINT_PROFILE) defining that archived personal data must be included in the search.
- Ensure that for archiving object CRM_PRVMA, there is at least one active archive information structure that contains the following selectable fields:
 - Field name (SOLD_TO_PARTY)

These fields are used by the IRF to determine which data should be fetched from the archive.

- Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection.
- You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Performing Application-Specific Configuration

Before you can use the CRM_PRVMA archiving object, you must first check the settings in Customizing under **SAP Customizing Implementation Guide** > **Service** > **Transactions** > **Data Archiving** .

Defining Preprocessing Variants

The preprocessing program CRM_ARC_PRVMA_CHECK for CRM_PRVMA will check the prerequisites discussed above so that BUS2000267 can be archived.

You can define variants for this program using the following parameters:

- Transaction Type
Transaction type/Process type of a document to be archived.
- Transaction ID
Object ID of a document to be archived.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction **SARA**.

A write variant contains the parameters for the <business objects> that you want to archive.

SAP delivers the following parameters:

The write program CRM_ARC_PRVMA for CRM_PRVMA writes the data to the archive file. The program contains the parameters for the master agreements that you want to archive.

SAP delivers the following parameters:

- Transaction Type
Transaction type/Process type of a document to be archived.
- Transaction ID
Object ID of a document to be archived.
- Changed On
Last changes made to the document.

Defining Delete Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction **SARA**.

The delete program CRM_ARC_PRVMA_DELETE_2 for CRM_PRVMA deletes the data from a write program successfully executed from the database tables. You must, therefore, select only a completed archiving session. Only then will the program delete the data already written to the archive.

Defining Read Program Variants

The read program CRM_ARC_PRVMA_READ for CRM_PRVMA reads data from the archive files and makes them available for a temporary display in the SAP S/4HANA system.

The read program only works if you have activated an infostructure that belongs to the archive object in the transaction **SARI**, step 'Customizing'. Here, you can select the parameters that you want to use for searching data in archive files.

i Note

The infostructure must be activated before the data is archived.

While SAP delivers the infostructure SAP_CRMS4_PRVMA and field catalog SAP_CRMS4_PRVMA, you can maintain your own infostructure and field catalogs.

Archiving Sharing Group Using SOM_SHGR

You can use archiving object **SOM_SHGR** to archive sharing groups.

Tables

SOM_SHGR archives data from several tables. To check which tables these are, call up transaction SARA, enter the archiving object, and choose [Database Tables](#). You can display the relevant tables in the lower part of the screen.

Archiving Classes

SOM_SHGR may trigger further archiving classes to write additional data, for example change documents, into the archive. To check which classes these are, call up transaction AOBJ, select your archiving object and choose [Archiving Classes Used](#).

Programs

To find out which programs this archiving object offers, call up transaction SARA enter the archiving object and press ENTER.

Prerequisites

The following prerequisites must be fulfilled before a sharing groups can be archived:

- No draft exists for the sharing group.
- The sharing group is not used in a subscription order, subscription contract, or a master agreement.

ILM-Related Information for the Archiving Object

You can use this archiving object with the SOM_SHGR ILM object as part of SAP Information Lifecycle Management. In transaction IRMPOL, you can create policies for residence or retention rules, depending on the available policy category. Here you can also see the available time references and which condition fields exist, and decide which of them shall be used in which order to define your rule structure.

The following condition fields are available:

- GROUP_ID: Sharing group ID

The following time references are available:

- CREATION_DATE: Date on which the sharing group was created
- LAST_CHANGE_DATE: Date on which the sharing group was last changed.

For more information, see [SAP Information Lifecycle Management](#).

Legal Holds

It is possible to set a legal hold on the data this archiving object archives/destroys. For more information, see [Setting Legal Holds](#).

Performing Application-Specific Configuration

There is no specific Customizing for the SOM_SHGR archiving object.

Defining Preprocessing Variants

There is no preprocessing program for SOM_SHGR.

Defining Write Variants

When you schedule the archiving run, you must enter an existing variant or create a new one. You can do so in transaction **SARA**.

A write variant contains the parameters for the sharing object that you want to archive. The program **CRMS4_SOM_SHGR_ARCHIVE_WRITE** is used to write the archive.

This report carries out the first step in archiving sharing groups and covers the following features:

- Analyzing sharing groups with respect to archivability
- Writing sharing groups to an archive file based on the results of the analysis

The following database tables are included:

- **CRMS4_SOM_SRG**
- **CRMS4_SOM_SRGT**

Additionally, change documents related to the archived sharing group are also archived.

SAP delivers the following parameters:

- Sharing Group ID

The write program is **CRMS4_SOM_SHGR_ARCHIVE_WRITE**

Defining Delete Variants

The program **CRMS4_SOM_SHGR_ARCHIVE_DELETE** is used to delete sharing groups based on an existing archive, and is the second step of the archiving process.

The report first reads the sharing groups belonging to an archiving run from the archive and then deletes the appropriate entries from the database tables.

Defining Read Program Variants

The read program **CRMS4_SOM_SHGR_ARCHIVE_READ** is used to display the archived and deleted sharing groups.

You must ensure the following prerequisites are fulfilled before executing the read program:

- Data is archived and deleted

You can select the archived data by using the following selection parameters:

- Sharing Group UUID
- Sharing Group ID
- Key for Archive File

The selected sharing groups are displayed using ALV. You can navigate to the details of each sharing group by double clicking on the relevant record.

Defining Postprocessing Variants

There is no postprocessing program for **SOM_SHGR**.

Defining Reloading Variants

There is no reloading program for SOM_SHGR.

Dependencies of SOM_SHGR

Subscription orders, subscription contracts, and master agreements using a specific sharing group ID must be archived before you can archive the sharing group.

Authorizations for SOM_SHGR

You need the following authorization objects:

Activity	Required Authorization Object
ACTVT: 24 - Archive	SOM_SHGR
ACTVT: 01 ARCH_OBJ: SOM_SHGR APPLIC: IS	S_ARCHIVE

Enhancements for SOM_SHGR

The SOM_SHGR archiving object offers the following Business Add-Ins (BAIs):

BAI / User Exit	Description	Used to
ARC_SOM_SHGR_CHECK	Archiving Sharing Group: Check Archivability	Check if an object can be archived
ARC_SOM_SHGR_WRITE	Archiving Sharing Group: Write Additional Table	Write additional tables for archiving

Displaying Sharing Groups Archived with SOM_SHGR

You can use the program CRMS4_SOM_SHGR_ARCHIVE_READ to display the archived and deleted sharing groups.

Blocking, Unblocking, and Deletion of Personal Data in Service

You can use SAP Information Lifecycle Management (SAP ILM) to control the blocking, unblocking, and deletion of business partner, customer, and supplier master data. According to the business partner approach, blocking and deletion of customer and supplier master data is part of the blocking and deletion of the business partner master data. This means that the business partner considers all residence periods of linked customer or supplier master data. Personal data collected in master data can be blocked as soon as the business activities for which this data was collected have been completed and the residence time for this data has elapsed.

After blocking, only users who are assigned additional authorizations can access this data.

After the retention period for data has expired, personal data can be deleted completely so that it can no longer be retrieved. In this case, customer and supplier master data must be archived and deleted from the database before the associated business partner

master data to prevent data inconsistencies.

You define residence and retention period policies for master data in ILM policies (transaction IRMPOL).

The following scenarios for blocking business partners are supported:

1. You want to block business partner master data for which the associated customer and supplier data is blocked. The business partner master data satisfies the EoP check criteria and therefore can be blocked.
2. You want to block business partner data for which the associated customer and supplier master data is not blocked. The associated customer and supplier master data satisfies their EoP checks as does the business partner master data. All of the associated customer and supplier master data as well as the business partner master data can be blocked.
3. You want to block business partner master data for which the associated customer and supplier master data is not blocked and where at least one customer or supplier does not meet the EoP check criteria. The business partner master data cannot be blocked until all of the associated customer and supplier master data meet the EoP check criteria and can also be blocked.
4. Independent blocking of the customer and supplier master data is possible. The corresponding roles of the business partner are also blocked.

After blocking, only users who are assigned additional authorizations can access this data.

After the retention period for data has expired, personal data can be deleted completely so that it can no longer be retrieved. In this case, associated customer and supplier master data must be archived and deleted from the database before the associated business partner master data to prevent data inconsistencies.

You define residence and retention period policies for master data in ILM policies (transaction IRMPOL).

Specifics Business Partner Master Data

The functions required for blocking and unblocking or viewing blocked business partner master data are provided through the data protection specialist roles:

- SAP_CA_BP_DP_DISPLAY
- SAP_CA_BP_DP_BLOCK
- SAP_CA_BP_DP_UNBLOCK
- SAP_CA_BP_DP_BLOCK_REQUEST

Specifics for Customer and Supplier Master Data

The functions required for blocking and unblocking or viewing blocked customer and supplier master data are provided through the data protection specialist roles:

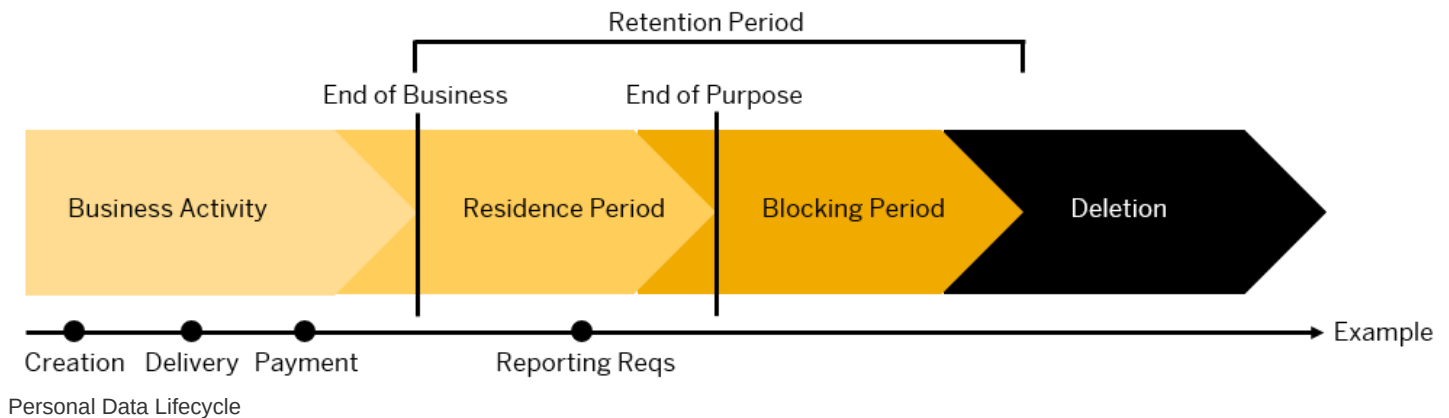
- SAP_LO_MD_CVP_DISPLAY
- SAP_LO_MD_CVP_BLOCK
- SAP_LO_MD_CVP_UNBLOCK

You can copy these template roles to custom roles, adjust them to meet your requirements, and assign them to your users.

Business Partner End of Purpose (EoP) Check in Service

An end of purpose (EoP) check determines whether data is still relevant for business activities based on the retention period defined for the data. The retention period is part of the overall lifecycle of personal data, which consists of the following phases:

- **Business activity:** The relevant data is used in ongoing business, for example contract creation, delivery or payment.
- **Residence period:** The relevant data remains in the database and can be used in subsequent processes related to the original purpose, for example reporting obligations.
- **Blocking period:** The relevant data must be retained for legal reasons. During the blocking period, business users of SAP applications are prevented from displaying and using the data. However, it can be processed, if so required due to mandatory legal provisions.
- **Deletion:** The data is deleted and no longer exists in the database.



Blocking of data impacts system behavior in the following ways:

- **Display:** Only users with specific auditing authorizations can display blocked data.
- **Change:** It is not possible to change a business object that contains blocked data.
- **Create:** It is not possible to create a business object that contains blocked data.
- **Copy or follow-up:** It is not possible to copy a business object or perform follow-up activities for a business object that contains blocked data.
- **Search:** Only user with specific auditing authorizations can search for blocked data or search for a business object using blocked data.

Customizing for the End of Purpose Check

Activity	Customizing Path
Define the settings for authorization management in Customizing for Cross-Application Components under:	► Data Protection ► Authorization Management ►
Configure the settings related to the blocking and deletion of business partner master data in Customizing for Cross-Application Components under:	► Data Protection ► Blocking and Unblocking of Data ► Business Partner ►
Configure the settings related to the blocking and deletion of customer and supplier master data in Customizing for Cross-Application Components under:	► Data Protection ► Blocking and Unblocking of Data ► Customer Master/Supplier Master Deletion ►
Define whether you want to use no blocking or dual control for blocking in Customizing for Cross-Application Components under:	► Data Protection ► Blocking and Unblocking of Data ► Business Partner ► Set Activation Status of No Blocking/Dual Control for Role Group ►

Activity	Customizing Path
Define whether you want to use dual control for unblocking in Customizing for Cross-Application Components under:	▮▮ Data Protection ▮ Blocking and Unblocking of Data ▮ Business Partner ▮ Define Reasons for Unblocking Business Partner ▮
	▮▮ Data Protection ▮ Blocking and Unblocking of Data ▮ Business Partner ▮ Define Registered Function Modules for Unblock BP ▮
	▮▮ Data Protection ▮ Blocking and Unblocking of Data ▮ Business Partner ▮ Define Dual Control Setting for Unblocking ▮
	▮▮ Data Protection ▮ Blocking and Unblocking of Data ▮ Business Partner ▮ Set Status for Automatic Unblocking ▮
Configure the settings for transactional data specific to service solution in the Customizing for Service under:	▮▮ Cross-Application Components ▮ Data Protection ▮

For more information about the possible Customizing settings, refer to the system help of the respective activity.

Process Flow

Activity	Additional Information
Enable the end of purpose check function for the central business partner by running transaction BUPA_PRE_EOP	System help for the Blocking of Business Partner Data report
Enable the end of purpose check function for the customer and supplier master data by running transaction CVP_PRE_EOP	System help for the Block Customer and Supplier Master Data report
Request the unblocking of blocked data	Request Unblocking of Master Data
Unblock master data	Unblocking Master Data
Delete master data	Deletion of Master Data
Display application logs for the business partner, customer, and supplier	Logging

ILM-enabled Objects in Service

Service uses the central business partner CA_BUPA.

Authorizations

Users who have a role based on the SAP_CA_BP_DP_DISPLAY or SAP_LO_MD_CVP_DISPLAY template roles, can display blocked business partner master data; however, they cannot create, change, copy, or perform follow-up activities on blocked data.

Users who have a role based on the SAP_LO_MD_CVP_DISPLAY template role with the B_BUP_PCPT authorization object including the assignment of activity 03, DISPLAY can display blocked customer and supplier master data; however, they cannot create, change, copy, or perform follow-up activities on blocked data.

Business Partner Master Data in Multisystem Landscapes

In a multisystem landscape, the EoP for business partner master data is checked for an application in the local system as well as the remote systems. The EoP check is used to ensure data integrity in case of potential blocking. The EoP determines whether there are any data dependencies, that is, whether the data is still required for business activities. If there is such a dependency, the

system does not block the data. If you still want to block the data, the dependent data must be blocked or deleted using archiving and deletion tools. If the EoP criteria are met, the business partner master data is blocked in the local and remote systems.

Blocking Master Data

You can use the **Blocking of Business Partner Data** report (transaction BUPA_PRE_EOP) to block data related to a business partner including associated customer and supplier master data or the **Block Customer and Supplier Master Data** report (transaction CVP_PRE_EOP) to block data related to customer and supplier master data. These reports determine whether data can be blocked using the End of Purpose (EoP) check. If the data is still in use and business activities have not been completed, the system does not block the data. If business is complete and the residence period has expired, the reports block the business partner, customer or supplier master data as defined in the report parameters. To block business partner master data, any associated customer or supplier master data must first be blocked in your system.

Business Partner Master Data

Business partner master data is blocked completely in one step.

Customer and Supplier Master Data

Data is blocked by the system in the following order:

1. Block contact persons and/or the company codes
2. Block general data

You can perform both steps at the same time. However, performing the first step separately allows the contact person and/or company code data to be blocked and the residence time of the general data to be reset. The general data can then be blocked when the revised residence time is met.

More Information

- Report documentation for the **Blocking of Business Partner Data** report (transaction BUPA_PRE_EOP)
- System help for the **Block Customer and Supplier Master Data** report (transaction CVP_PRE_EOP)

Request Unblocking of Master Data

You can request the unblocking of business partner, customer or supplier master data using the **Request Unblocking** report (transaction BUP_REQ_UNBLK). You may do this because new business starts with a historic business partner, customer or supplier and their master data has not yet been deleted.

More Information

System help for the **Request Unblocking** report (transaction BUP_REQ_UNBLK).

Unblocking Master Data

You can use the **Request Unblocking** report (transaction BUP_REQ_UNBLK) to unblock business partner data. If the dual control setting for unblocking data has not been set, you can unblock data from your own unblocking request. If your system has dual

control for unblocking set-up, another user must process the request to unblock data.

You can unblock customer and supplier master data with the [Unblock Customer and Supplier Master Data](#) report (transaction CVP_UNBLOCK_MD).

Blocked data can only be unblocked, if the master data has not been archived and it is still within the retention period.

Unblocking is performed in the opposite order of the blocking process. That means that business partner master data is unblocked in one step.

Blocked customer or supplier master data that extends across several company codes and has several contact persons, is unblocked as follows:

1. Unblock the general data
2. Unblock the contact persons and the company codes

More Information

- Report documentation for the [Blocking of Business Partner Data](#) report (transaction BUPA_PRE_EOP)
- System help for the [Request Unblocking](#) report (transaction BUP_REQ_UNBLK)
- System help for the [Unblock Customer and Supplier Master Data](#) report (CVP_UNBLOCK_MD)
- For more information on unblocking, see [Request Unblocking of Master Data](#)
- For more information on dual control and unblocking, see the product assistance on SAP Help Portal and search for *Unblocking*.

Deletion of Master Data

After the residence period has been completed and data is blocked, it either remains in the database or is stored in an external archive. If the blocked data remains in the database, the data manager can delete the blocked business partner, customer, and supplier master data from the database using the corresponding archiving objects. The business partner master data can be deleted once all associated customer and supplier master data has been deleted from the database.

To destroy data from the database or archives, you can use the [Data Destruction](#) report (transaction ILM_DESTRUCTION). Subsequently, the data can no longer be retrieved.

Logging

You can use the [Analyze Application Log](#) report (transaction SLG1) to view the centralized business partner log. To do so, use the BDT_DATAARCHIVING object with the BUPA_EOP subobject.

You can use the [Display Stored Application Logs for Customer and Supplier](#) report (transaction CVP_DISPLAY_LOG) to display the application log for one or various subobjects including:

Subobject	Subobject Description
CUSTOMER_BLOCKING	Customer blocking
CUSTOMER_UNBLOCKING	Customer unblocking

Subobject	Subobject Description
SUPPLIER_BLOCKING	Supplier blocking
SUPPLIER_UNBLOCKING	Supplier unblocking

This report provides more contextual details at the level of the log items compared to the [Analyze Application Log](#) report.

More Information

- System help for the [Display Stored Application Logs for Customer and Supplier](#) report (transaction CVP_DISPLAY_LOG)
- System help for the [Analyze Application Log](#) report (transaction SLG1)

Blocking and Deleting Personal Data in Subscription Order Management

The processes and functions of Subscription Order Management use personal data of the SAP business partner. If, from the point of view of these processes and functions, there is no longer any reason to store the data (for example, for a possible tax audit), you can block this data from the perspective of Subscription Order Management and ultimately destroy it.

For destroying personal data, use the functions of Information Lifecycle Management (ILM). Using the ILM functions, you first block the personal data of business partners for whom there is no further business. Then you delete this data as soon as it is no longer needed.

All archiving objects in Subscription Order Management support data protection notifications (see [Deletion of Personal Data in SAP BW/4HANA and SAP BW](#)).

Related Information

[Deletion of Personal Data in SAP BW/4HANA and SAP BW Data Protection](#)

ILM Objects in Subscription Order Management

Use

In Subscription Order Management, the following archiving objects are enabled for use with SAP Information Lifecycle Management (SAP ILM):

ILM-Enabled Archiving Objects			
Technical Name	Description	Available Fields of Retention Period	Available Start Times
SOM_PROD	Subscription product-specific data	PRODUCT - Material ID	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On

Technical Name	Description	Available Fields of Retention Period	Available Start Times
SOM_ADG	Allowance definition group	GROUP_ID - Allowance definition group ID	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On
CRM_PRVO	Subscription order	BS_CRM_COUNTRY_OF_ORGUNIT - Standard field <i>Country</i> from Org Unit OBJECT_TYPE - Business Trans. Cat. PROCESS_TYPE - Business Transaction Type	LAST_CHANGE_DATE - Last Changed On
CRM_PRVC	Subscription contract	BS_CRM_COUNTRY_OF_ORGUNIT - Standard field <i>Country</i> from Org Unit OBJECT_TYPE - Business Trans. Cat. PROCESS_TYPE - Business Transaction Type	LAST_CHANGE_DATE - Changed On
CRM_PRVMA	Master agreement	OBJECT_TYPE - Business Trans. Cat CRMT_SUBOBJECT_CATEGORY_DB PROCESS_TYPE - Business Transaction Type CRMT_PROCESS_TYPE_DB	CONTRACT_END_DATE - Contract End Date LAST_CHANGE_DATE - Last Changed On
SOM_SHGR	Sharing group ID	GROUP_ID – Sharing Group ID	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On
CRMS4_SOM_MR_DESTRUCTION	Mass Runs	SERVICEMASSRUNTYPE – Mass Run Type SERVICEMASSRUNSTATUS – Mass Run Status	CREATION_DATE - Created On LAST_CHANGE_DATE - Last Changed On

Prerequisites

To use the blocking and delete functions, you must:

- Set up SAP Information Lifecycle Management. For more information, see [SAP Information Lifecycle Management](#).
- Activate the business function ILM. For more information, see [Activating SAP ILM](#).

ILM Notifications for Data Protection

The archiving objects in Subscription Order Management support data protection notifications. The table below highlights the specific ILM objects and the relevant fields that are included in the notifications.

ILM Object	Source Fields
Subscription Order (CRM_PRV0)	OBJECT_ID OBJTYPE_H
Subscription Contract (CRM_PRVC)	OBJECT_ID OBJTYPE_H NUMBER_INT CI_CONTRACT_ID
Master Agreement (CRM_PRVMA)	OBJECT_ID OBJTYPE_H
Allowance Definition Group (SOM_ADG)	ALLOWANCEDEFINITIONGROUPID ALLOWANCEDEFINITIONGROUPUUID
Subscription Product-Specific Data (SOM_PROD)	PRODUCT SUBSCRPNPRODSPECUUID
Sharing Group (SOM_SHGR)	SUBSCRPNCONTRACTSHARINGGROUPID SUBSCRPNCONTRACTSHARINGGROUPUUID

Notification Modes

ILM notification logs the following processes categorized as notification modes:

Notification Mode	Process Step
01	Archiving Write
02	Archiving Delete
03	Data Destruction Run Using Archiving Write Report
04	Data Destruction of Archive Files
05	Destruction Run Using Data Destruction Object
*	All

Authorization Objects

The authorization object **S_ILM_NOTI** is created with the Display, Delete, and Distribute activities. This object is used for the following authorizations required for performing actions on the notification logs:

- **S_ARCHIVE** is checked for creation of log entries during archive process.
- **S_ARCD_RUN** is checked for ILM destruction mode during deletion of archive files.

- S_ARCD_OBJ is checked when the Create API (SO Create SOAP API?) Is called from the destruction object program.
- S_ILM_NOTI with activity = Distribute is required to extract the notification log data.

Veto Check for HR Master Data in Service

Perform a veto check for HR master data in Service.

Use

Service provides a veto check for the deletion of the HR master data. The veto check enables the application to prevent HR master data being deleted for as long as transactions requiring HR master data information exist.

The veto check is relevant to the deletion of personnel numbers. This reference indirectly extends the residence and retention policy of the HR master data. As long as such references exist, HR master data is not being deleted.

Application Name	Function Module Type	Function Module for Check
S4CRM	For Deleting Personnel Numbers	CRMS4_HR_PER_VETO_CHECK

You have to activate all applications that should participate in the veto check for blocking data destruction

If not already available, create an entry for S4CRM in Customizing under ► [Personnel Management](#) ► [Personnel Administration](#) ► [Tools](#) ► [Data Privacy](#) ► [Data Destruction](#) ► [Block Data Destruction \(Veto\)](#) ► [Activate Applications for Veto Check](#) ►. To activate the veto checks, set the entry with S4CRM to [Needs to be checked](#) ([Application Active](#) checkbox).

If necessary, enter the logical system for S4CRM as the destination.

Technical Details

ILM Objects

The veto check evaluates data for the following ILM objects:

- CRM_SERORD (service orders)
- CRM_SRVCN (service confirmations)
- CRM_SRCONT (service contracts)
- CRMS4_REPA (in-house services)

Veto Check Implementation

Service provides the following functionality for the veto check of the HR master data:

- The application searches for the following data in relation to the HR master data:
 - Service orders, service contracts, and in-house services that do not have the status [Completed](#).
 - Service confirmations that have the status [Completed](#) but have not been released for billing.
- In the case of deletion of the personnel numbers, the application returns the personnel number to be deleted.

Based on this information, the function modules can return information at personnel number level about whether the data in the applications is still required. The **Veto** checkbox is selected at personnel number level.

If the **Veto** checkbox is selected, the system prevents both the destruction of infotype and subtype data and the deletion of personnel numbers.

Analysis

In the **ILM Data Destruction** app, once the job that performs the veto check has finished, the results of the check are written to the application log.

If the veto check has failed, the log provides information about any service transactions that triggered the veto and are therefore preventing the deletion of HR master data.

The log also provides the personnel numbers for the affected employees.

Related Information

- [Destruction of Data in HR](#)
- [Destroying Personnel Numbers Using HRPD_PERNR](#)

Veto Check for HR Master Data in Subscription Order Management

Use

Subscription Order Management provides a veto check for the deletion of the HR Master Data. The veto check enables the application to prevent the deletion of the HR Master Data as long as transactions requiring HR Master Data information exist.

The veto check is relevant for the following situations:

- For deletion of employee data. A veto check is required to ensure that the employee data remains accessible for users (for audit and revision reasons, for example) with special authorization to the application and its processes.

These references indirectly extend the residence/retention policy of the HR Master Data. As long as such references exist, HR Master Data will not be deleted.

Application Name	Application Description	Function Module Type	Function Module for Check
Register the application and implementation class in view cluster VC_WFD_EOP_APP. Application Name: S4CRM Application Component: CRM-BF-AR	Subscription Order Management	Remote Enabled Function Module	CRMS4_HR_PER_VETO_CHECK (Veto check for HR Master - S4CRM)

You have to activate all applications that should participate in the veto check for blocking data destruction

For this, if not already available, create an entry for S4CRM in Customizing under **Personnel Management > Personnel Administration > Tools > Data Privacy > Data Destruction > Block Data Destruction (Veto) > Activate Applications for Veto Check**. To activate the veto checks, set the entry with S4CRM to **Needs to be checked** (**Application Active** checkbox).

If needed, enter the logical system for S4CRM as the destination.

Technical Details

ILM Objects

The veto check evaluates data for the following ILM objects:

- HRPAPERNR (Destruction Object for Personnel Number)

Veto Check Implementation

Subscription Order Management provides the following functionality for the veto check of the HR master data:

- The application searches for open transactions with relation to the personnel number. If open transactions exist, the application will not allow data related to the personnel number to be deleted.
- The end of business is reached when it reaches the end of the retention period.
- The application returns the following information, depending on the scenario:
 - For data destruction, the type of data (infotype/subtype) to be destroyed as well as a list of personnel numbers and the date on which the data for each personnel number is to be destroyed
 - For deletion of personnel numbers, the personnel number to be deleted.

Based on this information, the function modules can return information on personnel number level about whether the data in the applications is still required. If the veto checkbox is selected, the system prevents the destruction of infotype and subtype data as well as the deletion of personnel numbers.

Analysis

Analysis Question	Answer
What is the purpose of the HR Master Data?	The fields from CRM tables always refer to the Business Partner ID, which is associated with (replicated from) an employee. In Service, it is mandatory to have an associated business partner.
Would a deletion in HR Master Data impact the integrity of your records?	Yes
Describe the impact.	The veto check against the blocking of the HR Master Data person is needed in case a 1Order is not completed as the data is retrieved from HR Employee and not stored in parallel in the BP.

Data Destruction Objects in Subscription Order Management

You use data destruction objects to delete data from the database that is no longer relevant to your business processes, and that you aren't legally required to keep.

SAP has defined a minimum residence time for some data objects. You can also define your own residence time. If more than one residence time is defined, the longer residence time applies.

You destroy data by using [Schedule Data Destruction Run](#). Execute a data destruction run for each data destruction object.

Destroying Mass Runs Using CRMS4_SOM_MR_DESTRUCTION

You can use the data destruction object CRMS4_SOM_MR_DESTRUCTION for destroying Subscription Order Management mass runs.

For more information, see:

- [Processing Audit Areas](#)
- [Editing ILM Policies](#)
- [Editing Retention Rules](#)

ILM-Based Information for the Data Destruction Object

You create policies and rules for the related ILM object of this data destruction object.

The following fields for the ILM object are defined in the ILM policy and are visible when creating or processing the **ILM Policies**:

- Available **Time References**
 - CREATION_DATE - Creation date of the mass run
 - LAST_CHANGE_DATE - Last change date of the mass run
- Available **Condition Fields**
 - SERVICEMASSRUNSTATUS - Status of the mass run
 - SERVICEMASSRUNTYPE - Type of the mass run

Enablement for Information Retrieval

The ILM object CRMS4_SOM_MR_DESTRUCTION assigned to this data destruction object is enabled for use with the Information Retrieval Framework (IRF). The IRF allows you to search for and retrieve personal data of a specified data subject, based on a data model you define in your system. To help you get started with the data modeling process, SAP delivers an implementation (BADI_CRM_MRUN_ILM_OBJ_TABLES) of BAdI: Tables Used by an ILM Object (BADI_DTINF_ILM_OBJ_TABLES) for CRMS4_SOM_MR_DESTRUCTION. For more information, see SAP Note [2939146](#)  (S/4HANA).

Before you can start a search for personal data related to this ILM object, you must assign it to a purpose for which the data was collected and processed using customizing activity **Maintain Purposes** (transaction DTINF_MAINTAIN_PURP).

Once the profile has been created, it can be used for data collection using the transaction DTINF_START_COLL - Start Data Collection. You can view the collected data using the transaction DTINF_PROC_COLL - Process Data Collection Results.

For more information, see [Information Retrieval Framework \(IRF\)](#).

Compatibility Scope

With this compatibility pack, SAP provides a limited use right to SAP S/4HANA customers to run certain classic SAP ERP solutions on their SAP S/4HANA installation. Use the compatibility scope to understand the compatibility packages offered.

For more information about the compatibility scope and the compatibility packages offered for SAP S/4HANA, see SAP Help Portal at <http://help.sap.com/s4hana>  **<Choose an on-premise release>** **Discover** **Compatibility Packs** .

i Note

- The features described in this entire compatibility scope are only available in SAP S/4HANA.
- The system behavior described in this entire compatibility scope is only valid in SAP S/4HANA.

Related Information

[SAP Note 2269324](#) 